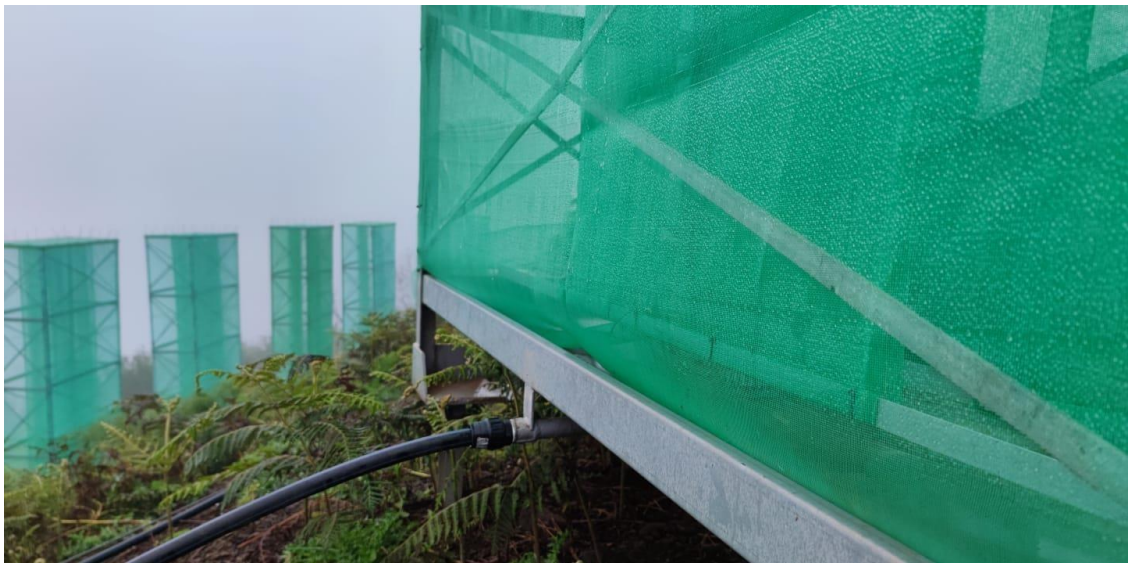
Action: D3. Monitoring and evaluation of the reforestation execution and results

Reforestation execution and results monitoring report from the beginning of the project until 9/2024

Compiled by: CREAF, UAB, GESPLAN, ITC, CIM VDL



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1. Introduction

This report presents the monitoring results of the plantations carried out in El Pinillo (Gran Canaria, Spain), and Carregal do Sal and Vouzela (Vizeu, Portugal) in the context of C2 actions (see Execution status report 2020-2024). Restoration monitoring began according to the protocol established in Action A2 (see Environmental, economic and social indicators monitoring protocol). According to this protocol, prior to planting, a detailed soil physicochemical characterization was conducted for the main units of the areas where the planting systems were placed. From each unit, a composite sample of topsoil (first 20 cm) was obtained by mixing sub-samples homogeneously distributed along the longest linear transect in each unit. Additionally, soil profiles for the most representative soil types in each area were described. Soil sampling was repeated in 2024 to evaluate the evolution of some soil properties 2-3 years after planting.

Important efforts for seedling monitoring have been done in El Pinillo plantations, with more than 5000 seedlings labelled and monitored during the 2022-24 period. Despite the heterogeneity in the soils and the variability in the meteorological conditions between the planting years and areas, the high number of seedlings allows us to take robust conclusions about the effectiveness of the treatments. However, in order to provide even more robust conclusions, an experimental area was established to reduce this heterogeneity and improve the comparison between treatments.

The monitoring campaign in the Carregal do Sal plantations started in September 2022, after the summer drought. More than 700 seedlings were labelled and monitored in Carregal up to September 2024, when unfortunately, a forest fire affected the area destroying all the plantations. Regarding the Vouzela reforestation, the initial plantation took place from February to March 2023, and more than 400 seedlings were labelled and monitored in 2023 and 2024 field campaigns.

In addition of providing results about the growth and vigour of the plantations, this report presents an assessment of the evolution of vegetation based remote sensing data, alongside with meteorological data collected from the stations installed by the project (El Pinillo) or from public institutions in the area (Vouzela and Carregal do Sal).

The results in this deliverable are of relevance, since they provide robust evidence of the effectiveness of the different planting systems and restoration strategies, even detected in the first years after planting, which are considered the most critical for the establishment of the

plantations. However, since the evolution of plantations must be evaluated in the midterm to take more robust conclusions, a monitoring scheme has been planned in the context of Post-LIFE plan (see deliverable Post-LIFE Conservation and Communication Plan).

2. Reforestation methods description

In LIFE Nieblas project, four primary reforestations methods have been employed: (i) the traditional (TR) method utilizing hose irrigation; (ii) the Cocoon (CCN) system; (iii) the Individual Fog Water Collector (IFWC); and (iv) the Autonomous Fluid Discharge System (AFDS), with drip irrigation. These methods were implemented in the reforestations carried out at El Pinillo (Gran Canaria, Spain), Carregal do Sal (Portugal), and Vouzela (Portugal).

2.1 Traditional method (TR)

TR involves planting trees in a conventional manner by digging a hole and placing the seedling. This method serves as a control group for comparison against other methodologies. In addition, in the restoration projects carried out in Gran Canaria over the last decade, there has been a recognized need to provide water to newly planted seedlings, particularly during their first drought season, to prevent high mortality rates (Carabassa et al., 2021)[1]. This practice aligns with the traditional method, involving an initial settlement irrigation and a maintenance irrigation during the next dry season. Exceptionally an emergency irrigation in case that severe drought affects the plantations could be done.

2.2 Cocoon system (CCN)

The Cocoon system is a self-contained and cellulose-based biodegradable nurturing system individually installed and buried beneath each tree. This system not only creates a humid and cool environment but also offers protection against weather and herbivore protection (Figure 1-Figure 2). The Cocoon's deposits can hold up to 30L of water, slowly infiltrating into the soil with minimum evaporation, thanks to their insulated underground. This irrigation and nursery method was developed in a previous Life project, LIFE The Life Green Link, that has undergone successful testing in high drought-risk areas in the southern part of Gran Canaria[1].

[1] Carabassa, V., Alba-Patiño, D., García, S., Campo, J., Lovenstein, H., Van Leijen, G., Castro, A., González,

One of the most significant advantages of this system is that it does not demand constant monitoring or regular irrigation, thereby reducing the need for recurrent human intervention and minimizing economic investment (lowers water consumption, absence of irrigation systems, etc.). Additionally, the biodegradable materials used in these systems are more cost-effective compared to synthetic alternatives used previously in similar systems.

In terms of the irrigation process, seedlings initially receive a settlement irrigation, with 20-25l inside the Cocoon and some liters more in the central hole, in order also to stabilize the soil.

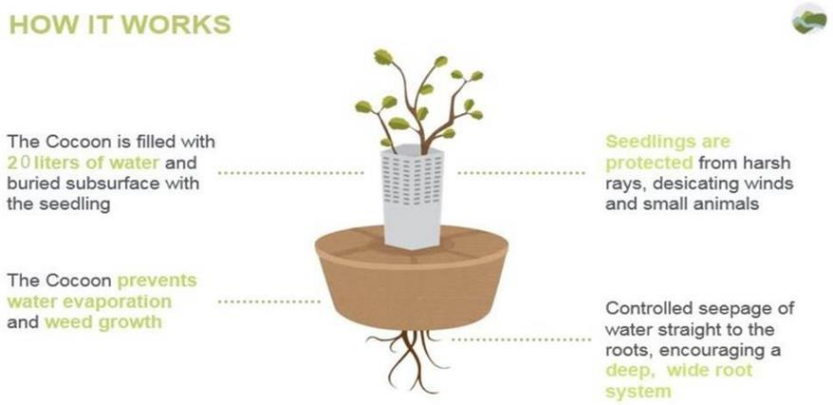


Figure 1 Diagram of a Cocoon system and its advantages. Source: Land Life Company.

F., Viera, G., Kourkoumpas, D.S., Zioga, A.A., Papadelis, C.E., Andreu, A., Gimeno-García, E., Kallen, S., Alcañiz, J.M. (2021) Water-saving techniques for restoring desertified lands: some lessons from the field. Land Degradation and Development 33, 133-144. https://doi.org/10.1002/ldr.4134

Figure 2 Diagram Two years old faya (Mirica faya) seedling planted using the Cocoon technology.

2.3 Individual Fog Water Collectors (IFWC)

Based on the insights from a prior study, the Life-GuGuy project (https://www.lifeguguy.com/), scientific observations indicated that individual protection metal mesh can reach the dew condensation temperature under suitable conditions. Consequently, when this phenomenon occurs, condensed dew directly drips onto the root area of seedlings by gravity, establishing a direct, natural, and regular irrigation system. Thus, Individual Fog Water Collectors (IFWC) use a metal mesh to capture fog and dew condensation. Moreover, an innovative enhancement has been introduced to improve water collection capacity, involving the installation of a plastic net around the metallic shelter (Figure 3) to maximise water collection. The IFWC collectors not only protect against herbivores but also provide shade and shield against wind, resulting in healthier plant development during the early stages.

Figure 3 IFWC with plastic net installed in the experimental plot of LIFE Nieblas, in El Pinillo (Gran Canaria).

2.4 Autonomous Fluid Discharge System (AFDS)

The Autonomous Fluid Discharge System (AFDS, Figure 4) is an automatic device developed by the Instituto Tecnológico de Canarias (ITC) engineers, designed to irrigate an entire reforestation area without relying on a power supply or human action. The system operates on Archimedes' principle and the Gaussian curve, facilitating to the automatic, autonomous, and controlled discharge of fluids into a deposit without requiring any energy supply or electronic control device. It combines efficiency, autonomy, and regularity characteristics without electrical consumption or CO_2 emissions. Specifically designed for connection to a drip irrigation network, the AFDS should be installed only when this type of irrigation method is available or necessary.

Figure 4 AFDS connected to a water deposit, in El Pinillo planting area (Gran Canaria).

3. Reforestations execution

3.1 Portugal plantations

Plantations in Portugal were carried out in the protected areas of Vouzela and Carregal do Sal (see Product A1. *Analysis of the status of the area of operation*). The main objective in Carregal do Sal is to restore habitat 9230, "Galician-Portuguese oak woods with *Quercus robur* and *Quercus pyrenaica*". To achieve this goal, these species were planted in the area. The Vouzela plot is located in the Vouga-Caramulo Local Natural Park (Figure 5 and Figure 6), whose conservation strategy includes the use of species of native flora. In addition to *Q. robur* and *Q. pyrenaica*, *Quercus suber* was also introduced for its even higher resistance to fire, making it especially suitable for reforestation in these fire-prone areas.

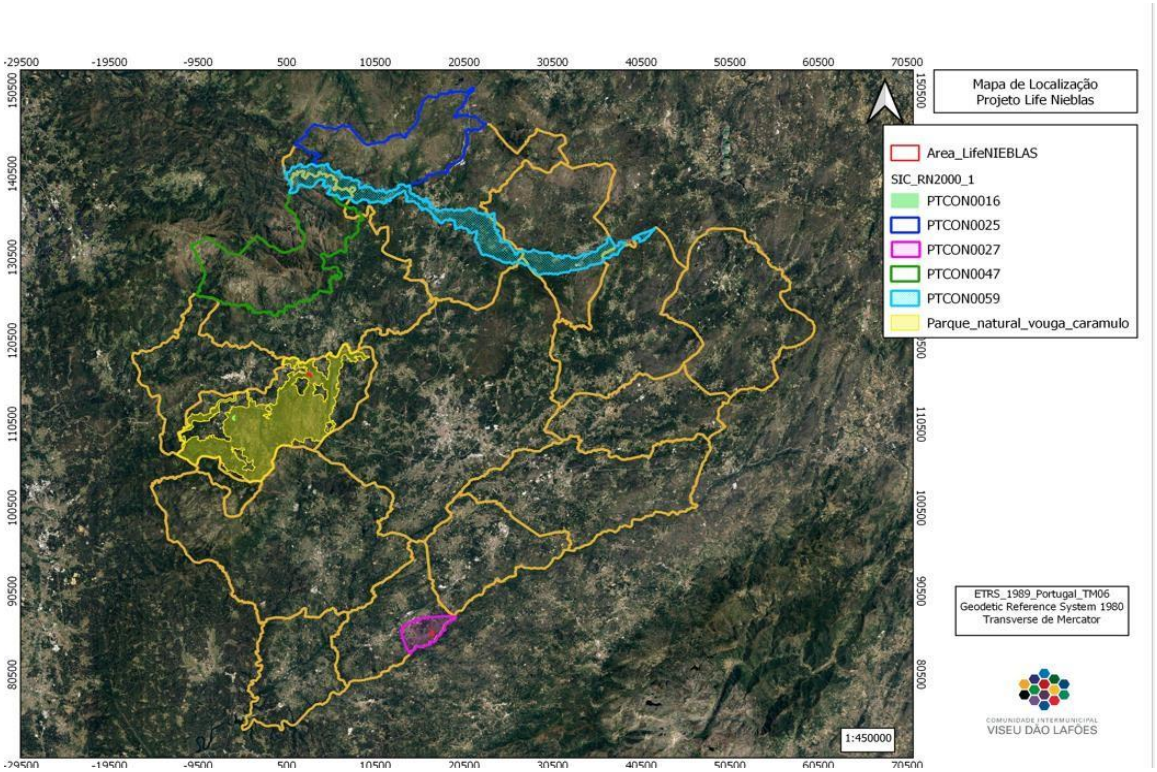


Figure 5 Vouzela (higher red polygon) and Carregal do Sal (lower red polygon) planting areas in Viseu dao Lafoes region.

3.1.1 Carregal do Sal plantations

In Carregal do Sal, ground conditioning took place in February and March 2022 before the plantations were made. Ground preparation works were carried out by the Brigades of Forest Sappers from the Viseu Dão Lafões Intermunicipal Community. The plot was cleared in strips along the terrain's contour lines. Given the slope of the land, some vegetation was left intact to help to retain soil moisture, to protect plants during the hottest periods, and prevent soil from erosion. Reforestation activities were conducted between February and March 2022. 1658 seedlings equally distributed among the selected species were planted (see Table 1). In addition, to improve the comparison among treatments, in March 2023, 50 seedlings per treatment were planted in the upper part, in a flat area, with the same soil and shadowing conditions. This was called the experimental area.

Table 1 Number of seedlings planted in Carregal do Sal planting area in 2022-24 period, per planting system and species and water used for irrigation.

Planting system	Species – Number of plants (March 2022)	Species – Number of plants (March 2023)	Number of seedlings per planting system	Watering 2022 (l)	Watering 2023 (l)	Watering 2024 (l)
Traditional (TR)	(2.3ha) *Quercus pyrenaica* – 430 *Quercus robur* – 430	(0.45ha) *Quercus robur* – 172	1032			
Cocoons (CCN)	(1.25ha) *Quercus pyrenaica* – 100 *Quercus robur* – 98	(0.1ha) *Quercus robur* – 52	250			
Individual Fog Water Collectors (IFWC)	(2.4ha) *Quercus pyrenaica* – 300 (250 plants with hexagonal net; 50 with hexagonal net+ mosquito net) *Quercus robur* – 300 (250 plants with hexagonal net; 50 with hexagonal net+ mosquito net)	(0.1ha) *Quercus robur* (with hexagonal net+ mosquito net) – 37 *Quercus robur* (with hexagonal net) – 13	650	4000	6740	4700
Total	1658	274	1932			

In terms of water use for settlement irrigations, 250 Cocoons were filled with a total of 4000l of water, equivalent to 16l per Cocoon. Following reforestation using the Individual Water Fog Collector (IFWC) and traditional (TR) methods, technicians considered the subsequent rainfall sufficient for the tree's settlement, saving water in the process. Additionally, during Spring 2023 and 2024, Cocoons (CCN) with live trees have been refilled. In 2023 and 2024, the plants in the TR reforestation method and IFWC sectors were also watered during summer with 5l per plant.

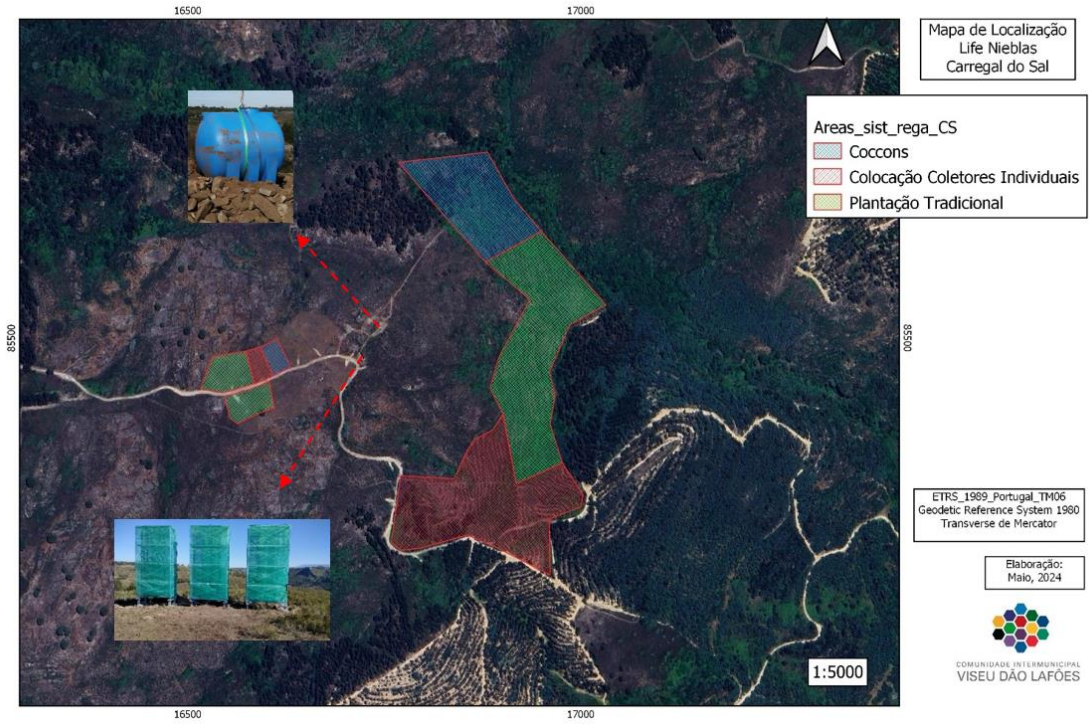


Figure 6 Location of plots and equipment in Carregal do Sal.

As for the water collected by the collectors, it was only possible to install the water meters at the entrance to the tanks in September 2023, so data are only available from that date onwards. Between November 2023 and June 2024, the collectors collected 5087 litres of water, which amounts to around 212 litres per collector per month. The following Figure illustrates the volume of water entering the tank (Figure 7).

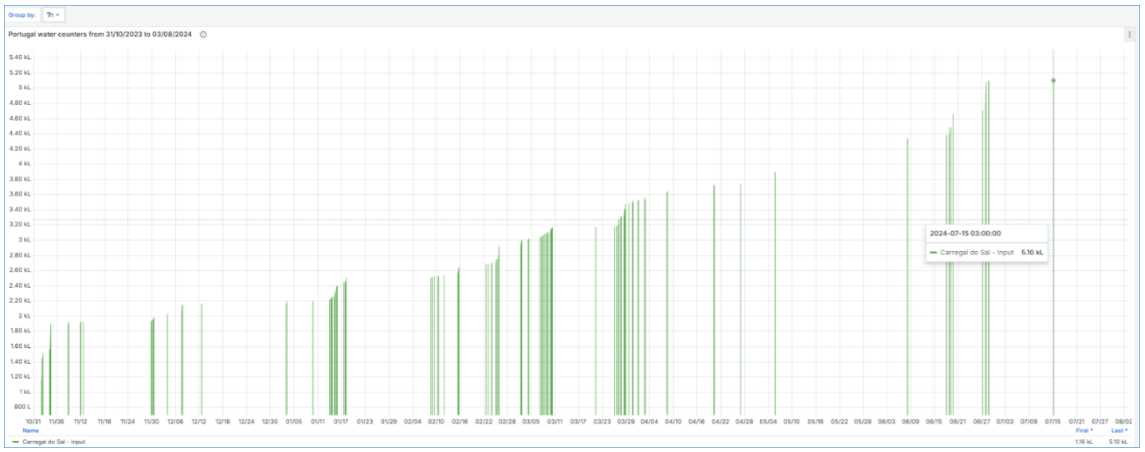


Figure 7 Amount of water that entered the tank between 10/30/2023 and 07/25/2024

3.1.2 Vouzela plantations

Cleaning and conditioning works were carried out in Vouzela during the spring of 2022, followed by planting in February and March 2023. A total of 1143 seedlings were planted over an area of around 3.2ha (Figure 8), with equal distribution by species and planting systems (see Table 2).

Figure 8 Location of planting areas and equipment in Vouzela planting area.

After planting, the Cocoons received around 20l of water. TR and IFWC did not receive a settlement watering due to the wet conditions of the terrain after rains in winter 2023. In 2024, in addition to the Cocoons, plants in the traditional sectors and Individual Water Fog Collectors received around 3l of water each.

Table 2 Number of seedlings planted in Vouzela planting area in 2023, planting system and species.

Planting system	Species – Number of plants	Number of seedlings per planting system
Traditional (TR)	(1.6ha) *Quercus pyrenaica* – 80 *Quercus robur* – 110 *Quercus suber* – 120	310
Cocoons (CCN)	(0.6ha) *Quercus pyrenaica* – 80 *Quercus robur* –75 *Quercus suber* – 78	233
Individual Fog Water Collectors (IFWC)	(1ha) *Quercus pyrenaica* – 200 *Quercus robur* – 208 *Quercus suber* – 192	600
Total		1143

As for the water collected by the collectors, the water meters could only be installed at the deposits installed in September 2023, so data are available only from that date onwards. Between October 2023 and October 2024, the collectors registered 4128 litres of water, amounting to approximately 172 litres per collector per month.

3.2 El Pinillo plantations

3.2.1 Reforestation actions 2021-2023

The habitats targeted in Gran Canaria align with Annex I of the EU Habitats Directive with respect to Natura 2000 Habitat restoration species. Reforestation actions are planned in Gran Canaria within the priority habitats 4050, "Macaronesian endemic heaths." The primary objective of these restoration efforts is to establish, protect, and expand these protected habitats by introducing xeric Laurisilva species. Additionally, the area serves as a case study for implementing four distinct reforestation methods, enabling a thorough assessment of each methodology's practicality and benefits within the study area.

3.2.2 Seeds selection

To achieve these reforestation objectives, thirteen different species have been selected (Figure 9), each contributing to the ecosystem's diversity and resilience. Among these species are: brezo (*Erica arborea*), laurel *(Laurus novocanariensis*), acebiño (*Ilex canariensis*), faya (*Morella faya*), palo blanco (*Picconia excelsa*), barbusano (*Apollonias barbujana*), mocán (*Visnea mocanera*), madroño canario (*Arbutus canari*ensis), bencomia (*Bencomia caudata*), follao (*Viburnum rigidum*), peralillo (*Maytenus canariensis*), tajinaste azul (E*chium callithyrsum*), cedro (C*edrus sp*.), and viñatigo (*Persea indica*). To ensure a continuous supply of plants, seeds and seedlings are collected for local nurseries responsible for their propagation. The seedlings then undergo an acclimatisation phase in a dedicated nursery area, allowing them to adapt before being transplanted to reforestation sites.

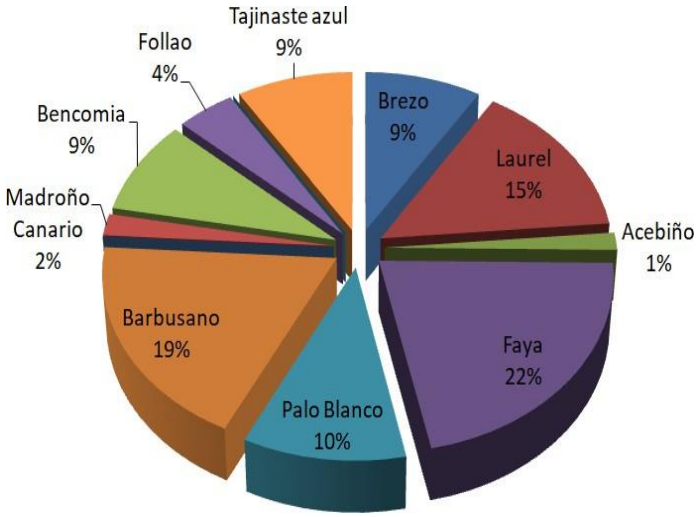


Figure 9 Proportion of each plant species planted in El Pinillo.

3.2.3 Ground conditioning

Before undertaking reforestation works, in addition to the installation of safety measures and the set up the water storage system, land was cleared of vegetation. The vegetation clearing was thoughtfully planned by the technical staff, considering the terrain characteristics, erosion processes, (such as runoff), weather conditions (heavy rainfall, drought, and heat waves), and the orographic profile (e.g., hilly areas).

To mitigate erosion and preserve natural vegetation, the clearing of the vegetation was done in strips of 0.5 to 1 m width, perpendicular to the slope, and 0.5 to 1 m strips of intact vegetation were left in between the cleared strips (Figure 10). This design allows to the remaining vegetation to anchor the soil and prevent soil erosion. Additionally, the already established vegetation provides shelter and shade for young plants, protecting them from excessive sunlight and prolonged exposure.

Figure 10 Vegetation clearing in strips previously to the plantation.

3.2.4 Reforestation execution

The reforestation goal at El Pinillo was to plant 20000 trees over three seasons (2021-2022, 2022-2023, and 2023-2024). In pursuit of this goal, Life Nieblas has implemented four different planting systems, including Autonomous Fluid Discharge System (AFDS), CCN, TR, and IFWC across separate plots of land. More information about the choice of systems and their functioning can be found in report 'A3: Equipment evaluation and selection - Monitoring and evaluation protocol'. All the equipment was tested during the action 'C2: Execution of traditional and innovative reforestations', where a detailed description of the methods used can be found.

Reforestation activities began in October 2021 and plantations were executed along the 2021-22, 2022-23 and 2023-24 campaigns, planting one plot for each planting system in each campaign (Figure 11). In this regard, a total of 18.3 hectares have been planted in El Pinillo by December 2024. So far, the project has successfully planted 15000 trees, which represents 74% of the target (20000 trees). The difference between the planned and the executed reforested surface and number of seedlings is due to the fact that some areas remain inaccessible, while others are made up of existing roads or forests.

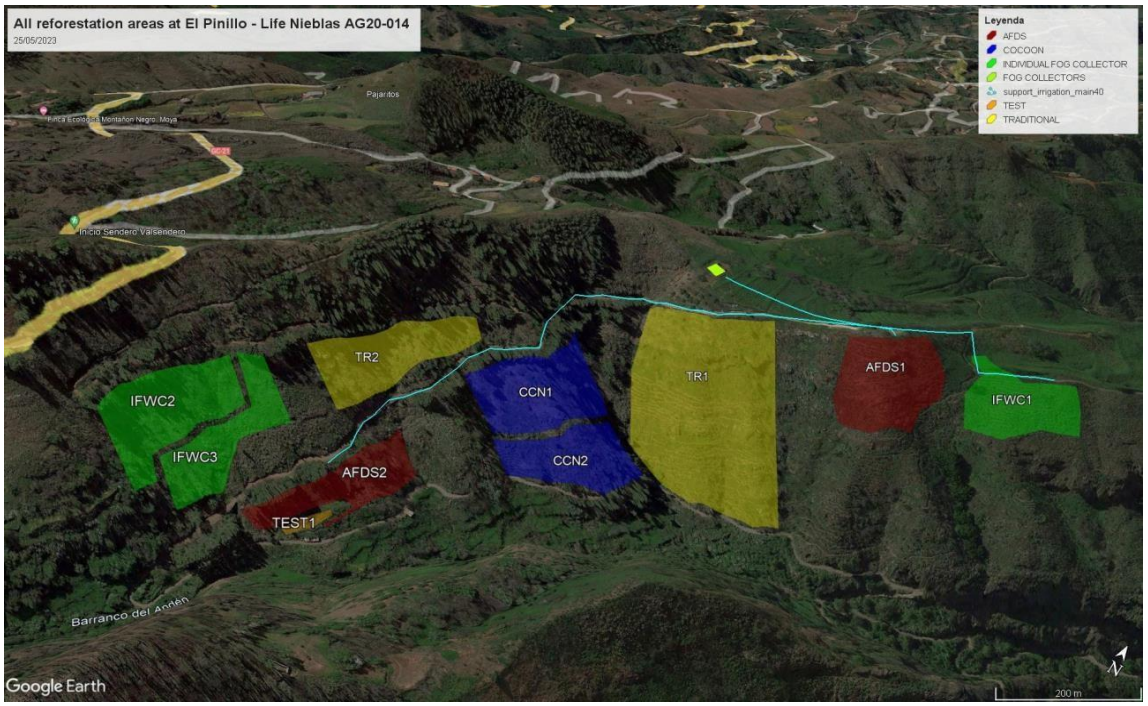


Figure 11 Distribution map of plantations according to the reforestation method applied. Note - ADFS1 area contains three tanks (AFDS1, AFDS2 & AFDS3) and AFDS2 area contains (AFDS4, AFDS5 & AFDS6).

Traditional method (TR)

In the TR method each tree was planted with the addition of 1l compost and receives 10l of water during settlement. In emergency situations, extra 10l irrigation was provided, followed by one instant 10l irrigation per month during the summer for the first few years. From 2021, a total of 4575 plants were planted using the traditional method. These plantations were carried out over an area of 5.5ha.

Autonomous Fluid Discharge System (AFDS)

In the ADFDS system, trees were planted with the addition of 1l of compost and receive 10l of water during settlement, and afterwards, the system discharged water regularly to the plants. In particular, a part of the seedlings under the AFDS received controlled discharges from May to September (Strategy 3), while others were irrigated from June to September (Strategy 2). The remaining AFDS seedlings were connected to the fog water collectors and only received the initial settlement irrigation (10l), and two emergency irrigations -one in July and one in August (10l per plant, Strategy 1). Since 2021, a total of 2082 plants have been planted using the AFDS method, covering an area of 2 ha.

Cocoon (CCN)

The trees in the CCN system were planted with the addition of 1l compost, and the containers were filled with 25l of water, followed by 10l irrigation per month during the summers of the first years. Under extreme dry conditions, additional irrigation can be provided. Particularly in El Pinillo, some Cocoons were refilled once (25l) during the first summer (Strategies 2 and 3), while some seedlings were not watered during summer (Strategy 1) to test the device's performance without additional watering. Since 2021, a total of 2165 plants have been planted using the CCN method, covering an area of 4.5 hectares.

Individual Fog Water Collectors (IFWC)

The trees in IFWC were planted with the addition of 1l compost and received 10 litres of water during settlement. Under drought emergencies, additional 10-litre irrigation was provided, followed by one event of 10l irrigation per month during the summer for the first few years. However, some of the seedlings were not watered during summer (Strategy 1) to test the functioning of the device without additional watering. Since 2021, a total of 5916 plants have been planted using the cocoon method, covering an area of 6 ha.

Further information about the seedlings distribution per planting system and tree species can be found in Table 3 and Figure 12.

Table 3 Description of the reforestation works carried out at El Pinillo, described by the area covered, the number of trees planted, the number of devices installed, and the implementation days (time required for reforestation) per planting system.

Area per method	Area (ha)	Trees planted	Systems installed (devices)	Irrigation network (m)	Implementation time (days)
IFWC	6	5916	5916	2,000	142
AFDS	2	2082	6	6700	48
TR	5.5	4575	N/A	750	62
CCN	4.5	2165	2165	380	84
TOTALS	18	14738	8087	7830	336

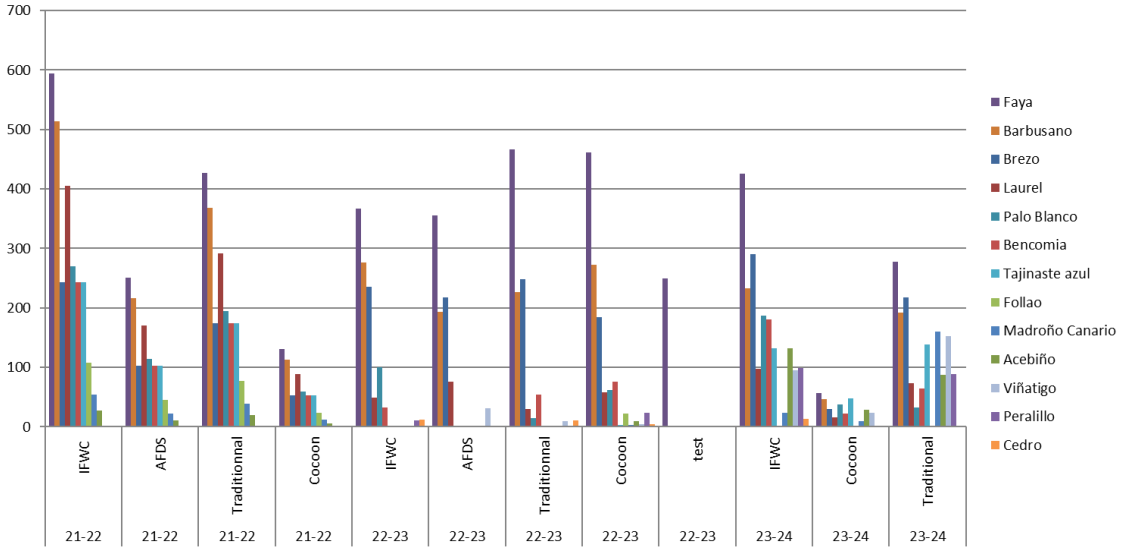


Figure 12 Number of seedlings planted per species, planting system and season in El Pinillo, between 2021 and 2024.

3.3 Experimental plot

An experimental plot was established on a terrace near *Barranco del Andén* (Figure 13) over skeletal soils Figure 14 with high stone content (Figure 14) this soil is similar to that found in the surrounding area (see section 4.3.1.1 Skeletal soils (profile ZE)). The plot was designed to better compare the effectiveness of the different treatments and to incorporate new techniques, such as the DeepDrop system and the addition of biochar as organic amendment in two different doses (1 and 2 kg per tree). It is important to highlight that this experimental plot was an addition to the project, which was not initially in the proposal. However, this extra effort done by the project partners, both in terms of installing the equipment and carrying out the reforestation,

responds to the need to reduce, as much as possible the uncertainties of the reforestation area. The idea was to enable the comparison of traditional and innovative reforestation systems under the most similar soil and shading conditions possible, in terms of both soil moisture retention and survival rates.

In this context, the experimental design consisted of parallel rows for each irrigation system. A random distribution of treatments was deemed unnecessary, as the plot maintained uniform soil type and consistent shading conditions throughout the area. Additionally, organizing the treatments in rows ensured proper management in each case, as for instance water application.

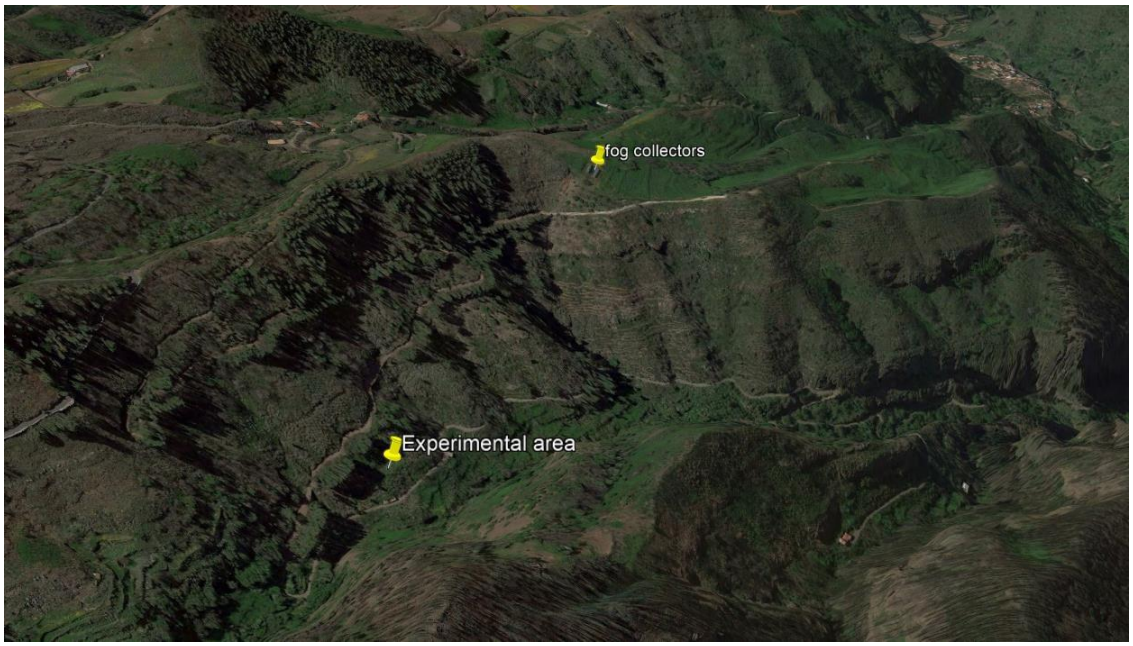


Figure 13 Localization of the experimental area in El Andén ravine, at the lower part of the southern slope.

Figure 14 Soil profile of the experimental area, which might be considered very similar to the ZE profile.

Planting systems in the experimental plot included:

- Cocoon (CCN).
- An improved version of the innovative water fog collectors (IFWC): The improved version of the IFWC consists in adding a 1 m high polyethylene mesh around the metal mesh to enhance fog water collection and capitalize on environmental humidity condensation (Figure 15).
- DeepDrop (DD): DeepDrop (Figure 16) is a drip irrigation system that optimises water use by reducing evaporation losses since it delivers water directly to the root system (https://deepdrop.es/). This system It has been tested successfully in agricultural plantations across the Iberian Peninsula and the Canary Islands. However, the deep Drop system has not been tested in reforestation projects yet.
- Traditional+biochar: Biochar was produced by another European project, VERCOCHAR (https://www.vercochar.com/) that uses *Pinus radiata* charcoal. Biochar was obtained

from the pyrolysis of pine residues (obtained through pruning of *Pinus radiata* trees by Cabildo de Gran Canaria after wildfires in 2019) by the pot method (Figure 17). Biochar is an organic amendment known for its high stabilised organic carbon and high water and nutrient retention capacity, that could help establish seedlings in drought environments. The treatments included the the addition of 1 and 2 kg respectively (BC_1, BC_2):

- Traditional (TR).

Figure 15 IFWC and Cocoon lines in the experimental area plot

Figure 16 Detail of a DeepDrop dispositive before (right) and after (left) being installed at field.

Figure 17 Biochar pot used for the pyrolization of pine remains.

Fifty seedlings were planted per treatment, resulting in a total of 250 seedlings (Figure 18). Each seedling was labelled, and basal diameter and height were measured. To avoid including additional variables while ensuring enough replicates, only *Morella faya* - the most representative species of the target habitat (fayal-brezal)- was planted. Comparison among species could be done in the general plantations. All planting systems (except biochar) were fertilized with compost at a rate of 1l per tree, mixed directly with the soil. In the Traditional+biochar treatment the two doses tested (1 kg and 2 kg of biochar per planting hole) were applied on 25 individuals each. The biochar was also mixed with the soil (Figure 19).

Figure 18 General view of the experimental area (left), and Morella faya seedling planted amended with compost.

Figure 19 Biochar incorporation at the planting hole.

The compost used (Figure 20) was obtained from the organic fraction of municipal solid waste, separated at source composting carried out in Ecoparque Gran Canaria Sur (San Bartolomé de Tirajana). According to the compost producer, the compost composition was mainly plant remains (70-80%), expired food (5-10%), and digestate from household organic waste (15-20%) (see Figure S1). Also, according to the producer the heavy metals content was relatively low (class B according to the RD 506/2013 of fertiliser products), and its stability is also low (as indicated by an acute ammonia smell).

However, the analysis of the soils after the compost application revealed important increase in heavy metals content, especially Cr (100 mg/kg), which suggests that the compost may have a higher heavy metal concentration than reported by the supplier. Nonetheless, heavy metal concentrations in all treatments were below the generic reference level for ecosystems protection in Canary soils, as recently established by the Cabildo de Gran Canaria (*ORDEN de 17 de febrero de 2022*). Soils amended with compost also presented higher carbon content and a slight increase in electrical conductivity, mainly due to the salinity present in the compost (Table 4).

Figure 20 Compost pile located close to the main road, in the workers warehouse (left), and compost detail where plant remains could be observed, jointly with impurities (plastics).

Table 4. Chemical characterization of the control soils and soils amended with compost in the experimental area.

	Control soil	Soil + compost
pH	6.9	7
Elect. Cond. (ext. 1:5; dS/m)	0.085	0.14
Carbonates (%)	Detec. (<3)	Detec. (<3)
Kjeldahl N (%)	0.26	0.26
P Olsen (mg/kg)	44.9	43.3
Organic C (%)	2.24	3.06
K (mg/kg)	1241	1007
Ca (mg/kg)	3518	3027
Mg (mg/kg)	1071	905
Na (mg/kg)	85	98
Cd (mg/Kg)	<0.5	<0.5
Cu (mg/Kg)	81	55
Cr (mg/Kg)	21	100
Hg (mg/K)	<0.4	<0.4
Ni (mg/Kg)	36	150
Pb (mg/kg)	<5	Detec. (<5)
Zn (mg/kg)	77	120

Regarding the irrigation regime, all plants received a settlement watering of 10l per plant at the time of planting. Moreover, all plants received 10l watering per plant twice a month from July to September both inclusive (60l per plant), except the Cocoon system which will requires two tank refills during summer (50l per plant). Additionally, emergency watering will be done if necessary, according to the drought regime to guarantee the plantation survival.

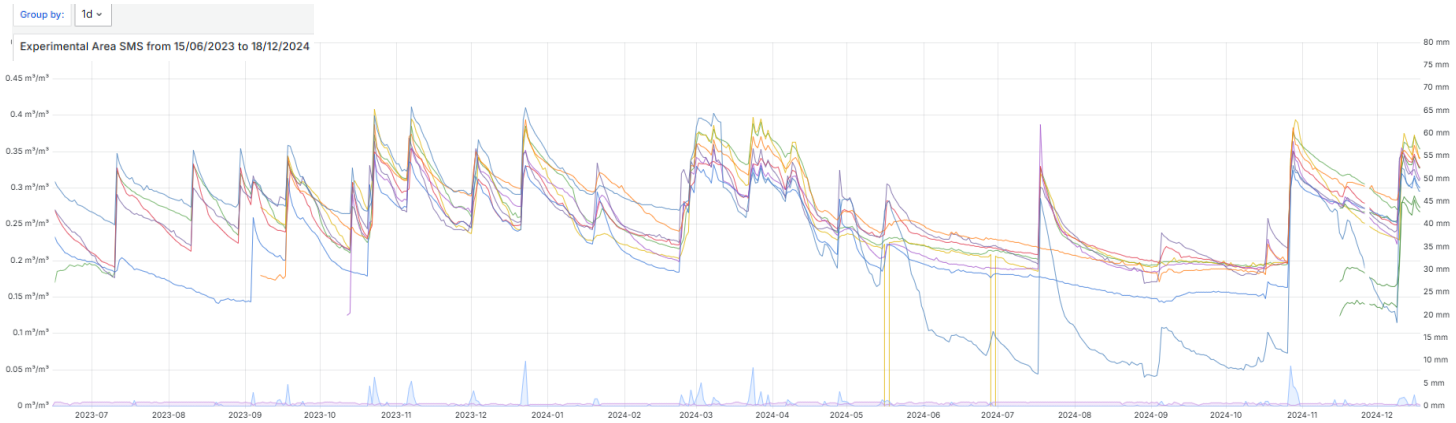
Starting on June 13, 2023, soil moisture content sensors (SMS) were installed in the experimental plot. Specifically, moisture sensors associated with Cocoon, traditional reforestation, individual collector, traditional reforestation with Biochar (first dose), and Deep Drop were installed. On September 6th, new sensors were installed: one as a reference for the non-intervention area (No Plant) and another at a traditional reforestation site with Biochar (second dose). A second measurement point for Deep Drop was installed on October 11th, 2023. During the period from May 10th, 2024, to July 3rd, 2024, data from the Biochar 2 sensor was lost due to misalignment with the sun of the pulse emitter and battery depletion.

Figure 21 Experimental plot during the installation of soil moisture sensors (June 13th, 2023).

The graph below shows the soil water content measurements of the full period up to September 13, 2024. A more detailed analysis is performed in Section 4.3.3.

Name	Max	Min	Mean
Experimental Area Biochar	0.390 m³/m³	0.169 m³/m³	0.266 m³/m³
Experimental Area Biochar 2	0.408 m³/m³	-889 m³/m³	-3.53 m³/m³
Experimental Area Biochar 2 Bis	0.344 m³/m³	0.254 m³/m³	0.284 m³/m³
Experimental Area Biochar Bis	0.284 m³/m³	0.124 m³/m³	0.177 m³/m³
Experimental Area Cocoon	0.354 m³/m³	0.187 m³/m³	0.258 m³/m³
Experimental Area Deep Drop	0.347 m³/m³	0.141 m³/m³	0.222 m³/m³
Experimental Area Deep Drop 2	0.387 m³/m³	0.124 m³/m³	0.252 m³/m³
Experimental Area Individual	0.368 m³/m³	0.169 m³/m³	0.259 m³/m³
Experimental Area Individual Bis	0.289 m³/m³	0.164 m³/m³	0.205 m³/m³
Experimental Area No Plant	0.393 m³/m³	0.171 m³/m³	0.270 m³/m³
Experimental Area Traditional	0.411 m³/m³	0.0396 m³/m³	0.234 m³/m³
Accumulated Rain (right y-axis)	9.80 mm	0 mm	323 µm

Figure 22 Experimental plot soil moisture data.



4. Reforestations monitoring

4.1 Carregal do Sal

4.1.1 Soil characterization

The soil in the Carregal do Sal (Figure 23) planting area is characterised by a shallow depth (<40 cm) and high stoniness (>25% in volume in superficial horizons and >50% in volume in subsuperficial horizons), which substantially reduces the water retention. Additionally, its sandy texture further decreases the water-holding capacity, limiting the water available to plants. Moreover, despite having a high organic matter content (>9% in superficial horizons and >5% in subsuperficial horizons), the acidic pH, high infiltration capacity, and relatively high to medium precipitation make this soil prone to nutrients leaching (Table 5).

Figure 23 Profile CS-IND, characteristic of the planting area in Carregal do Sal.

Macromorphological description of horizons

A1 (0 to 10 cm)**. Moisture state**: dry. **Redox state**: oxidied. **Staining**: none observed. **Coarse fragments content**: frequent (39% in weight), granite, 100 mm diameter, rounded, medium spherical, irregular distribution. **Texture**: sandy loam. **Structure**: weak, granular, 1-2 mm diameter. **Organic matter**: very high (9.62 %), decomposed plant debris and humus. **Fauna activity**: not observed. **Roots**: abundant, 1 mm diameter, living. **Field tests**: null reaction to HCl. **Accumulations**: not observed. **Cementations**: not observed. **Coatings**: not observed. **Horizon lower limit, transition**: no soil discontinuities, wavy, gradual. **Additional observations:** charcoal fragments in the soil matrix.

A2 (5 to 25 cm)**. Moisture state**: dry. **Redox state**: oxidised. **Staining**: **Horizon lower limit** observed. **Coarse fragments content**: frequent (42% in weight), granite, 200 mm diameter, rounded, plains, irregular distribution. **Texture**: sandy loam. **Structure**: weak, granular, 1-2 mm diameter. **Organic matter**: high (5.56 %), decomposed plant debris and humus. **Fauna activity**: not observed. **Roots**: abundant, 3 mm diameter, living. **Field tests**: null reaction to HCl. **Accumulations**: not observed. **Cementations**: not observed. **Coatings**: not observed. **Horizon lower limit, transition**: no soil discontinuities, plain, abrupt. **Additional observations:** none.

Table 5. Physicochemical characterization of the horizons of CS-IND profile.

	A1 (0-10 cm)	A2 (10-40 cm)
>2 mm (%)	38.9	42.1
Sand (%)	77.3	75.3
Silt (%)	15.5	19.5
Clay (%)	7.2	5.1
Textural class	Sandy loam	Sandy loam
pH H2O	4.8	5
Elect. Cond. (ext. 1:5; dS/m)	<0.05	<0.05
Carbonates (%)	Undetec. (<3)	Undetec. (<3)
N Kjeldahl (%)	0.4	0.29
P Olsen (mg/kg)	13.2	6.5
Organic C	5.58	3.22
K (mg/kg)	66	27
Ca (mg/kg)	131	45
Mg (mg/kg)	20	8
Na (mg/kg)	29	19

4.1.2 Soil monitoring

4.1.2.1 Physicochemical parameters

In the Carregal do Sal restoration areas, three distinct soil units were identified, each corresponding to a different planting system applied (Figure 24). All soil units exhibited a high organic matter content (7.6-11.1%) and low conductivity (Table 6). However, the acidic pH (4.6-5.2) combined with the sandy texture and very low clay content (<5%) could promote nutrient leaching. Since no severe degradation processes have been detected, we considered that the use of organic and/or calcic amendments unnecessary, despite the recommendations of analytical bulletins, with benchmark values referred to agricultural soils. In any case, this decision will be re-evaluated considering ongoing monitoring of passive restoration parameters, such as vegetation structure measures.

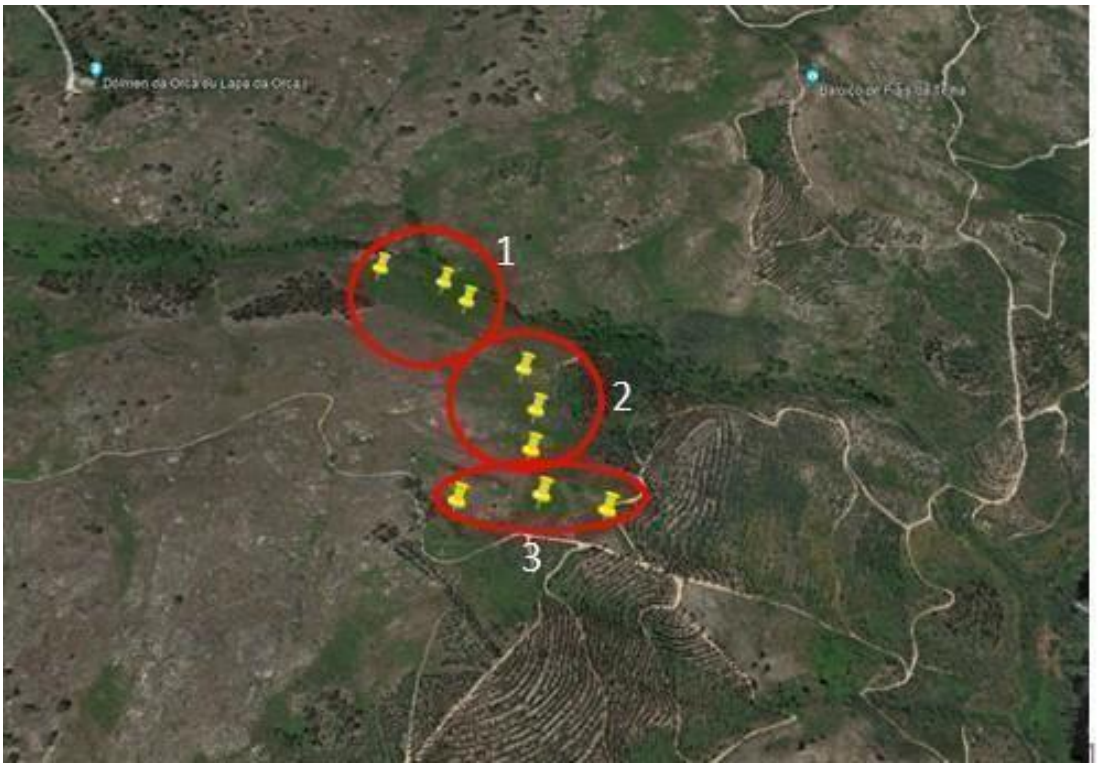


Figure 24 Soil sampling units in Carregal do Sal planting area. 1: CCN; 2: TR; 3: IND.

Table 6 Physicochemical characterization of the topsoil at Carregal Do Sal planting areas.

	Unit 1 (CCN)	Unit 2 (TR)	Unit 3 (IND)
Org. Matter (%)	5.2	11.1	7.6
pH	4.6	4.8	4.6
Elect. Cond. (mS/cm)	0.06	0.05	0.07
P (mg/kg)	12	10	17
K (mg/kg)	47	116	122
Kjeldahl N (%)	0.23	0.37	0.4
Coarse sand (%)	59	52	57
Fine sand (%)	33	33	27
Silt (%)	6	10	11
Clay (%)	2	5	5
Textural class	Sandy	Sandy-loam	Sandy-loam

4.1.3 Soil moisture monitoring and meteorological indicators

According to the meteorological station of the municipality of Carregal do Sal (Figure 25), 2022 was a normal year regarding total precipitation. However, humidity was relatively low, especially during summer, but also during January 2023, with 37 days with less than 50% relative humidity. From January to December 2023 the accumulated precipitation was 820l. However, from January to May 2023, accumulated rain was only 180l, that means a especially dry spring. 2024 was again a normal year regarding precipitations, with abundant precipitations during spring and fall. Summer 2022 was especially hot, with more than 85 days with minimum temperatures higher than 20ºC.

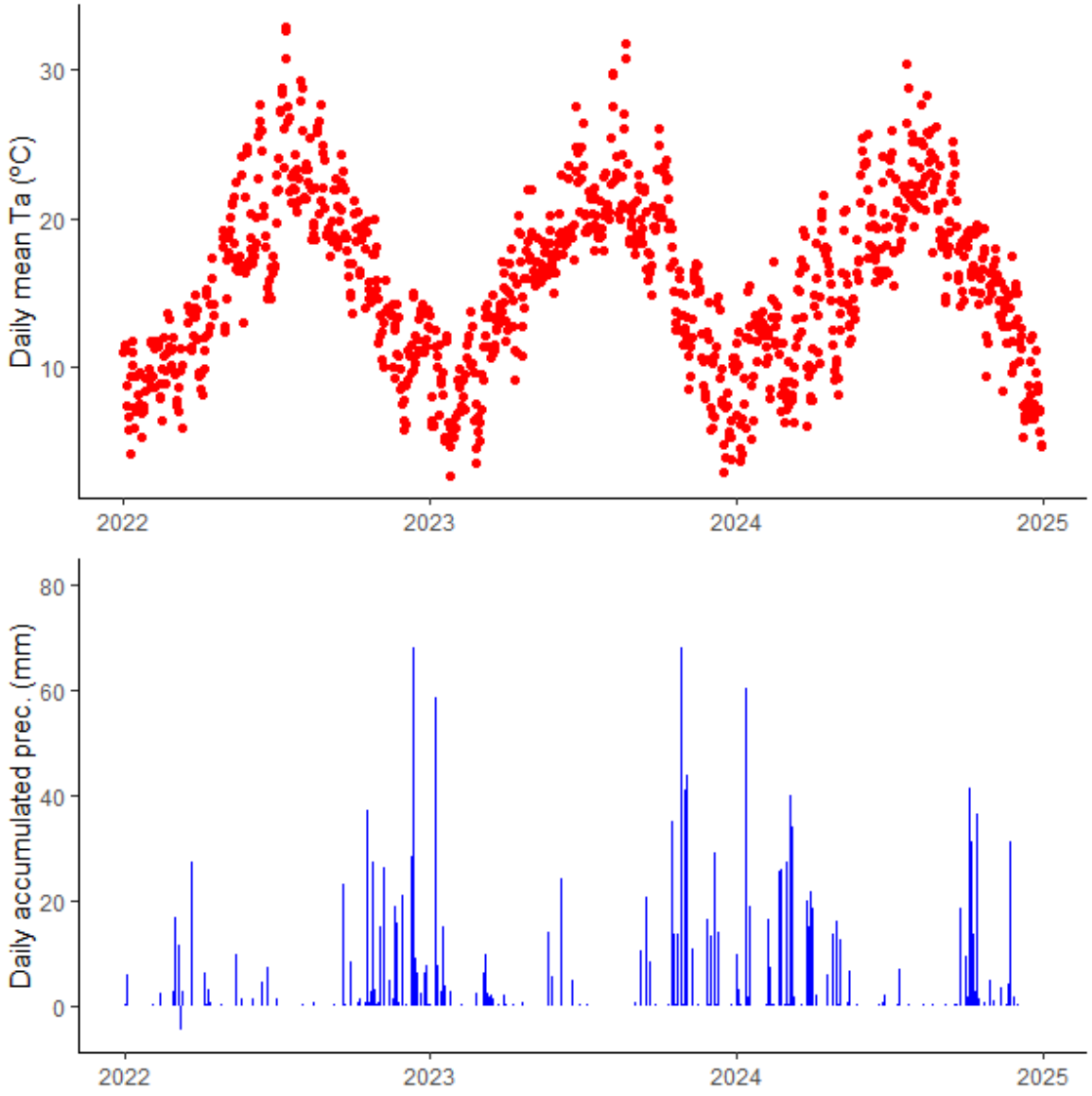


Figure 25 (top) Daily average air temperature and (bottom) daily accumulated precipitation in Carregal do Sal observatory between 2022 to 2024.

Capacitive sensors were chosen for monitoring the soil moisture (see Figure 26), since this method is widely used and, through calibration, can be applied to estimate the soil matrix potential. This kind of sensor measures the dielectric permittivity of soil, which varies proportionally with its water content. The permittivity of a mixture of materials is directly related to the permittivity of its individual components, with water having particularly high permittivity compared to other soil elements.

These sensors typically consist of three parts: the electrodes (probe), the transmission circuit, and the electronics. Miniaturisation due to microelectronics allows the probes to incorporate all

necessary components for generating, reading, interpreting, and encoding the reading at their end, providing a standard analogue, digital or wireless output signal. If the electronic part processing the signal is separate from the probe, the transmission will be analogue, often via a coaxial cable.

The use of capacitive sensors for measuring soil moisture is convenient due to their ability to provide accurate results and require low maintenance. Moreover, these sensors are suitable for telemetry, facilitating remote soil monitoring and optimization of irrigation management and other aspects of agricultural management.

Figure 26 Decagon ECH2O 10HS (left) provided with the Hobolink RXW-SMD (right).

Regarding installation, sensors were placed vertically in the soil which provides an average moisture value along the probe; it could also be installed horizontally so the moisture is measured at the specific depth of the probe. It is important to consider that burying the probe can modify the soil characteristics, potentially affecting the representativeness of the reading. Despite differences among dielectric methods, none are affected by soil texture, temperature, or salinity. However, these sensors have certain limitations related to frost, soil conductivity, gaps, initial calibration, temperature, and agricultural mechanisation.

To estimate the dielectric permittivity and, consequently, soil moisture, techniques such as time domain reflectometry (TDR), capacitive method, and frequency domain reflectometry (FDR) are used. The Hobolink RXW-SMD sensors acquired for the project integrates an ECH2O 10HS sensor that consists of a 10 cm probe manufactured by Decagon which uses FDR technique at a 70 MHz frequency.

Capacitive techniques rely on creating a capacitor from two electrodes, using the soil as a dielectric material. The capacitance of this capacitor will depend on the dielectric constant of the soil. The FDR technique incorporates this capacitor into an oscillating circuit, so that by sweeping the frequency, its resonant frequency can be determined and, in turn, its capacitance.

The frequency of 70 MHz minimises the effects of salinity and texture, making it suitable for most types of soils.

According to the manufacturer the sensor accuracy is ±3.3% typical 0 to 50°C (32° to 122°F) for mineral soils up to 10 dS/m and ±2% with soil specific calibration. This is a system level accuracy specification and is comprised of the probe's accuracy of ±0.03 m^3/m^3 typical (±0.02 m^3/m^3 soil specific) plus the mote accuracy of ±0.003 m^3/m^3 at 25°C (77°F). There are additional temperature accuracy deviations of ±0.003 $m^3/m^3/°C$ maximum for the probe across operating temperature environment, typical <0.001 $m^3/m^3/°C$. (The temperature dependence of the mote is negligible.)

Two methods could be used for calibrating soil moisture sensors: the standard oven-drying method and the weighing lysimeter method. The oven-drying method is considered the most accurate, but it is also destructive, labour-intensive, and time-consuming. The weighing lysimeter method, on the other hand, is a reliable approach for estimating crop evapotranspiration, but it is also quite complex. However, both the standard oven-drying method and the weighing lysimeter method may not substantially improve accuracy in certain soil conditions, such as those with an electrical conductivity of less than 10dS/m (Qin et al., 2021).

Sensors installed in the field to measure moisture levels have been found to exhibit a similar behaviour pattern, which is closely correlated with the occurrence of precipitation. This trend has been observed across different geographical locations and irrigation systems (Figure 27).

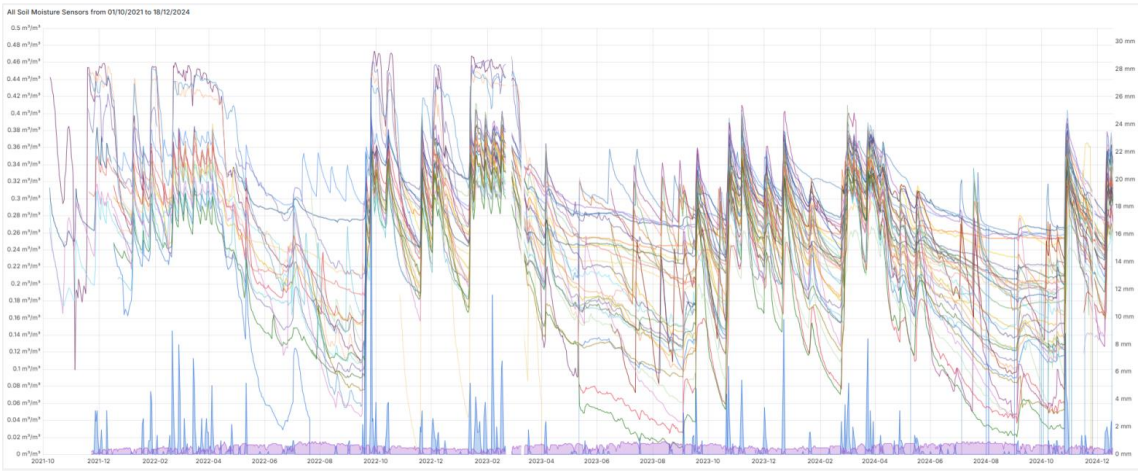


Figure 27 All soil moisture sensors data from October 2021 to December 2024 represented with daily evapotranspiration and daily accumulated rain.

When it comes to soil moisture, there are several factors that can affect their readings, even under the same conditions. These factors can include:

Soil composition: The composition of the soil can affect the dielectric constant, which is the measure of a material's ability to store electrical energy in an electric field. Soil with high clay content, for example, will have a higher dielectric constant than soil with a low clay content, which can affect the readings of a capacitive soil moisture sensor.

Soil temperature: The temperature of the soil can affect the readings of a capacitive soil moisture sensor, as it can affect the dielectric constant of the soil. Cooler soil will have a lower dielectric constant than warmer soil, which can affect the accuracy of the sensor's readings.

Soil compaction: Soil compaction can affect the readings of a capacitive soil moisture sensor, as it can affect the electrical conductivity of the soil. Compacted soil will have a higher electrical conductivity than loose soil, which can affect the sensor's readings.

Sensor placement: The placement of the sensor in the soil can affect its readings, as it can affect the area of soil that the sensor is measuring. If the sensor is placed too close to a rock or other obstacle, for example, it may not be measuring the true moisture content of the soil.

Sensor calibration: The calibration of the sensor can also affect its readings. If the sensor is not calibrated correctly, it may produce inaccurate readings, even under the same conditions.

As the manufacturer indicated, the devices installed in Carregal do Sal require manual downloading (via Bluetooth or USB). The manufacturer also highlighted the importance of clearing the devices' memory after data is downloaded. This is because if the memory becomes full, they will stop recording new data, and this issue cannot be avoided.

Considering the manual nature of the download process, ITC propose the implementation of an upload page could be enabled within Grafana, similar to the image shown below (Figure 28). This approach should result in a semi-automatic process that not only safeguards the logs but also imports the data into the Grafana visualisation system, which will be further discussed later in this document.

Figure 28. Proposed datalog upload page.

In particular, in Carregal do Sal plantation 5 soil moisture sensors have been installed since May 2022 (Table 7).

Table 7. Soil moisture sensors installed in Carregal do Sal plantations.

Sector	Sensor code 1	Sensor code 2	Sensor location
IND	21490994	IND_sms_1	https://maps.app.goo.gl/o6WKhAT6BMK7Kgtu9
IND	21490986	IND_sms_2	https://maps.app.goo.gl/36y4DcafhzgtV4vD9
TR	21490992	TR_sms_1	https://www.google.com/maps/place/40%C2%B026'20.0%22N+7%C2%B056'03.0%22W/@40.4389439,-7.9343001,131m/data=!3m1!1e3!4m4!3m3!8m2!3d40.4388889!4d-7.9341667?entry=ttu
CC	21490998	Cocoon_sms_1	https://www.google.com/maps/place/40%C2%B026'20.0%22N+7%C2%B056'03.0%22W/@40.438893,-7.9367416,880m/data=!3m2!1e3!4b1!4m4!3m3!8m2!3d40.4388889!4d-7.9341667?entry=ttu
CC	21490993	CC	https://www.google.com/maps/place/40%C2%B026'17.0%22N+7%C2%B056'03.0%22W/@40.4380596,-7.9390376,880m/data=!3m2!1e3!4b1!4m4!3m3!8m2!3d40.4380556!4d-7.9341667?entry=ttu

However, due to a problem on the internal software of the sensors no data is available for CCN and TRD sensors. Regarding individual fog water captors (IFWC), the same pattern has been observed in both sensors, which demonstrates that they are correctly installed (Figure 24). However, soil in individual_sms_1 presents a lower moisture content probably due to a slightly higher stoniness in the soil or differences in soil texture. It is important to note that retrieving data from the sensors requires traveling to the measurement site, and issues such as missing sensors, incomplete data downloads, or damage to cables and transmitters caused by wildlife—particularly wild boars—have been encountered.

Figure 29 shows the evolution of two measurement points associated with Individual Collectors. Both points reached peak soil moisture content between 0.432 and 0.444 m^3/m^3, with average values ranging from 0.264 to 0.291 m^3/m^3. Compared to the data collected in Vouzela from February 16, 2023, to August 1, 2024, for the Individual Collector, the measurement points in Carregal do Sal appear to show overall better results. However, it is important to note that these datasets correspond to different measurement periods, so comparisons should be made with caution.

Figure 29 Individual collectors Soil Moisture in Carregal do Sal, 26th of October 2022 to 12th of January 2023.

4.1.4 Seedlings vigour

Monitoring campaign in Carregal do Sal started in September 2022, after the first summer drought, which could be considered as representative of survival rates after the first dry season after planting (see number of seedlings monitored in Table 8). Monitoring campaigns were repeated in 2023 and spring 2024, according to the monitoring protocol. The monitoring campaign in September 2024 had to be cancelled after the fire that affected the area. In total more than 1500 vigour measures were taken between 2022 and 2024.

Due to the dense growth of spontaneous vegetation, some seedlings could not be located during the spring 2024 survey. However, to ensure an accurate assessment of seedling physiological status over time, those previously recorded as dead but not found in afterwards to confirm their condition were classified as dead in subsequent samplings. This correction was also applied in also Vouzela.

Table 8 Number of monitored seedlings per species and planting system, in 2023 and 2024, in Carregal do Sal plantations.

2023 Campaign	**TR**	**CCN**	**IFWC**	**TOTAL**
Quercus pyrenaica	76	146	63	285
Quercus robur	131	182	115	428
Total	207	328	178	713
2024 Campaign	TR	CCN	IFWC	TOTAL
Quercus pyrenaica	33	35	7	75
Quercus robur	78	111	11	200
Total	111	146	18	275

4.1.4.1 Physiological state: global results

Plantations made in Carregal do Sal presented a relatively high mortality ratio at the beginning of the experiment (Figure 30). In 2022 almost half of the seedlings were affected or in bad condition, such as presenting symptoms of hydric stress (loss of leaves and chlorotic leaves). In 2023 almost half of the seedlings were dead (~40%), and although some of the seedlings that were affected in 2022 had resprouted by 2023, mortality rates kept increasing. Only around a 35% of the seedlings were in good condition by 2024 (i.e. with all leaves living and with active apexes).

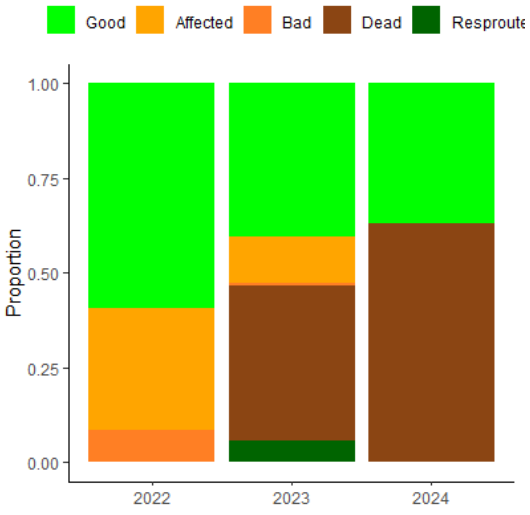


Figure 30 Distribution of the seedlings according to their physiological state per year in Carregal do Sal.

4.1.4.2 Physiological state per planting system

When we analyse the results per planting system, a positive influence of the CCN method was found in comparison to IFWC and the TR plantation (Figure 31). In 2023, CCN was the planting system with the best results, with a mortality ratio of ~20%, which contrasted with the high mortality ratio (above 50%) of TR. In 2023 the IFWC treatment also presented a quite high mortality ratio (44%), but still lower than TR. In 2024, the mortality rate increased a bit more, especially in TR, but interestingly the CCN system kept the mortality rates at a relatively low rate, with survival rates ~ 60%.

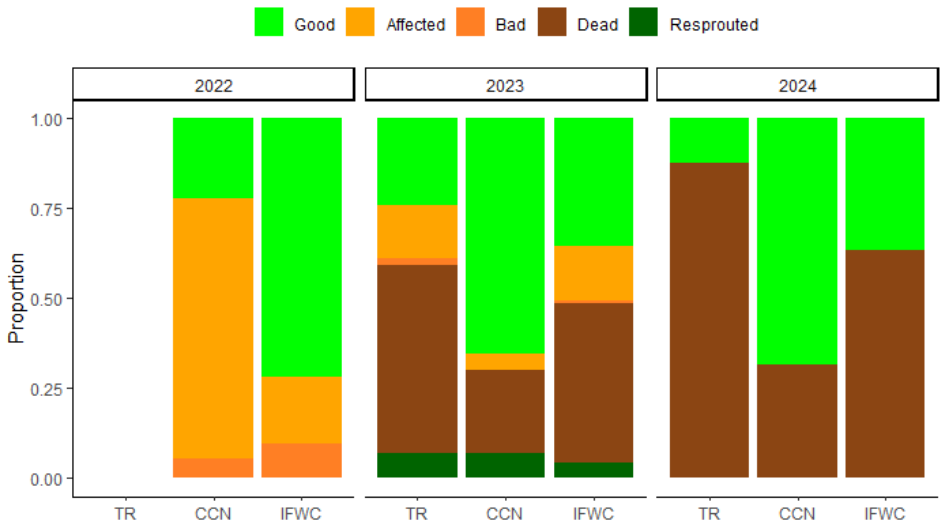


Figure 31. Distribution of the seedlings according to their physiological state per planting system and year in Carregal do Sal.

4.1.4.3 Physiological state per species

Regarding the results per species, CCN keeps highlighting as the planting system with the highest survival ratio for both *Quercus robur* and *Quercus pyrenaica*, both in year 2023 and 2024. Interestingly, the IFWC system improved survival rates compared to TR in *Q. robur* but not in *Q. pyrenaica* (Figure 32). This could be due to an innovation that was introduced to the IFWC system, putting a plastic net of around 1 m height (**Figure 33**) to improve water capture and reduce evapotranspiration.

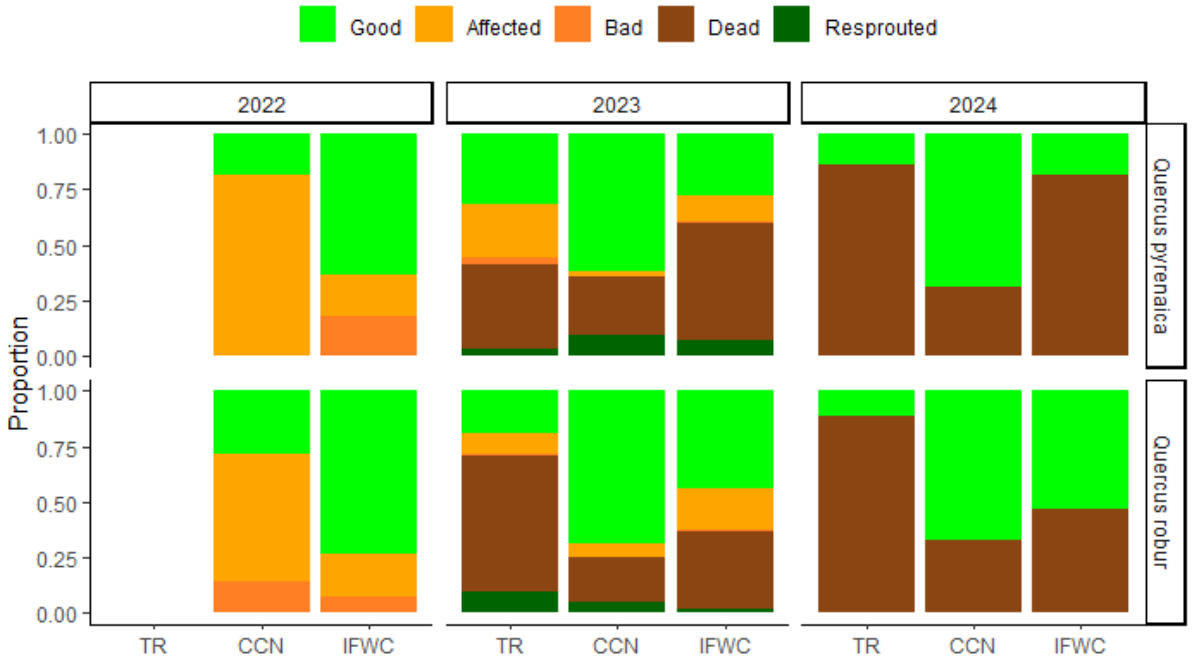


Figure 32 Distribution of the seedlings according to their physiological state per tree species, planting system and year in Carregal do Sal.

Thus, *Q. robur,* which had a higher proportion of individuals of the IFWC system with plastic net, presented lower mortality rates with plastic net than without (Figure 30), while this effect on survival rates was less pronounced in *Q. pyrenaica* (Figure 34).

Figure 33 *IFWC with plastic net in Portugal plantations. Wasps took advantage of the plastic net for hive establishment.*

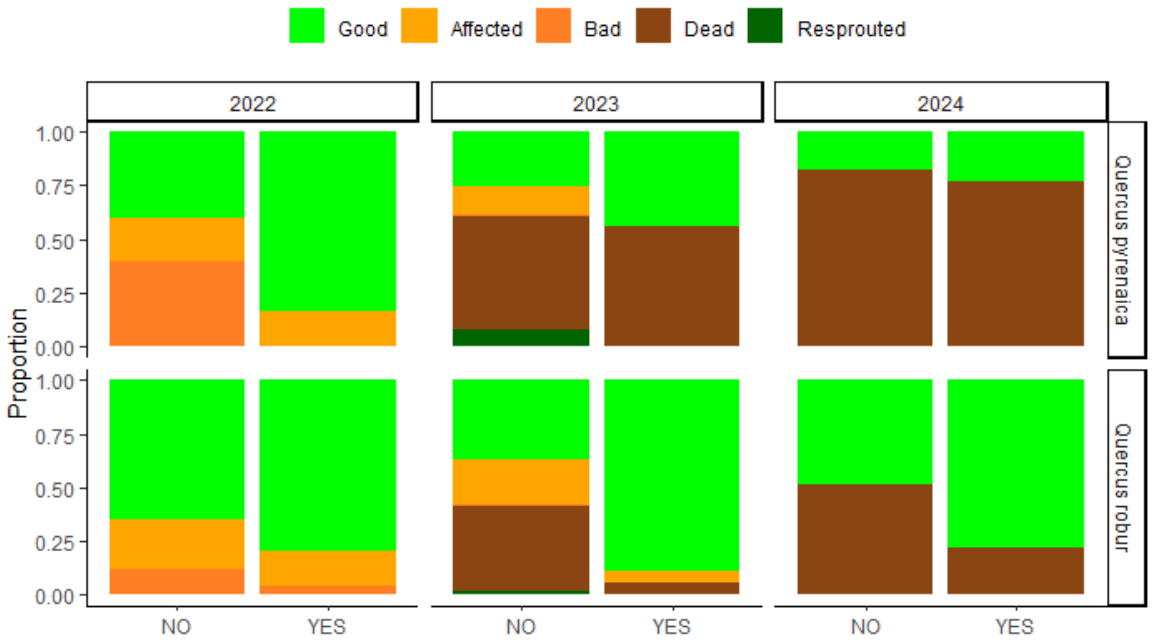


Figure 34 Physiological state of Carregal do Sal seedlings from the IFWC planting method (with and without plastic net) per species and year.

4.1.4.4 Seedlings growth

The CCN method increased the height of the seedlings compared to the TR method, especially in the case of *Q. robur*. The IFWC method did not present an improvement of height. On the other hand, both the IFWC and the CCN method increased seedlings width compared to TR (Figure 35), and seedlings of *Q. robur* were slightly wider than *Q. pyrenaica* (Figure 36).

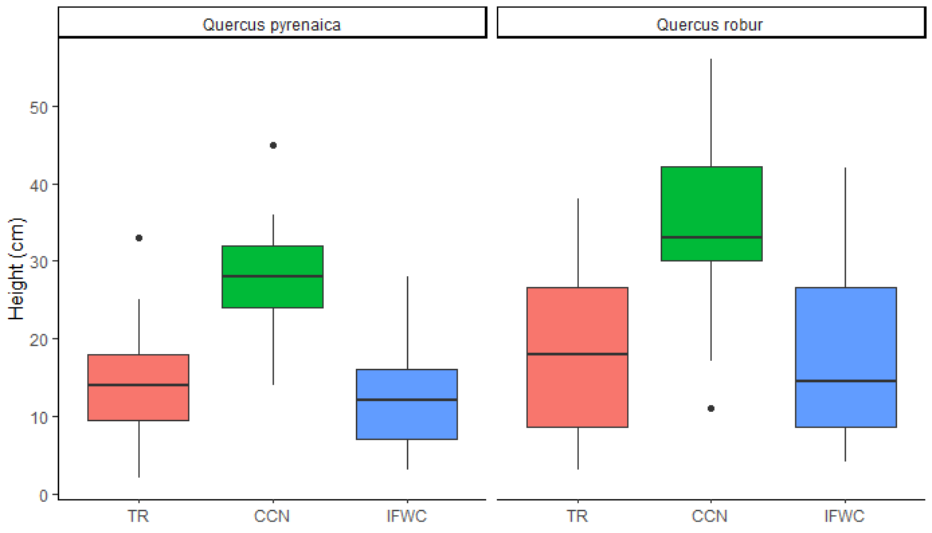


Figure 35 Height of the seedlings per tree species, planting system and year in Carregal do Sal.

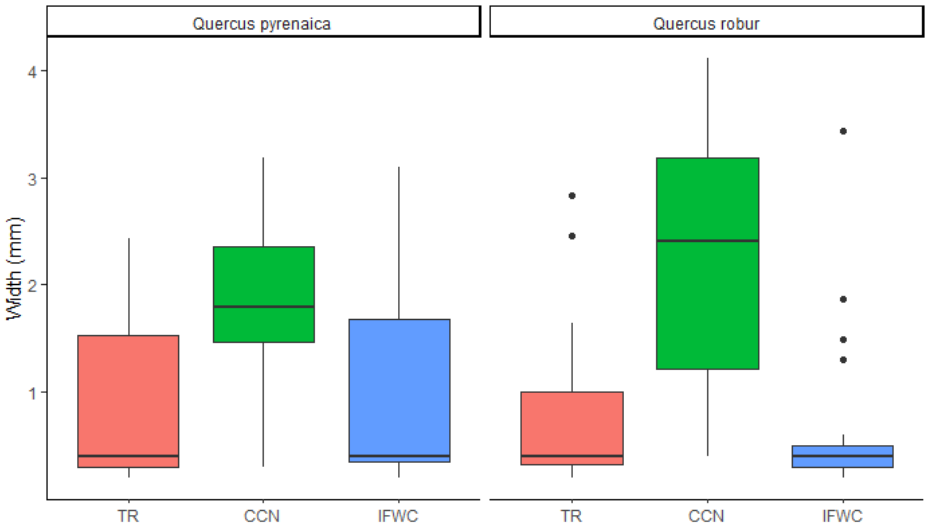


Figure 36 Width of the seedlings per tree species, planting system and year in Carregal do Sal.

Regarding the vigour monitoring in the experimental area of Carregal do Sal planting systems clearly increased survival rates, with CCN performing the best rates followed by IFWC (Figure 37). While all the individuals under the TR treatment were dead by 2024, CCN and IFWC systems kept the mortality reasonably low, without much increase between 2023 and 2024.

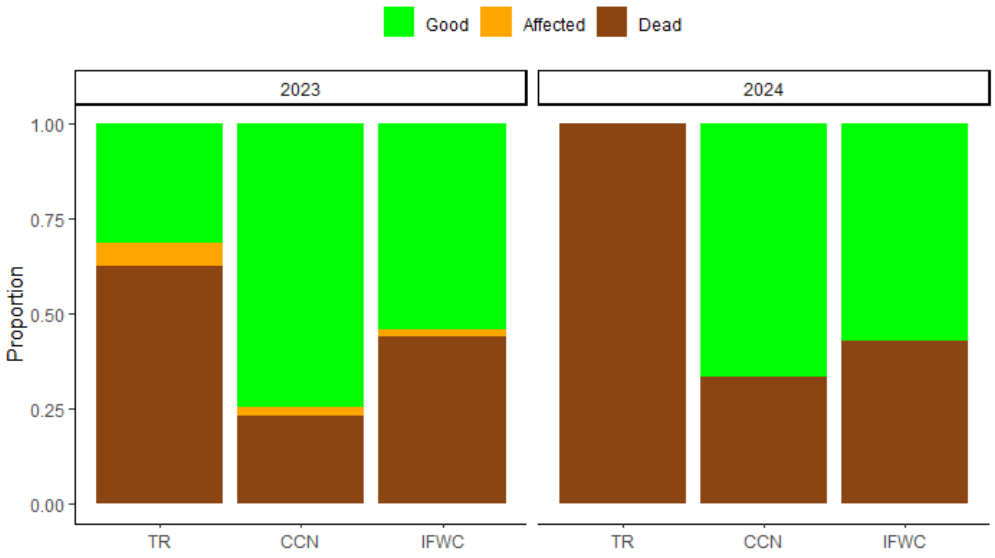


Figure 37 Distribution of the seedlings according to their physiological state per planting system and year in Carregal do Sal experimental area.

Due to the extremely high mortality of TR in the experimental zone of Carregal, we were unable to assess the effect of the planting systems CCN and IFWC on height and width in relation to TR. However, our results indicate a tendency for greater height and width in CCN compared to IFWC (Figure 38). This aligns with the positive effect of CCN in vigour (Figure 34) and the observed enhancement of height and width in CCN across the general plantation area (Figure 35-Figure 36).

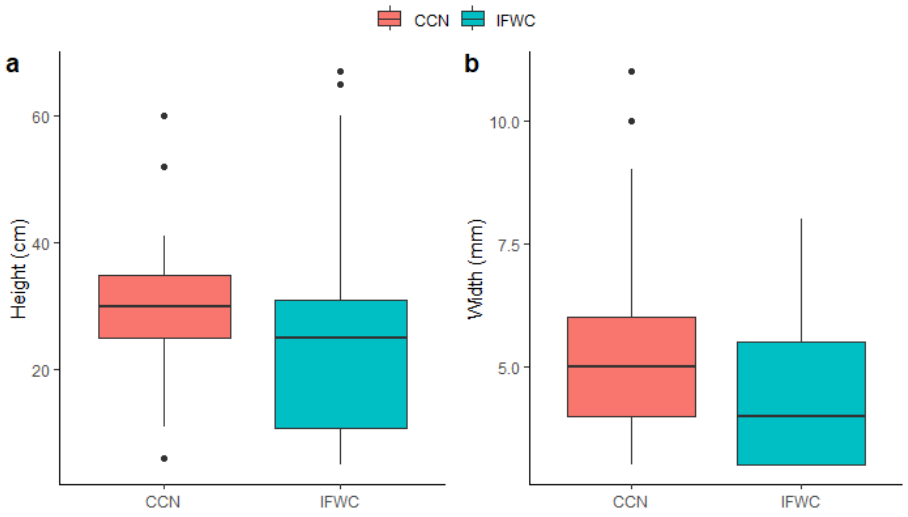


Figure 38. (a) Height and (b) width of the seedlings per planting system in Carregal do Sal experimental area.

4.1.5 Passive restoration

Before the forest fire in September 2024 natural regeneration of Carregal do Sal planting area was working well. Vegetation cover was very high (88 %), dominated by grasses and shrubs (see species abundance in Table 9), with the exception of the west sector, where poor vegetation and tiny soils existed (Figure 39-Figure 40).

Figure 39 Pictures of Carregal do Sal planting area. Herbaceous and shrubs cover almost all the soil (right), despite some patches with big granite blocks and outcrops (left).

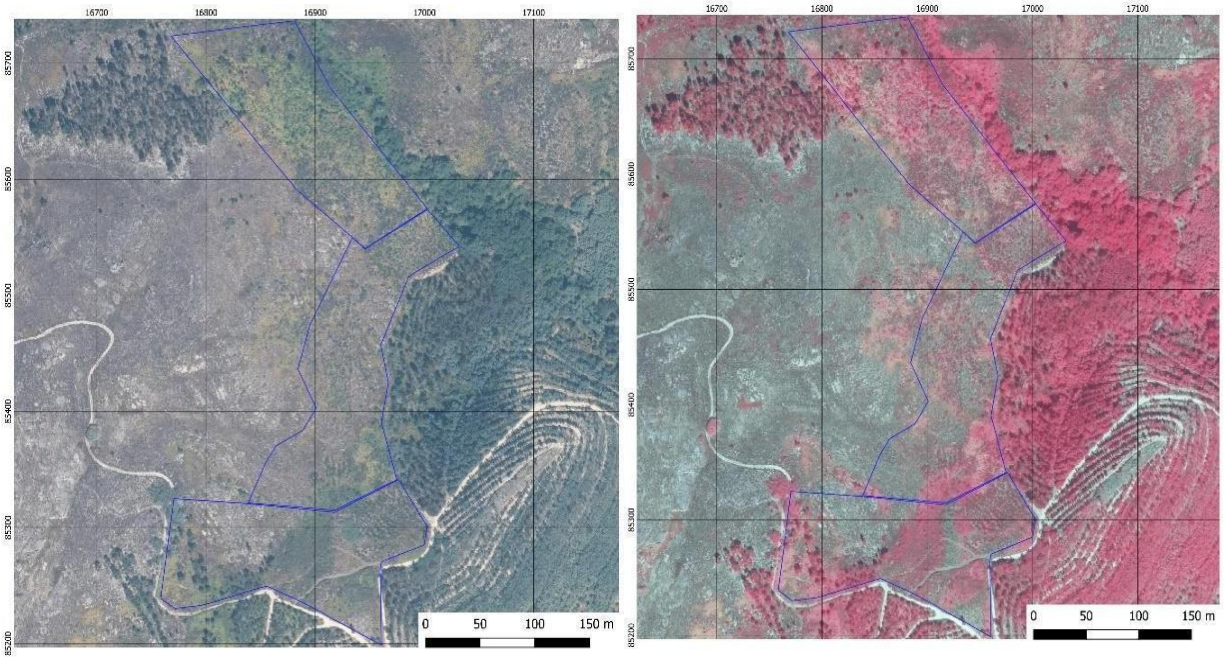


Figure 40 Aerial view of the planting area in Carregal do Sal (blue poligons) in 2021. RGB (left) and false color (right) Source: Sistema Nacional de Información Geográfica.

Table 9 Species inventoried in Carregal do Sal planting area in May 2023. Abundance ranges: 4, dominant (>75% vegetation cover); 3, high frequency (25-75% vegetation cover); 2, low frequency (<25% vegetation cover); 1, testimonial.

SPECIES	ABUNDANCE
Erica australis	4
Graminoids	4
Citysus grandiflorus	4
Cytisus multiflorus	3
Ulex europaeus	3
Pteridium aquilinum	3
Crepis capillaris	3
Acacia dialvata	3
Eucalyptus globulus	3
Rubus sp.	2
Campanula sp.	2
Quercus robur	2
Quercus pyrenaica	2
Cistus lasianthus	2
Glandora prostata	2
Lavandula stoechas	2
Thapsia villosa	1
Crataegus sp.	1
Hakea sericea	1

In addition, to assess the passive restoration evolution, Sentinel-2 satellite images (10 m/pixel) from 2016, 2019, and 2023 were analysed. The images were processed using a semi-automatic classification method (K-nearest neighbor, KNN) and categorized into different land cover types, including trees, shrubs, grass, and bare soil (Figure 38). As expected, the image analyses revealed a decrease in the shrub cover after the forest fire in year 2017, as shows the decrease in the shrub cover, and the increase in the bare soil and grass cover from 20216 to 2019, and by 2023 the tree cover had already started to increase.

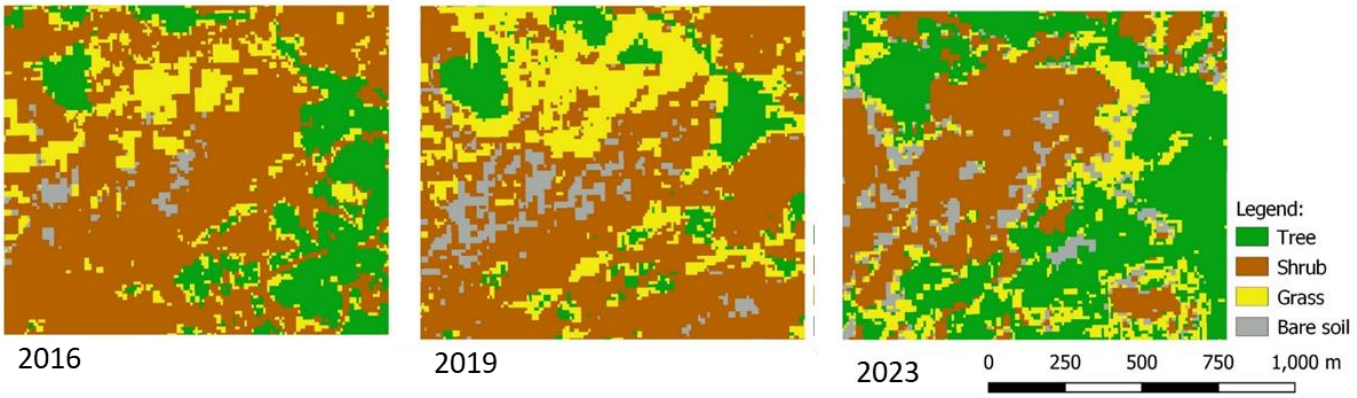


Figure 41 Land cover evolution in Carregal do Sal planting area in the period 2016 (before the forest fire), 2019 (just after the forest fire) and 2023, using Sentinell-2 images and knn classifier (semi-automatic classification)

However, in September 2024 an intense forest fire affected the area, completely destroying all the plantations carried out in the context of the project (Figure 42). The forest fire severity index was calculated using Sentinel images taken just after the fire. Heavily affected areas could be observed in some areas, despite in others some patches of unaffected vegetation remain (Figure 43).

Figure 42 Images of the planting area in Carregal do Sal after the forest fire in September 2024.

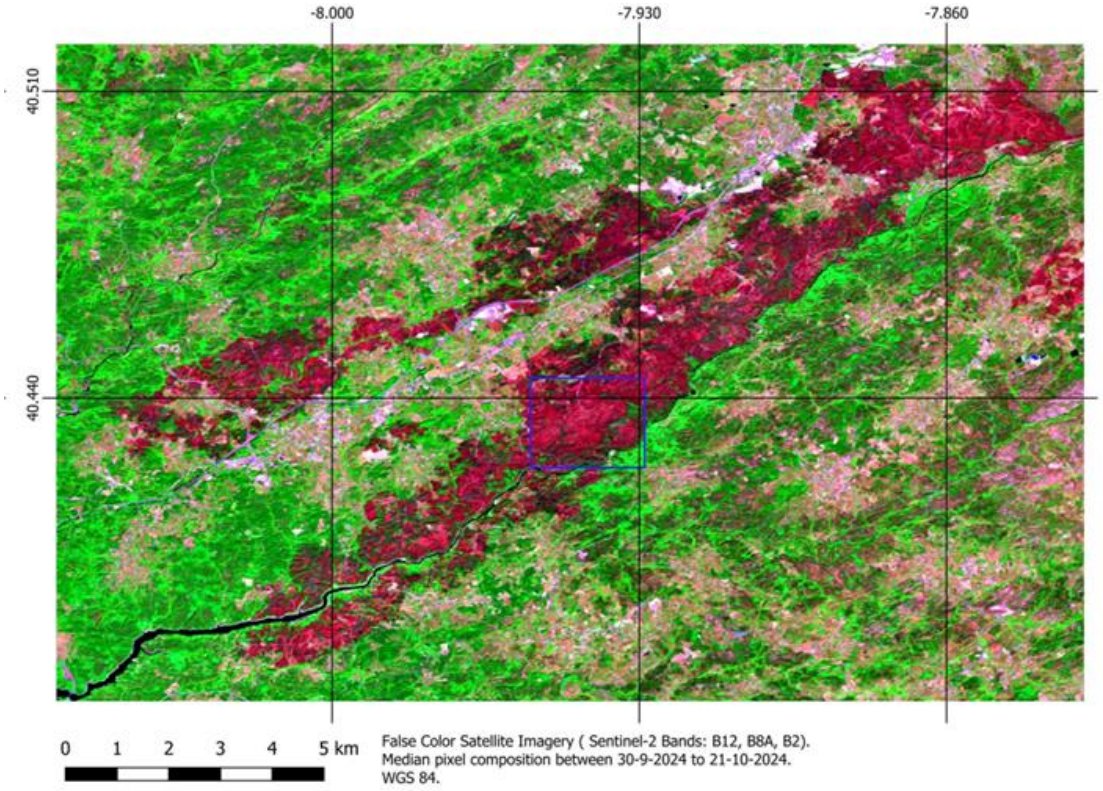


Figure 43 Burnt Area Index map of the area affected by the September 2024 forest fire in Carregal do Sal. Blue polygon shows the area where plantations are located.

4.2 Vouzela

4.2.1 Soil characterization

The soil in the Vouzela planting area (Figure 43) is characterised by a high heterogeneity due to the presence of large granite blocks (40% of the surface occupied by rocky heats). This results in highly variable soil depths, ranging from less than 20 cm to over 100 cm, although deeper soils are most common. Across all horizons, the stoniness is medium and relatively homogeneous (29-32%), mainly comprising fine gravel fragments (5-2 mm). The sandy texture of the soil reduces its water-holding capacity, consequently limiting the water available to plants. Moreover, despite the high organic matter content (>6% in superficial horizons and >3% in subsuperficial horizons), the acidic pH (Table 10), high infiltration capacity, and relatively high to medium precipitation led to nutrient leaching.

Figure 44 Profile VOU, characteristic of the planting area in Vouzela.

Macromorphological description of horizons

O_I (-2 to -5 cm). Dry leaves and plant remains from clearing works.

A1 (0 to 20 cm)**. Moisture state**: dry. **Redox state**: oxidised. **Staining**: not observed. **Coarse fragments content**: medium (30% in weight), granite, 2-5 mm diameter, angular, medium spherical, regular distribution. **Texture**: Loamy sand. **Structure**: moderate, angular blocks, 50 mm diameter, soft, friable, non-plastic. **Organic matter**: high (6.08 %), decomposed debris and humus. **Fauna activity**: galleries. **Roots**: abundant, 2 mm diameter, living. **Field tests**: null reaction to HCl. **Accumulations**: not observed. **Cementations**: not observed. **Coatings**: not observed. **Horizon lower limit, transition**: no soil discontinuities, wavy, gradual.

A2 (20 to 50 cm)**. Moisture state**: dry. **Redox state**: oxidised. **Staining**: not observed. **Coarse fragments content:** medium (29% in weight), granite, 2-5 mm diameter, angular, medium spherical, regular distribution. **Texture**: Loamy sand. **Structure**: moderate, angular blocks, 30 mm diameter, soft, friable, non-plastic. **Organic matter**: moderate (3.01 %), decomposed debris and humus. **Fauna activity**: galleries. **Roots**: abundant, 2 mm diameter, living. **Field tests**: null reaction to HCl. **Accumulations**: not observed. **Cementations**: not observed. **Coatings**: not observed. **Horizon lower limit, transition**: no soil discontinuities, wavy, gradual.

B (50 to 100 cm)**. Moisture state**: dry. **Redox state**: oxidised. **Staining**: scarce, oxidation, 5 mm, net limit, irregular form, in the soil matrix. **Coarse fragments content:** medium (32% in weight), granite, 2-5 mm diameter, angular, medium spherical, regular distribution. **Texture**: Loamy sand. **Structure**: weak, granular, 2 mm diameter, soft, friable, non-plastic. **Organic matter**: low (0.89 %), inappreciable. **Fauna activity**: not observed. **Roots**: scarce, 5 mm diameter, living. **Field tests**: null reaction to HCl. **Accumulations**: not observed. **Cementations**: not observed. **Coatings**: not observed.

Table 10 *Physicochemical characterization of the horizons of VOU profile. *in weight.*

	A1 (0-20 cm)	A2 (20-50 cm)	B (50-100 cm)
>2 mm (%)	30.2	29	31.6
Sand (%)	70.3	64.2	69.5
Silt (%)	22.9	28.9	21.5
Clay (%)	6.8	6.9	9
Textural class	Loamy-sand	Loamy-sand	Loamy-sand
pH H2O	5.3	5.3	5.3
Elect. Cond. (ext. 1:5; dS/m)	<0.05	<0.05	<0.05
Carbonates (%)	Undetect. (<3)	Undetect. (<3)	Undetect. (<3)
N Kjeldahl (%)	0.2	0.15	0.06
P Olsen (mg/kg)	10.3	5.19	5.83
Organic C	3.53	1.75	0.52
K (mg/kg)	64	35	29
Ca (mg/kg)	51	48	166
Mg (mg/kg)	15	10	17
Na (mg/kg)	20	20	26

4.2.2 Soil monitoring

4.2.2.1 Physicochemical parameters

In Vouzela restoration areas, three distinct soil units have been defined per area (Figure 45). Vouzela samples present a lower organic matter and N contents compared to those in Carregal do Sal, and they show higher stoniness (Table 11). As in Carregal do Sal, the use of organic and/or calcic amendments was considered unnecessary. Moreover, given the acidophilic nature of the species selected for planting, no toxicity issues are anticipated.

Figure 45 Soil sampling units in the Vouzela planting area. 1: CCN; 2: TR; 3: IND.

Table 11 *Physicochemical characterization of the topsoils in Vouzela planting areas.*

	Unit 1 (CCN)	Unit 2 (TR)	Unit 3 (IND)
Org. Matter (%)	5.3	2.1	3.2
pH	5.2	4.6	4.9
Elect. Cond. (mS/cm)	0.13	0.04	0.04
P (mg/kg)	80	52	42
K (mg/kg)	52	33	34
Kjeldahl N (%)	0.21	0.04	0.12
Coarse sand (%)	52	58	58
Fine sand (%)	34	31	29
Silt (%)	10	6	10
Clay (%)	4	5	3
Textural class	Sandy	Sandy-loam	Sandy-loam

Among the various soil chemical properties studied at Vouzela (Figure 46-Figure 47), organic matter, total nitrogen (N), P Olsen, and extractable potassium (K+) all showed an increase over time, similarly to what is observed at El Pinillo (Section 3.3.2). This suggests that the repeated vegetation clearings increased organic matter input, which in turn, facilitated the release of soil nutrients, particularly P and K+. Moreover, it must be considered that the important increase in soil organic matter in some samples could be partially attributed to the fact that some organic debris (O horizon) were sampled jointly with the soil mineral fraction (A horizon), overestimating the organic matter in these samples.

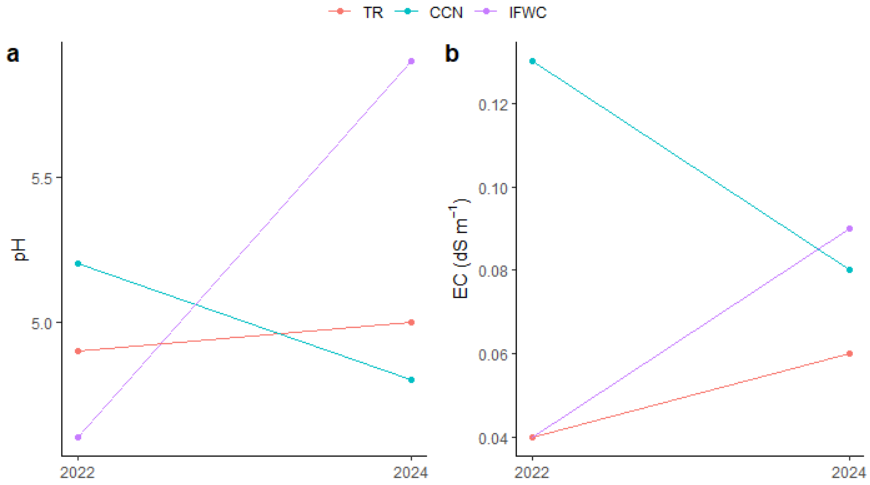


Figure 46 (a) pH, and (b) electrical conductivity (EC) per planting system: traditional (TR), cocoon (CCN), and individual fog water collectors (IFWC).

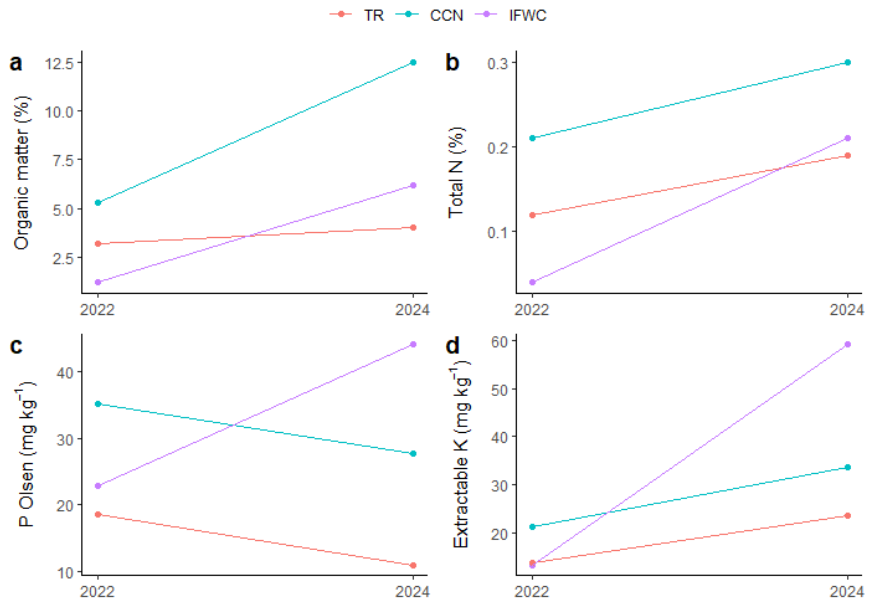


Figure 47 (a) Organic matter, (b) total nitrogen, (c) phosphorus (P) Olsen, and extractable potassium (K) per planting system: traditional (TR), cocoon (CCN), and individual fog water collectors (IFWC).

4.2.3 Soil moisture monitoring and meteorological indicators

According to the meteorological station of the municipality of Vouzela (Figure 48), 2022 was an especially dry and hot year, with less than half of the mean annual precipitation. From January to December 2023 only 300l accumulated in the village. Summer 2022 was especially hot, with more than 40 days with minimum temperatures higher than 20ºC. Relative humidity was also very low, especially during summer 2022, but also during January 2022, with 69 days with less than 50% in 2022. However, precipitations in year 2023 followed a more normal pattern. From January to May 2023, accumulated rain was 340l, followed by a relatively dry summer and a rainy autumn. The station suffered a breakdown in October 2023 and was not repaired, for which no data exist from this period.

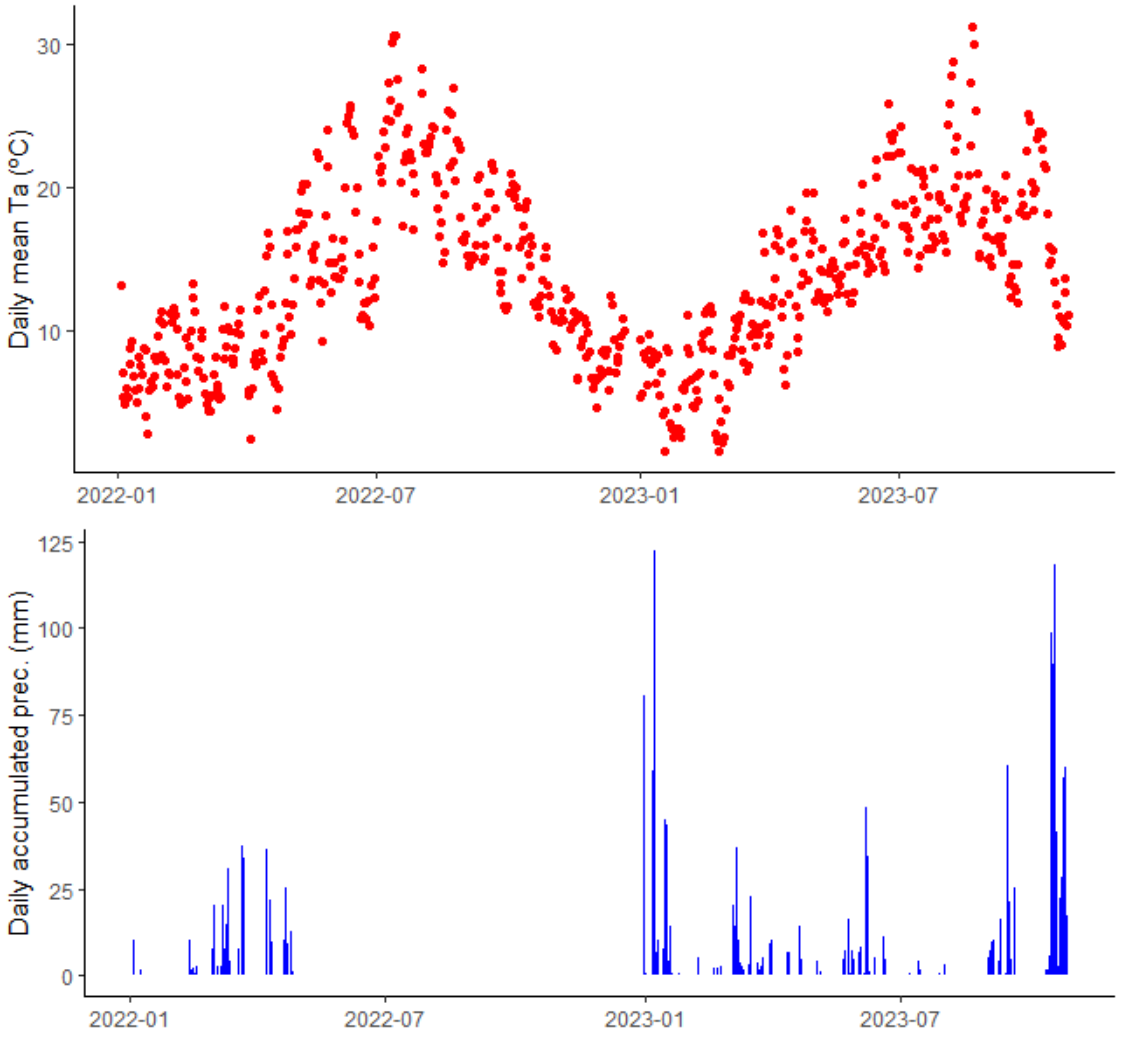


Figure 48 (top) Daily average air temperature and (bottom) daily accumulated precipitation in Vouzela observatory between 2022 to 2024.

Moreover, in March 2023, six soil moisture sensors were installed in Vouzela plantations (two per planting system (Table 12).

Table 12. Soil moisture sensors installed in Vouzela planting area.

Sector code	Sensor code	Sensor location
IND	21491000	https://maps.app.goo.gl/563cu99dUd6GYPPe9?g_st=com.microsoft.Office.Outlook.compose-shareextension
IND	21491002	https://maps.google.com?q=40.7155553,-8.0947100&hl=pt-PT&gl=pt&entry=gps&lucs=47062720&g_st=com.microsoft.Office.Outlook.compose-shareextension
TR	21490990 (VZ TR baixo)	https://maps.google.com?q=40.7172351,-8.0966378&hl=pt-PT&gl=pt&entry=gps&lucs=47062720&gst=com.microsoft.Office.Outlook.compose-shareextension
TR	21490987	https://maps.app.goo.gl/Mrfg48e8vLMhz6Jc7?g_st=com.microsoft.Office.Outlook.compose-shareextension
CCN	21490991 (VZL Cocoon cima)	https://maps.google.com?q=40.7173024,-8.0960407&hl=pt-PT&gl=pt&entry=gps&lucs=47062720&gst=com.microsoft.Office.Outlook.compose-shareextension
CCN	21490995 (VZL cocoon baixo)	https://maps.app.goo.gl/EyDzBJgrM4WHjSv39?gst=com.microsoft.Office.Outlook.compose-shareextension

A clear effect of CCN is evident (Figure 49), with the soil moisture in the central hole of the CCN consistently higher than in the IFWC almost all time. This may be attributed to both the CCN's ability to retain runoff, resulting in a more pronounced increase in soil moisture after rain and their capacity to reduce evaporation losses at both the root system level and at the surface level inside the planting pot (Carabassa et al., 2021)[2].

[2] Carabassa, V., Alba-Patiño, D., García, S., Campo, J., Lovenstein, H., Van Leijen, G., Castro, A., González, F., Viera, G., Kourkoumpas, D.S., Zioga, A.A., Papadelis, C.E., Andreu, A., Gimeno-García, E., Kallen, S., Alcañiz, J.M. (2021) Water-saving techniques for restoring desertified lands: some lessons from the field. Land Degradation and Development 33, 133-144. https://doi.org/10.1002/ldr.4134

Figure 49. Soil humidity in Vouzela plantations, per planting system.

Here we provide an overview of the quarterly monitoring of soil moisture for the project area (Figure 50-Figure 55). The data collected for this section covers from February of 2023 to the August of 2024 and provides insight into how soil moisture levels fluctuate during this period. The results of this monitoring will be used to inform future decisions regarding irrigation, planting, and maintenance, with the goal of determining the performance of different reforestation techniques with respect to soil moisture.

2023 Q1: January to March 2023

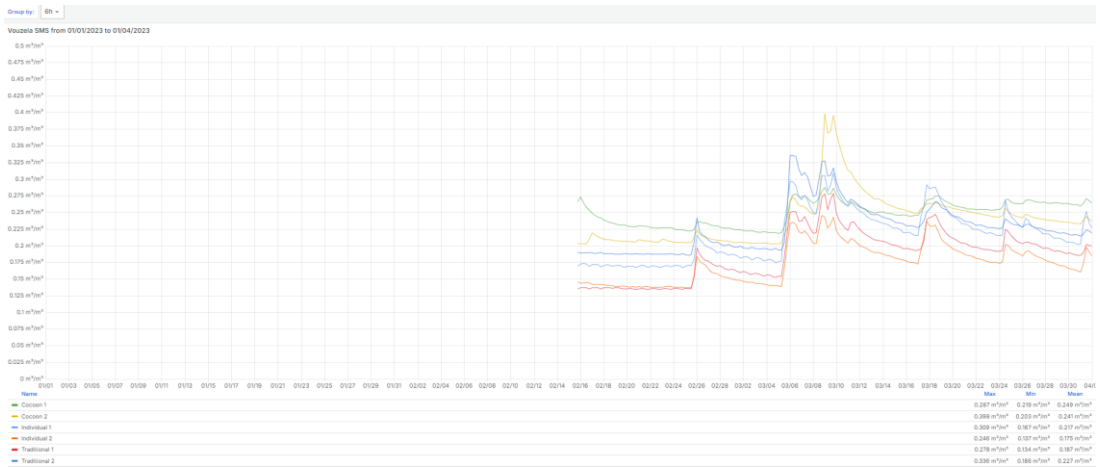


Figure 50 Soil Moisture in Vouzela, January to March 2023.

2023 Q2: April to June 2023

Figure 51 Soil Moisture in Vouzela, April to June 2023.

2023 Q3: July to September 2023

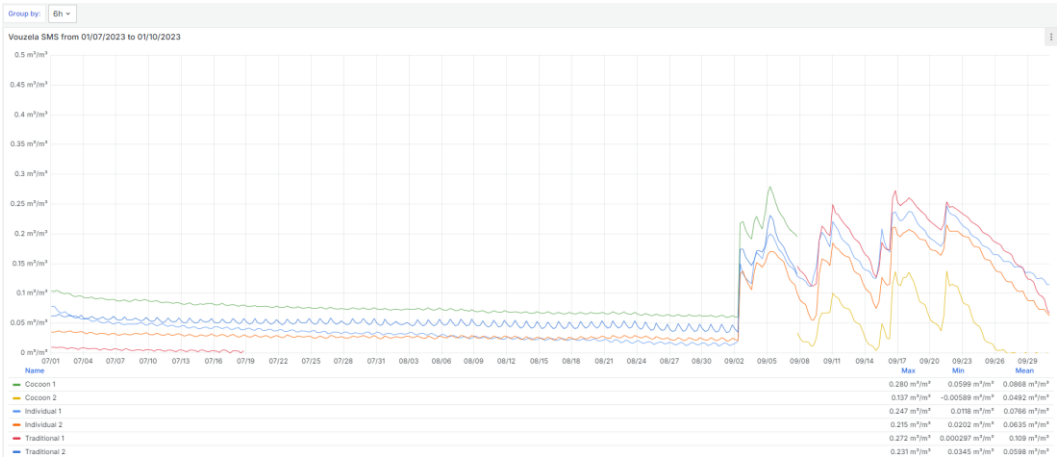


Figure 52 Soil Moisture in Vouzela, July to September 2023.

2023 Q4: October to December 2023

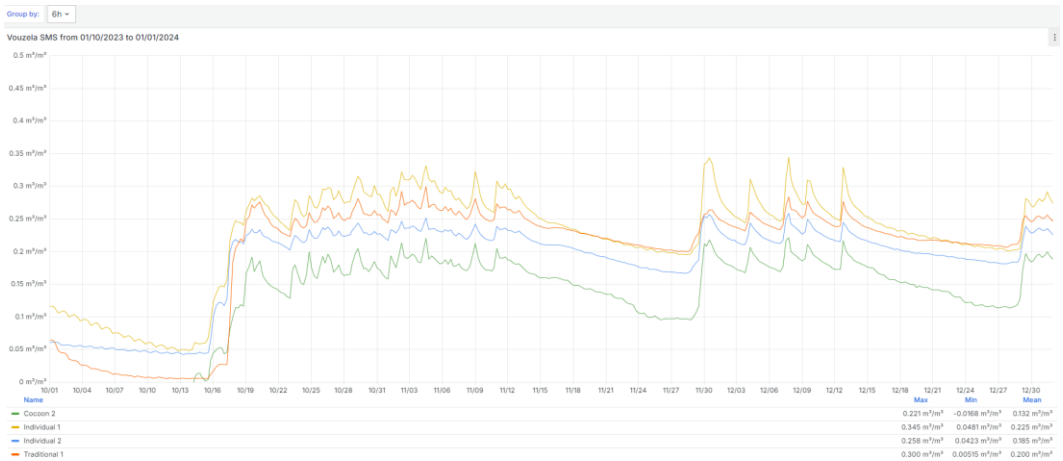


Figure 53 Soil Moisture in Vouzela, October to December 2023.

2024 Q1: January to March 2024

Figure 54 Soil Moisture in Vouzela, January to March 2024.

2024 Q2: April to June 2024

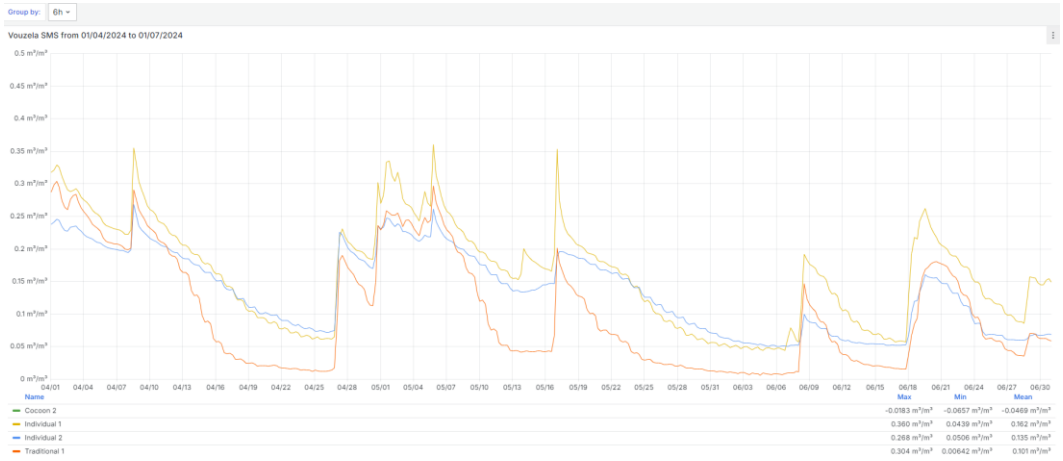


Figure 55 Soil Moisture in Vouzela, April to June 2024.

2024 Q3: July to September 2024

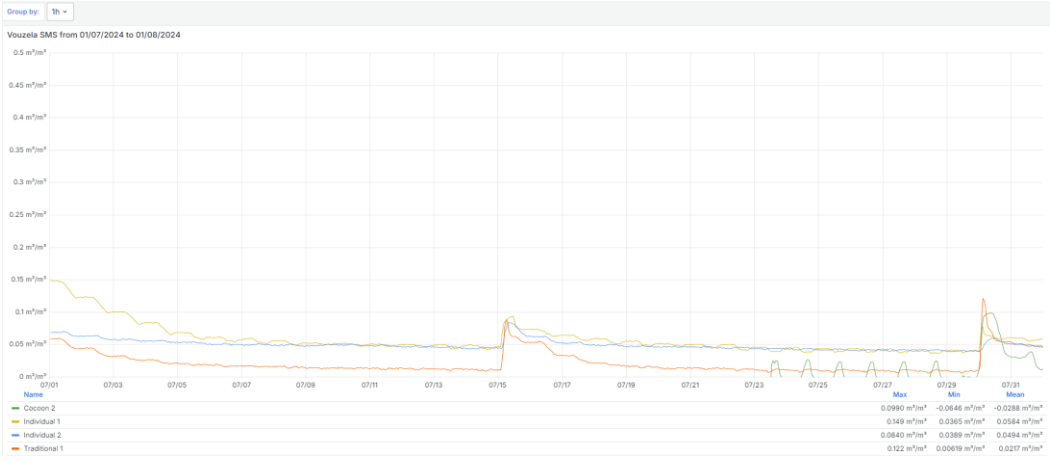


Figure 56 Soil Moisture in Vouzela, July to September 2024.

4.2.3.1 Results and conclusions from soil moisture monitoring in Vouzela

Cocoon Systems:

Cocoon 1: Functioned from 14th February 2023 to 7t h September 2023, reaching a maximum soil moisture level of 0.398 m^3/m^3 and an average of 0.177 m^3/m^3. This demonstrates moderate water retention capability during its operational period.

Cocoon 2: Operated from 15th February 2023 to 4th May 2023, also reaching a maximum of 0.398 m^3/m^3, but with a higher average moisture level of 0.232 m^3/m^3. Cocoon 2 maintained a stronger moisture retention compared to Cocoon 1 within its shorter operational window.

Traditional Reforestation Systems:

Traditional 1: Functioned properly during two periods: from 16th February 2023 to 1st July 2023, with a maximum moisture of 0.378 m^3/m^3 and an average of 0.162 m^3/m^3, and again from 6th September 2023 to 1st August 2024, where it reached 0.385 m^3/m^3 maximum and 0.168 m^3/m^3 on average. These values suggest that traditional systems maintained stable, though lower, moisture levels compared to Cocoon.

Traditional 2: Operated from 14th February 2023 to 7th September 2023, with a maximum of 0.350 m^3/m^3 and an average of 0.147 m^3/m^3, indicating the lowest retention among the monitored reforestation techniques.

Individual Collectors:

Individual 1: Monitored from 16th February 2023 to 1st August 2024, achieving the highest maximum moisture level of 0.404 m^3/m^3 and an average of 0.176 m^3/m^3. This suggests that individual collectors were particularly effective at capturing and retaining moisture over an extended period.

Individual 2: Functioned during the same period as Individual 1 but had the lowest maximum moisture level of 0.285 m^3/m^3 and an average of 0.138 m^3/m^3. This variance could be attributed to environmental or placement differences.

In general, Vouzela Cocoon systems demonstrated moderate moisture retention but were more efficient than traditional methods. Traditional systems showed lower average moisture levels overall, indicating less efficient water retention compared to the Cocoon and Individual collector

systems. Individual collectors proved to be the most effective at capturing moisture, particularly in Individual 1, which maintained the highest moisture levels across the monitored period. These findings highlight the efficiency of individual collectors in retaining moisture, followed by the Cocoon systems, with traditional reforestation techniques showing the lowest performance in moisture retention.

4.2.4 Seedlings vigour

4.2.4.1 Physiological state: global results

A total of 436 seedings from the tree species planted during winter 2023 (*Q.robur, Q. pyrenaica* and *Q. suber*) were been marked with plastic tags, in May 2023, and monitored during the project (Table 13). In 2023, the vast majority of the seedlings were in good condition (79%), despite there being a relatively high proportion (11%) affected by the severe drought that happened during spring 2023. Considering those results and the high mortality in spring 2023 of the plantations made in Carregal do Sal during the 2021-22 campaign, emergency waterings were carried out during summer 2023-24, for all the treatments.

Table 13 Number of seedlings monitored per species, treatments and monitoring campaign in Vouzela planting area.

	2023				2024			
	TR	**CCN**	**IFWC**	**TOTAL**	**TR**	**CCN**	**IFWC**	**TOTAL**
Quercus pyrenaica	17	47	34	98	11	42	30	83
Quercus robur	28	55	66	149	24	50	69	143
Quercus suber	72	57	60	189	60	48	52	160
Total	117	159	160	436	95	140	151	386

The good performance of the seedling decreased a bit in 2024, with seedlings that were in good condition decreasing to ~ 75% and dead seedlings increased up to ~25 % (Figure 57). But, in any case, these must be considered excellent results, taking into account the vulnerability of the species planted at the seedling phase.

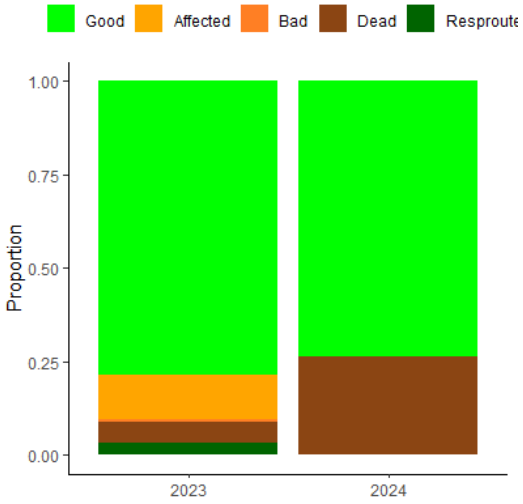


Figure 57 Distribution of the seedlings according to their physiological state per year in Vouzela.

4.2.4.2 Physiological state per planting system

In 2023, CCN and IFWC systems improved seedlings survival compared to TR. In 2024, CCN again highlighted with the lowest mortality and the best survival rates (Figure 58), but IFWC 2024 mortality rates were similar to those of TR. We hypothesise that this could be explained by the characteristics of the plastic net used in IFWC in Portugal plantations, with low density and poor shadowing effect, which probably reduced the effectivity of IFWC compared to those tested in El Pinillo.

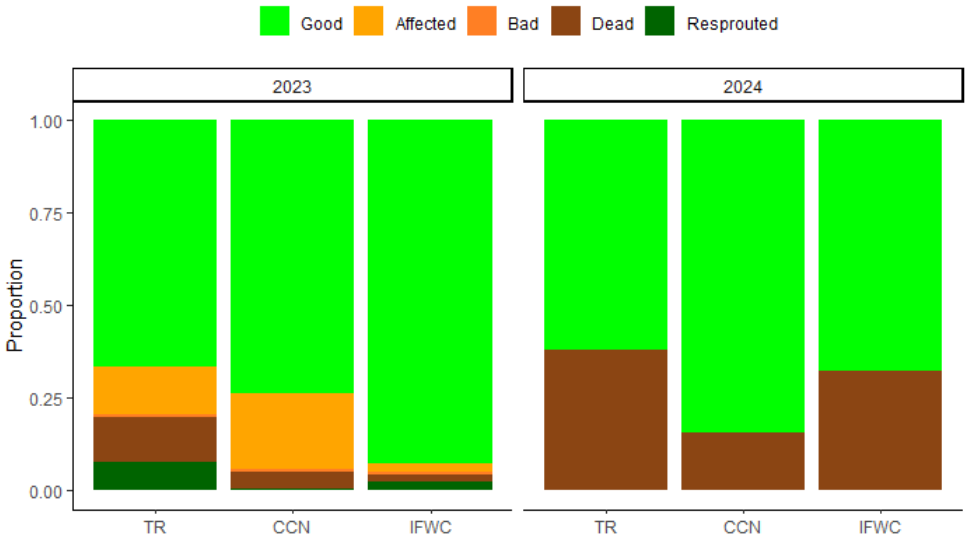


Figure 58 Distribution of the seedlings according to their physiological state per planting system and year in Vouzela.

4.2.4.3 Physiological state per species

When assessing differences among species, we detected that in 2023 *Q. suber* was the most vulnerable species, and the seedlings in the CCN method experienced a higher proportion of physiological stress than other species, with 43% of seedlings being affected (Figure 59). Possibly, this could be due to the installation of new CCN without protective shelters (Figure 60), which may have reduced their protective effect and increased water loss through evapotranspiration. In contrast, by 2024, the CCN system survival rates stabilized among species, and *Q. suber* presented good survival rates under all three treatments (TR, CCN and IFWC), while on the contrary *Q. pyrenaica* and *Q. suber* higher mortality rates were markedly higher under TR than under CNN or IFWC.

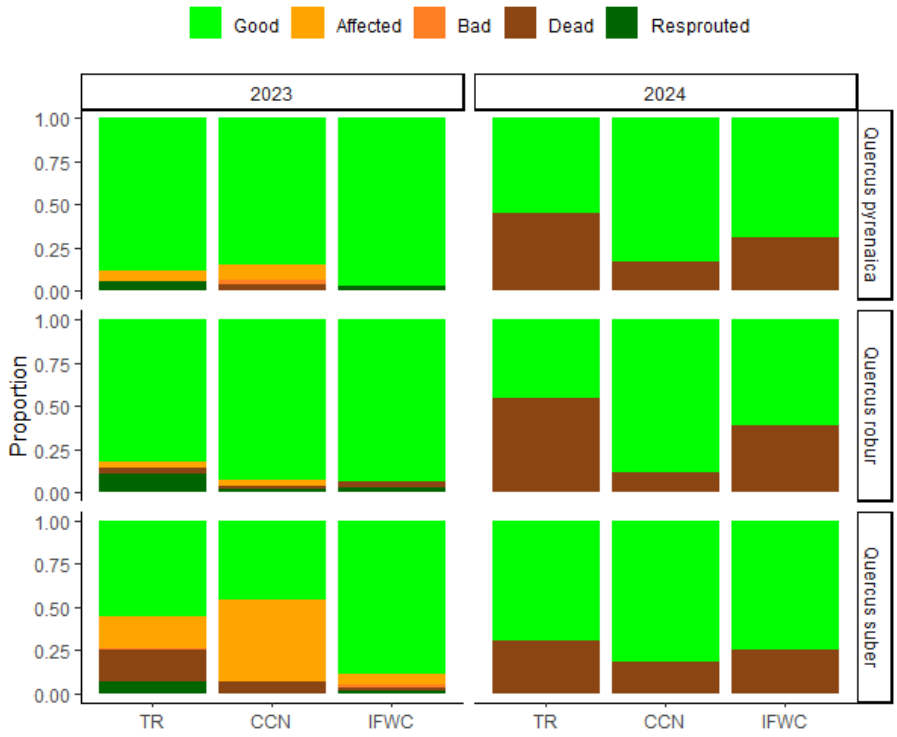


Figure 59 Distribution of the seedlings according to their physiological state per planting system, tree species and year in Vouzela.

Figure 60 Cocoons were installed without the shelter in Vouzela, which reduces the protective effect of the device.

When comparing the survival rates of the IFWC system with and without a plastic net in Vouzela, we did not observe an improvement (Figure 61), unlike the positive effect detected in Carregal (Figure 34). This could be attributed to the less restrictive conditions in Vouzela, which may diminish the net's potential benefits.

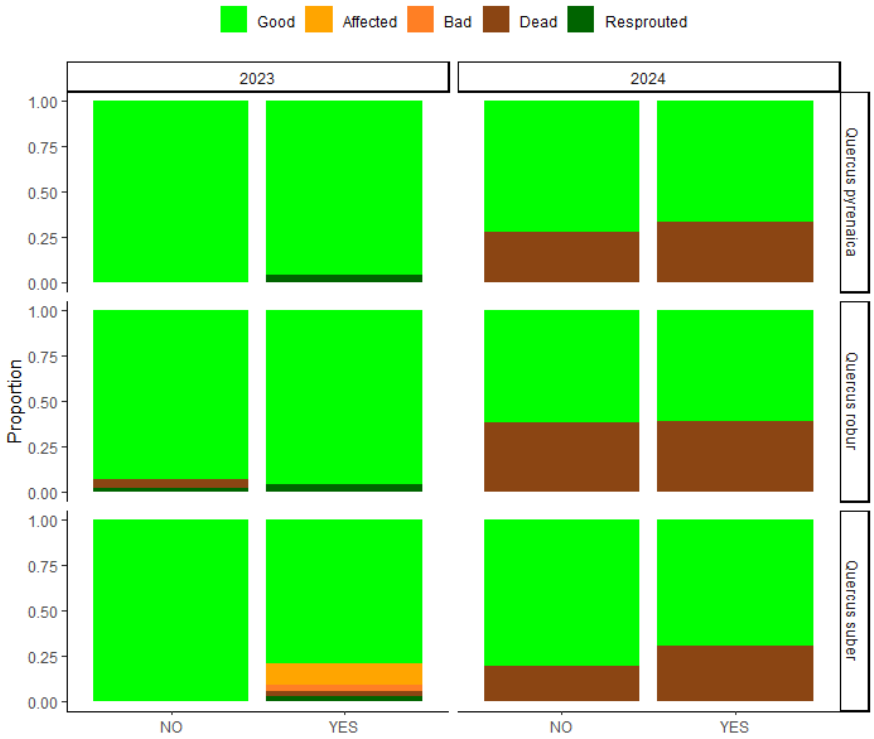


Figure 61 Physiological state of Vouzela seedlings from the IFWC planting method (with and without plastic net) per species and year.

4.2.4.4 Seedlings growth

Unfortunately, the height and width of the TR system was not recorded in Vouzela in 2024 (around a year after planting), and we were unable to assess the effect of different planting systems on height and width in relation to TR. However, similarly to Carregal, our results suggest a tendency for greater height and width in CCN compared to IFWC (Figure 62).

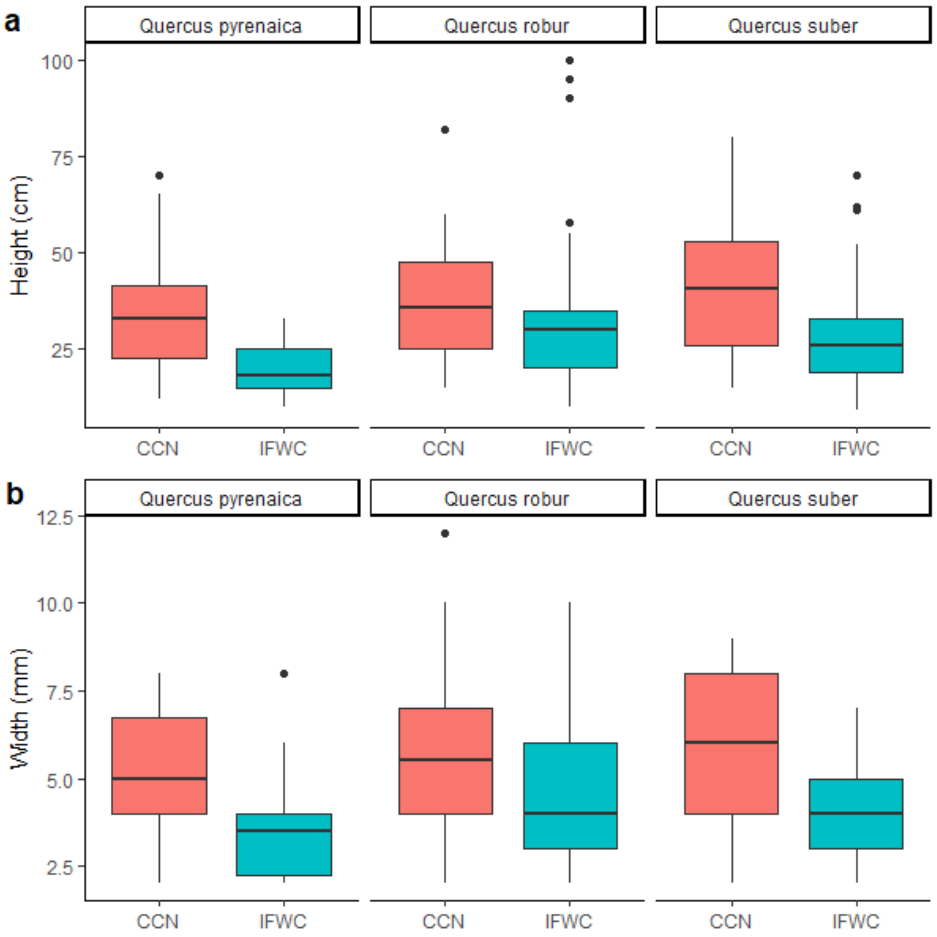


Figure 62 (a) Height, and (b) width of the seedlings per planting system in Vouzela experimental area. 2024 data.

In addition to the height and width measurements, three seedlings form each species and planting system were unburied and process for the determination of shoot and root length and width (see some examples of the process in Figure S1). CCN and IFWC planting systems fairly improved shoot/root growth in Vouzela across the different tree species, increasing shoot and total length compared to TR. On the contrary, differences among treatments on root length were less pronounced (Figure 63).

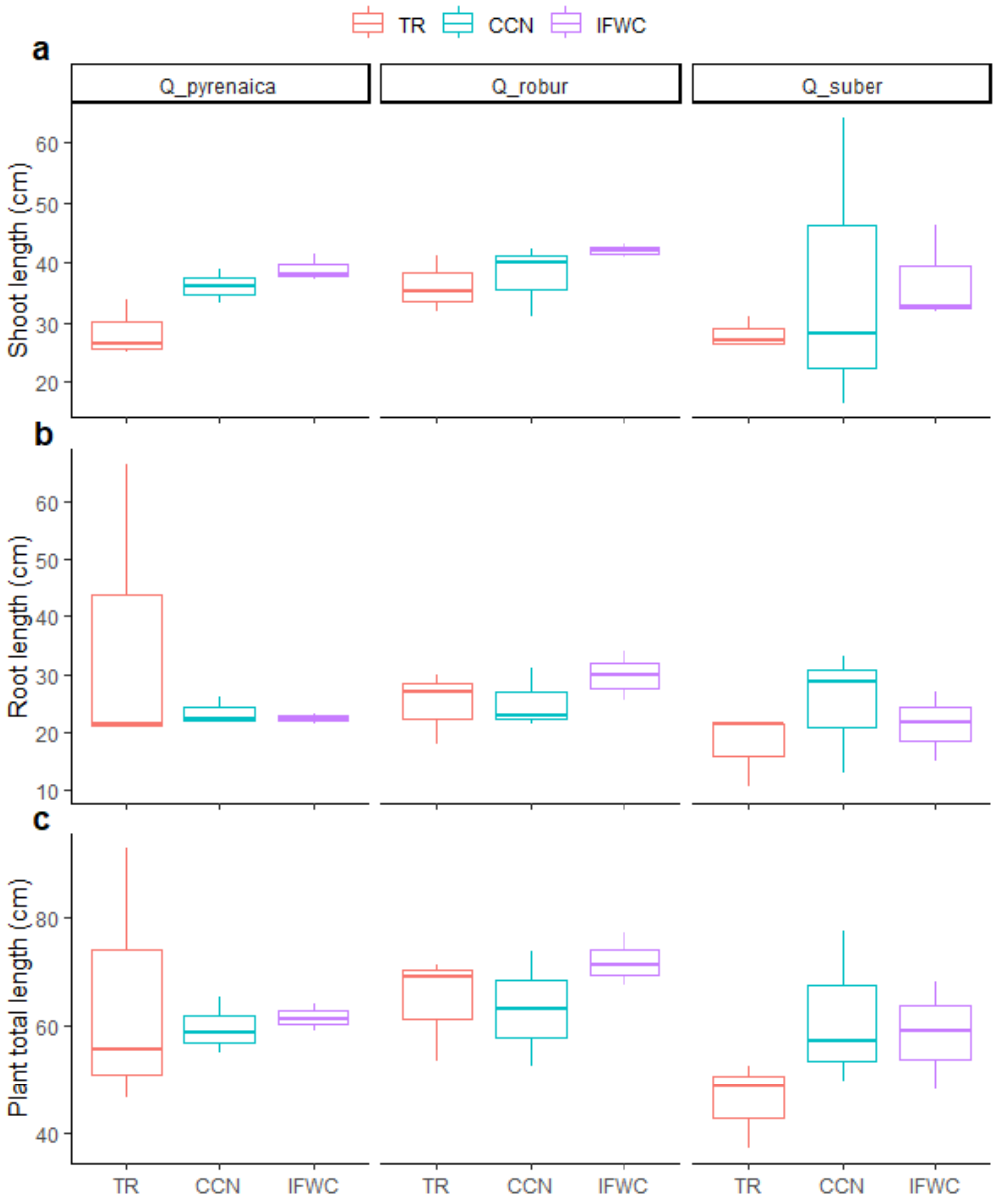


Figure 63 (a) Shoot length, (b) root length, and (c) plant total length per tree species (Q. pyrenaica, Q. robur, and Q. suber) and planting system: traditional (TR), biochar (BC), cocoon (CCN), and individual fog water collectors (IFWC). Boxplots show the interquartile range (IQR: Q1-Q3), whiskers extend to the farthest data points within 1.5 times that IQR, and dots display outliers outside this range.

In terms of weight, the positive effect of innovative planting systems was less notable than on height. The only positive effect was detected on the shoot/root ratio of *Q. suber* (Figure 64), which may suggest that plants in CCN and IFWC treatments could allocate more resources to the aboveground biomass than belowground production compared to TR. This may be indicative of less stressful environmental conditions. However, the small sample size and data dispersion make it difficult to draw robust conclusions.

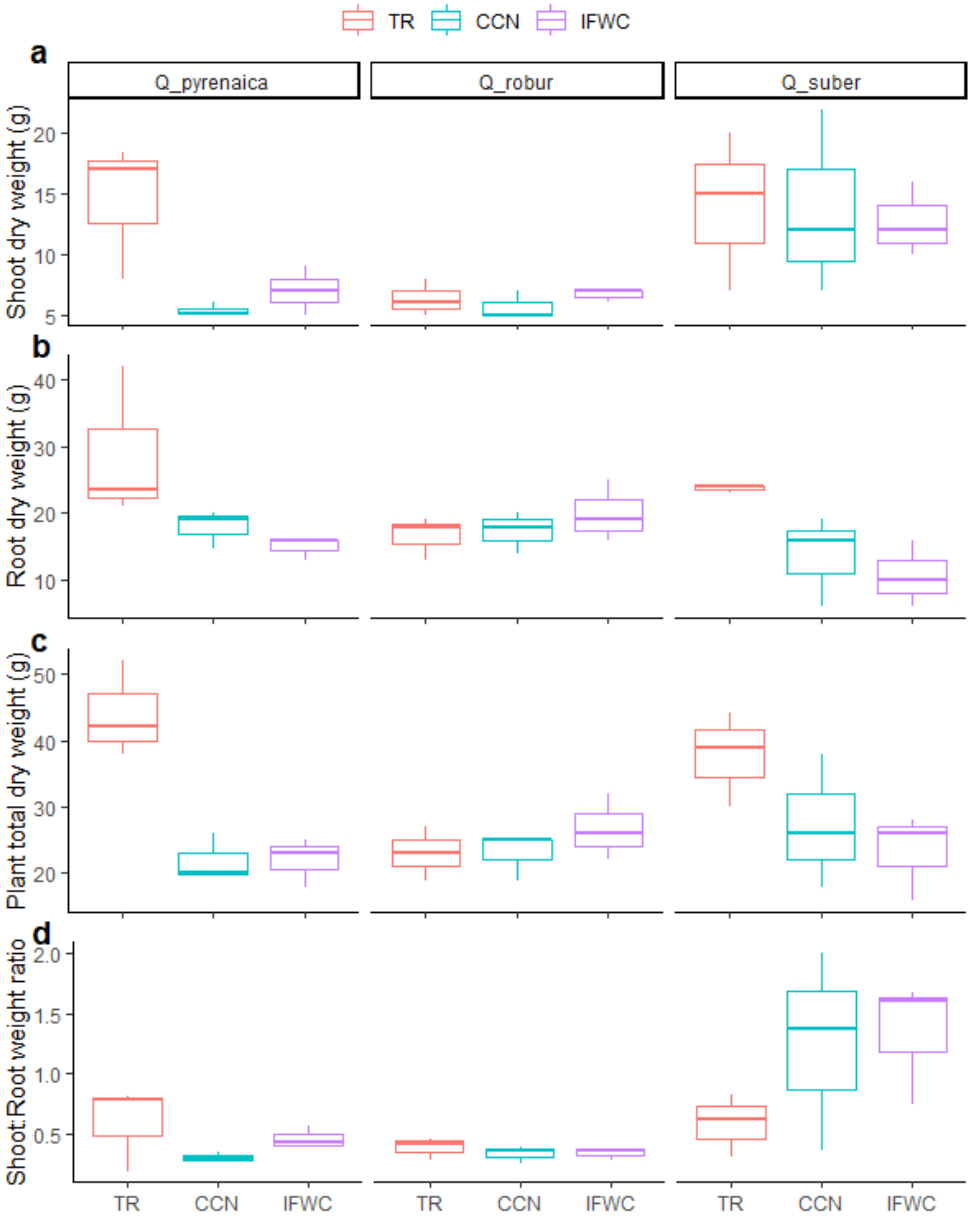


Figure 64 (a) Shoot dry weight, (b) root dry weight, (c) plant total dry weight, and (d) shoot: root dry weight ratio per tree species (Q. pyrenaica, Q. robur, and Q. suber) and planting system: traditional (TR), biochar (BC), cocoon (CCN), and individual fog water collectors (IFWC). Boxplots show the interquartile range (IQR: Q1-Q3), whiskers extend to the farthest data points within 1.5 times that IQR, and dots display outliers outside this range.

4.2.5 Passive restoration

Similar to Carregal do Sal, Sentinel-2 satellite images (10 m/pixel) from 2017, 2019, 2023, and 2024 were analysed in order to assess the passive restoration evolution in Vouzela. Following the forest fire (beginning of 2018) the passive restoration (i.e., natural regeneration) in Vouzela was significant (Figure 65-Figure 66). However, the regenerating vegetation was largely dominated by invasive species, such as eucalyptus (*Eucalyptus globulus*) and acacia (*Acacia melanoxylon*, *Acacia caudata*), which covered the entire area and as show the increase in the shrub cover between 2017 (21.5%) and 2019 (33.4%). For this reason, the planting area needed to be cleared before reforestation started, which reduced shrub cover, from 33.4% in 2019 to 28.5% in 2023 (Table 14). Yet, after clearing eucalyptus, acacia and other invasive species resprouted again (Figure 67-Figure 69) for which hand eradication and localized herbicide treatment were required to control these species. By 2024 the cover of shrubs had increased again, but after all the efforts the presence of invasive species was much lower than before the clearing (Table 15).

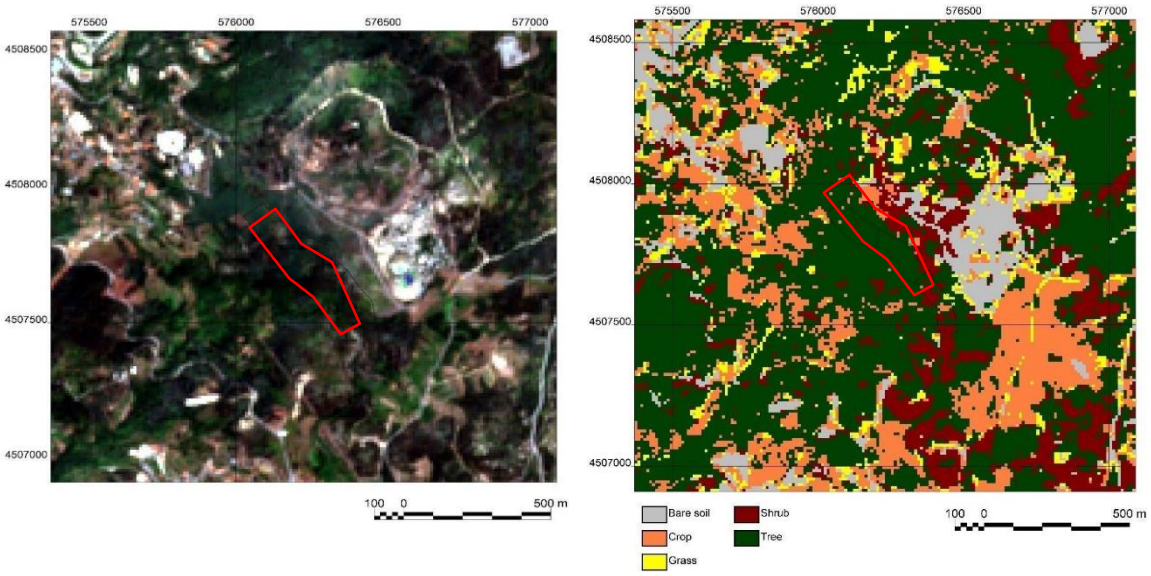


Figure 65 Satellite image of spring 2021 and land cover classification of Vouzela area, using Sentinel-2 images. Tree species mainly composed of eucalyptus and acacia invasive species dominate the planting area (red polygon).

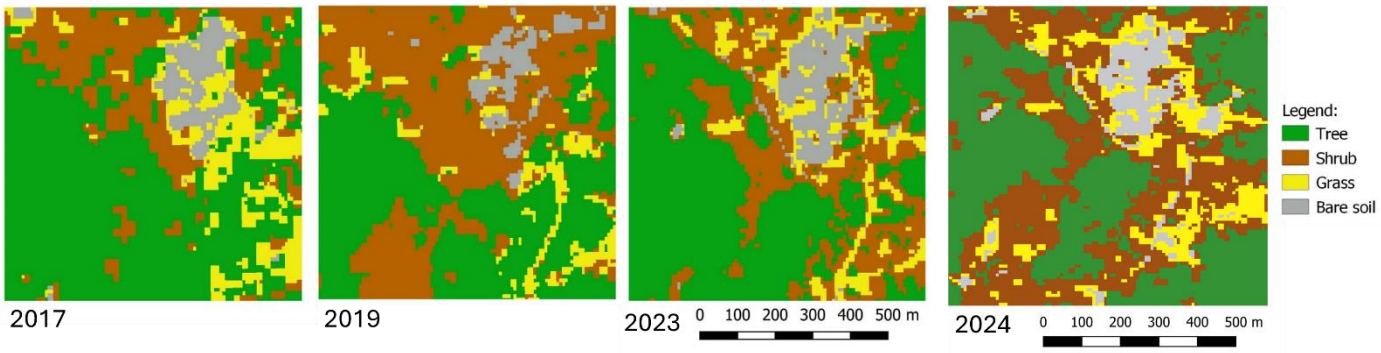


Figure 66 Land cover classification of Vouzela area using Sentinel-2 images, spring 2017-2024.

Table 14. Distribution of land covers maps (Figure 57).

Year	Trees (m^2)	Shrubs (m^2)	Grass (m^2)	Bare soil (m^2)	TOTAL (m^2)	Trees (%)	Shrubs (%)	Grass (%)	Bare soil (%)
2017	874461	296912	146267	62258	1379898	63.4	21.5	10.6	4.5
2019	804393	460477	70506	44522	1379898	58.3	33.4	5.1	3.2
2023	847748	355886	97876	78388	1379898	61.4	25.8	7.1	5.7
2024	651193	512371	123932	92402	1379898	47.2	37.1	9.0	6.7

Thus, after clearing, main invasive plant species were not fully eradicated but noticeably decreased (Table 15) as the result of the weakening of the plant, delaying its sprouting/development. *Eucalyptus globulus* and *Acacia caudata* declined from abundant (4) to low frequency (2). Meanwhile, other alien species, including *Acacia melanoxylon*, and *Phytolacca americana* were maintained at a low (2) to testimonial abundance (1). On the other hand, the abundance of native species remained fairly stable. Note, that these species hold very significant ecological value and are particularly relevant for achieving the target ecosystem.

These results show how regular clearings highlight as a strategic management technique to promote ecosystem stabilization. In particular, cutting or clearing at the end of summer or beginning of autumn is particularly effective, as plants may be under water stress during this time, which helps to reduce their regrowth. In addition, other physical methods used for the control of *Acacia* sp. are the peeling of adult specimens and manual extraction of juveniles, which has been applied with good results in the context of LIFE Ribermine in Lousal.

Table 15 Plant species abundance in Vouzela planting area after clearing in 2023 and 2024. Species abundance is categorized as follows: 1, testimonial; 2, low frequency (<25% soil cover); 3, high frequency (25%-75% soil cover); and 4, dominant (>75% soil cover).

SPECIES	2023	2024
Euclyptus globulus	4	2
Acacia caudata	4	2
Pteridium aquilinum	4	2
Pinus pinaster	3	3
Quercus robur	3	2
Cytisus grandiflorus	3	1
Cistus psilosepalus	2	2
Quercus suber	2	2
Ulex europaeus	2	2
Graminoids	2	3
Acacia melanoxylon	2	2
Rubus albiflorus	2	2
Erica sp.	1	2
Senecio sylvaticus	1	2
Phytolacca americana	1	1
Geranium purpureum	1	1
Centaurea seridis	1	1
Digitalis thapsi	1	1
Juncus conglomeratus	1	1
Jasione montana	1	1
Sesamrides purpurasceus	1	1

Another effective management technique for controlling alien species is grazing, which has been recognized as a suitable strategy in other projects. This approach not only reduces the abundance of invasive species but also promotes the growth of native plants by decreasing resource competition. A notable example is the LIFE PRIMED project, which the CREAF team visited in spring 2024 to explore networking opportunities and learn about planting and management strategies. LIFE PRIMED employs grazing to control *Amorpha fruticosa* in the Nestos Delta, yielding promising results. However, as the project coordinators emphasized, grazing must be restricted in areas with individual plant protectors to prevent damage to seedlings.

Additionally to the land cover classification done by image processing, soil cover types were assessed in the field in 2023 and 2024 (Table 16). Woody species were cleared prior to estimating soil cover in 2024. As a result, the remnants of woody species are classified in 2024 as organic debris. The herbaceous cover notably increased in all treatments, on average from 13 to 38%. However, we did not detect an increase in the herbaceous cover in the treatments IFWC and CCN compared to TR. Rock cover varied slightly between treatments from 2023 to 2024, but it remained consistently low and stable overall, averaging between 6% and 8%. Bare soil cover decreased substantially in 2024 compared to 2023, likely due to the notable increase in herbaceous cover over time and the added input of organic debris.

After clearing, plant remains were left covering the soil surface (Figure 71), which helps reduce soil erosion vulnerability and increase soil organic matter (51% of soil cover by plant remains, 12% of bare soil). However, in some areas plant remains were piled following the slope, which reduces the soil retention potential. In any case, no erosion problems have been detected, despite the soils in the area are vulnerable to erosion due to the sandy texture and low aggregation (see soil characterization section).

Table 16 Soil cover types in transects made in the plantation area of Vouzela after clearing in May 2023 and October/November 2024

Sub-site	Woody species cover (%)		Woody species height (cm)		Herbaceous species cover (%)		Herbaceous species height (cm)		Organic debris (%)		Rocks cover (%)		Vegetation cover (%)		Bare soil cover (%)	
	2023	2024	2023	2024	2023	2024	2023	2024	2023	2024	2023	2024	2023	2024	2023	2024
IFWC	14	0	150	0	16	38	29	4	73	57	2	6	28	38	10	0
IFWC	11	0	20	0	14	34	17	2	45	65	11	0	25	34	19	1
CCN	7	0	32	0	1	52	14	12	56	38	5	11	8	52	14	0
CCN	25	0	25	0	1	46	31	3	51	56	15	0	26	46	8	0
TR	18	0	29	0	15	26	15	3	39	55	13	18	33	26	13	1
TR	40	0	37	0	30	31	14	2	39	68	2	1	31	31	6	0
Average	19	0	49	0	13	38	20	4	51	57	8	6	25	38	12	0

Figure 67 Dense resprouting layer appeared after the fire in the Vouzela planting area.

Figure 68 General view of Vouzela planting area six months after clearing.

Figure 69 (top) Eucalyptus resprouts appeared in the Vouzela planting area after clearing, (bottom-left) Eucalyptus resprouts around a Q. suber seedling planted with Cocoon, (bottom-right) ferns grow after clearing in winter 2023, also pines that germinated after the fire and were not cut during clearing work.

Figure 70 Branches piled following the slope, which reduces the potential for reducing erosion risk.

4.3 El Pinillo

4.3.1 Soil characterization

The plantations made in El Pinillo during 2021-22 occupy and area of around 10 ha, mainly corresponding to steeped slopes (sometimes with a >100% steep), with skeletal (shallow) soils, despite other soil types are present in the abandoned fields and the slopes of the main road of the property. To better analyse the results of the plantations and erosion/instability problems detected in El Pinillo plantations during the 2021-22 campaign, a description of the main soil types existing in the area was done in 2022.

4.3.1.1 Skeletal soils (profile ZE)

Skeletal soils are the predominant ones in the planting areas, corresponding to the south slope of El Pinillo property (Figure 71-Figure 73). These soils have favourable pH (around 7.2) and a relatively high organic carbon content in the first 5 cm (A horizon), jointly with high N and P contents (Table 17). However, those soils are shallow, with a depth below 40 cm, and have a high stoniness (56%) and sandy loam textured, and therefore have low water retention capacity and difficulties for plant roots development, which represents a very challenging situation. Moreover, the steep slope, sometimes >100%, makes the planting process and maintenance. This is also why the physiological state of the seedlings heavily depend on the microclimatic conditions created by the vegetation patches present (*codeso* shrubs, canary pines, and chestnut tree resprouts), that reduce insolation and evapotranspiration. Similarly, the microtopographic conditions, with intercalation of slopes and banks, also positively condition the evolution of the planted seedlings.

Figure 71 Areas with skeletal soils (semi-transparent polygons) in El Pinillo planting area.

Macromorphological description of horizons

A (0 to 5 cm)**. Humidity state**: dry. **Redox state**: oxidised. **Stains**: not observed. **Thick elements**: frequent (28% in weight), basalt, 5 mm diameter, rounded, irregular distribution. **Texture**: sandy loam. **Structure**: weak, granular, 5 mm diameter, soft, friable, plastic. **Organic matter**: high (7.37 %), identifiable debris and humus. **Fauna activity**: not observed. **Roots**: not observed. **Field tests**: null reaction to HCl. **Accumulations**: not observed. **Cementations**: not observed. **Coatings**: not observed. **Lower limit, transition**: no soil discontinuities, plain, net.

B (5 to 25 cm)**. Humidity state:** dry. **Redox state:** oxidised. **Stains:** abundant, 5-10 mm diameter, clear limit, mineral alteration, in the matrix. **Thick elements:** very frequent (56% in weight), basalt, 50 mm diameter, rounded, irregular distribution. **Texture:** sandy loam. **Structure:** massive. **Organic matter:** poor (0.72%). **Fauna activity:** not observed. **Roots:** frequent, fine (2-3 mm). **Field tests:** null reaction to HCl. **Accumulations:** not observed. **Cementations:** not observed. **Coatings**: not observed. **Lower limit, transition**: no soil discontinuities, plain, net.

Table 17 *Physicochemical characterization of the horizons of ZE profile.*

	A (0-5 cm)	B (5-25 cm)
>2 mm (%)	28.2	55.7
Sand (%)	71.6	73.5
Silt (%)	20.7	16.7
Clay (%)	7.8	9.9
Textural class	Sandy loam	Sandy loam
pH H_2O	7.3	7.2
Elect. Cond. (ext. 1:5; dS/m)	0.14	0.071
Carbonates (%)	Undetect. (<3)	Undetect. (<3)
N Kjeldahl (%)	0.42	0.05
P Olsen (mg/kg)	66.5	20.9
Organic C	4.27	0.42
K (mg/kg)	1529	978
Ca (mg/kg)	3870	2154
Mg (mg/kg)	1064	812
Na (mg/kg)	68	63

Figure 72 Profile ZE, characteristic of the skeletic soils of El Pinillo.

Figure 73 General view of the slopes of El Pinillo property where skeletic are mostly represented. This corresponds to the lower area of the IND-1 plantation.

4.3.1.2 Deep volcanic soil (profile IND-1)

According to the sampling campaign made before summer 2022 we observe a high percentage of living seedlings, from different species (*Erica canariensis, Apollonias barbujana, Bencomia caudata, Morella faya*), in the areas with deep volcanic soils (see Preliminary report about plantations monitoring. September 2022). Those soils are found in a small area (1 ha) located mainly in the old fields close to the fog collectors, where mainly individual fog water collectors (IFWC) have been installed (Figure 74-Figure 76). They correspond to deep soils, with more than 1 m depth, which compensates for their moderate stoniness (21-25%) and sandy loam texture in terms of water retention, which is moderate. These soils have an acid pH but have a relatively high fertility (high organic matter, N and P contents, Table 18), as shown by its covering with a dense layer of 1 m high ferns, plausibly adapted to these acidic conditions. The high percentage of living trees in those soils confirms the relatively high-water availability, already explained, and promoted by the flat topography that improves infiltration ratios and minimises surface runoff. Those soils could increase the organic carbon content with the advance of the afforestation process.

Figure 74 Areas with deep volcanic soils (white polygons).

Macromorphological description of horizons:

Oi (-10 to 0 cm). Dry fern leaves not burned.

A1 (0 to 5/10 cm)**. Humidity state:** dry. **Redox state:** oxidised. **Stains:** not observed. **Thick elements:** low (26% in weight), basalt, 5 mm diameter, rounded, irregular distribution. **Texture:** sandy loam. **Structure:** weak, granular, 5 mm diameter, soft, friable, plastic. **Organic matter:** high (5.20 %), identifiable debris and humus. **Fauna activity:** not observed. **Roots:** very frequent, 10 mm, living, from ferns. **Field tests:** null reaction to HCl. **Accumulations:** not observed. **Cementations:** not observed. **Coatings**: not observed. **Lower limit, transition:** no soil discontinuities, plain, gradual.

A2 (5/10 to 60 cm). **Humidity state:** dry. **Redox state:** oxidised. **Stains:** not observed. **Thick elements:** low (22% in weight), basalt, 5 mm diameter, rounded, irregular distribution. **Texture:** sandy loam. **Structure:** weak, granular, 5 mm diameter, soft, friable, plastic. **Organic matter:** high (4.42 %), identifiable debris and humus. **Fauna activity:** galleries. **Roots:** frequent, fine (5 mm). **Field tests:** null reaction to HCl. **Accumulations:** not observed. **Cementations:** not observed. **Coatings**: not observed. **Lower limit, transition:** no soil discontinuities.

Table 18 *Physicochemical characterization of the horizons of IND-1 profile.*

	A1 (0-5/10 cm)	**A2 (5/10-60 cm)**
>2 mm (%)	25.6	21.6
Sand (%)	69.3	65
Silt (%)	26.7	29
Clay (%)	4	6
Textural class	Sandy loam	Sandy loam
pH H_2O	5.1	5.2
Elect. Cond. (ext. 1:5; dS/m)	0.072	0.063
Carbonates (%)	Undetect. (<3)	Undetect. (<3)
N Kjeldahl (%)	0.35	0.25
P Olsen (mg/kg)	105	70
Organic C	3.02	2.56
K (mg/kg)	134	102
Ca (mg/kg)	428	326
Mg (mg/kg)	212	144
Na (mg/kg)	37	30

Figure 75 Profile IND-1, characteristic of the deep soils in higher areas of El Pinillo.

Figure 76 One of the abandoned agricultural fields of the higher area of El Pinillo, where deep soils. This corresponds to the higher area of the IND-1 plantation.

4.3.1.3 Road slopes (profile IND-2)

The areas surrounding the main road of the property along the south slope, are very unstable for corresponding to the dumping of spoils associated with the road construction works (Figure 77-Figure 79). Those materials (A1 and A2 horizons) bury the original soil (2A) and are unstructured materials highly vulnerable to water erosion that slips away easily. A1 and A2 have a proper pH (7) and texture (loamy) but have an excessive stone content (50%) that strongly limit water retention and low organic carbon, N and P contents (Table 19). These last limitations together with their unstructured nature explain why the seedlings plantations carried out on those soils present a high mortality. This is also shown by the poor natural vegetation cover in these soils, dominated by the invasive species *Oxalis pes-caprae* and the sparse presence of some canary pine and chestnut individuals that have resprouted after the fire.

Figure 77 Area with steep unstable road slopes (semi-transparent polygons).

Macromorphological description of horizons

A1 (0 to 15/20 cm). **Humidity state:** dry. **Redox state:** oxidised. **Stains:** not observed. **Thick elements:** very frequent (48% in weight), basalt, 5 mm diameter, rounded, irregular distribution. **Texture:** loamy. **Structure:** moderate, granular, 1-10 mm diameter, soft. **Organic matter:** moderate (2.30 %). **Fauna activity:** not observed. **Roots:** frequent, fine (2-3 mm), living, *Oxalis pes-caprae*. **Field tests:** null reaction to HCl. **Accumulations:** not observed. **Cementations:** not observed. **Coatings**: not observed. **Lower limit, transition:** no soil discontinuities, wavy, gradual.

A2 (15/20 to 35/40 cm). **Humidity state:** dry. **Redox state:** oxidised. **Stains:** abundant, 5mm diameter, clear limit, irregular, oxido-reduction and humic material, in the matrix. **Thick elements:** very frequent (49% in weight), basalt, 10 mm diameter, rounded, irregular distribution. **Texture:** loamy. **Structure:** moderate, granular, 5 mm diameter, soft. **Organic matter:** low (1.79 %), identifiable debris and humus. **Fauna activity:** not observed. **Roots:** frequent, fine (2-3 mm), living, *Pinus canariensis*. **Field tests:** null reaction to HCl. **Accumulations:** abundant, 10 mm, planes, concretions, in the soil matrix, iron oxide. **Cementations:** not observed. **Coatings**: not observed. **Lower limit, transition:** soil discontinuity, wavy, gradual.

2A (35/40 to 50/55 cm). **Humidity state:** wet. **Redox state:** oxidised. **Stains:** not observed. **Thick elements:** frequent (37% by weight), basalt, 5 mm diameter, rounded, irregular distribution. **Texture:** sandy loam. **Structure:** moderate, granular, 5 mm diameter, soft. **Organic matter:** high (3.79 %), identifiable debris and humus. **Fauna activity:** not detected. **Roots:** rare, 5 mm, living, *Pinus canariensis*. **Field tests:** null reaction to HCl. **Accumulations:** abundant, 10 mm, planes, concretions, in the soil matrix, iron oxide. **Cementations:** not observed. **Coatings**: not observed. **Lower limit, transition:** no soil discontinuities, wavy, gradual.

Table 19 Physicochemical characterization of the horizons of IND-2 profile.

	A1 (0-15/20 cm)	**A2 (15/20-35/40 cm)**	**2A (35/40-50/55 cm)**
>2 mm (%)	47.9	49.1	36.8
Sand (%)	50.9	49.2	61.1
Silt (%)	33	34.7	33
Clay (%)	16.1	16	6
Textural class	Loamy	Loamy	Sandy loam
pH H_2O	7.1	6.8	6.6
Elect. Cond. (ext. 1:5; dS/m)	<0.05	0.064	<0.05
Carbonates (%)	undetec. (<3)	undetec. (<3)	undetec. (<3)
N Kjeldahl (%)	0.14	0.11	0.31
P Olsen (mg/kg)	14.9	13.3	17.4
Organic C	1.33	1.04	2.2
K (mg/kg)	297	157	49
Ca (mg/kg)	2943	2675	1556
Mg (mg/kg)	1624	1516	830
Na (mg/kg)	97	128	133

Figure 78 Profile IND-2, representative of unstable road slopes. We can observe the bury 2A horizon (35-40 cm depth), with higher organic matter content than the superior horizons A1 and A2.

Figure 79 General image of IND-2 area, with unstable slopes. We can observe the distribution of the invasive species Oxalis pes-caprae.

To stabilise the road slopes, the application of compost at 25 Mg/ha (dry matter) is recommended. Compost could be applied on the surface without mixing with the soil to reduce soil alteration. In this case, compost will act as mulch, reducing the drop impact.

Additionally, after compost application, sowing with an endemic grass is recommended, to guarantee the soil cover of the soil and reduce the erosion risk. In case no endemic grass seeds are available, fast growing commercial species could be used instead whenever potentially invasive species are not used. In this case, *Hordeum vulgare* or *Avena sp.*, which are planted in most of the agricultural fields in the island, could be used for this purpose. Seeds could be applied by mixing them with the compost or after compost application, spreading directly at the soil surface. Revegetation actions in those areas should help in reducing the fast growing of *Oxalis pes-caprae*, a well-established invasive species. All the actions carried out on those slopes must be done trying to disturb the soil as little as possible. After compost application and sowing, entering at the slopes must be avoided to minimise the impacts on vegetation establishment.

4.3.2 Soil monitoring

4.3.2.1 Physicochemical parameters

Four different areas have been identified in the 2021 planting area according to the different planting treatments (Figure 80). Topsoil (20 cm) of those areas has been sampled taking composite samples (4-5 samples per area), in 2022 and 2024. The soils of El Pinillo present a good aptitude for planting, with a loamy texture, slightly acidic pH, and low conductivity (Table 20). However, the low nutrient and organic matter content in some soils (AFDS1 and CCN1) could represent a limitation for the development of an herbaceous cover. Moreover, Mg levels are relatively high which could affect Ca assimilation in CCN1. However, since no severe degradation processes have been detected in the area and considering the orography and the objectives of the restoration actions, the use of organic and/or calcic amendments is not necessary. In any case, this consideration will be re-evaluated according to the monitoring of the passive restoration parameters (vegetation structure measures). The use of calcium sulphate as liming agent could increase the problems of potassium availability, for which it would be inadvisable (it is a treatment indicated for sodium- affected soils). Regarding the planting process, it could be recommendable adding at the planting hole 0.5 kg of compost, or a stabilised organic amendment (a.e. biochar), to improve plant establishment.

Figure 80 Soil sampling units at El Pinillo restoration area.

Table 20 Physicochemical characterization of the topsoils in 2021 planted areas of El Pinillo property.

	IFWC1	AFDS1	CCN	TR
pH H2O	6.1	6.67	6.92	5.97
EC (dS/m)	0.16	0.08	0.08	0.12
CaCO3 (%)	0.4	0.3	0.3	0.4
Org. Mat (%)	5	2.44	2.73	5.57
Total N (%)	0.26	0.14	0.14	0.28
P (ppm)	39	18	16	15
NO3 (ppm)	80	31	41	75
Sand (%)	39.3	31.8	31	25.9
Lime (%)	29	31.5	30.3	31.8
Clay (%)	31.7	36.7	38.6	42.4
Texture	Loamy-clay	Loamy-clay	Loamy-clay	Loamy-clay
Na (meq)	0.5	0.5	0.7	0.5
K (meq)	0.8	0.3	0.9	0.8
Ca (meq)	8.3	7	17.6	7.1
Mg (meq)	3.8	3.3	11.8	3.5

Among the various soil chemical properties (Figure 81-Figure 83), our results highlight the positive impact of vegetation clearings on soil organic matter and total nitrogen (N), both of which increased significantly over the two-year period (Figure 72). Similarly, electrical conductivity (EC) also increased over time, indicating greater mineralization and nutrient release. However, similarly to Vouzela, the high soil organic matter contents in some samples must be taken with caution as during the sampling process some of the organic horizon fractions could be mixed with mineral fractions. In addition, although we observed some differences between treatments—such as an increase in organic matter in AFDS and CNN compared to a decline in TR—we cannot directly attribute these effects to the planting systems, as soil samples were collected from the broader treatment area rather than from the soil immediately surrounding the seedlings. Future soil sampling in the coming years may clarify the effects associated with planting systems.

On the other hand, it is noteworthy the increase in the available P overtime (P Olsen, Figure 82), possibly as result of several factors. On one hand, the higher P Olsen may be related to the greater organic matter input (Figure 82), as the decomposition of organic matter can release P. Additionally, better-developed vegetation and root system, which increased along time after reforestation, may have allowed plants to access more P from the soil. Furthermore, in recent years, frequent and intense Saharan dust storms (calima) — such as a "super calima" event in December 2024 — have resulted in extremely high concentrations of airborne particles, including significant amounts of phosphorus from phosphate mines, those contributing to increased phosphorus deposition.

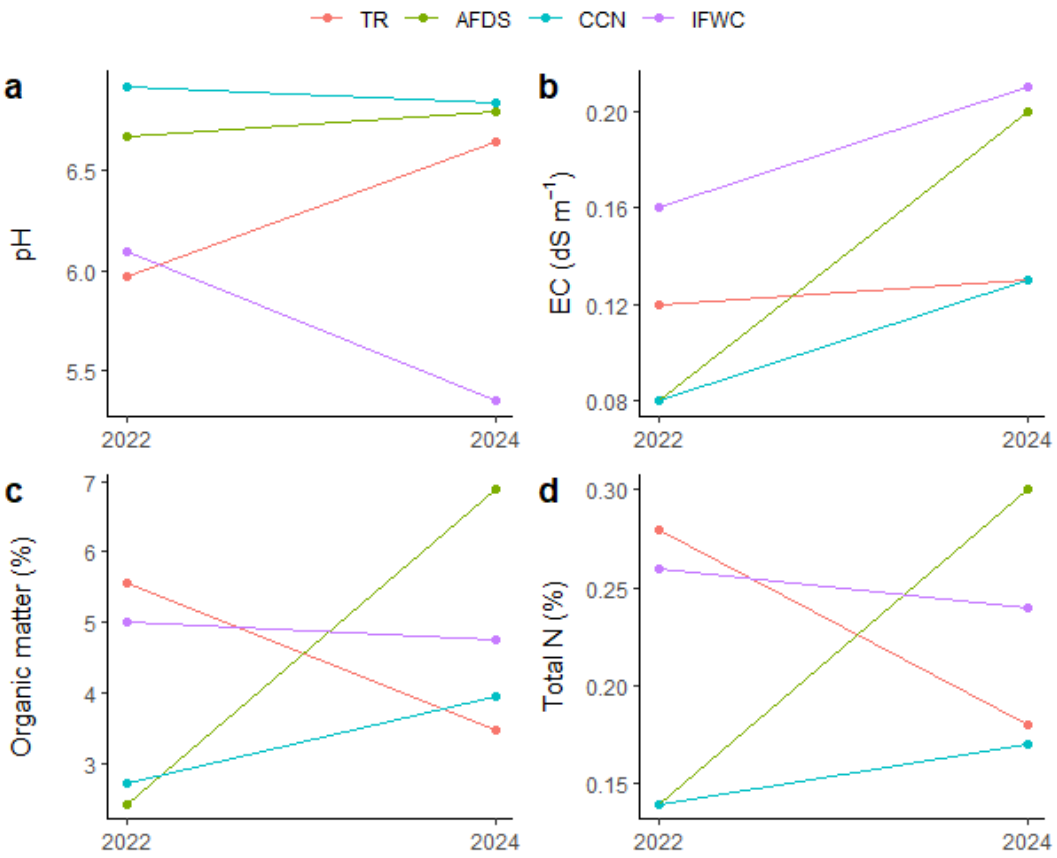


Figure 81 (a) pH, (b) electrical conductivity (EC), (c) organic matter, and (d) total nitrogen per planting system: traditional (TR), autonomous fluid discharge system (AFDS), cocoon (CCN), and individual fog water collectors (IFWC).

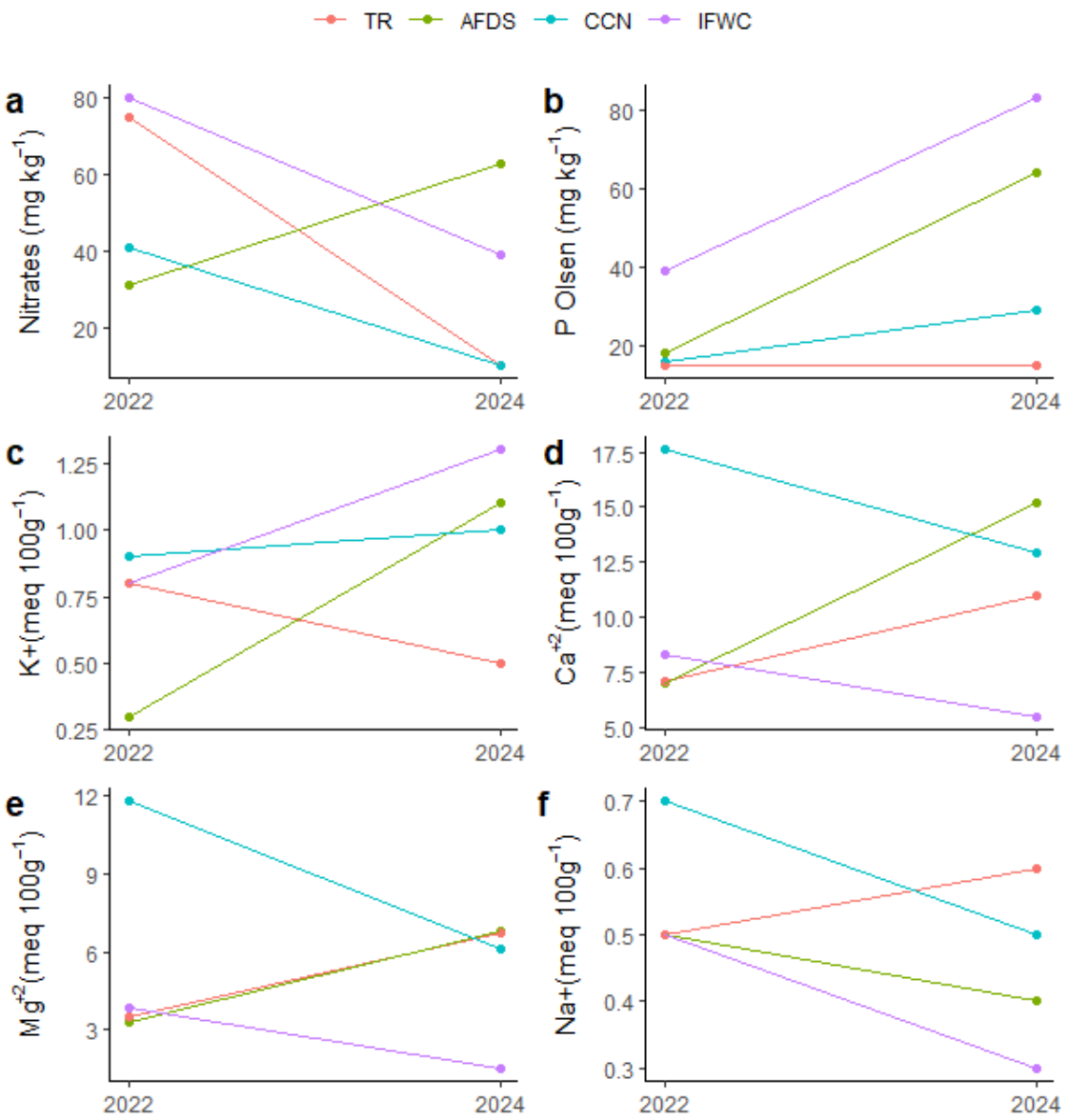


Figure 82 (a) Nitrates, (b) phosphorus (P) Olsen, and soil cations (c) potassium (K+), (d) calcium (Ca2+), (e) magnesium (Mg2+), and (f) sodium (Na+) per planting system: traditional (TR), autonomous fluid discharge system (AFDS), cocoon (CCN), and individual fog water collectors (IFWC).

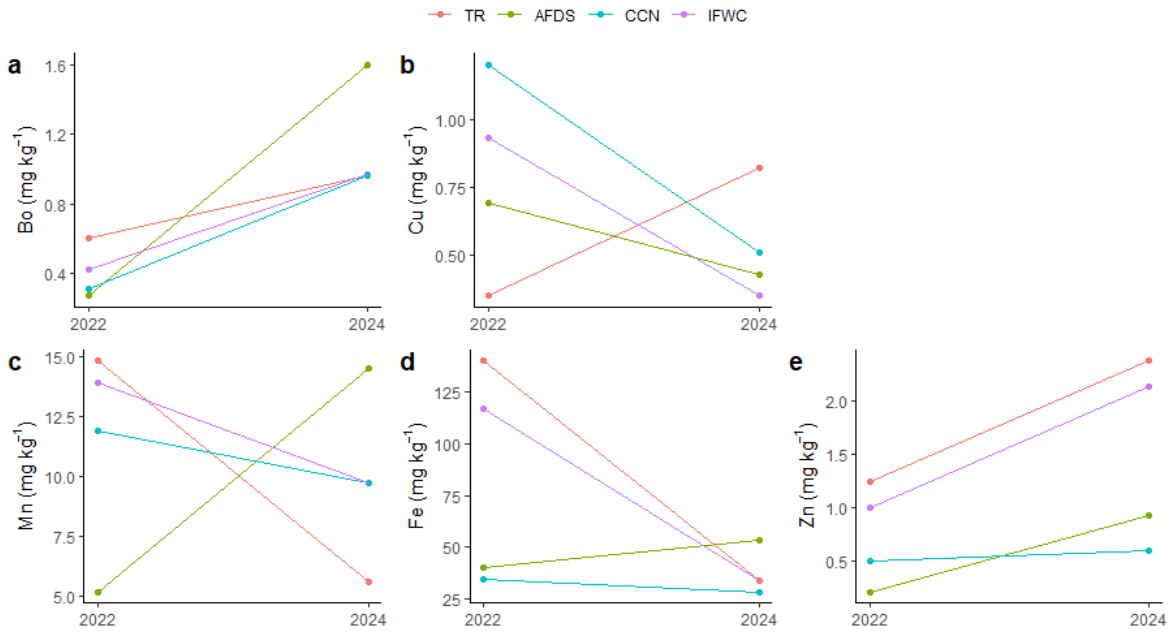


Figure 83 Bioavailable metals (DTPA extract) (a) Boron (Bo), (b) copper (Cu), (c) manganese (Mn), (d) iron (Fe), and (e) zinc (Zn) per planting system: traditional (TR), autonomous fluid discharge system (AFDS), cocoon (CCN), and individual fog water collectors (IFWC).

4.3.3 Soil moisture monitoring and meteorological indicators

The soil moisture sensor data is collected by the datalogger of the meteorological station and transmitted to the manufacturer's cloud. As we stated before, this data is incorporated to an InfluxDB database to be visualised in a Grafana server. This allows us to ensure that soil moisture sensors effectively monitor the impact of precipitation and irrigation on the soil, demonstrating good performance.

4.3.3.1 Reference areas soil moisture sensors

Four reference soil moisture sensors (SMS) were placed in four separate areas to establish a foundational standard for comparison. Next to the meteorological station, one sensor was installed in a shallow soil area without vegetation cover (though in practice, the area is covered with ferns), referred to as *No Grove*. Another was placed in a lightly shaded area under a chestnut tree canopy, named *Chestnut*. A third sensor was installed in a flat area with elm trees, called *Elms*. The last sensor was positioned on a steep slope within a pine forest, designated as *Pines* (Figure 84-Figure 85). The aim of those sensors is to provide a baseline for comparison with other sensors deployed in the surrounding areas and analyse their evolution (Figure 86).

Figure 84 Reference Soil Moisture Sensors map.

Figure 85 (top left) SMS in Chestnut area, (top right) SMS in Elms area, (bottom left) SMS in No Grove area. Finally, ferns covered the area, (bottom right) SMS in Pines area.

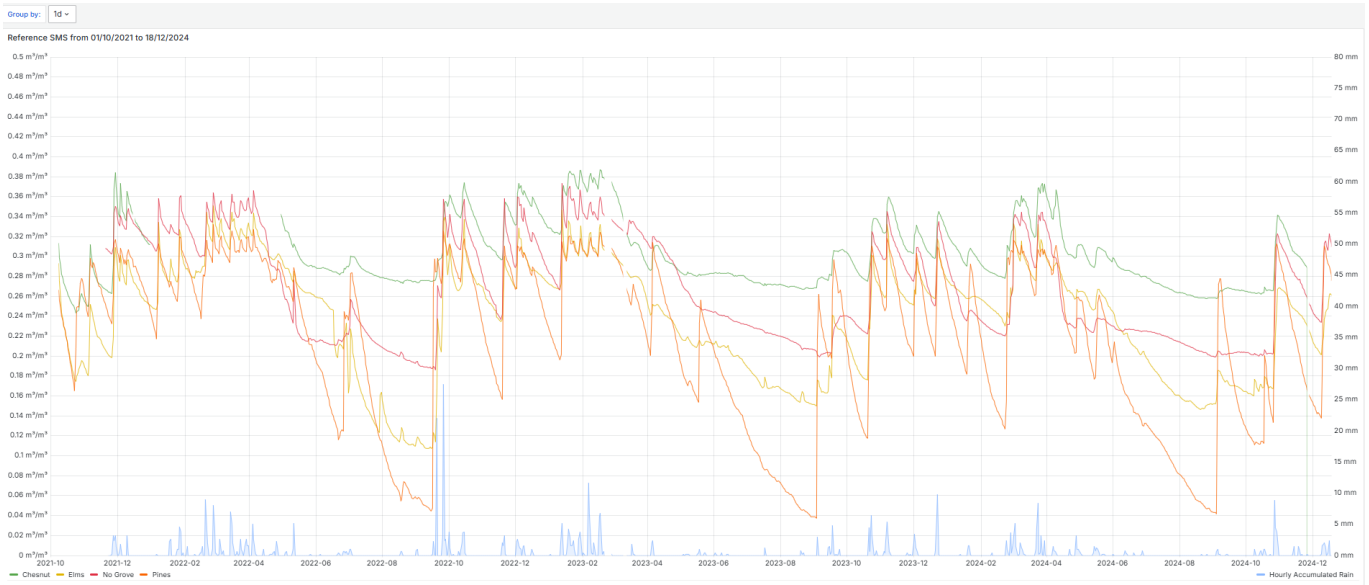


Figure 86 Reference Soil Moisture Sensors graph (lines) and precipitation (blue areas) within the October 2021 – December 2024 period.

4.3.3.2 Reforestation areas soil moisture sensors

To monitor the water stress that plants in different areas and reforestation techniques may be subjected to, at least two sensors were installed for each distinct area (Figure 87). One or two sensors are placed within the trench, near the plant, while another is positioned in the vicinity of the same plant, but at a distance from the irrigation system. These sensors aim to assess the effectiveness of each reforestation method. The areas where the 32 soil moisture sensors where placed are collected in Table 21.

Figure 87 Reforestation areas and initial soil moisture sensors map.

Table 21 Soil moisture sensors in El Pinillo reforestation areas.

Area	Code	Planted / Unplanted area	Serial number	Commission date
Automatic Fluid Discharge Systems Area 1	AFDS03_no_plant	Unplanted	21084778	**16/12/2021**
	AFDS03_plant	Planted	21084780	**16/12/2021**
	AFDS01_plant	Planted	21084781	**16/12/2021**
	AFDS01_no_plant	Unplanted	21084782	**16/12/2021**
	AFDS02_no_plant	Unplanted	21084784 21490960	**16/12/2021-11/05/2023**
	AFDS02_plant	Planted	21084785 21490973	**16/12/2021-11/05/2023**
	AFDS02_plant_bis	Planted	21490966	**26/10/2022**
	AFDS03_plant_bis	Planted	21490967	**26/10/2022**
	AFDS01_plant_bis	Planted	21490968	**26/10/2022**
Automatic Fluid Discharge Systems Area 2	AFDS05_plant	Planted	21084787	**11/05/2023**
	AFDS06_plant	Planted	21084795	**11/05/2023**
	AFDS04_plant	Planted	21084800	**11/05/2023**
	AFDS_area_2_no_plant	Unplanted	21298128	**11/05/2023**
Cocoon	cocoon_plant	Planted	21084786	**21/07/2023**
	cocoon_no_plant	Unplanted	21084796	**21/07/2023**
	cocoon_plant_bis	Planted	21490956	**26/10/2022**
Individual Collectors Area 1	individual_collectors_plant	Planted	21084788	**21/07/2023**
	individual_collectors_no_plant	Unplanted	21084789	**21/07/2023**
	individual_collectors_plant_bis	Planted	21490971	**26/10/2022**
Individual Collectors Area 2	individual_collectors_2_plant	Planted	21490959	**26/10/2022**
	individual_collectors_2_no_plant	Unplanted	21490964	**26/10/2022**
Traditional reforestation	traditional_no_plant	Unplanted	21084790	**21/07/2023**
	traditional_plant	Planted	21298130	**21/07/2023**
	traditional_plant_bis	**Planted**	**21490955**	**26/10/2022**

4.3.3.3 Monitoring soil moisture and meteorological conditions by Quarter

In this section, it is provided an overview of the quarterly monitoring of soil moisture for the project area. The data collected for this section covers from the last quarter of 2021 up to the las quarter of 2024 and provides insight into how soil moisture levels fluctuate during this period (Figure 88-Figure 211). The results of this monitoring are used to inform decisions regarding irrigation, planting, and maintenance, with the goal of determining the performance of different reforestation techniques with respect to soil moisture.

2021 Q4: October to December 2021

The Canary Islands are located within the trade wind belt, where the prevailing winds blow from the northeast year-round. As a result, the islands experience a high frequency of winds from the north and northeast, with occasional northwest winds. It is noteworthy that during this period of station operation, there has been a significant presence of south, southeast, and southwest winds, which are typically associated with greater meteorological instability and episodes of dust haze. These winds can bring warmer and less humid conditions, leading to reduce cloud cover, fog and occasional precipitations.

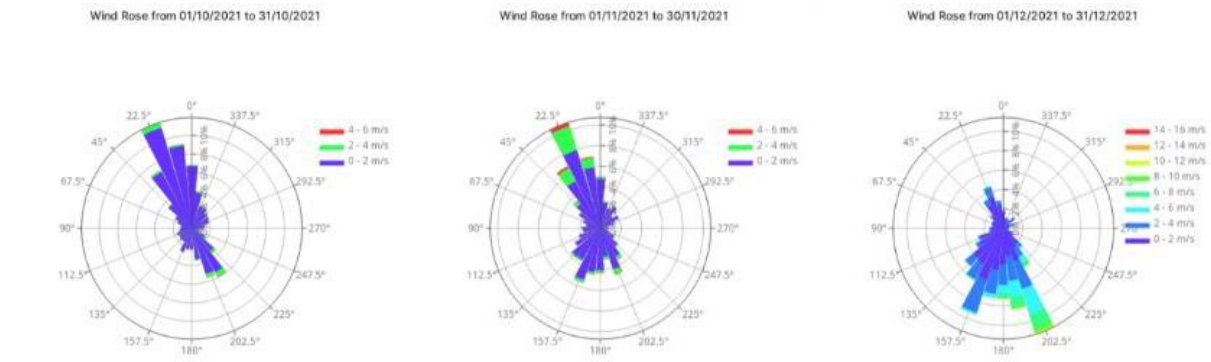


Figure 88 Wind in El Pinillo: October to December 2021 wind rose.

Based on the observation of wind roses during the last quarter of 2021 (Figure 88), we found moderate winds, mainly from the North mainly in the range from 0 to 2 m/s from October to November which is consistent with the dominant wind pattern previously mentioned. However, in December, there was registered a significant increase (mainly in the range from 0 to 6 m/s) in wind speed and a reversal in direction.

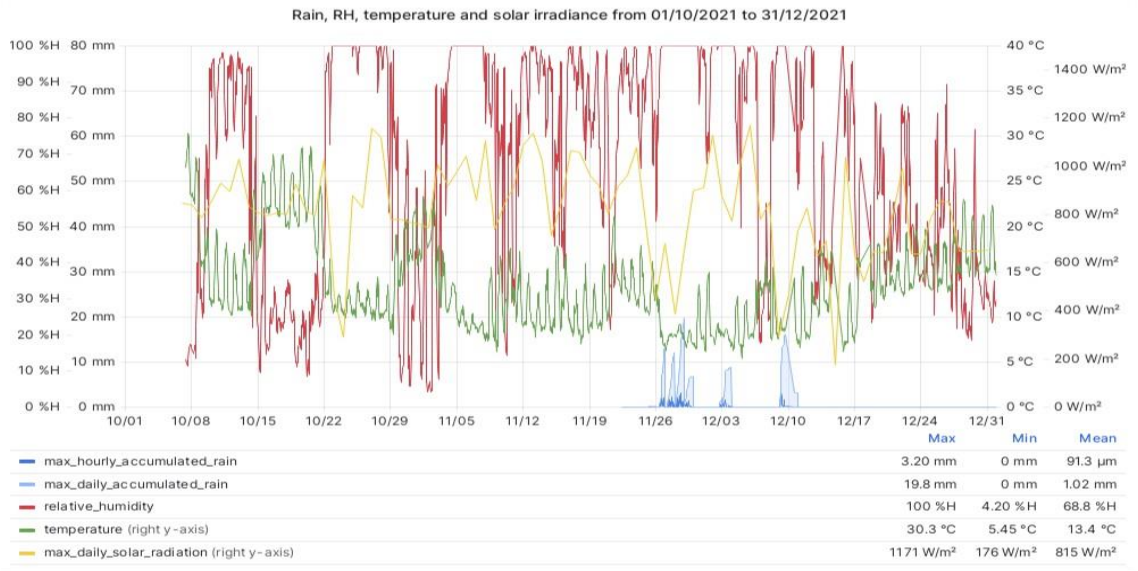


Figure 89 Meteorology in El Pinillo: October to December 2021 precipitation, relative humidity, temperature, and solar irradiance graph.

To enhance the clarity of our analysis, we've omitted the dew point, evapotranspiration and visibility from our graphical representations. The dew point is disregarded because it offers similar insights to those already provided by relative humidity and temperature, thus making it redundant. Furthermore, we have chosen to exclude reference evapotranspiration to showcase it separately in the soil moisture graphs. This approach enables us to spotlight the most pertinent data points, simplifying the visualisation process and easing the interpretation of individual elements and their interrelations.

Finally, visibility is not incorporated due to unforeseen practical complications during data collection. We experienced ongoing issues with the visibility metre installed during part of the project. It appeared that a malfunctioning power supply system was causing the night-time measurements to register as zero. Despite efforts to resolve the issue, including replacing the charge regulator and battery, the problem persisted until March 2024 when the issue was temporarily solved. Given these challenges, which had the potential to affect the precision and reliability of our data, it was deemed prudent to exclude visibility from our analysis until March 2024.

Analysing the main meteorological variables (precipitation, temperature, relative humidity, and solar radiation), we can observe the instability in the month of December 2021, where there was a decrease in relative humidity and an increase in temperature. Throughout the rest of the period, relative humidity generally remained at high levels, and the temperature stayed below

20°C except for two short periods in the middle and end of October 2021. The only rainfall that occurred during this period was recorded over 6 days, spread between November 26th and December 9th 2021, with a total precipitation of 121 mm and an average of 5.94 mm of rain per day.

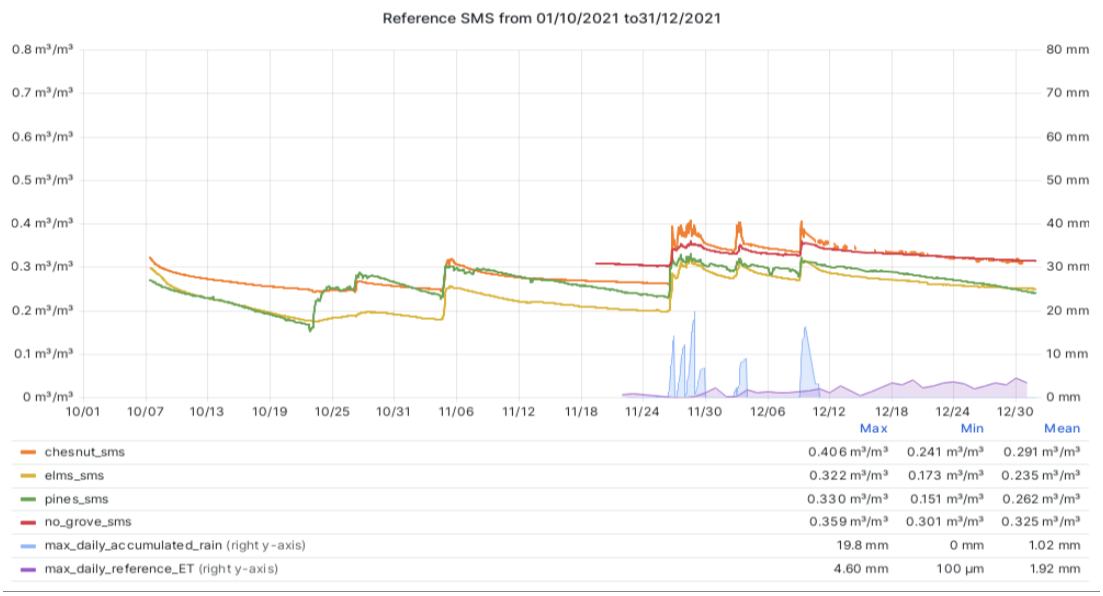


Figure 90 Soil moisture in reference sites (chestnuts, elms and pines) at El Pinillo from October to December 2021.[3]

Four reference soil moisture sensors were installed in three distinct zones with different characteristics: Pines, Elms, and Chestnut. As shown in Figure 90, Pines, Chestnut, and Elms went live on October 7th, while the No Grove zone did it on November 19th. After that, in 16th and 21st of December of 2021 the first set of soil moisture sensors were installed to monitor different reforestation zones and techniques: The first area designated for the use of individual collectors, and the first area designated for the use of Autonomous Fluid Discharge Systems (AFDS01, AFDS02, and AFDS03).

[3] Please note that the color codes of the charts from Figure 90 to Figure 125 are different for the following quarters, starting in April 2023. This is due to a system change in generating the graphs.

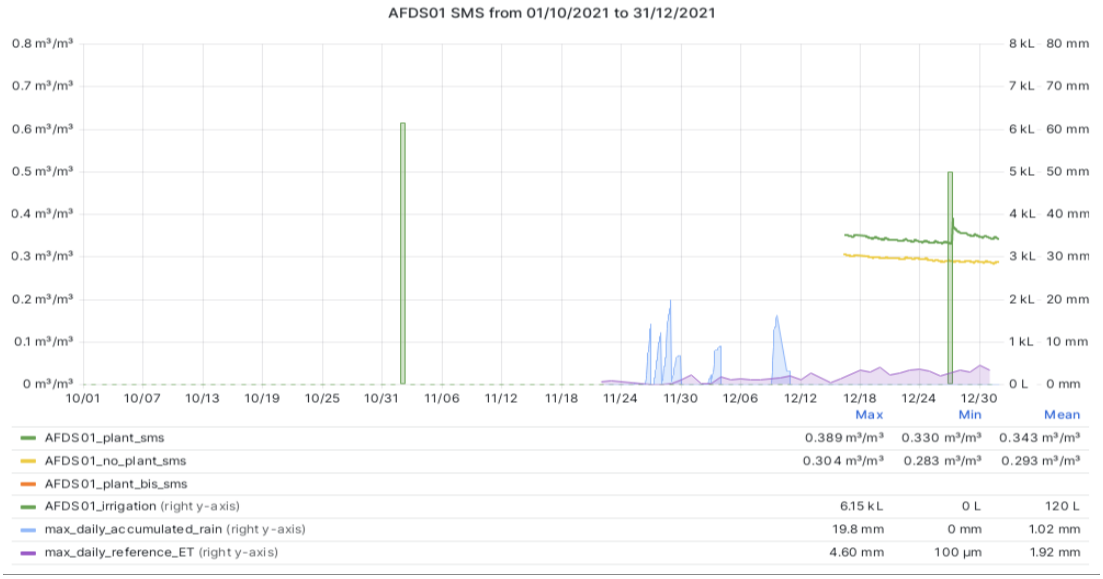


Figure 91 Soil Moisture, rain and evapotranspiration in El Pinillo: October to December 2021, AFDS01 areas. Green bars represent irrigation events.

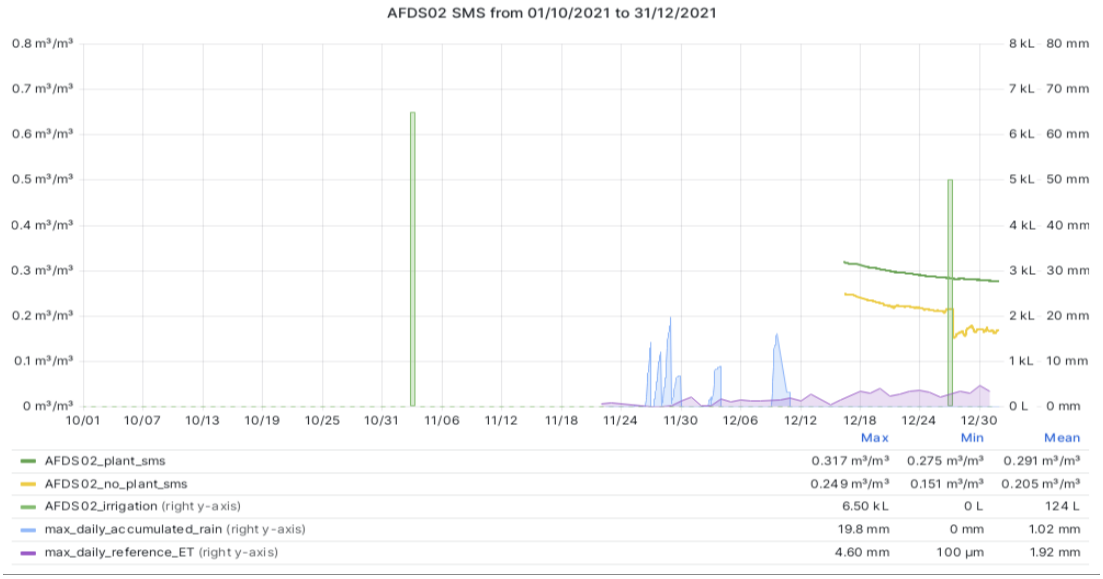


Figure 92 Soil Moisture, rain and evapotranspiration in El Pinillo: October to December 2021, AFDS02 areas. Green bars represent irrigation events

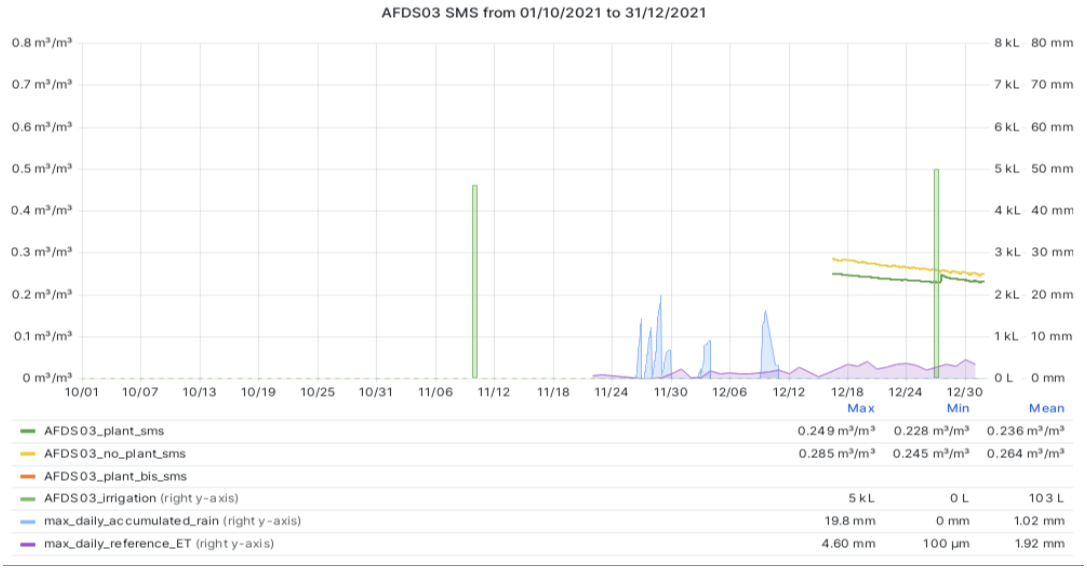


Figure 93 Soil Moisture, rain and evapotranspiration in El Pinillo: October to December 2021, AFDS03 areas. Green bars represent irrigation events.

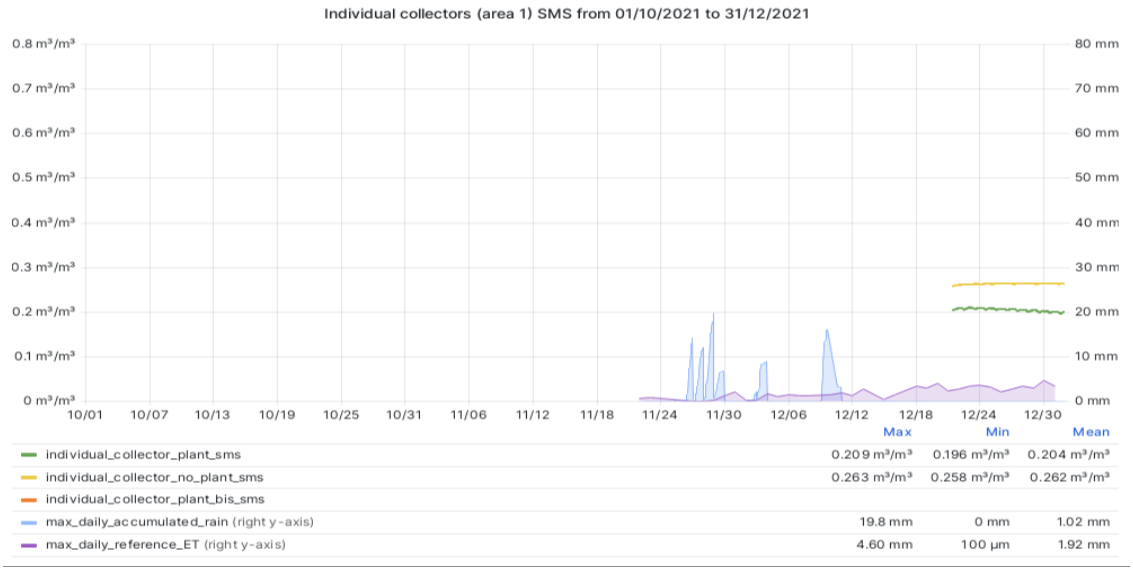


Figure 94 Soil Moisture, rain and evapotranspiration in El Pinillo: October to December 2021, in IFWC sectors.

During this period, we were able to verify the proper functioning of the reference sensors except for the sensor located in the chestnut area that stopped taking readings regularly due to insufficient solar irradiation to charge its batteries, receiving its last communication on December 30th. On December 27th, a manual discharge of each of the three AFDS tanks was carried out. However, the effects were only recorded for AFDS01 and AFDS03, which

necessitates verifying the proper functioning of AFDS02. The precipitation, although concentrated in a short time frame, provided moisture to the soil that can contribute to maintaining overall favourable conditions for the reforestation efforts.

2022 Q1: January to March 2022

In terms of wind, in the first quarter of 2022 (Figure 86) we observed a continuation in January of the trend from December 2021. However, this trend began to reverse in February and returned to the usual conditions of the region by March. Throughout this period, there are more precipitation events, which are mainly concentrated in the second half of the timeframe. These precipitation events are not particularly intense, maintaining an average value of 5.47 mm on the days when rain occurs, like the previous month, but reaching a total precipitation of 335 mm for the period. The temperature generally remained below 20°C (68°F) with an average temperature of 10.3°C (50.5°F), a relatively high average relative humidity (70.7%), and a solar radiation higher than the previous period.

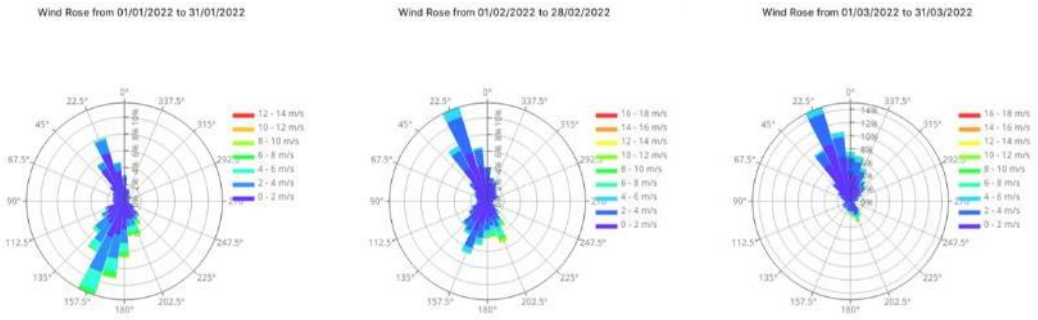


Figure 95 *Wind in El Pinillo: January to March 2022 wind roses.*

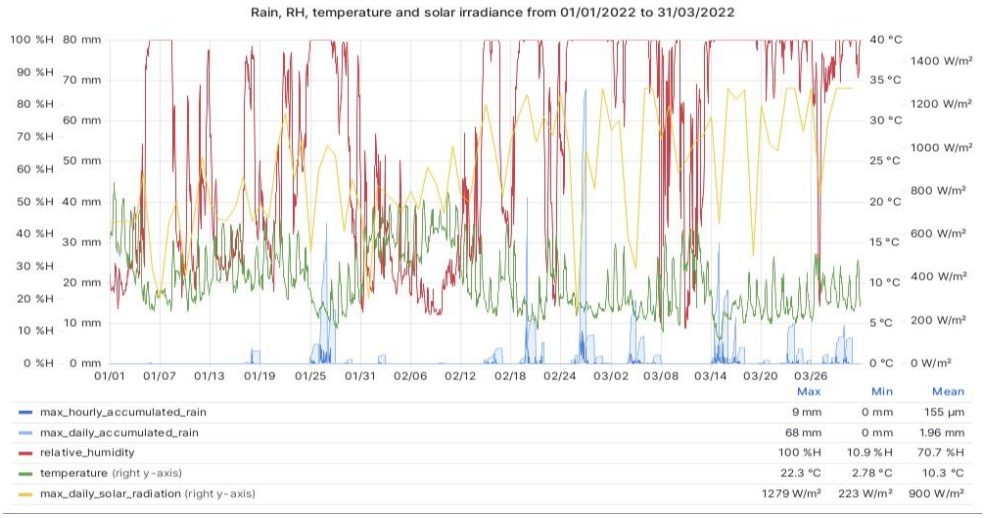


Figure 96 Meteorology in El Pinillo: January to March 2022 precipitation, relative humidity, temperature and solar irradiance graph.

Additionally, it has been observed that periods with the least favourable conditions as higher temperatures, lower humidity percentages, and absence of precipitation (beginning of January and early February) coincide with periods of southerly winds.

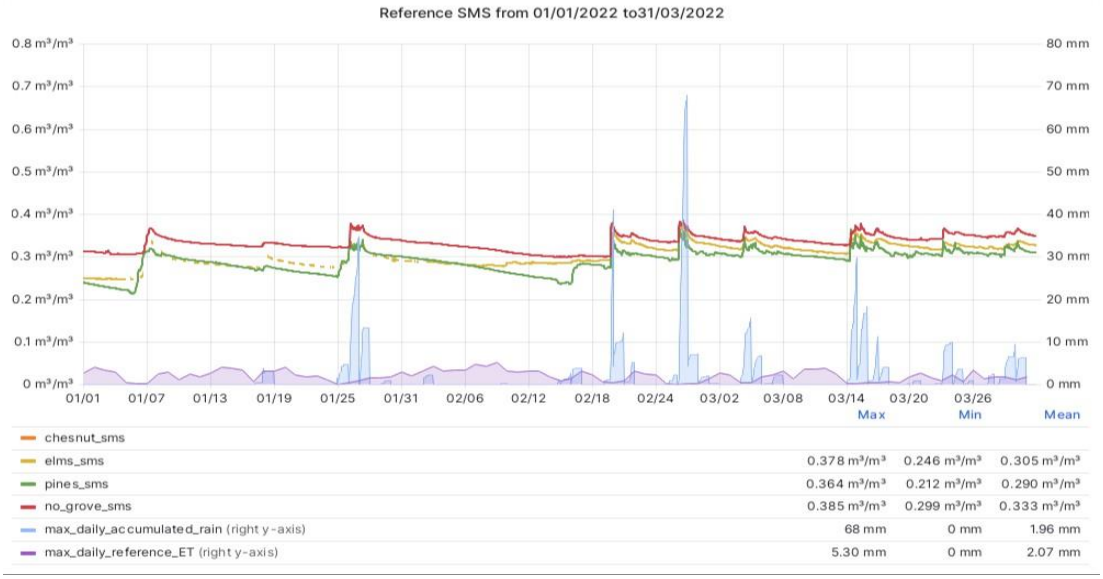


Figure 97 Soil Moisture, rain and evapotranspiration in El Pinillo: January to March 2022, in reference sites.

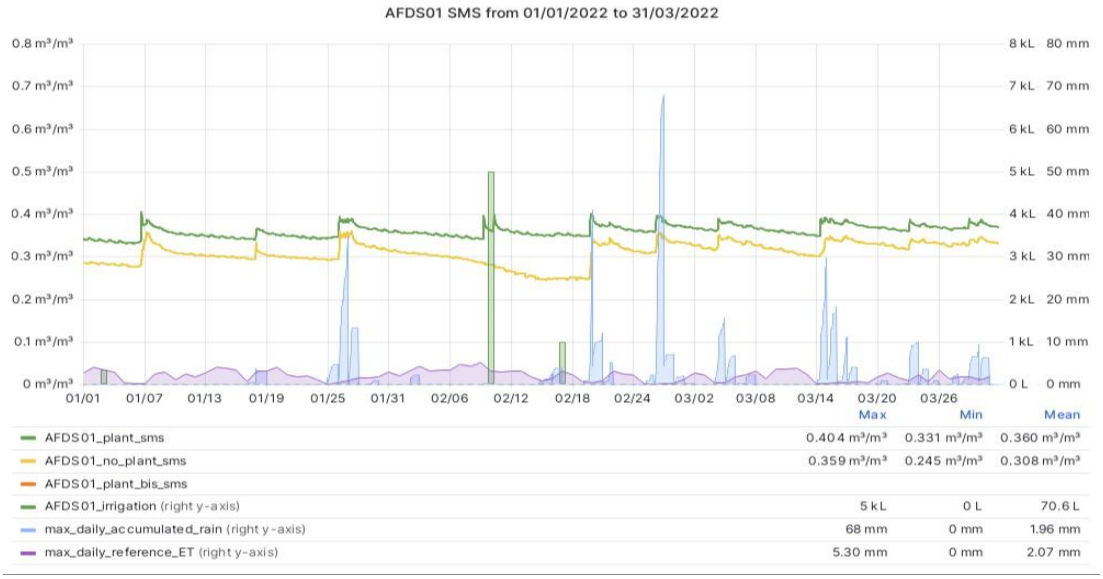


Figure 98 Soil Moisture, rain and evapotranspiration in El Pinillo: January to March 2022, in AFDS01. Green bars represent irrigation events.

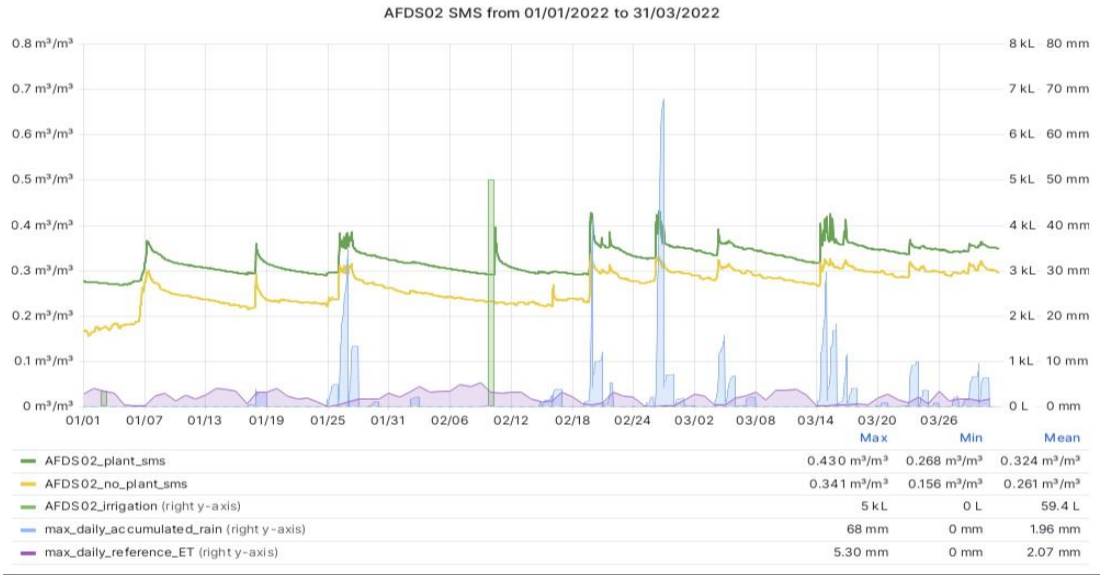


Figure 99 *Soil Moisture, rain and evapotranspiration in El Pinillo: January to March 2022, in AFDS02 sites. Green bars represent irrigation events.*

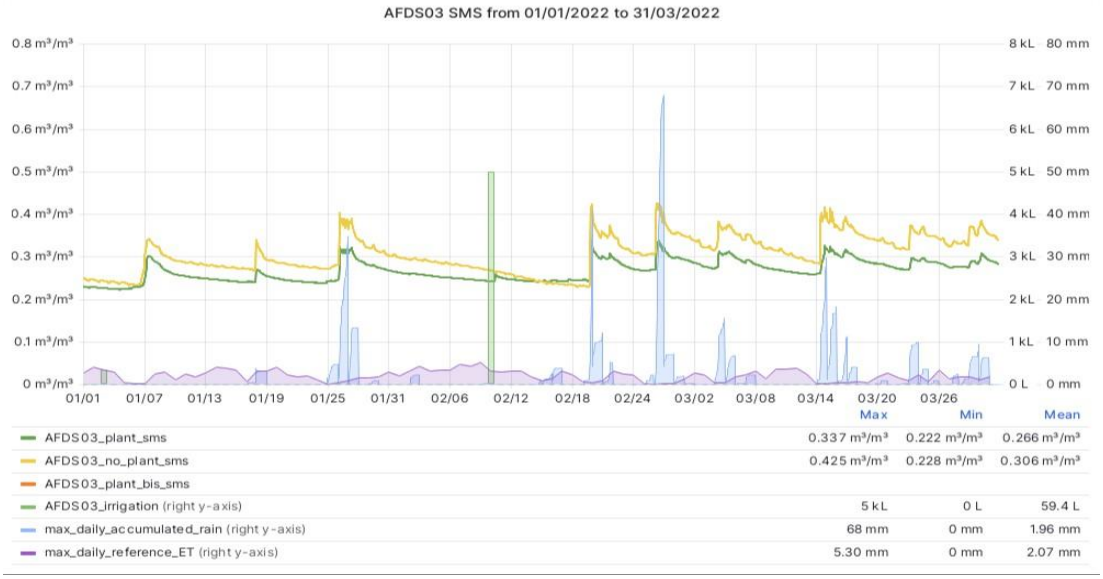


Figure 100 Soil Moisture, rain and evapotranspiration in El Pinillo: January to March 2022, in AFDS03. Green bars represent irrigation events.

During this period, the communication with the chestnut sensor went from being irregular to completely disappearing. Unfortunately, when we went to investigate the issue, we discovered that the sensor had disappeared. Additionally, there were also issues with the battery charging of the sensor located in the elms area. Corrective measures were implemented to ensure it received sufficient irradiation.

During the month of January, in accordance with the planned schedule and considering that weather conditions did not require it, no irrigation took place. However, during this month, the first moisture measurements were obtained, which showed the effects of both light rain (January 18) and moderate rain (January 25 to 27).

It can be observed that during the month of February, soil monitoring continued, and rain events as well as irrigation in the four controlled zones were recorded. The evapotranspiration calculated in this period is relatively low. Additionally, it can be observed that the corrective measures for AFDS03 had an effect, although the increase in moisture readings at the data collection point was not as significant as expected. It is worth noting that, due to the way the individual collector zones are irrigated, even if irrigation occurs and is recorded, it does not necessarily take place directly on the plant where the moisture sensor is installed. This can result in a lack of correspondence.

2022 Q2: April to June 2022

The usual wind pattern with a predominant northern direction was also observed during this quarter, although it exhibited a southern component, particularly during the month of April as can be seen in Figure 78. Regardless, the wind intensity remained mostly in the range of 0 to 2 m/s throughout the period.

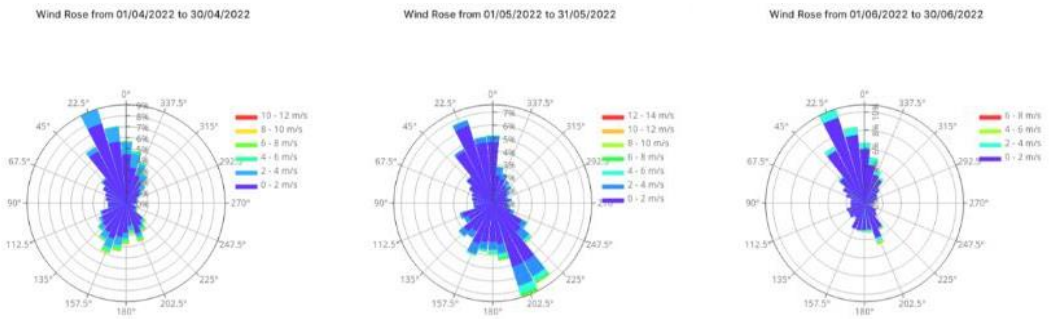


Figure 101 Wind in El Pinillo: April to June 2022 wind roses.

In the second quarter of 2022 it was registered the warmest period so far, with higher irradiance (averaging 1080 W/m^2) and higher temperatures than in previous periods (averaging 14.3°C with a maximum of 28.4°C). Likewise, precipitation is scarce, with a total of 61 mm, and only 34 mm if the first ten days of April are excluded.

The relative humidity remains at an average level of 67.3% but exhibits more variability. This time, the southern component of the wind, in addition to being associated with higher temperatures, is also linked to the 5.2mm of precipitation on May 11th, which lasted only an hour during the early morning hours.

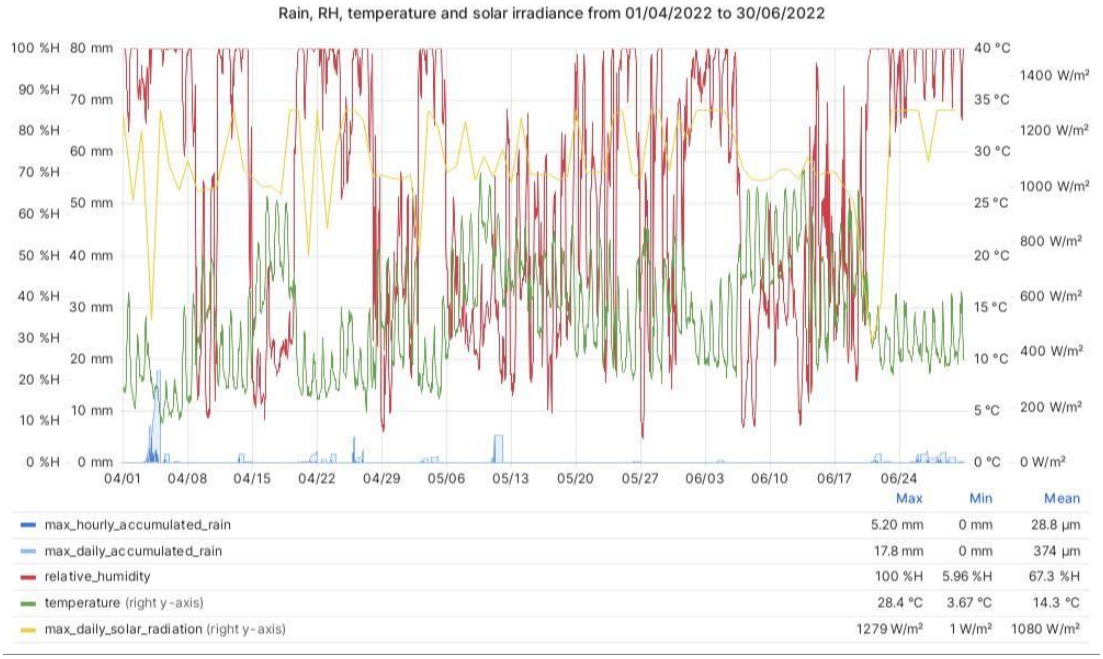


Figure 102 Meteorology in El Pinillo: April to June 2022 precipitation, relative humidity, temperature, and solar irradiance graph.

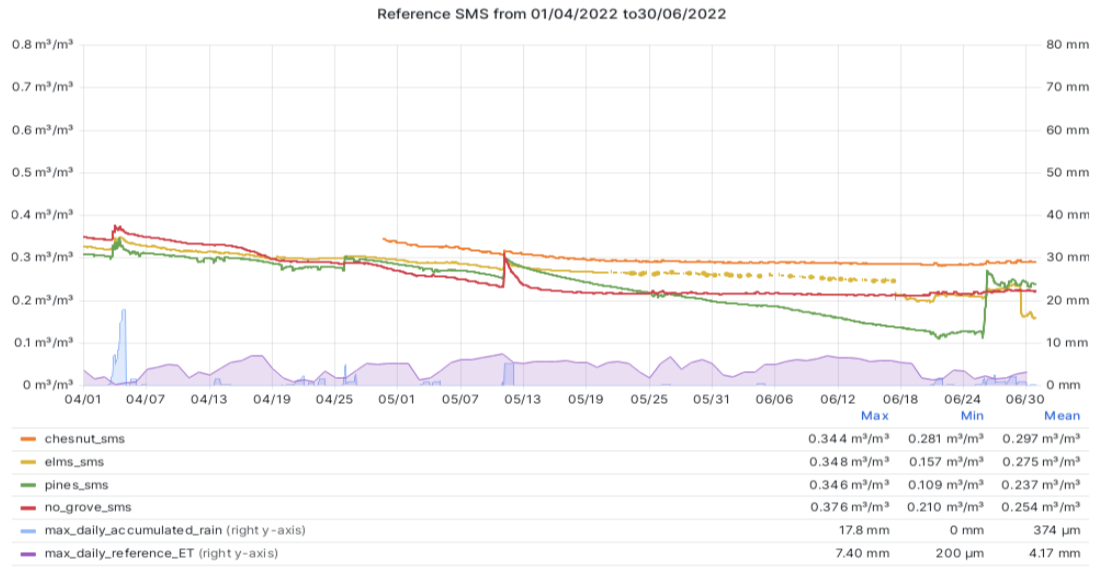


Figure 103 Soil Moisture, rain and evapotranspiration in El Pinillo: April to June 2022, in reference sites.

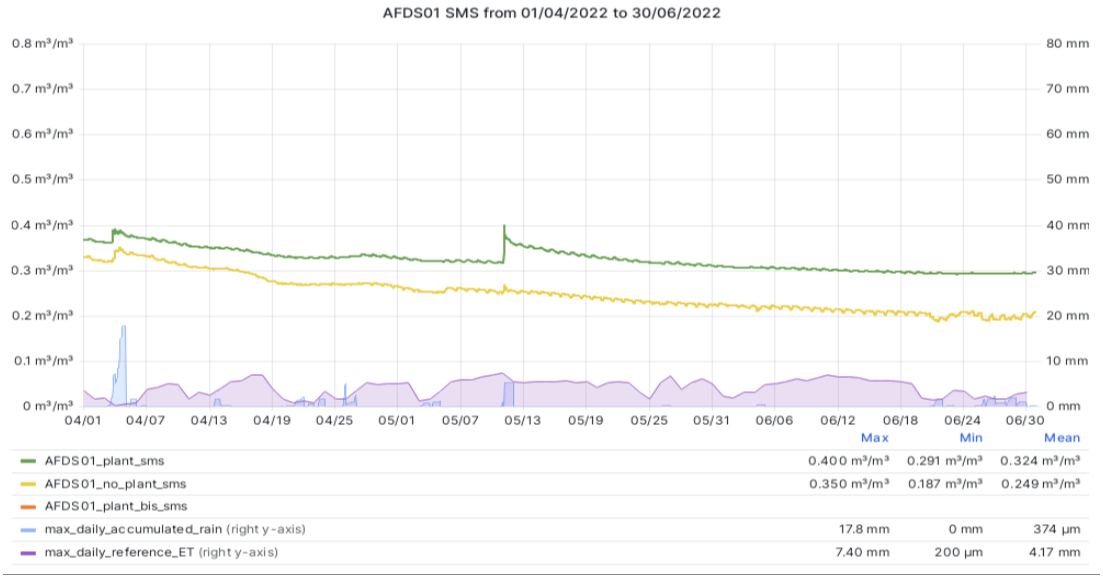


Figure 104 Soil Moisture, rain and evapotranspiration in El Pinillo: April to June 2022, in AFDS01 sites

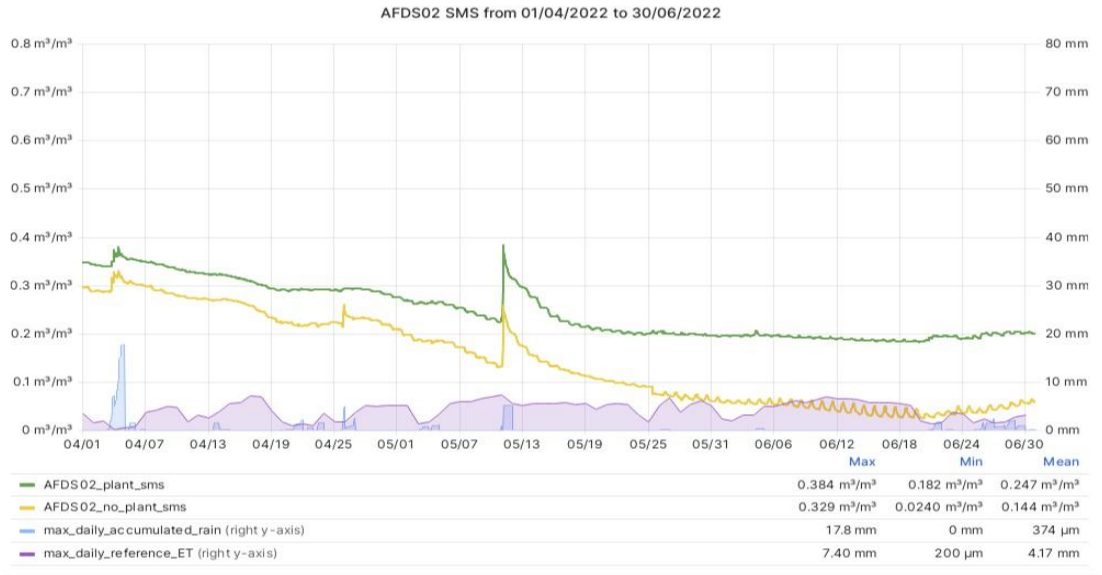


Figure 105 Soil Moisture, rain and evapotranspiration in El Pinillo: April to June 2022, in AFDS02 sites.

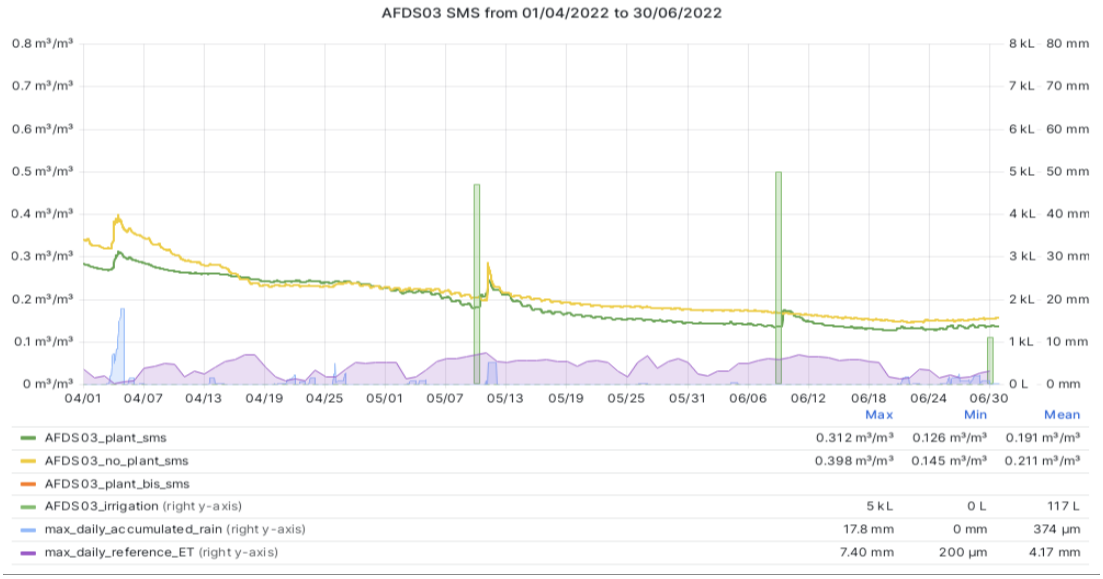


Figure 106 Soil Moisture, rain and evapotranspiration in El Pinillo: April to June 2022, in AFDS03 sites. Green bars represent irrigation events.

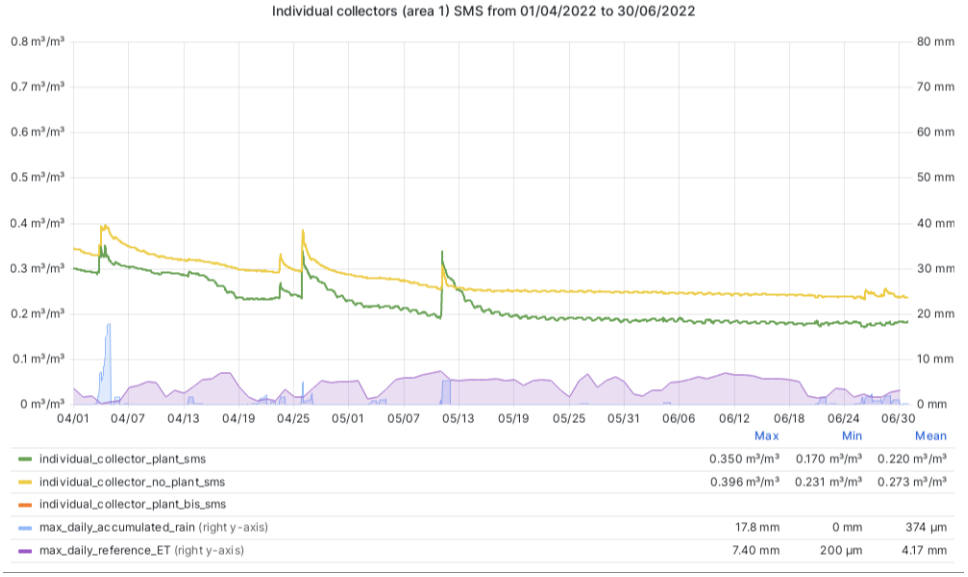


Figure 107 Soil Moisture, rain and evapotranspiration in El Pinillo: April to June 2022, in IFWC sites.

On April 29th, the moisture sensor that was lost from the chestnut's reference area was replaced. During the April to June period, the trend of soil moisture, regardless of the monitored area, is clearly decreasing as the evapotranspiration is higher during this time interval. It only happens three specific days in which slight precipitation provides some relief for the soil. However, this does not manage to fully restore the lost moisture during the period.

A notable observation from the reference sensors is that the light drizzle recorded at the end of June led to a significant increase in soil moisture in the pine forest. As for the other areas, the AFDS03 zone received two full discharges and one partial discharge, which, like the rain, temporarily halted the decreasing trend but did not succeed in restoring the soil to previous moisture levels.

2022 Q3: July to September 2022

In the third quarter of 2022, we observed a more unstable wind direction, which fluctuated between the predominant North-South direction and its opposite. This directional instability in the wind is accompanied by a wind speed that did not exceed 2 m/s in most cases.

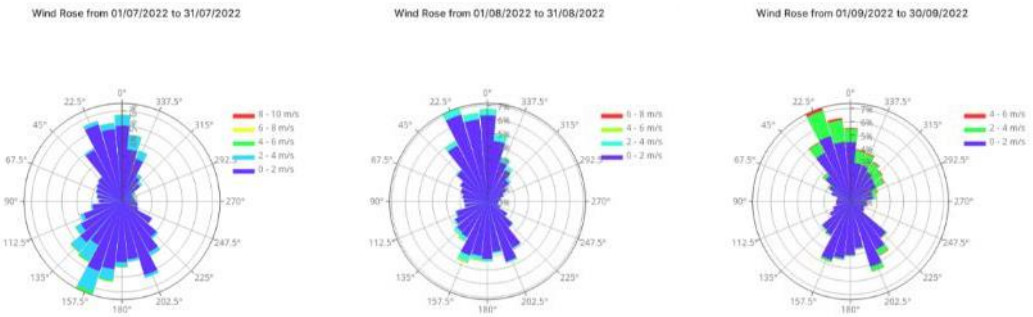


Figure 108 Wind in El Pinillo: July to September 2022 wind roses.

Regarding the rest of the meteorological variables, there was a rise to summer-like temperatures compared to the previous period. Specifically, in the months of July and August, the maximum temperature was 36.5°C, with an average temperature of 23.3°C. There were fewer periods of high relative humidity, with an average relative humidity below 40% and a total precipitation of 15mm. In contrast, in September, the trend reversed, especially in the latter half of the month. In this fortnight an average relative humidity of 89.5%, a total precipitation of 215mm, and an average temperature of 16°C was registered. This was mainly due to a tropical cyclone phenomenon between September 24 and 26, Storm Hermine, a rare occurrence in the area, which resulted in a single-day precipitation of 128 mm on September 25th.

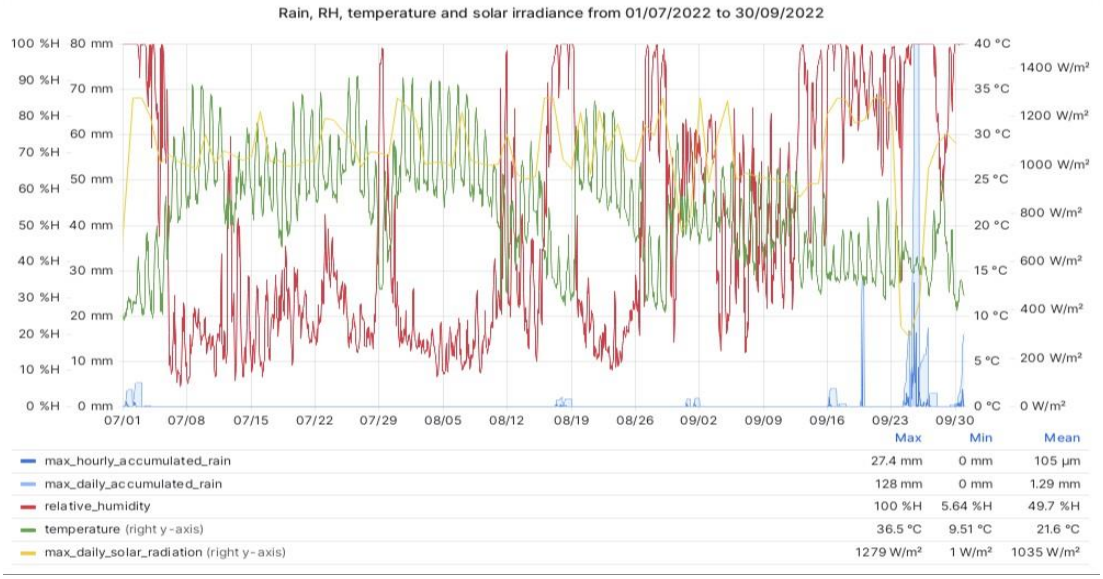


Figure 109. Meteorology in El Pinillo: July to September 2022 precipitation, relative humidity, temperature and solar irradiance graph.

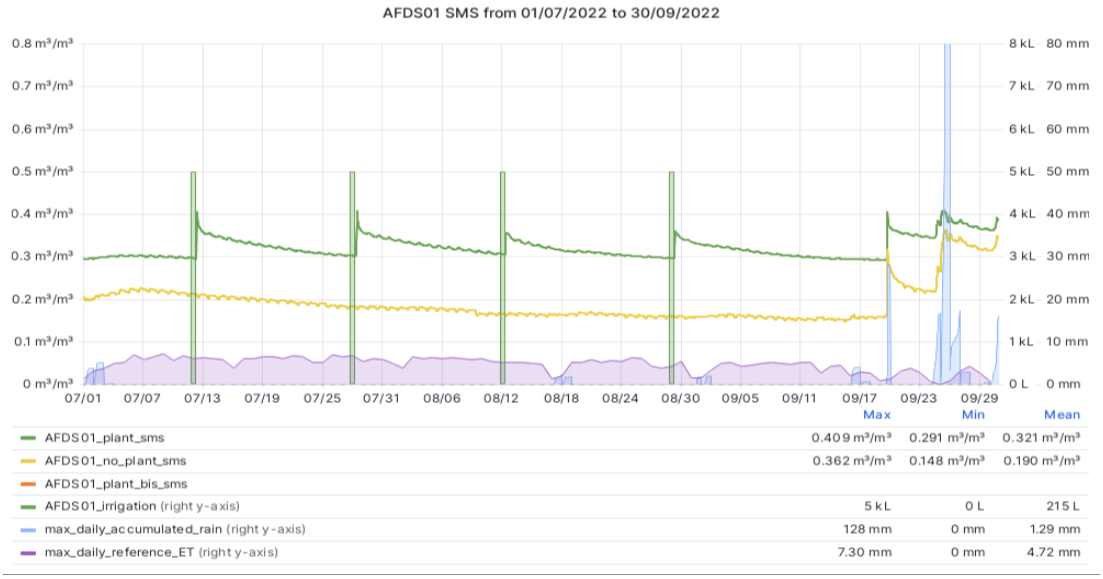


Figure 110 Soil Moisture, rain and evapotranspiration in El Pinillo: July to September 2022, in AFDS01 sites. Green bars represent irrigation events.

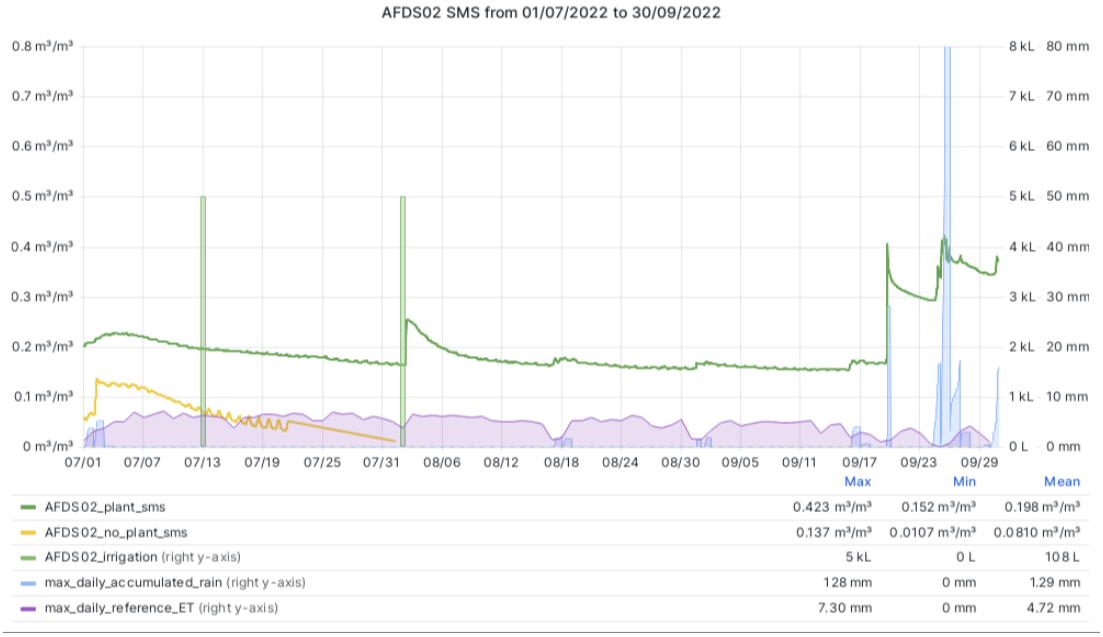


Figure 111 Soil Moisture, rain and evapotranspiration in El Pinillo: July to September 2022, in AFDS02 sites. Green bars represent irrigation events.

During this period, it is important to note the activation of the soil moisture sensors in both the traditional reforestation area and the cocoon area on July 21st. It should be emphasised that, as with individual collectors, the recorded irrigation does not necessarily imply that the trench where the sensors are installed are receiving this water. On August 1st it received the last communication from "AFDS02_no_plant_sms" sensor and during subsequent visits to the work area, due to the growth of weeds, it was not possible to locate the device to check the problem.

In this period, soil moisture is low for all installed sensors and evapotranspiration remained high (until mid-September), with the terrain undergoing the most severe water stress of the year. All AFDS underwent support discharges, four for AFDS01 and AFDS03, and only two for AFDS02. The readings from AFDS01 showed that the quantity and timing of the discharges were able to raise the soil moisture level so that the next discharge occurred when the minimum levels were reached again. However, this timing was not based on the soil moisture readings but on the experience of GESPLAN's reforestation experts and their on-site observations. So, these readings came to confirm that they employed good practices.

On the other hand, not all discharges on AFDS02 and AFDS03 were recorded as moisture in the same way, which led to a request for the on-site workers to check the condition of the irrigation network. It's worth noting that the traditional reforestation area seems to be under the greatest water stress. Finally, the tropical storm Hermine at the end of September provided a significant

recharge of moisture to all the soil, doubling its quantity and bringing it to its maximum levels ending the drought period.

2022 Q4: October to December 2022

The final period of 2022, in terms of wind, continues to exhibit a southern component, contrary to the prevailing winds of the islands. However, this is the time of year when such a phenomenon is historically more common. The wind speed was moderate throughout, except for the month of December, where a significant percentage of wind was recorded in the range of 2 to 6 m/s.

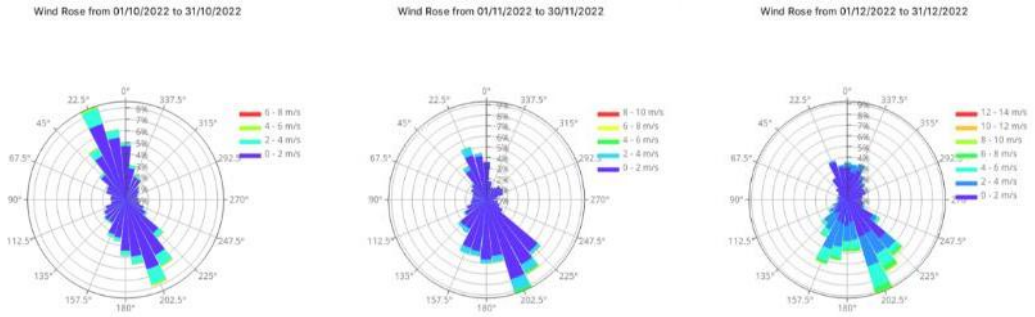


Figure 112 Wind in El Pinillo: October to December 2022 wind roses.

A generalised decrease in temperatures is observed, and both the second half of October and the first half of November prove to be dry due to the absence of rainfall and a relatively low humidity percentage (averaging at 40.9%). The rest of the quarter sees scattered, moderate rainfall amounting to a total of 136 mm. Figure 46: Meteorology in *El Pinillo*: October to December 2022 precipitation, relative humidity, temperature and solar irradiance graph. Figure 46 aids in better contextualising the impact of Storm Hermine, which over its three-day duration, discharged 161 mm of rain, 18% more than the total for this fourth quarter of the year. The relative humidity maintained an average level of 60.9%, and once again, it was noted that values around 100% are recorded with a northern wind component, and the lowest levels, around 30%, are recorded with a southern component.

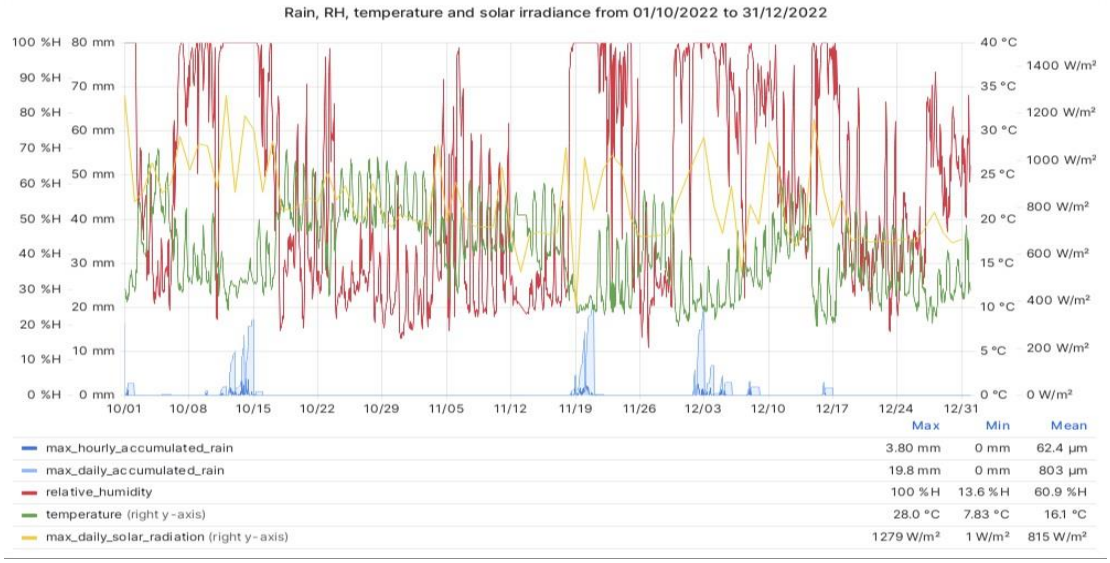


Figure 113 Meteorology in El Pinillo, from October to December 2022: precipitation, relative humidity, temperature and solar irradiance.

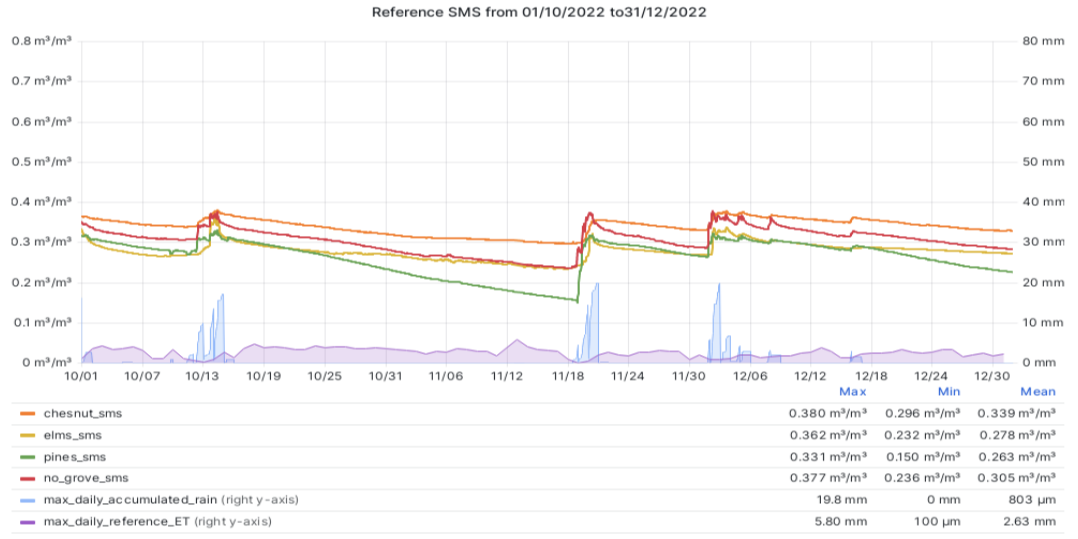


Figure 114 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2022, in reference sites.

During this period, on October 26th, a pair of sensors in the second reforestation area with individual collectors were activated (_plant and _no_plant), as well as additional sensors installed next to plants (referred to as _plant_bis) in areas where sensors were already deployed but at different locations: AFDS01, AFDS02, AFDS03, cocoon, traditional, and individual (area 1). All these sensors began to operate correctly. However, a new issue arose in the AFDS02 area, as the remaining operational sensor of the initially installed pair also ceased to function, sending its last communication on December 3rd.

Since late September, evapotranspiration levels have remained low that in combination with the rainfall in this quarter allowed for similar moisture levels to be maintained from start to finish in all reforestation areas.

Notably, the moisture sensors of the two individual collectors in area 1 showed an increase in humidity during rainfall periods greater than that recorded by the reference humidity sensor installed nearby (_no_plant). This was not observed in the area 2 of individual collectors nor in any of the other monitored reforestation areas. This could be due to the specific soil conditions in this area (see Soil characterization section), with a deeper soil and high organic matter content. In addition, the fact that vegetation (ferns) covers this entire area with remarkable height means that points with individual collector or without plant have no notable differences as to conditions of rainwater or fog collection.

Figure 115 (left) Individual collectors Area 2, (right) individual collectors Area 2 and SMS.

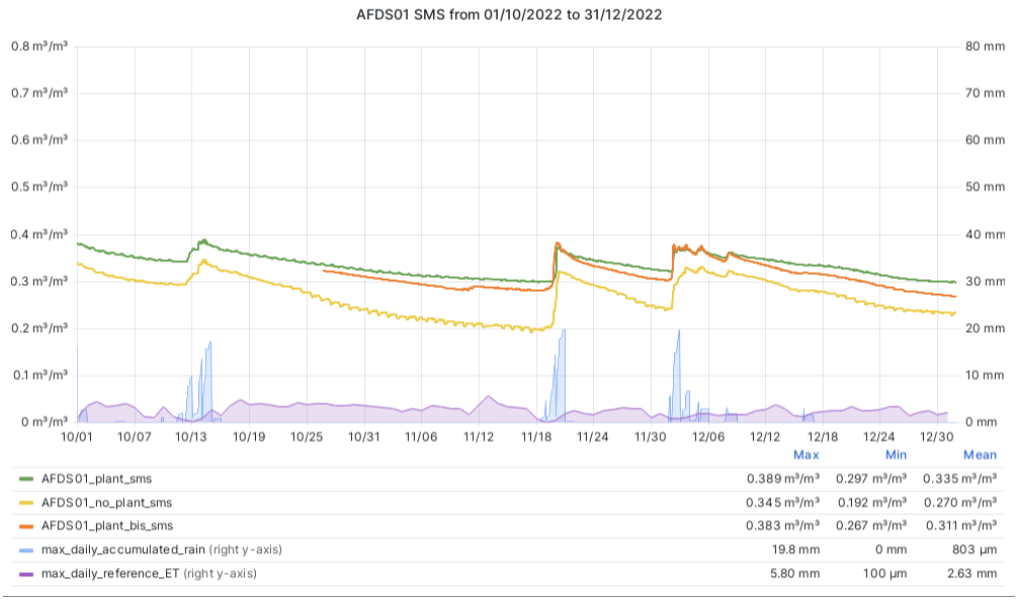


Figure 116 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2022, in AFDS01 sites.

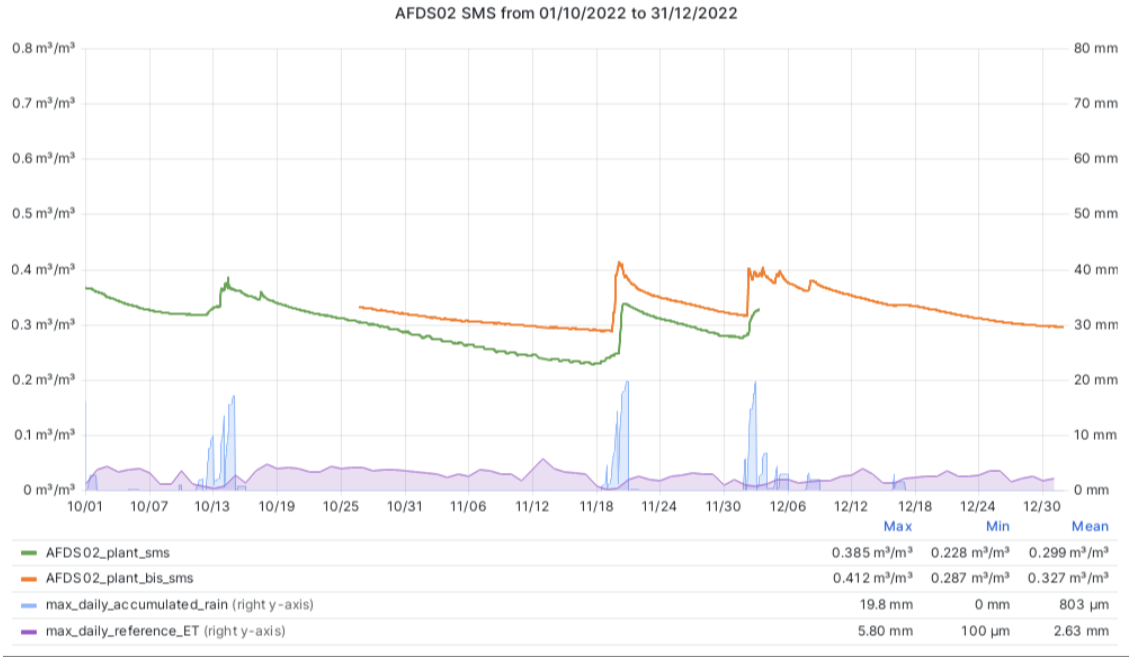


Figure 117 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2022, in AFDS02 sites.

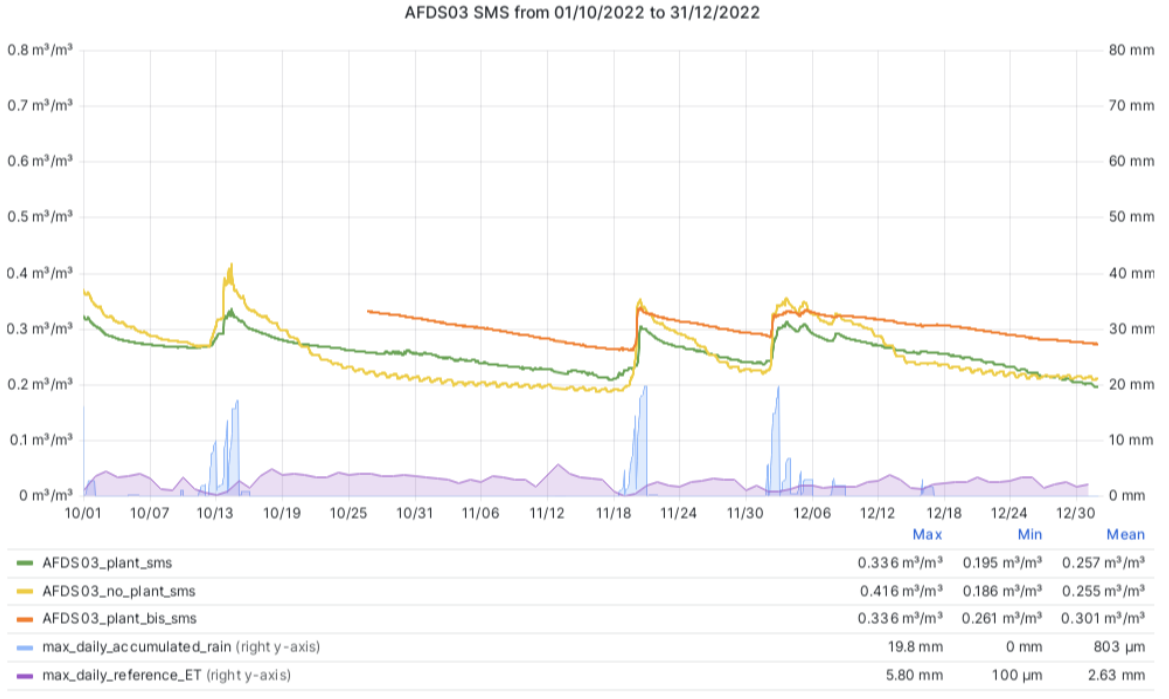


Figure 118 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2022, in AFDS03 sites.

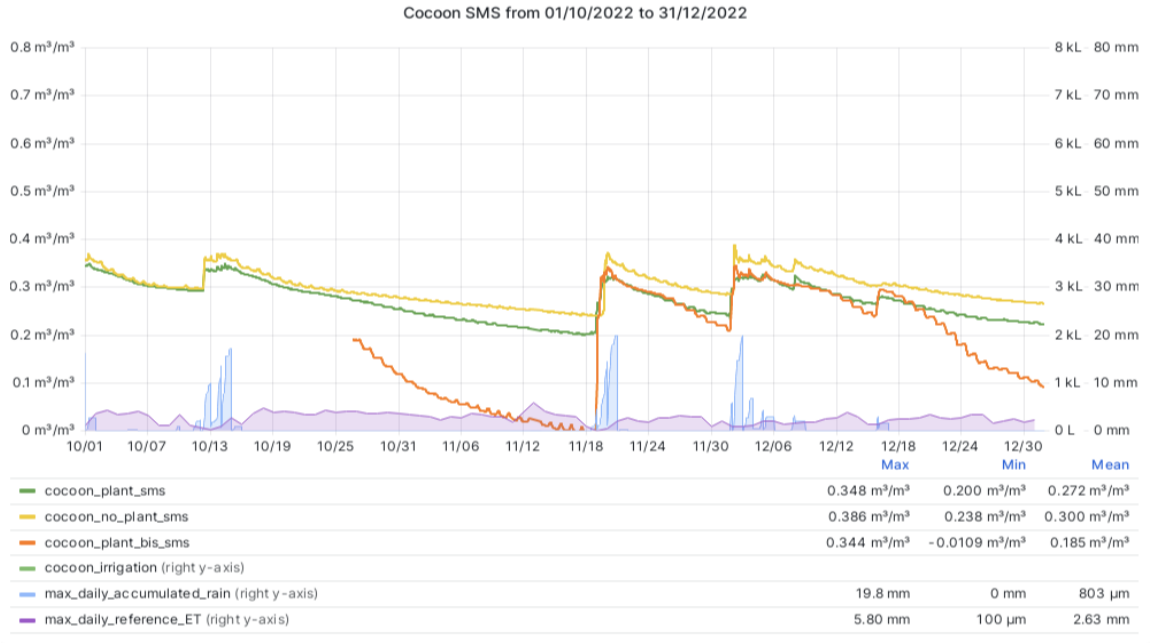


Figure 119 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2022, in ***Cocoon sites.***

Figure 120 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2022, in IFWC (area 1) sites.

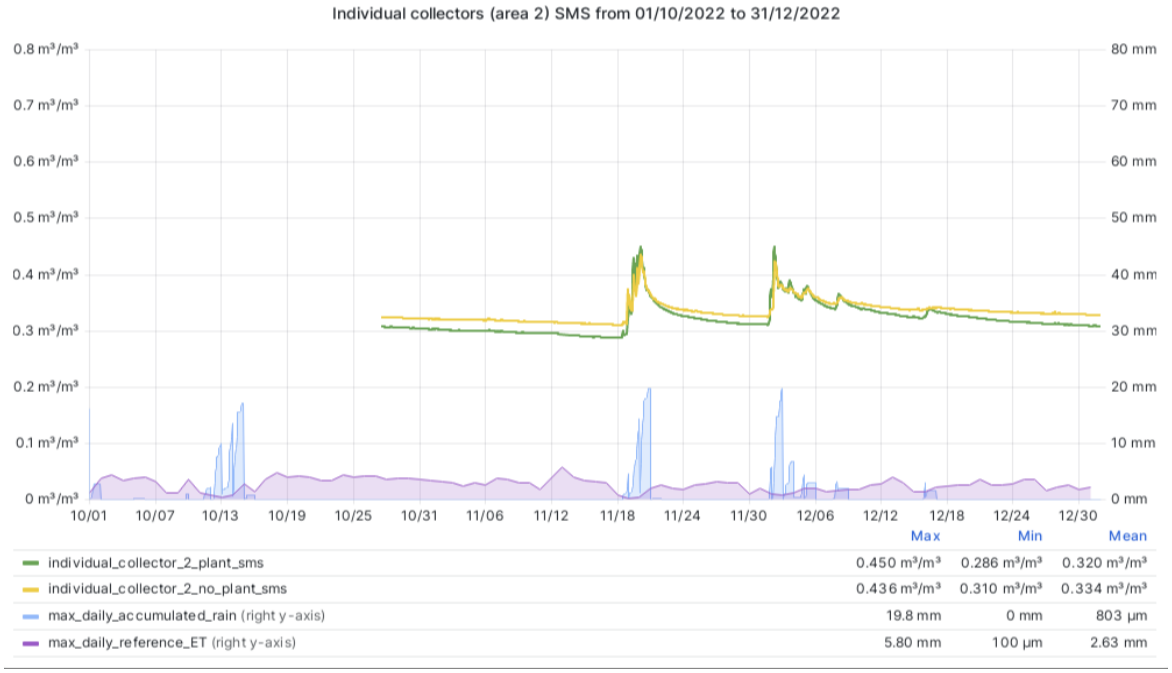


Figure 121 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2022, in IFWC (area 2) sites.

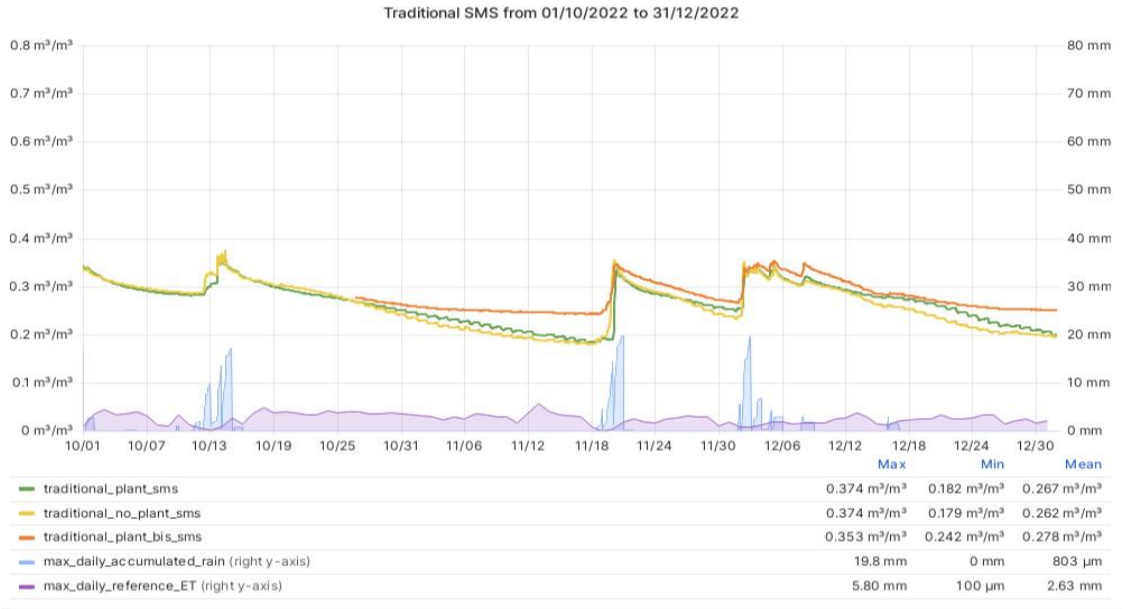


Figure 122 Soil Moisture, rain and evapotranspiration in El Pinillo: October to December 2022, TR sector.

2023 Q1: January to March 2023

In the first quarter of 2023, we observed a predominant north wind trend, with the exception of March, where the wind was weak and lacked a predominant direction. In the whole interval the main range of velocities was 0 to 2 m/s.

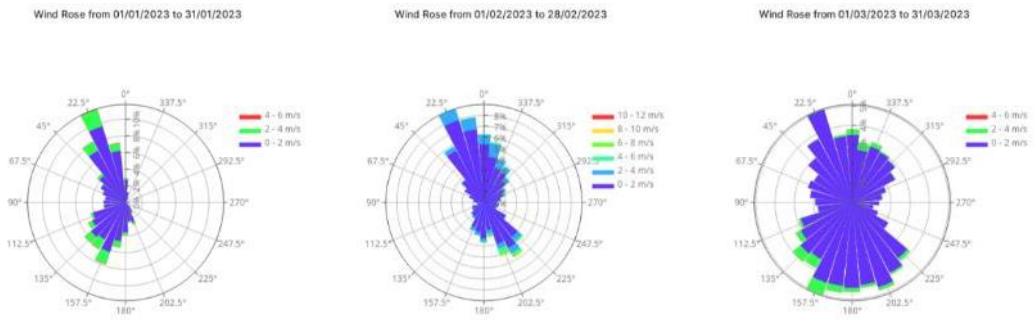


Figure 123 Wind in El Pinillo: January to March 2023 wind roses.

During this quarter, there was a significant amount of rainfall for the area, accumulating a total of 326 mm, with 318 mm concentrated between January 11th and February 20th. The months of January and February also experienced low temperatures (with a maximum of 18.5°C, and an average of 8.6°C), an average relative humidity above 80%, and low solar irradiance. However, in March, there was a rise in temperatures (with a maximum of 28.4°C, and an average of 16.2°C), slightly higher solar irradiance, less relative humidity (48.5% averaged), and only one day of rainfall, which amounted to 6.8 mm.

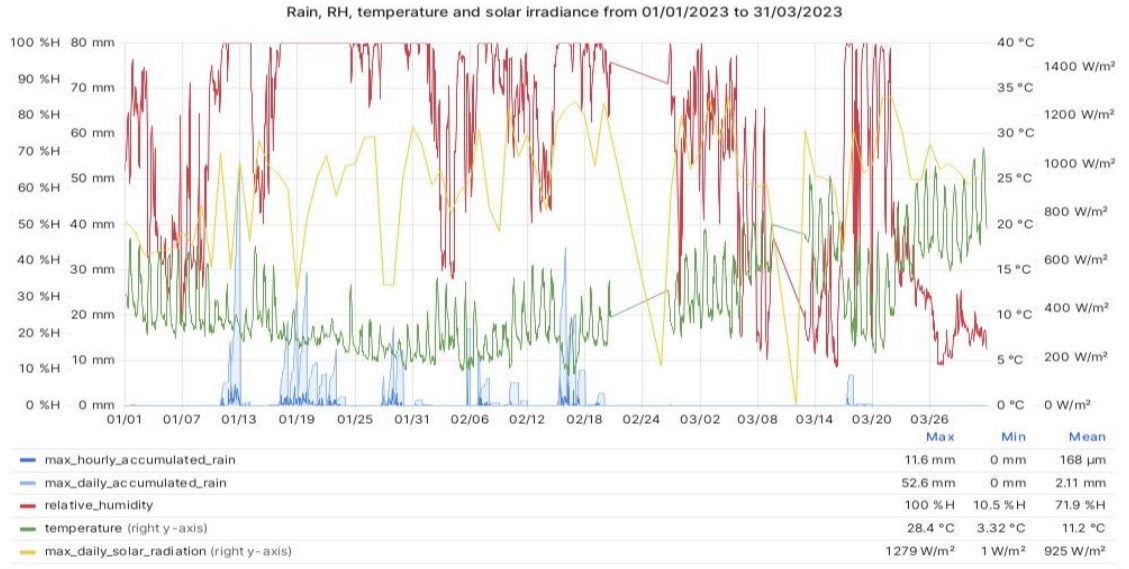
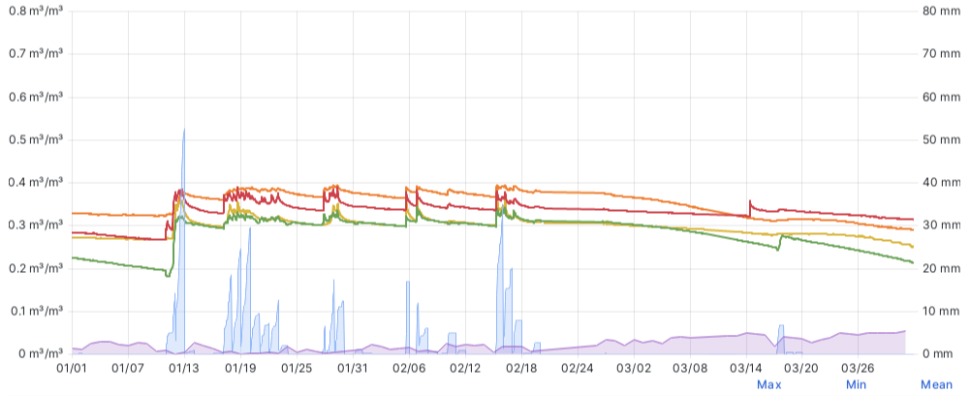


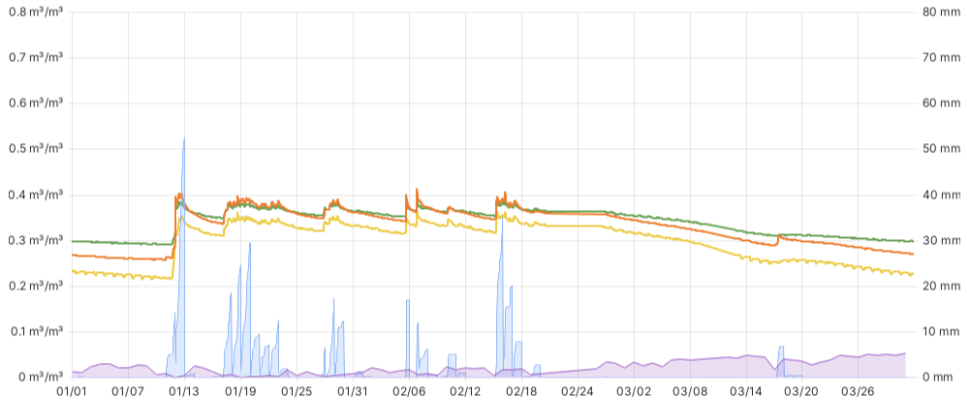
Figure 124 Precipitation, relative humidity, temperature and solar irradiance graph in El Pinillo, from January to March 2023.

Reference SMS from 01/01/2023 to31/03/2023



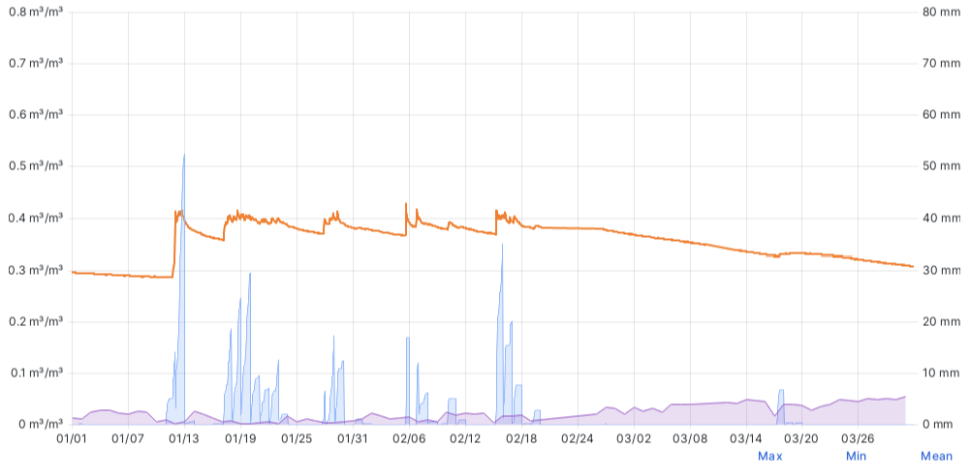
	Max	Min	Mean
chesnut_sms	0.394 m³/m³	0.289 m³/m³	0.350 m³/m³
elms_sms	0.366 m³/m³	0.250 m³/m³	0.296 m³/m³
pines_sms	0.343 m³/m³	0.181 m³/m³	0.279 m³/m³
no_grove_sms	0.389 m³/m³	0.266 m³/m³	0.331 m³/m³
max_daily_accumulated_rain (right y-axis)	52.6 mm	0 mm	2.11 mm
max_daily_reference_ET (right y-axis)	5.40 mm	100 µm	2.28 mm

AFDS01 SMS from 01/01/2023 to 31/03/2023



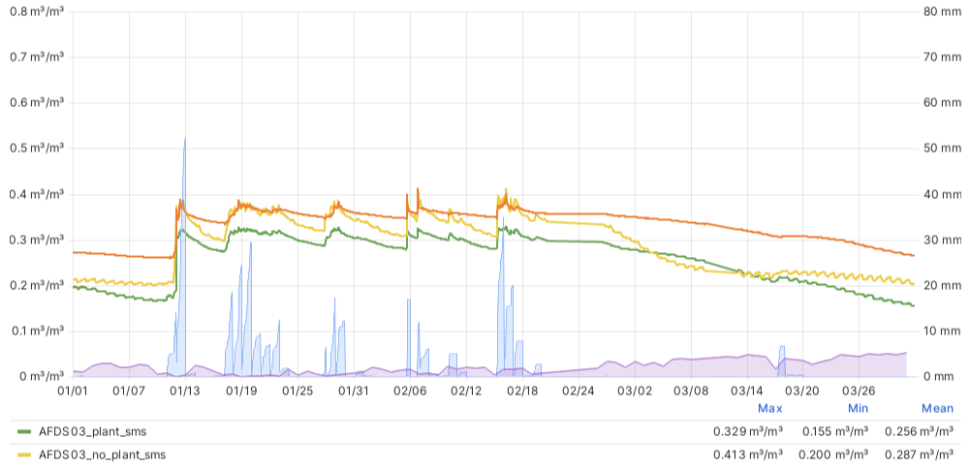
	Max	Min	Mean
AFDS01_plant_sms	0.400 m³/m³	0.291 m³/m³	0.339 m³/m³
AFDS01_no_plant_sms	0.366 m³/m³	0.214 m³/m³	0.294 m³/m³
AFDS01_plant_bis_sms	0.412 m³/m³	0.257 m³/m³	0.329 m³/m³
max_daily_accumulated_rain (right y-axis)	52.6 mm	0 mm	2.11 mm
max_daily_reference_ET (right y-axis)	5.40 mm	100 µm	2.28 mm

AFDS02 SMS from 01/01/2023 to 31/03/2023



	Max	Min	Mean
AFDS02_plant_bis_sms	0.427 m³/m³	0.285 m³/m³	0.354 m³/m³
max_daily_accumulated_rain (right y-axis)	52.6 mm	0 mm	2.11 mm
max_daily_reference_ET (right y-axis)	5.40 mm	100 µm	2.28 mm

AFDS03 SMS from 01/01/2023 to 31/03/2023



	Max	Min	Mean
AFDS03_plant_sms	0.329 m³/m³	0.155 m³/m³	0.256 m³/m³
AFDS03_no_plant_sms	0.413 m³/m³	0.200 m³/m³	0.287 m³/m³
AFDS03_plant_bis_sms	0.412 m³/m³	0.260 m³/m³	0.329 m³/m³
max_daily_accumulated_rain (right y-axis)	52.6 mm	0 mm	2.11 mm
max_daily_reference_ET (right y-axis)	5.40 mm	100 µm	2.28 mm

Cocoon SMS from 01/01/2023 to 31/03/2023

	Max	Min	Mean
cocoon_plant_sms	0.343 m³/m³	0.199 m³/m³	0.277 m³/m³
cocoon_no_plant_sms	0.377 m³/m³	0.246 m³/m³	0.317 m³/m³
cocoon_plant_bis_sms	0.365 m³/m³	0.0416 m³/m³	0.247 m³/m³
max_daily_accumulated_rain (right y-axis)	52.6 mm	0 mm	2.11 mm
max_daily_reference_ET (right y-axis)	5.40 mm	100 µm	2.28 mm

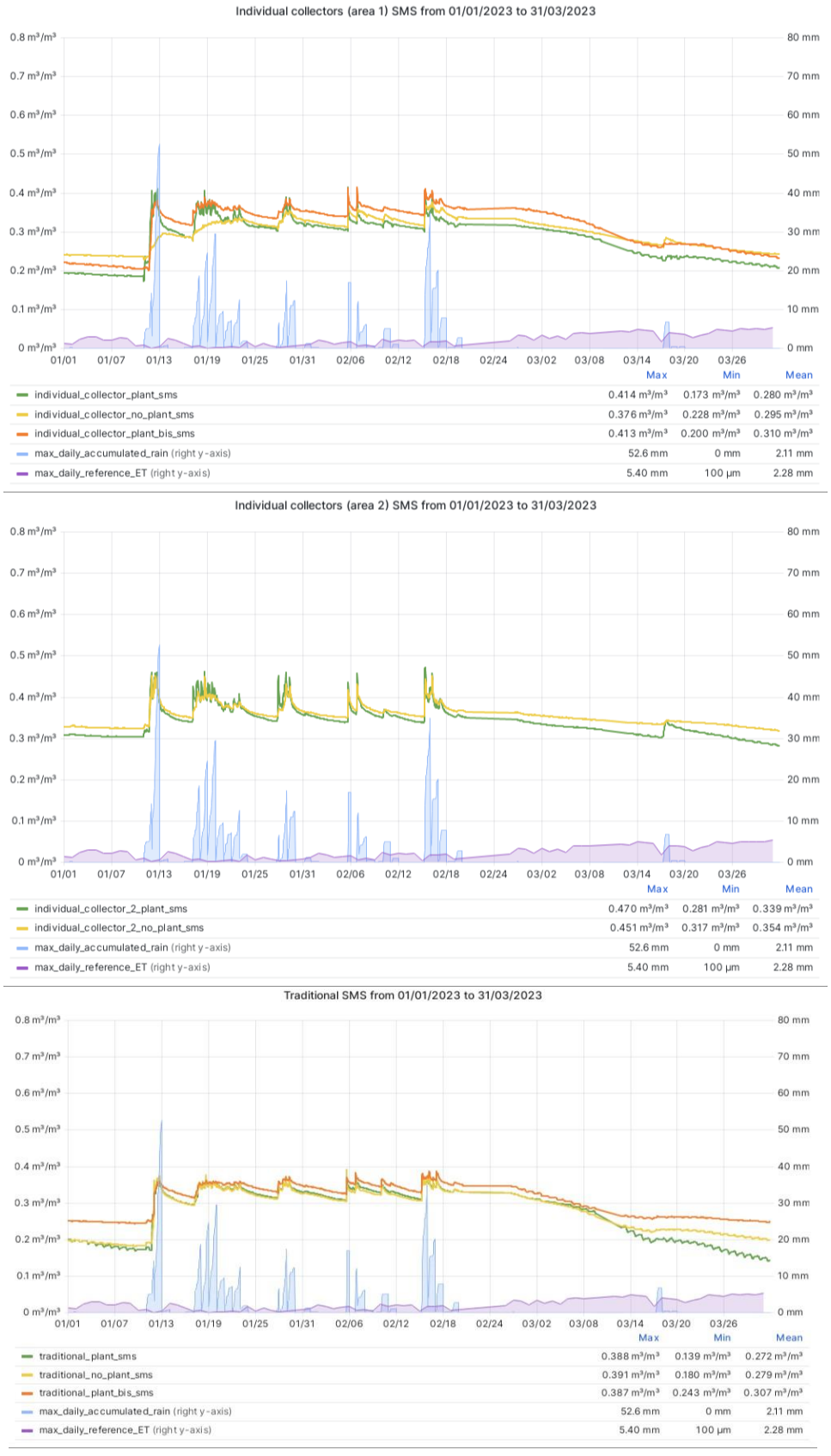


Figure 125. Soil Moisture, rain and evapotranspiration in El Pinillo: January to March 2023, all monitored areas graphs.

The decrease in soil moisture that occurred in December was reversed by the rainfall in this quarter. However, as March approached, evapotranspiration increased and soil moisture began to decline, particularly in the traditional reforestation area and in the cocoon zone.

2023 Q2: April to June 2023

In the second quarter of 2023, we observed a predominant north wind trend, with the exception of June, where the wind was weak and lacked a predominant direction. In the whole interval the main range of velocities was 0 to 4.38 m/s. During this quarter, there was not a significant amount of rainfall for the area, accumulating only a total of 23.8 mm. During the month of April, 15.2 mm of rain was collected and 6.4 mm in May. From June 26, temperatures rise significantly with an hourly maximum of 36.8 ºC and relative humidity very low up to 10%.

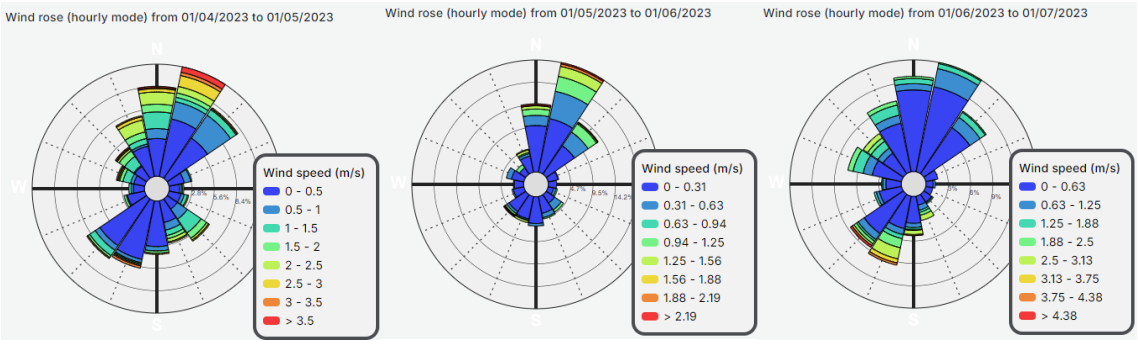


Figure 126 Wind rose (hourly mode) in El Pinillo: April to June 2023.

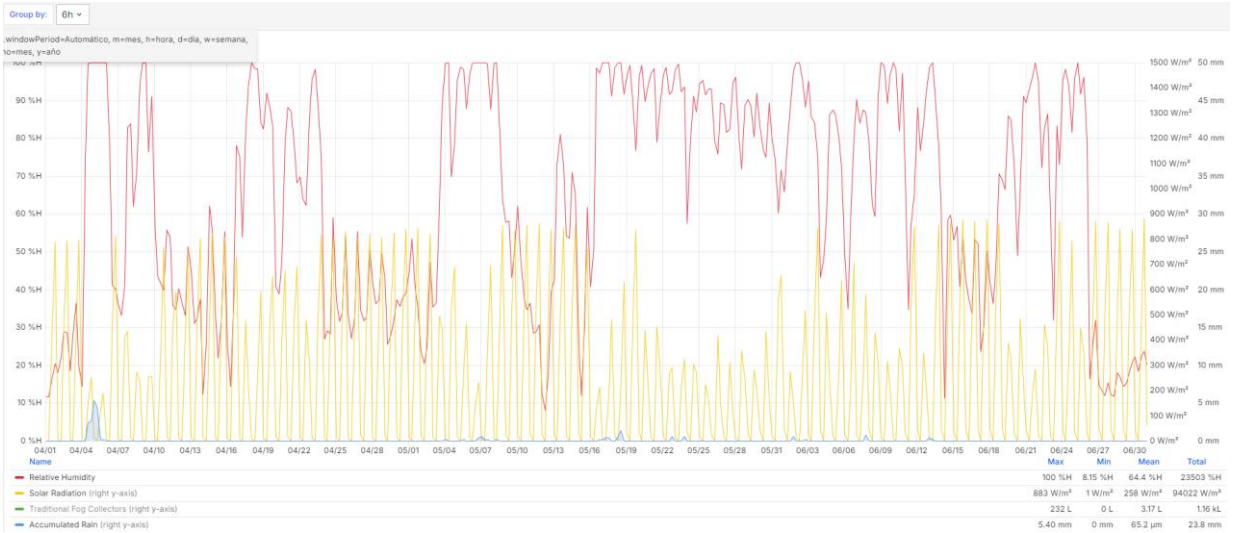


Figure 127 Precipitation, relative humidity, temperature and solar irradiance graph in El Pinillo, from April to June 2023.

During this period, the rain events of 4-5 April (15.2 mm) and the event of 18 May (3.2 mm) are to be highlighted. The soil moisture content in the pine area reacts quickly to these rain events, while areas of chestnut, elm and fern remain more stable. AFDS area 1, Traditional sector, Cocoon sector, IFWC 1 and 2 sectors are also influenced by the April rain event but not by the May rain event. In AFDS area 2, AFDS 4 discharged water on June 13th, so this area was the most water-rich soil of all measuring points at the end of this period.

It is important to note that the screens of the individual fog collectors in Area 1 began to be removed on June 19, 2023, and October 9, 2023. Therefore, measurements from these points should not be considered from this period of analysis onwards.

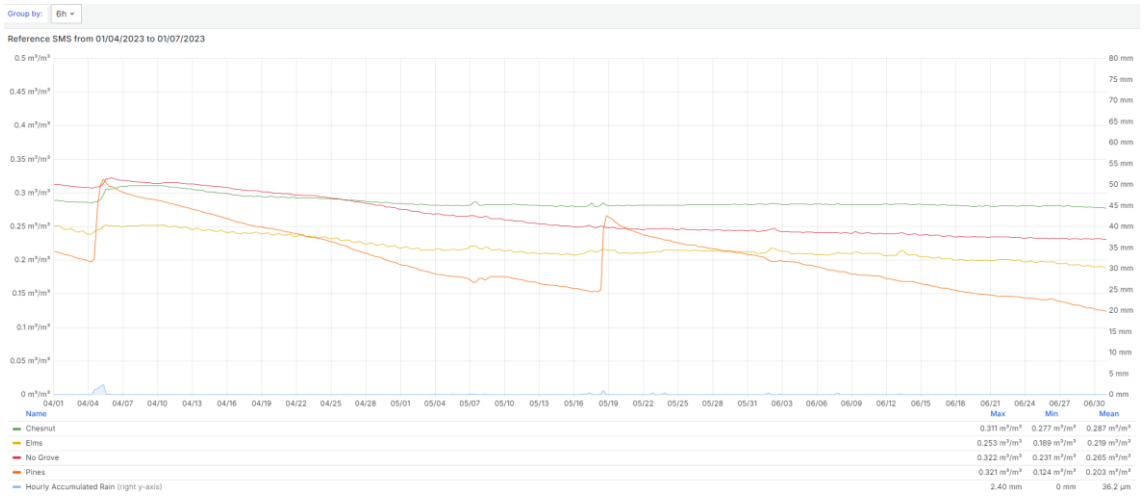


Figure 128 Soil Moisture references and rain in El Pinillo, from April to June 2023.

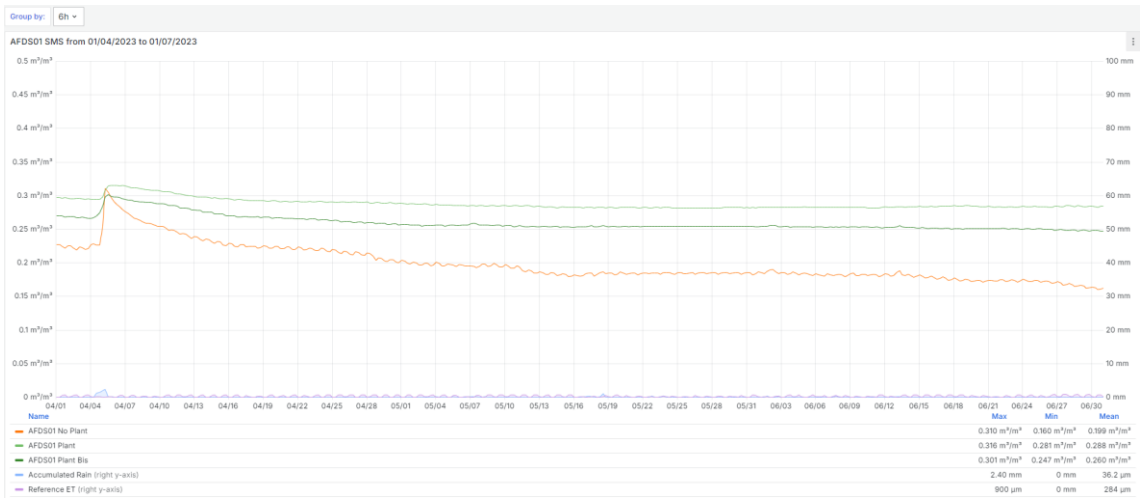


Figure 129 Soil Moisture, rain and evapotranspiration in El Pinillo, from April to June 2023, in AFDS01 sites.

Figure 130 Soil Moisture, rain and evapotranspiration in El Pinillo, from April to June 2023, in AFDS02 sites.

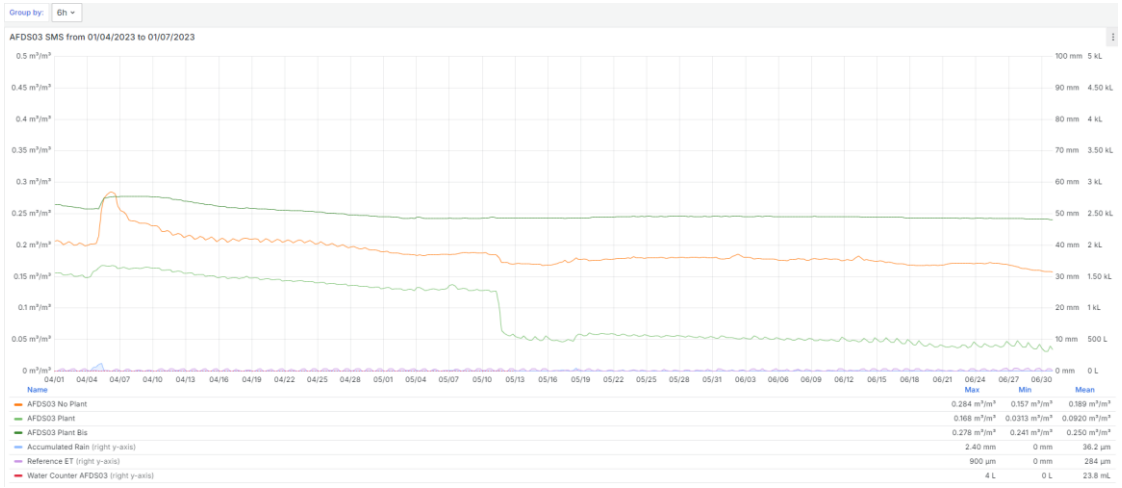


Figure 131 Soil Moisture, rain and evapotranspiration in El Pinillo, from April to June 2023, in AFDS03 sites.

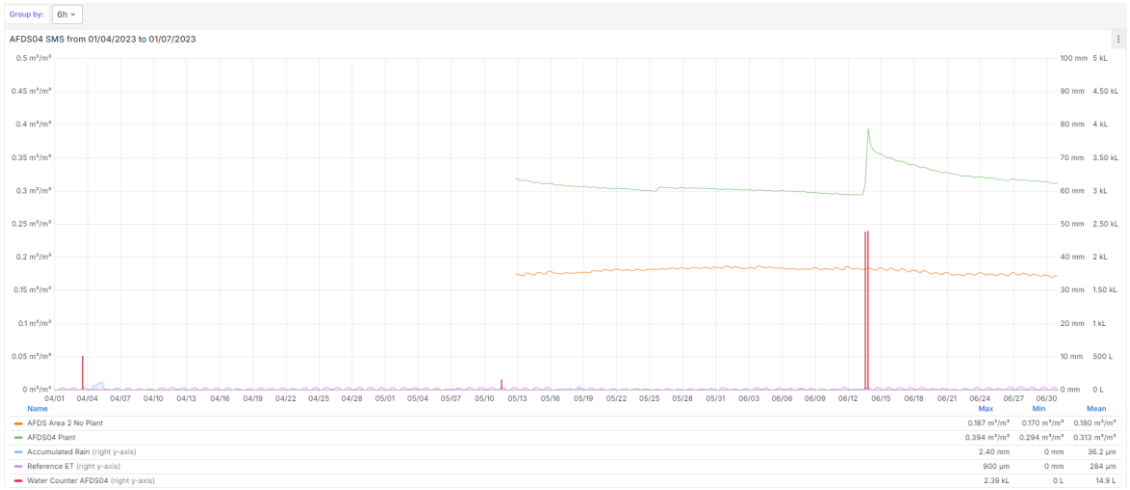


Figure 132 Soil Moisture, rain and evapotranspiration in El Pinillo, from April to June 2023, in AFDS04 sites.

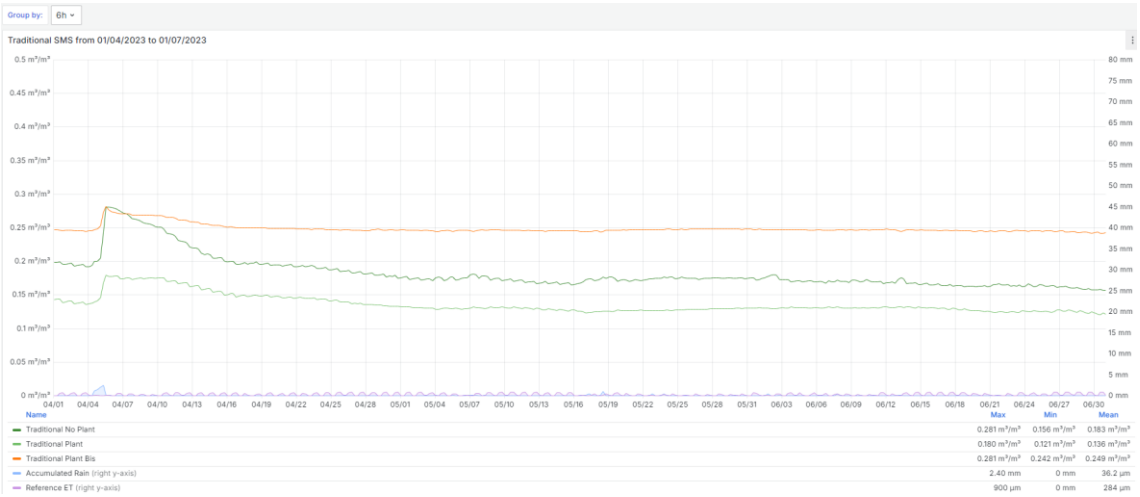


Figure 133 Soil Moisture, rain and evapotranspiration in El Pinillo, from April to June 2023, in Traditional sector.

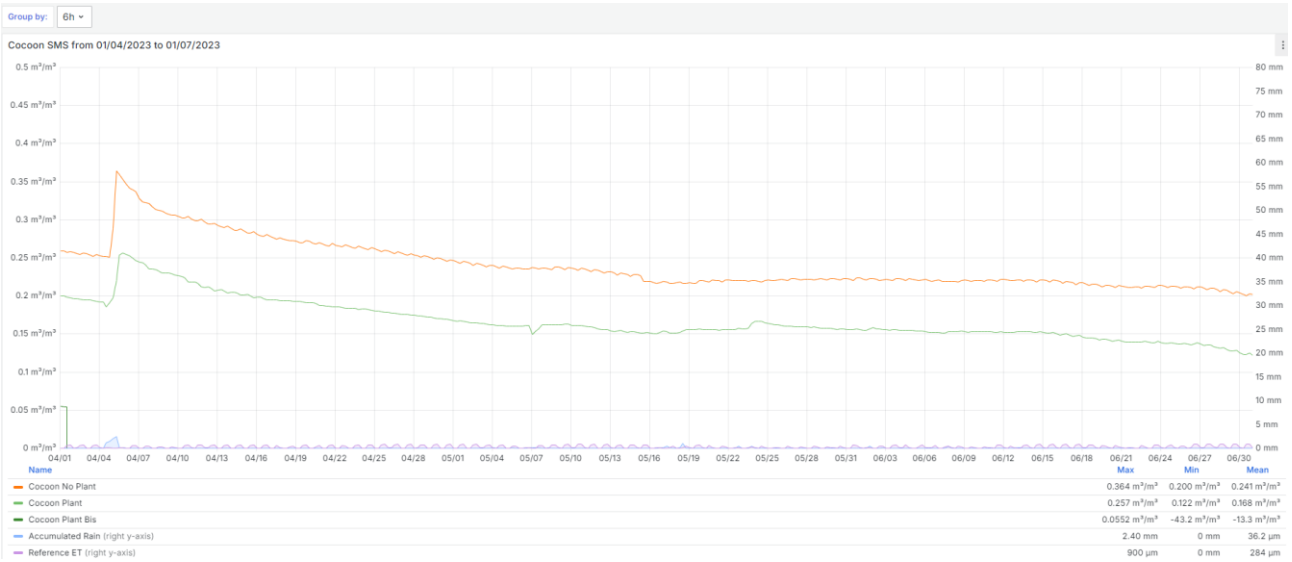


Figure 134 Soil Moisture, rain and evapotranspiration in El Pinillo, from April to June 2023, in Cocoon sector.

Figure 135 Soil Moisture, rain and evapotranspiration in El Pinillo, from April to June 2023, in IFWC 1 sector.

Figure 136 Soil Moisture, rain and evapotranspiration in El Pinillo, from April to June 2023, in IFWC 2 sector.

2023 Q3: July to September 2023

In the third quarter of 2023, we observed a predominant north wind trend, with the exception of August. In the whole interval the main range of velocities was 0 to 2.63 m/s. During this quarter, there was a significant amount of rainfall for the area in September, with a total of 56.8 mm in this month. The months of August experienced high temperatures (with a maximum of 39.4°C, and an average of 24.8°C), an average relative humidity below 37%, and high solar irradiance.

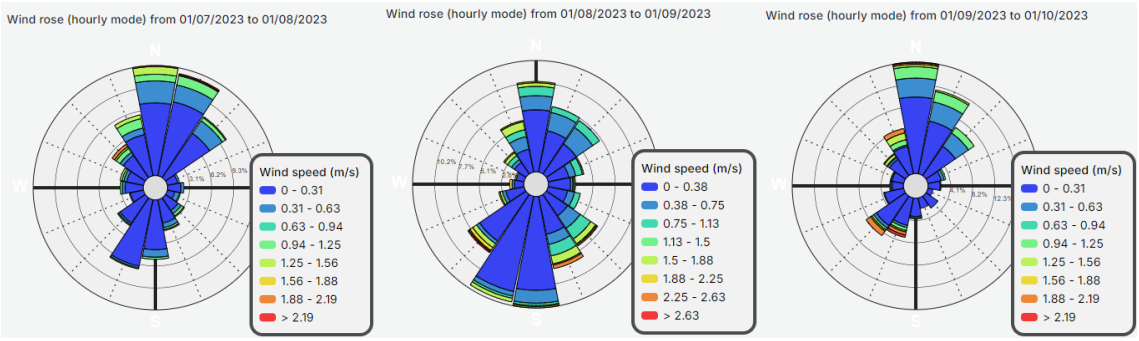


Figure 137 Wind rose (hourly mode) in El Pinillo: July to September 2023.

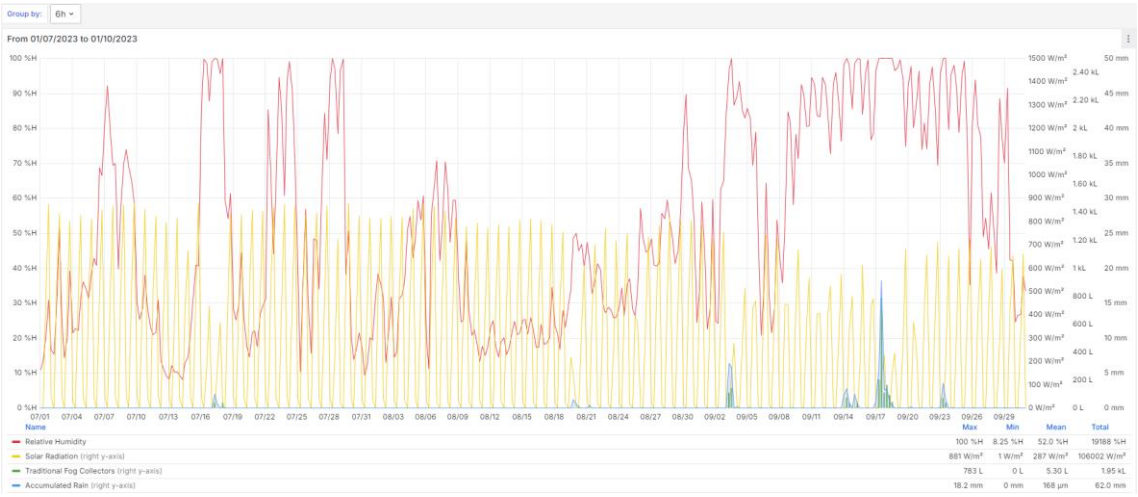


Figure 138 Precipitation, relative humidity and solar irradiance graph in El Pinillo, from July to September 2023.

During this period, the rain events of 3rd (12.4 mm) and 17-18 September (31.4 mm) are to be highlighted. The soil moisture content in the pine area reacts quickly to these rain events. AFDS area 1, Traditional sector, Cocoon sector, IFWC 1 and 2 sectors are mainly influenced by only the second rain event in September. In AFDS area 1, AFDS 2 was forced to discharge water on August 8th. In AFDS area 2, AFDS 4 and AFDS 6 had different discharges that improve soil water content in these areas.

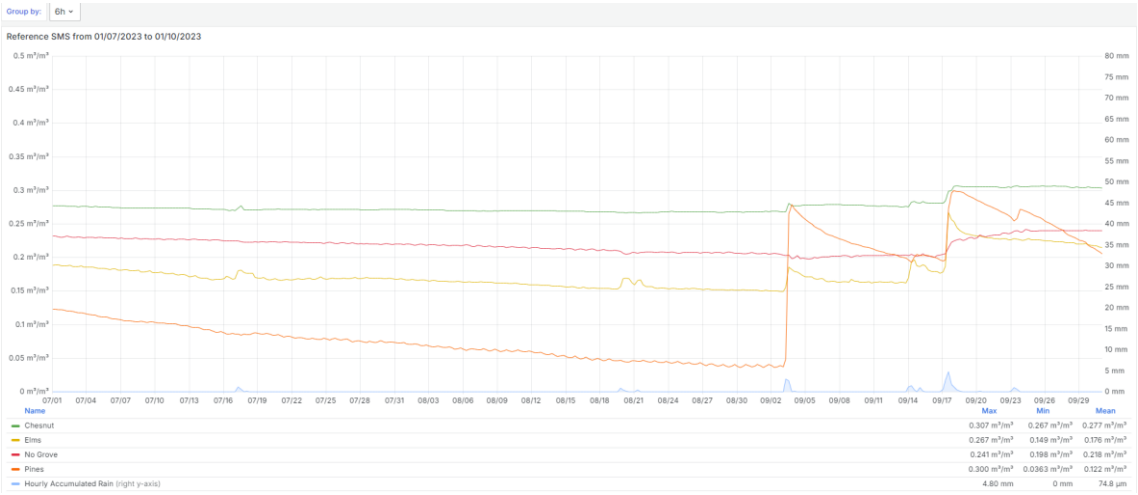


Figure 139 Soil Moisture references and rain in El Pinillo, from July to September 2023.

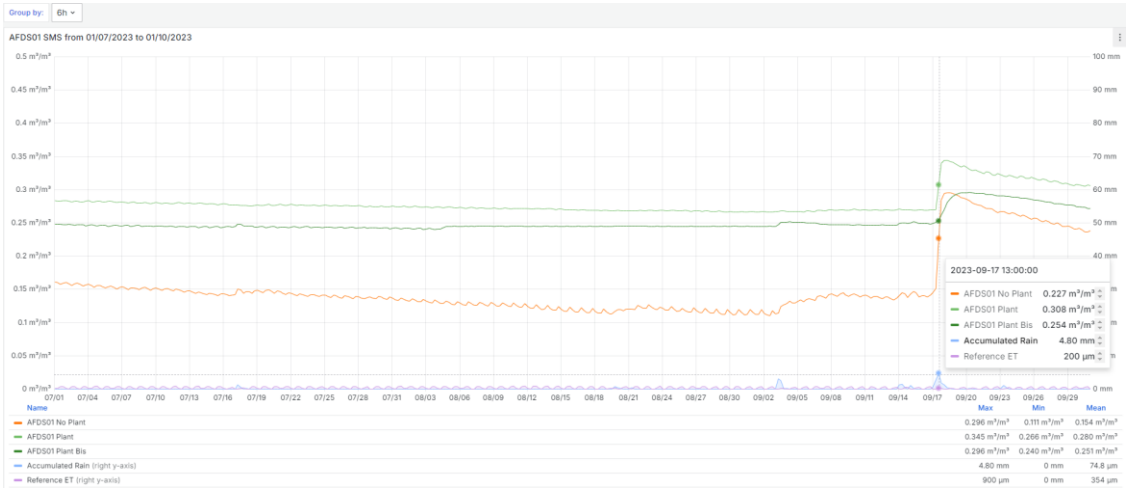


Figure 140 Soil Moisture, rain and evapotranspiration in El Pinillo, from July to September, in AFDS 1 sites

Figure 141 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September, in AFDS 2 sites.

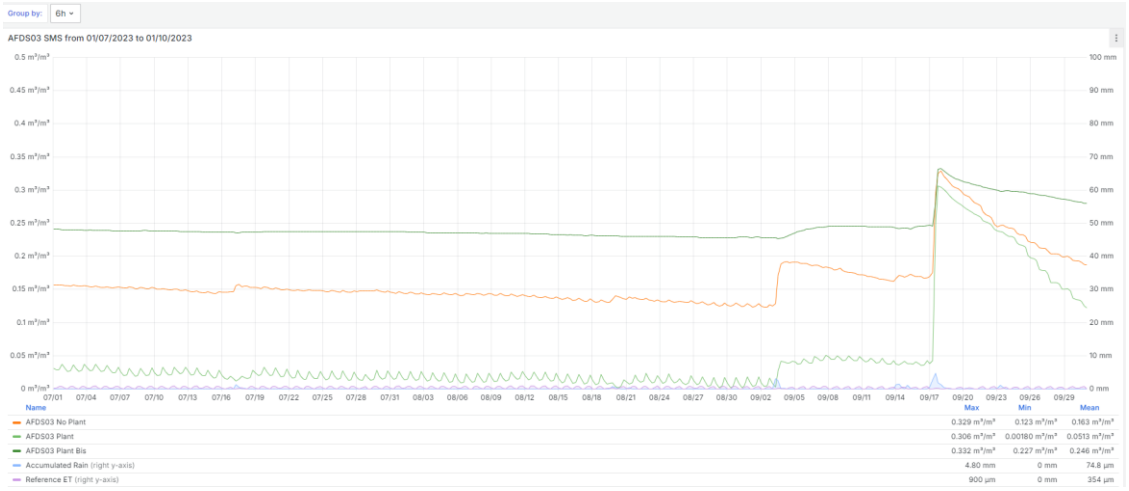


Figure 142 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September, in AFDS 3 sites.

Figure 143 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September, in AFDS 4 sites

Figure 144 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September, in AFDS 6 sites.

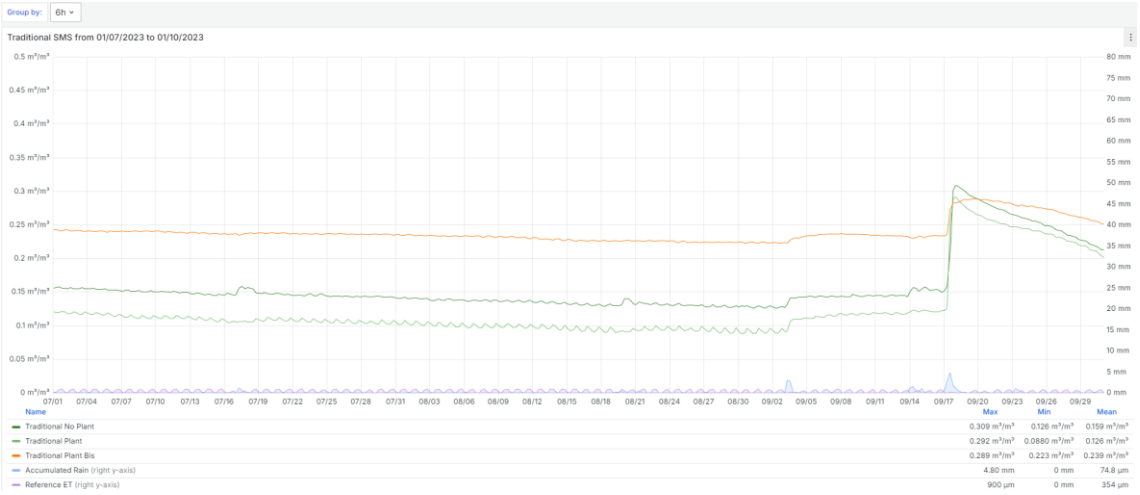


Figure 145 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September, in Traditional sector.

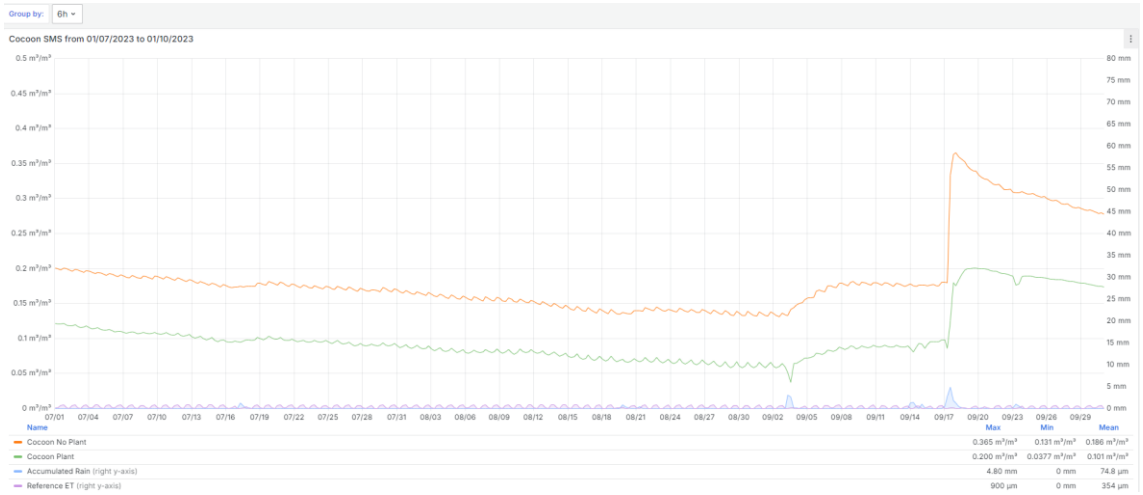


Figure 146 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September, in Cocoon sector.

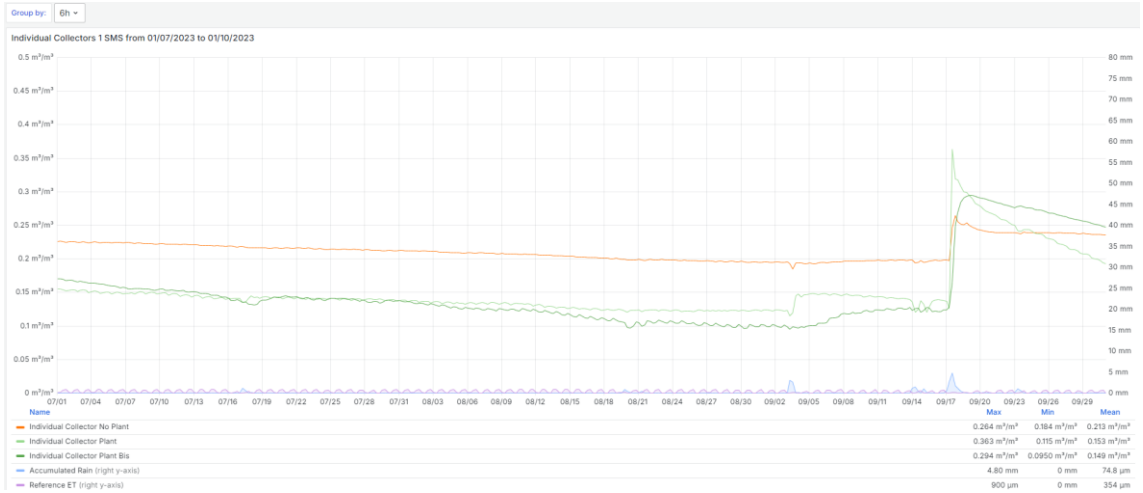


Figure 147 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September, in IFWC 1 sector.

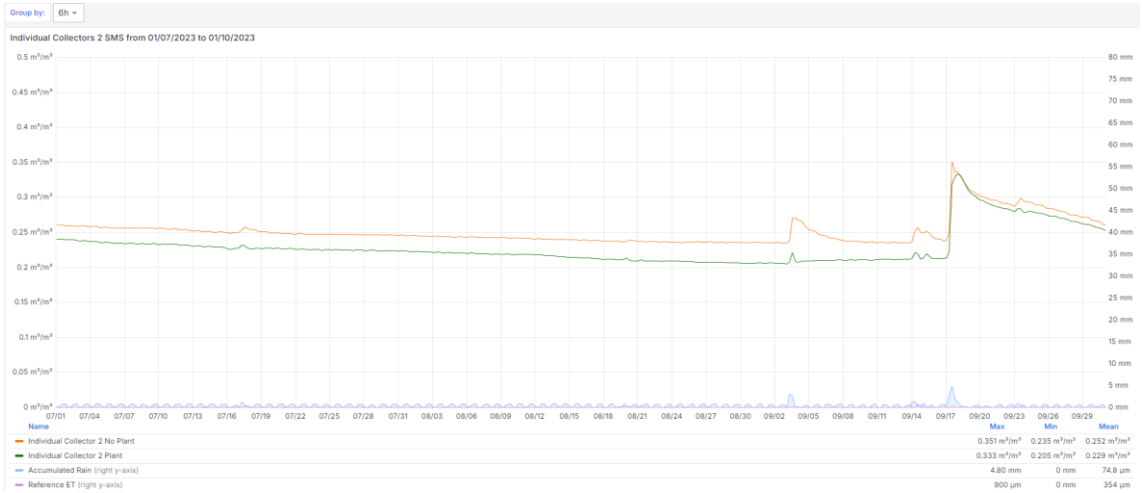


Figure 148 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September, in IFWC 2 sector.

Figure 149 Soil Moisture, rain and evapotranspiration in experimental plot of El Pinillo, July to September 2023.

During this period, the experimental plot is already operational with some of the soil moisture sensors installed. Forced irrigation events were observed on the plot on July 10th, August 10th, and August 28th. The influence of precipitation periods on September 3rd and September 17th is also noted. During the irrigation events, moisture was not detected in the Deep drop system, likely because the water was supplied at a depth below the measurement range of the soil moisture sensor. However, during the rainfall events, soil moisture content could be measured at this point due to water being supplied from the surface.

Starting on September 7th, a reference measurement point was installed in an area without reforestation and without soil disturbance. During the irrigation events, the highest moisture content was detected at the traditional reforestation point, followed by the one with a dose of biochar. The point with an individual collector retains less moisture (7% less) but reacts more quickly to rainfall events.

2023 Q4: October to December 2023

In the last quarter of 2023, we observed a predominant south wind trend. In the whole interval, the main range of velocities was 0 to 3.5 m/s and 7.5 m/s in November. During this quarter, there was a significant amount of rainfall for the area, accumulating only a total of 233 mm (83.6 mm in October, 79.8 mm in November and 69.6 mm in December).

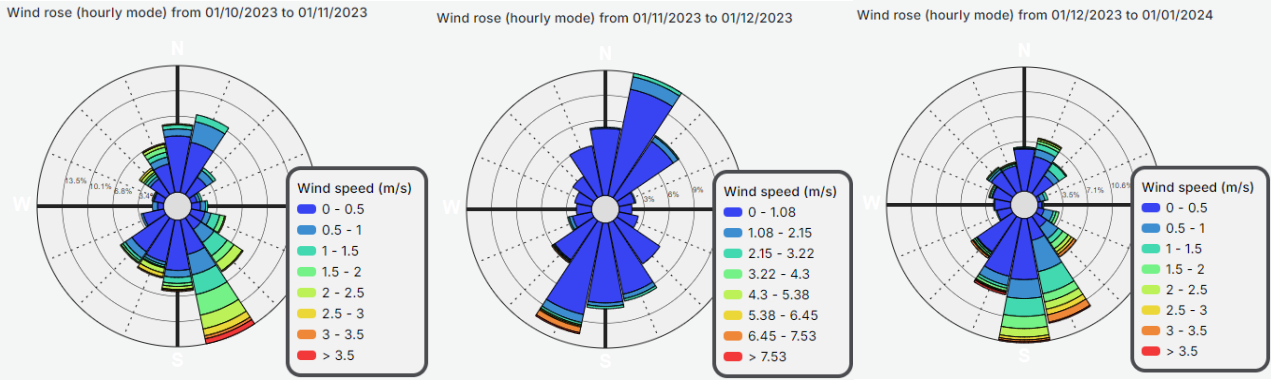


Figure 150. Wind rose (hourly mode) in El Pinillo: October to December 2023.

October experienced rainy days from the 20th to the 24th, with prevailing winds from the north and northwest at speeds of up to 2.6 m/s. In November, the rainy days were from the 6th to the 8th, accompanied by northeast winds. Similarly, December had rain from the 20th to the 22nd.

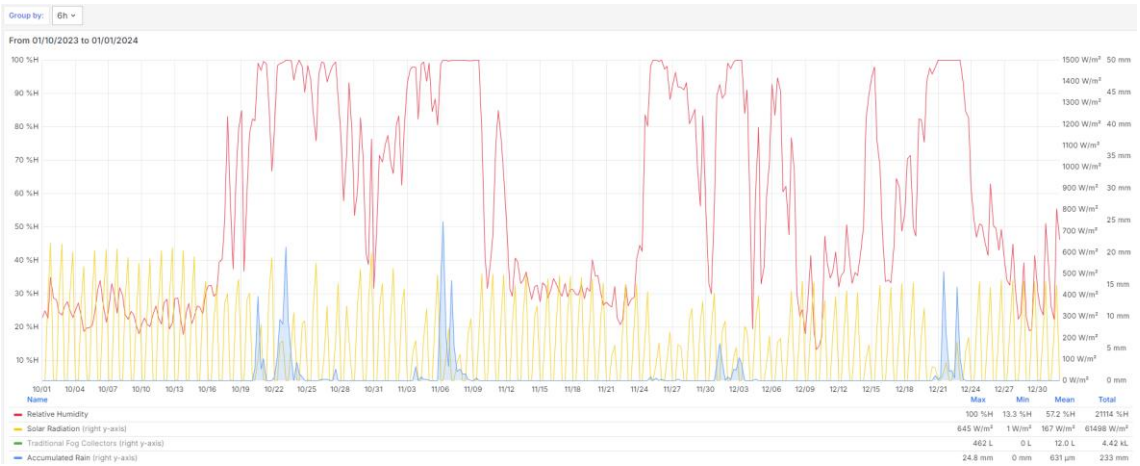


Figure 151 Precipitation, relative humidity and solar irradiance graph in El Pinillo, from October to December 2023.

In this period, the soil moisture content in all the areas reacts to rain events. In AFDS area 2, AFDS 4, 5 and AFDS 6 had different discharges that improve soil water content in these areas. The highest soil moisture levels (> 0.4 m3/m3) are obtained in individual collectors areas and at AFDS 2 sites. The highest average values of soil moisture (>0.3 m3/m3) correspond to the areas of AFDS and chestnut trees.

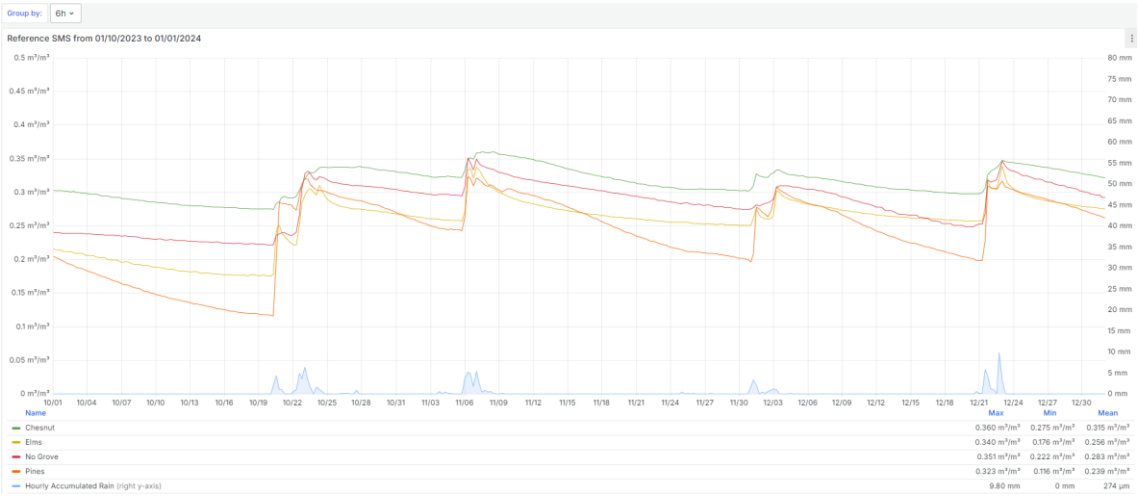


Figure 152 Soil Moisture references and rain in El Pinillo, from October to December 2023.

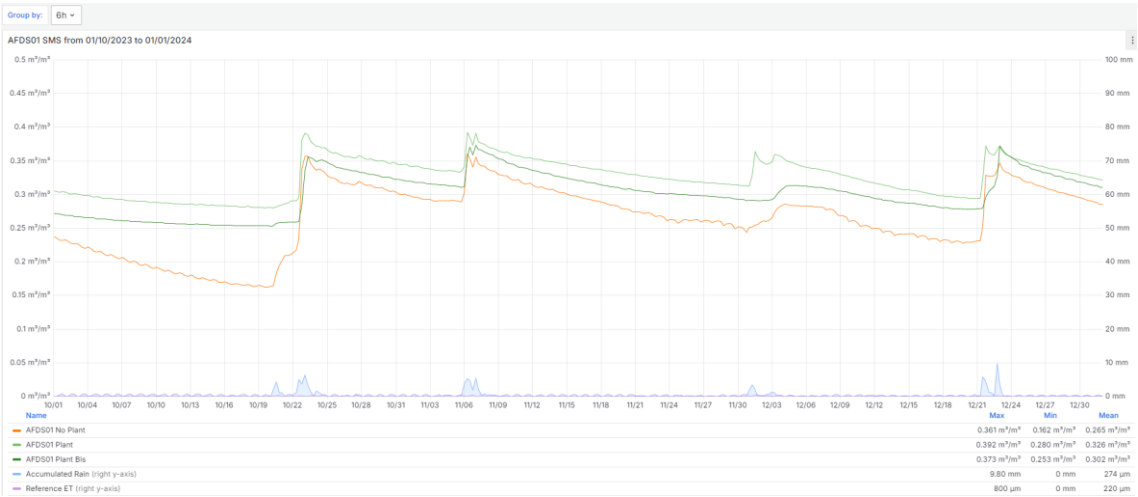


Figure 153 Soil Moisture, rain and evapotranspiration in El Pinillo, October to December, in AFDS 1 sites.

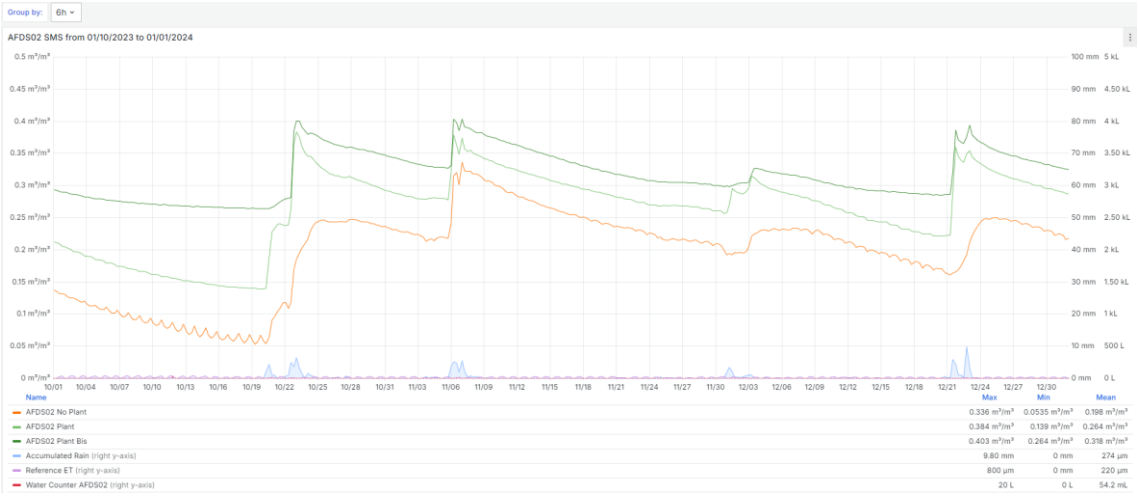


Figure 154 Soil Moisture, rain and evapotranspiration in El Pinillo, October to December, in AFDS 2 sites.

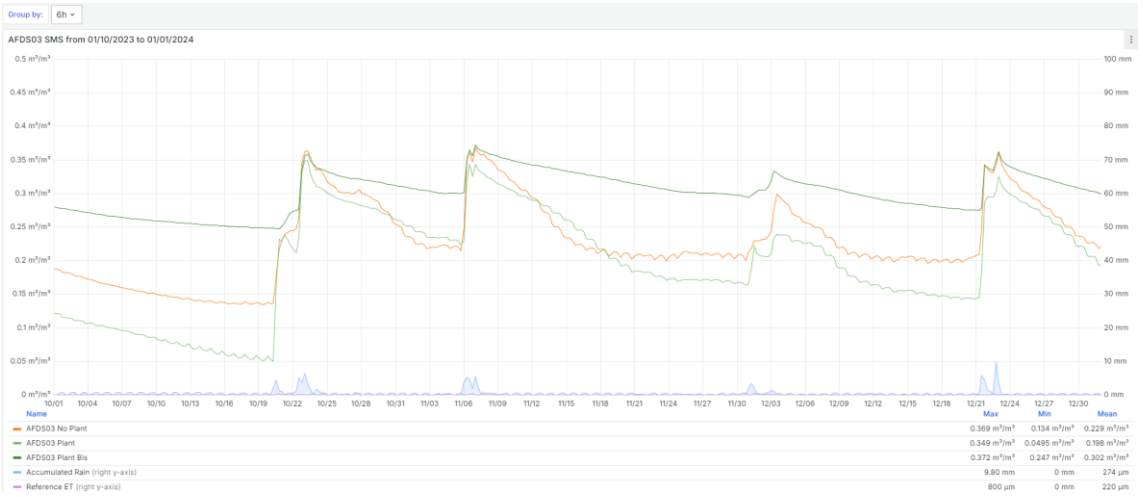


Figure 155 Soil Moisture, rain and evapotranspiration in El Pinillo, October to December, in AFDS 3 sites.

Figure 156 Soil Moisture, rain and evapotranspiration in El Pinillo, October to December, in AFDS 4 and 5 sites.

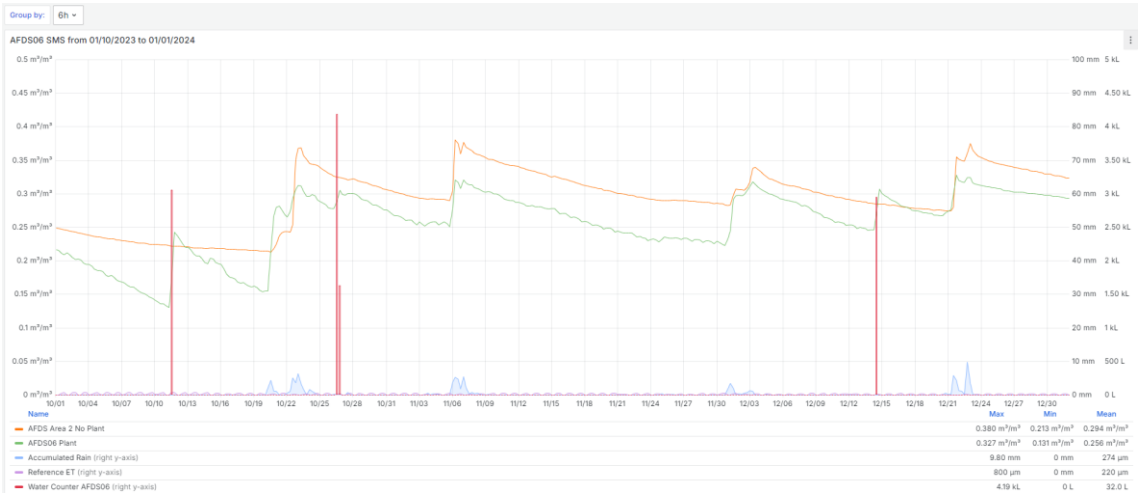


Figure 157 Soil Moisture, rain and evapotranspiration in El Pinillo, October to December, in AFDS 6 sites.

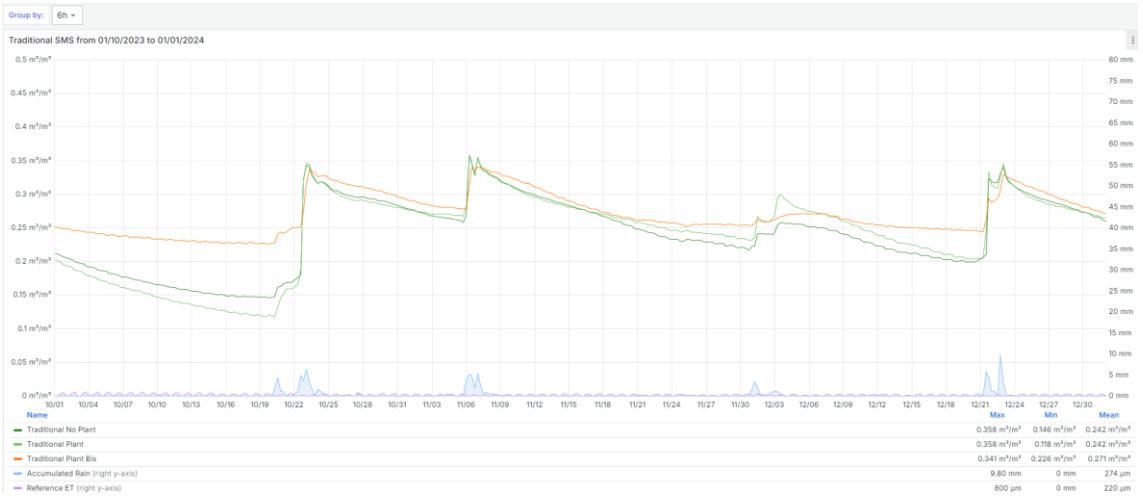


Figure 158 Soil Moisture, rain and evapotranspiration in El Pinillo, October to December, in Traditional area.

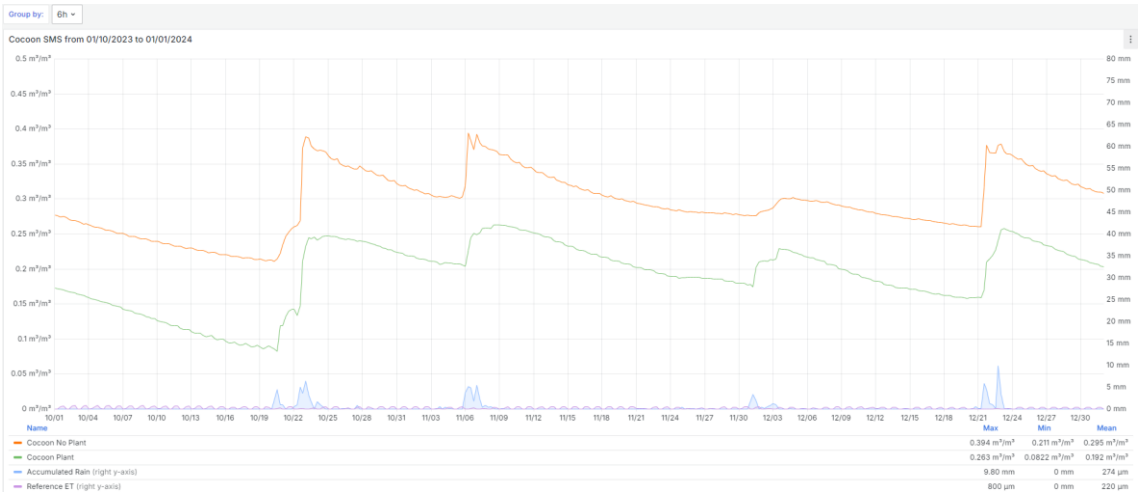


Figure 159 Soil Moisture, rain and evapotranspiration in El Pinillo, October to December, in Cocoon area.

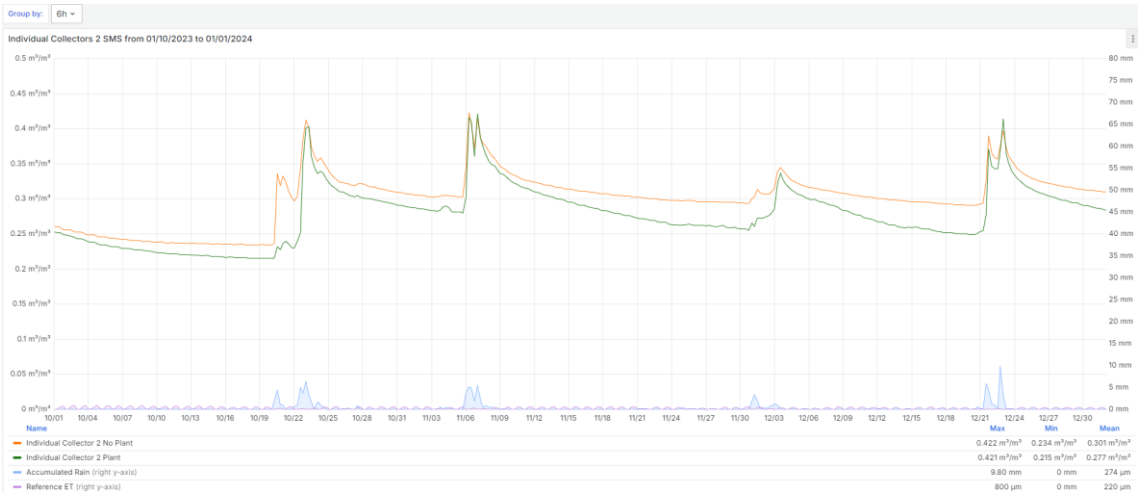


Figure 160 Soil Moisture, rain and evapotranspiration in El Pinillo, October to December, in IFWC 2 sector.

During this period, one forced irrigation event was observed on the experimental plot on October 13th. The influence of precipitation during this time was also noted. During the irrigation event, no moisture was detected in the Deep Drop system. A new soil moisture sensor, referred to as Deep Drop 2, was installed on October 11th and recorded both irrigation and rainfall events.

The highest soil moisture levels (> 0.4 m^3/m^3) were observed in the Traditional, No Plant, Biochar, Biochar 2, and Individual plots. The highest average soil moisture values (> 0.35 m^3/m^3) were recorded in the Traditional, No Plant, and Biochar 2 areas.

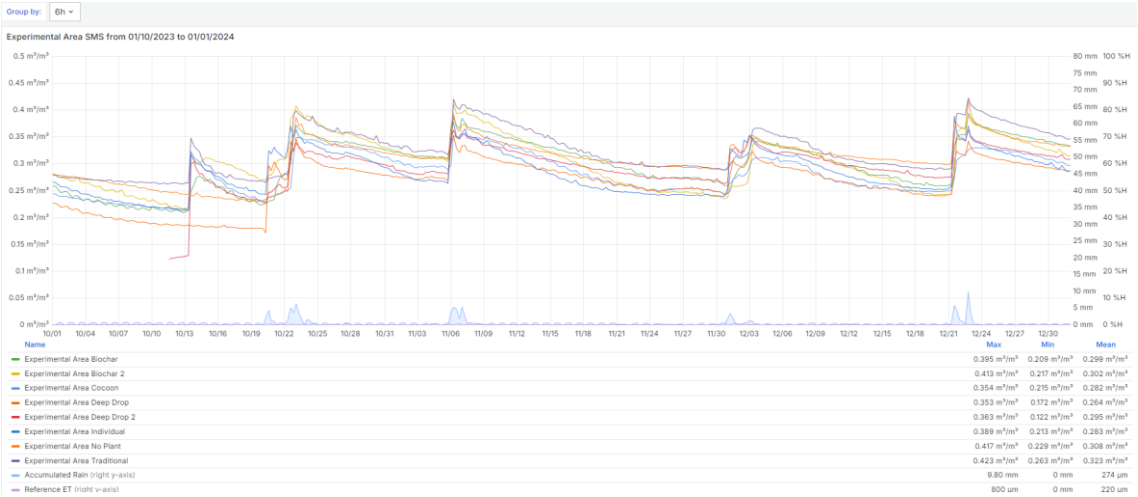


Figure 161 Soil Moisture, rain and evapotranspiration in experimental plot El Pinillo, from October to December 2023.

2024 Q1: January to March 2024

In the first quarter of 2024, we observed a predominant south wind trend, with the exception of March, where the wind trend was weaker and from the north. In January the wind reaches hourly periods with more than 7 m/s.

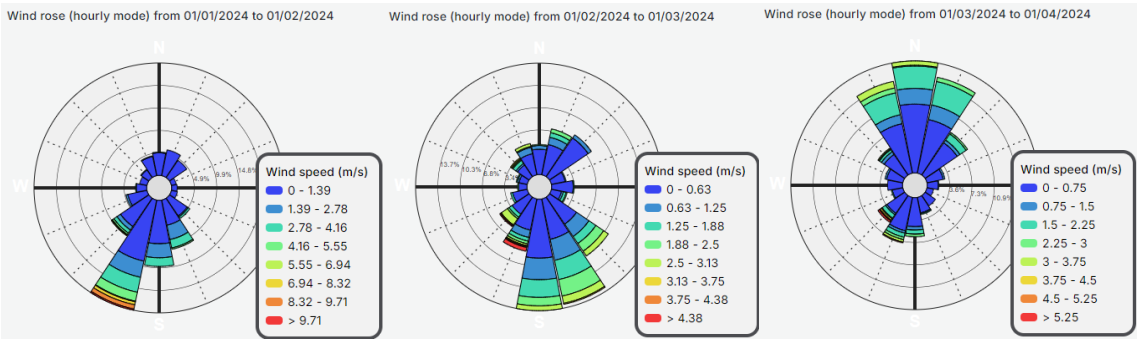


Figure 162. Wind rose (hourly mode) in El Pinillo: January to March 2024.

During this quarter, there was a total of 191 mm of rainfall for the area (15.2 mm in January, 43.6 mm in February and 132 mm in March). The months of January and February stand out due to the widespread low relative humidity levels. The month of January had a maximum of 22.9°C, and an average of 14.7°C, an average relative humidity below 50%. It contrasts with January 2023 with a maximum of 18.5°C, an average of 14.7°C and an average relative humidity over 80%. February 2024 was also dry, but March 2024 saw a decrease in mean temperature, with increased humidity and more rainfall.

Figure 163 Precipitation, relative humidity and solar irradiance graph in El Pinillo, from January to March 2024.

In this period, the soil moisture content in all the areas reacts to rain events. In AFDS area 1, AFDS 2 had a discharge at the beginning of March. It is important to highlight this water discharge since the AFDS2 obtains water from 3 fog water collectors with volumetric mesh.

In AFDS area 2, AFDS 4, 5 and AFDS 6 had different discharges that improve soil water content in these areas. As mentioned above, the AFDS in area 2 are supplied by the spring that collects water in the underground cave reservoir. The inlet flow rate to each AFDS tank is set at a very low level, and discharge occurs automatically when the pre-set water level is reached. It is important to note that these discharges occur at the end of January for AFDS 4 and 5, and in mid-March for AFDS 6, outside of rainy periods, thus improving soil moisture during times of greater water scarcity.

The highest soil moisture levels (> 0.4 m^3/m^3) was obtained only in individual collector area 2. The highest average values of soil moisture (>0.3 m^3/m^3) correspond to the areas of AFDS 1, 2 and 4, and chestnut trees.

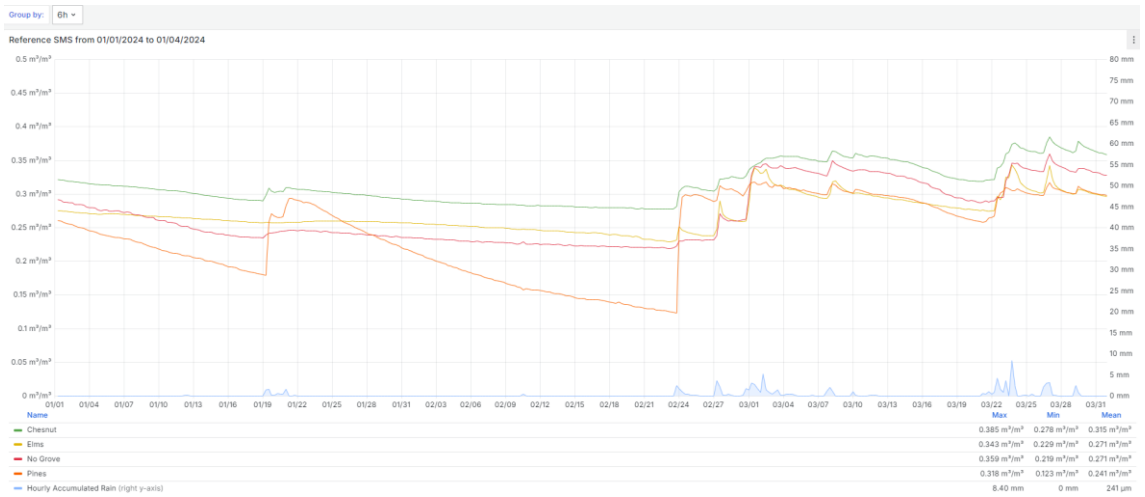


Figure 164 Soil Moisture references and rain in El Pinillo, from January to March 2024.

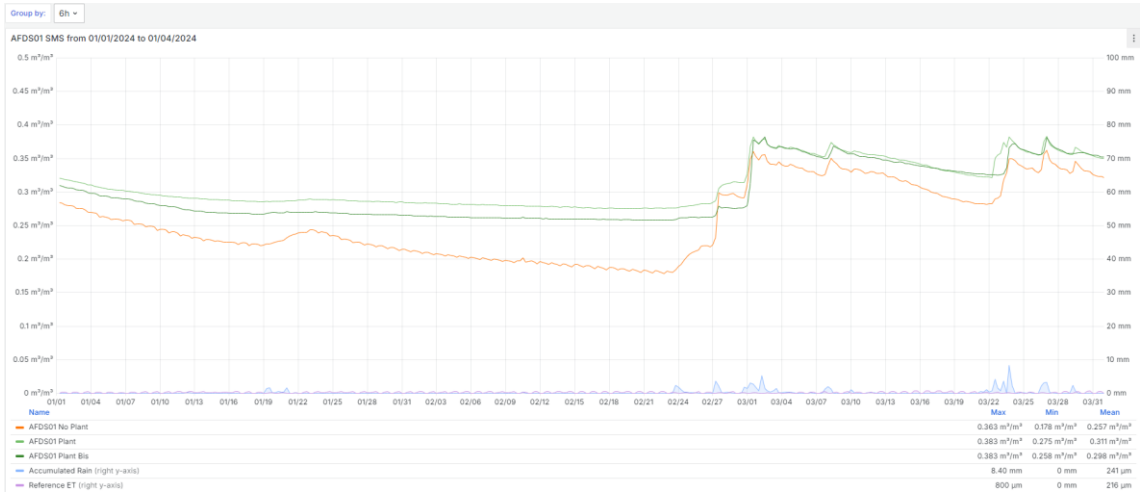


Figure 165 Soil Moisture, rain and evapotranspiration in El Pinillo, January to March 2024, in AFDS 1 sites.

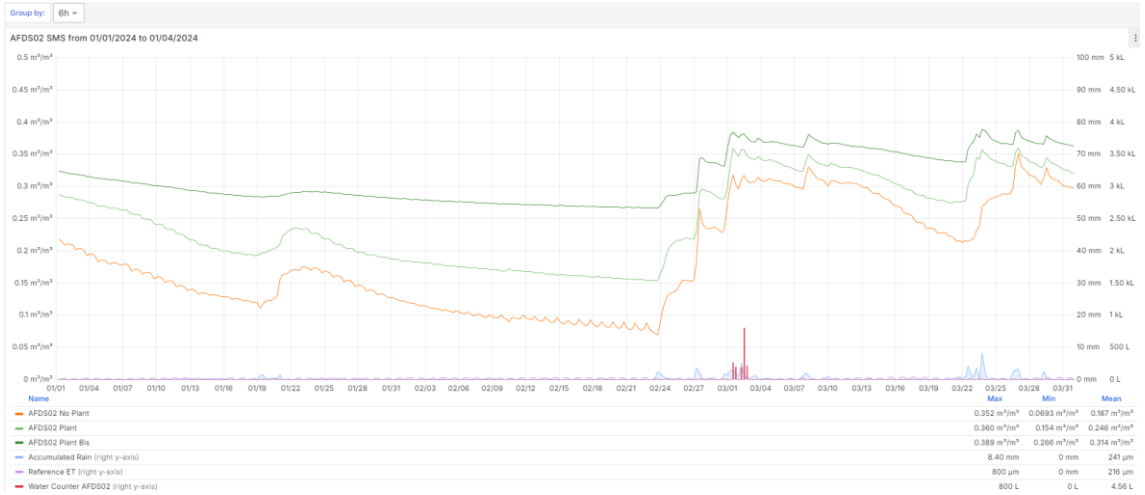


Figure 166 Soil Moisture, rain and evapotranspiration in El Pinillo, January to March 2024, in AFDS 2 sites.

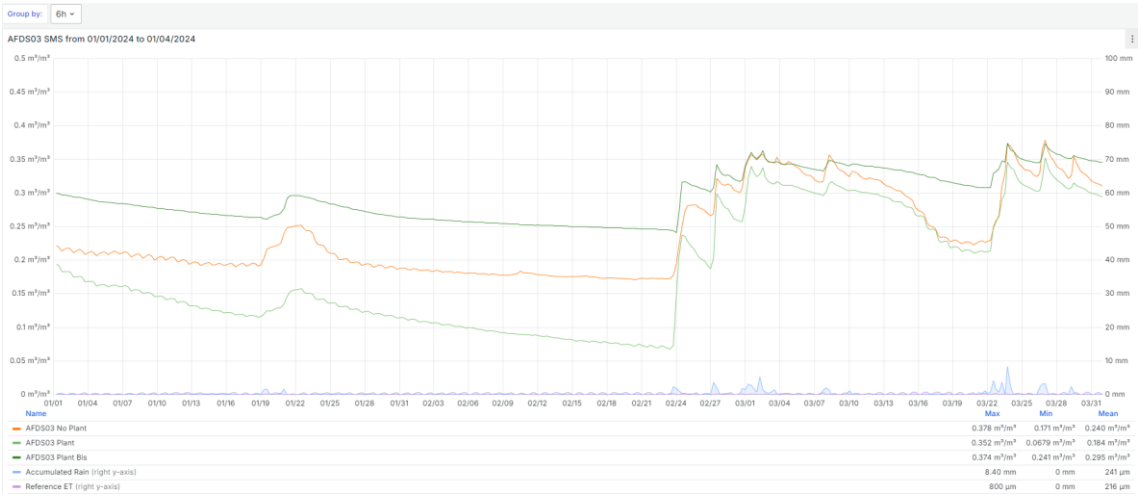


Figure 167 Soil Moisture, rain and evapotranspiration in El Pinillo, January to March 2024, in AFDS 3 sites.

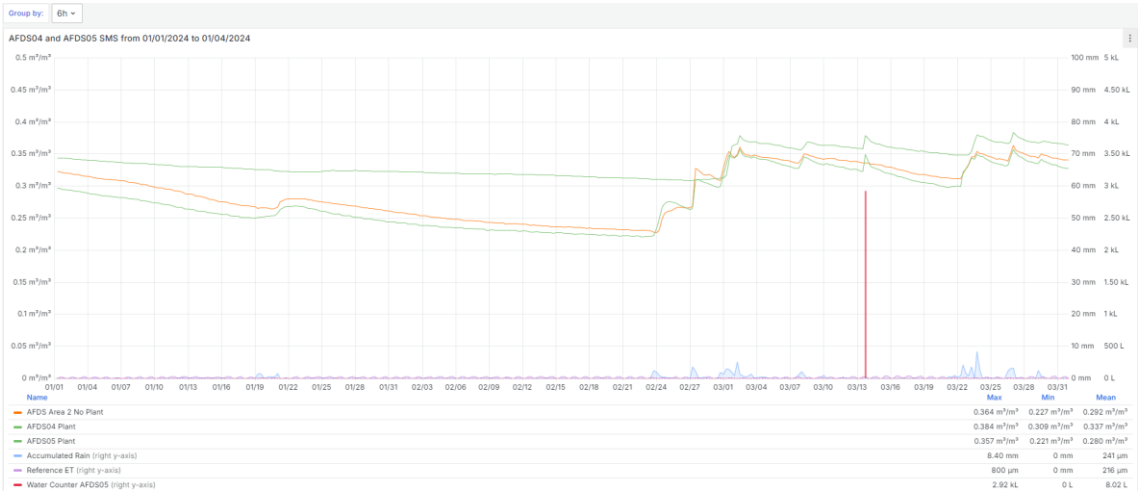


Figure 168 Soil Moisture, rain and evapotranspiration in El Pinillo, January to March 2024, in AFDS 4 an 5 sites.

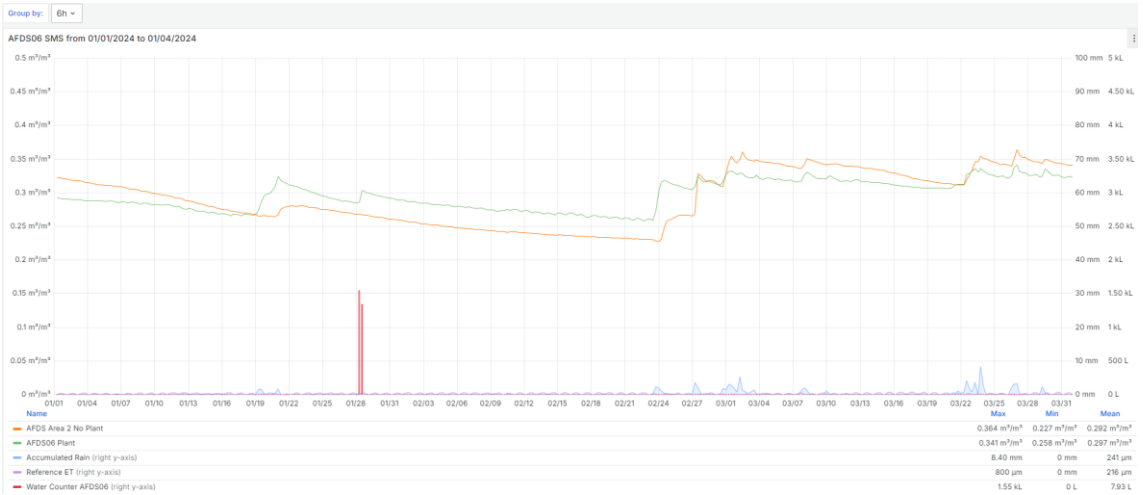


Figure 169 Soil Moisture, rain and evapotranspiration in El Pinillo, January to March 2024, in AFDS 6 sites.

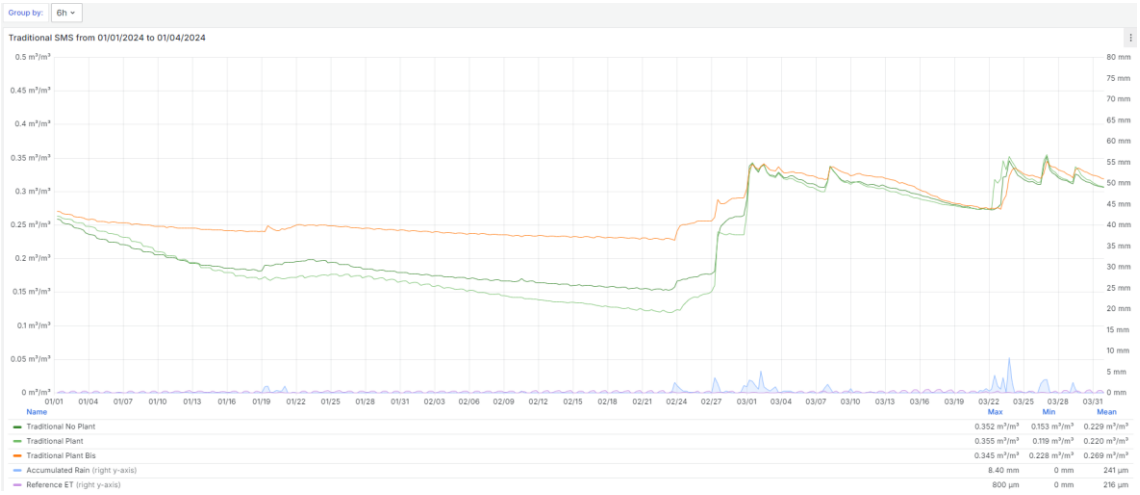


Figure 170 Soil Moisture, rain and evapotranspiration in El Pinillo, January to March 2024, in Traditional area.

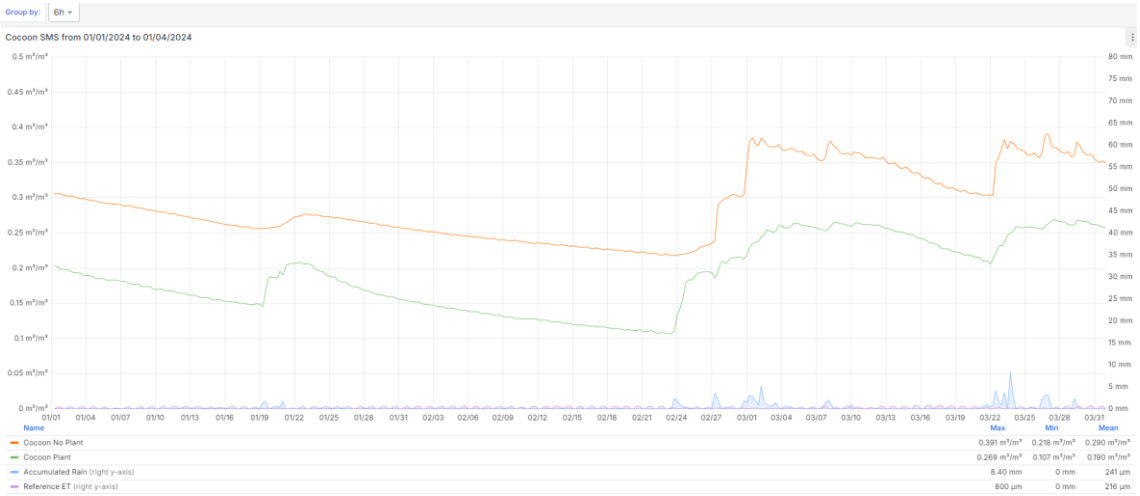


Figure 171 Soil Moisture, rain and evapotranspiration in El Pinillo, January to March 2024, in Cocoon area.

Figure 172 Soil Moisture, rain and evapotranspiration in El Pinillo, January to March 2024, in Individual colllector 2 area.

During this period, no irrigation events were observed on the experimental plot. The influence of precipitation during this time was noted clearly. We detected a sharp drop in the soil moisture content reading from the sensor associated with traditional reforestation (without Biochar) on March 11, 2024, at 9:50 AM. The reasons are still to be determined (as of September 6, 2024).

The highest soil moisture levels in this period (> 0.4 m^3/m^3) were also observed in the Traditional, Biochar, Biochar 2 and Individual plots (0.392 m^3/m^3). The highest average soil moisture values (> 0.28 m^3/m^3) were recorded in the Traditional, No Plant, Biochar an Individual collector points.

The highest soil moisture levels during this period (> 0.4 m^3/m^3) were observed in the Traditional, Biochar, Biochar 2, and Individual collector points (in this last case with a peak of 0.392 m^3/m^3). The highest average soil moisture values (> 0.28 m^3/m^3) were recorded at the Traditional, No Plant, Biochar, and Individual collector points.

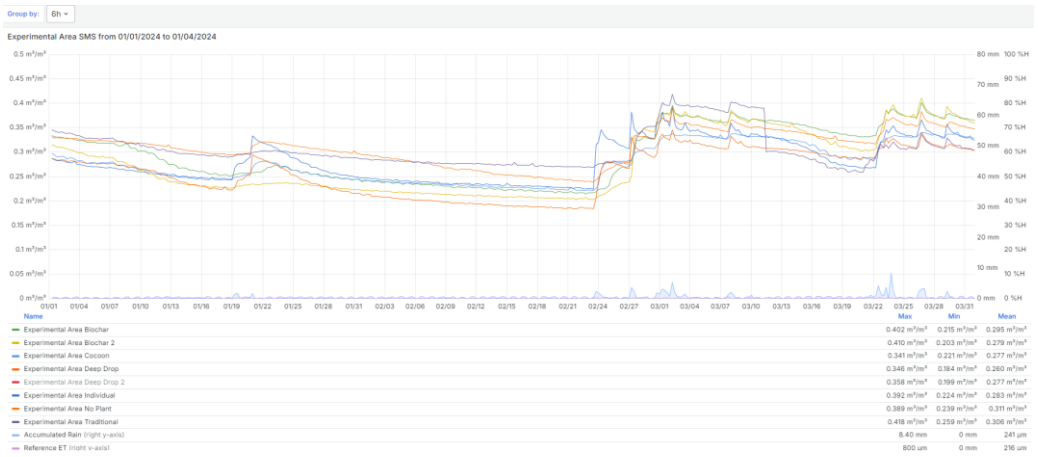


Figure 173 Figure 174 Soil Moisture, rain and evapotranspiration in experimental plot El Pinillo, from January to March 2024

2024 Q2: April to June 2024

In the second quarter of 2024, we observed a predominant northeast wind trend, with the exception of June, where the wind had a northwest to north eas predominant direction. In the whole interval the main range of velocities was 0 to 2 m/s.

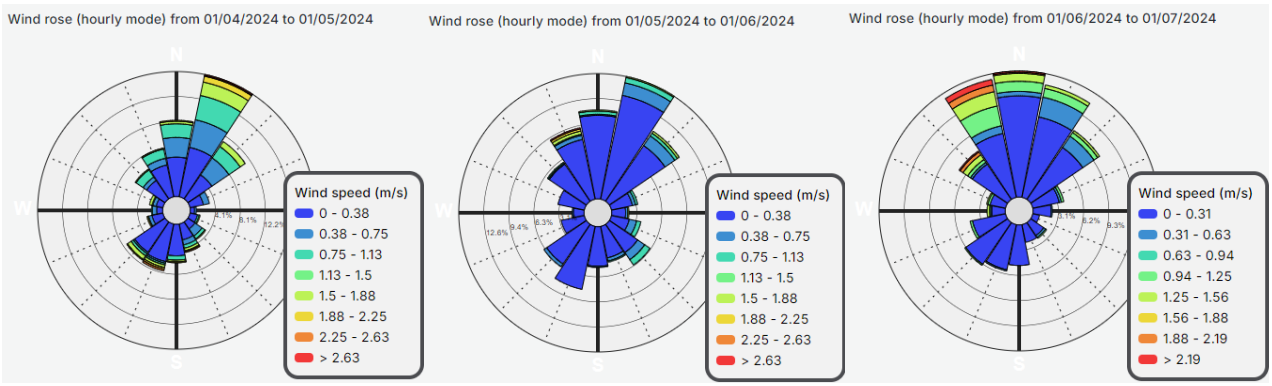


Figure 175. Wind rose (hourly mode) in El Pinillo: April to June 2024.

During this quarter, there was a total of 59.4 mm of rainfall for the area (28.8 mm in April, 23.4 mm in May and 7.2 mm in June). The quarter had mild temperatures with a mean value of 14.8 ºC and average relative humidity between 60 and 70%.

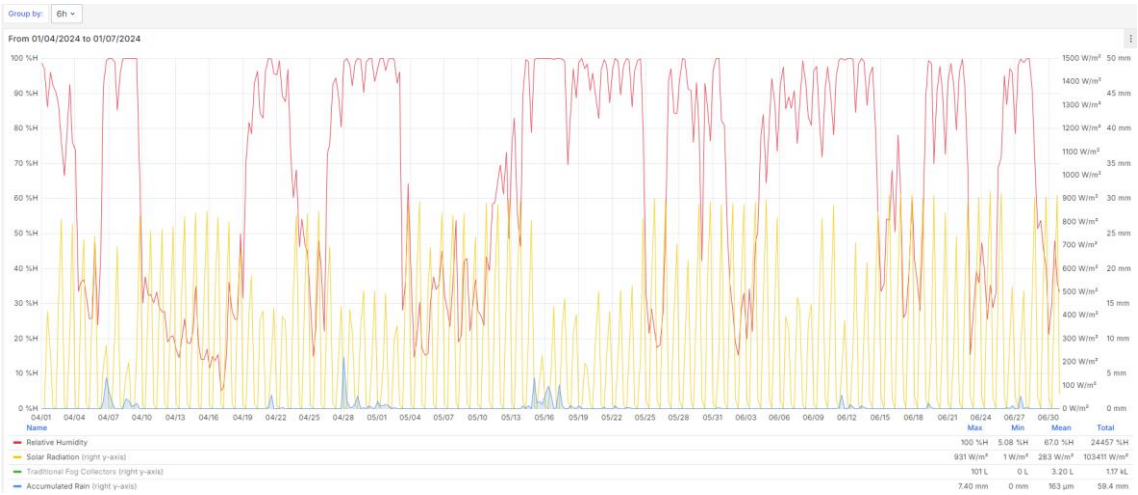


Figure 176 Precipitation, relative humidity and solar irradiance graph in El Pinillo, from April to June 2024.

In this period, the soil moisture content in all the areas reacts slightly to rain events. In AFDS area 1, AFDS 2 had a partial discharge at May 16th, 2:00 am. In AFDS area 2, AFDS 6 had an automatic complete discharge at April 3rd, 16:30 that improve soil water content in this point.

During this period, none of the measurement points exceeded 0.4 m^3/m^3, indicating generally drier soil conditions. The highest soil moisture levels (> 0.35 m^3/m^3) were obtained chestnut area, AFDS 4 (below chestnut forest too), AFDSs 2 and AFDS 1. The highest average values of soil moisture (> 0.3 m^3/m^3) correspond to the areas of AFDS 4, 2 and chestnut trees. Individual collectors continues to react rapidly to rainy events.

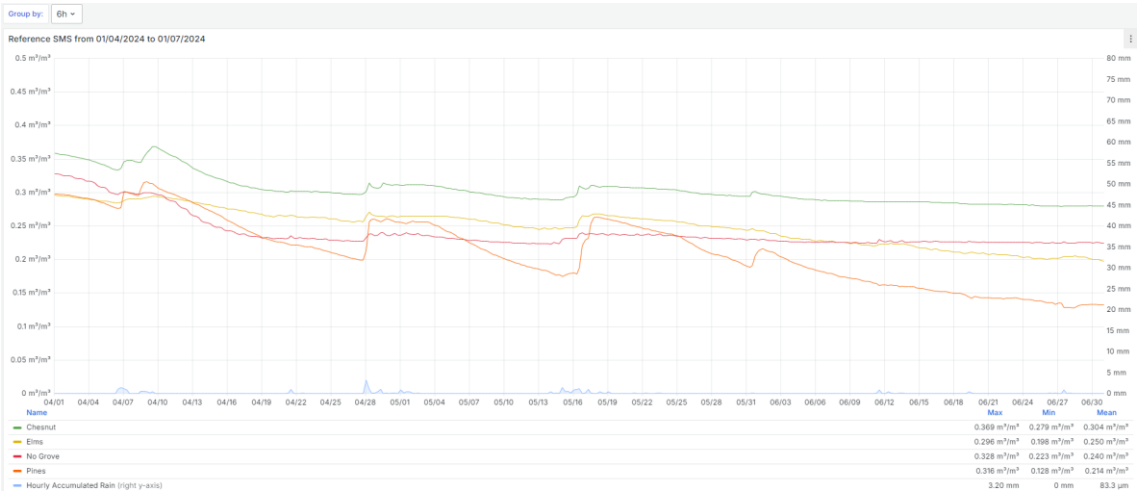


Figure 177 Soil Moisture references and rain in El Pinillo, from April to June 2024.

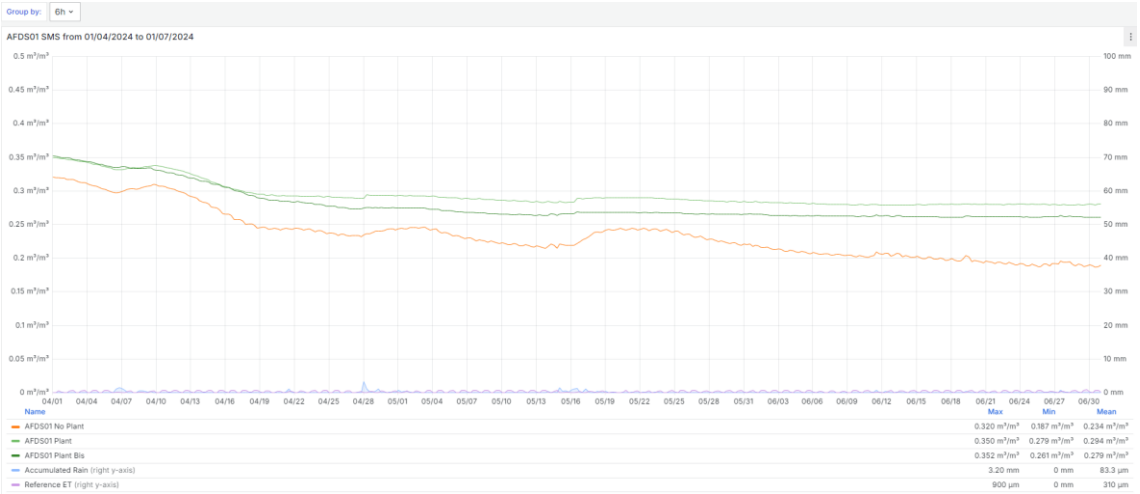


Figure 178 Soil Moisture, rain and evapotranspiration in El Pinillo, April to June 2024, in AFDS 1 sites.

Figure 179 Soil Moisture, rain and evapotranspiration in El Pinillo, April to June 2024, in AFDS 2 sites.

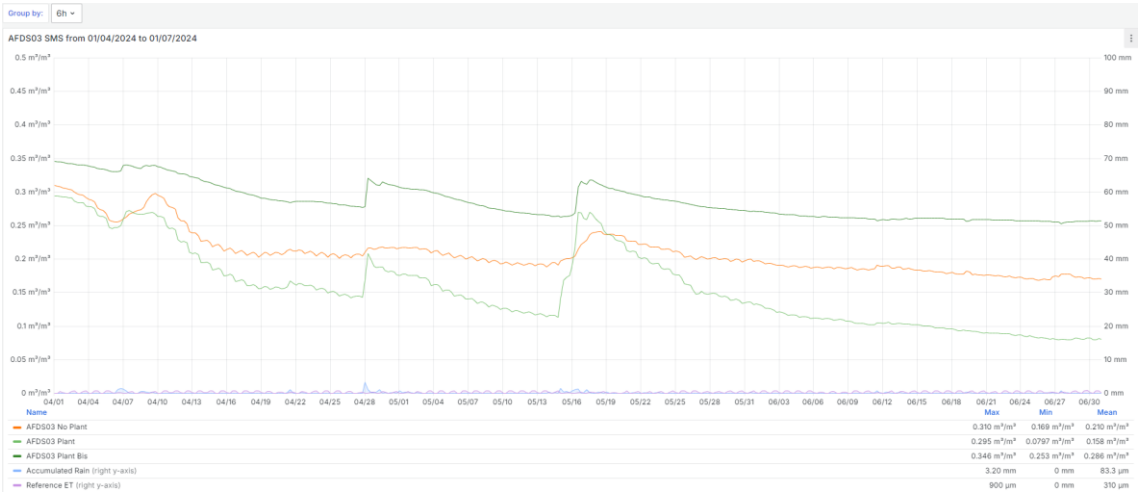


Figure 180 Soil Moisture, rain and evapotranspiration in El Pinillo, April to June 2024, in AFDS 3 sites.

Figure 181 Soil Moisture, rain and evapotranspiration in El Pinillo, April to June 2024, in AFDS 4 and 5 sites.

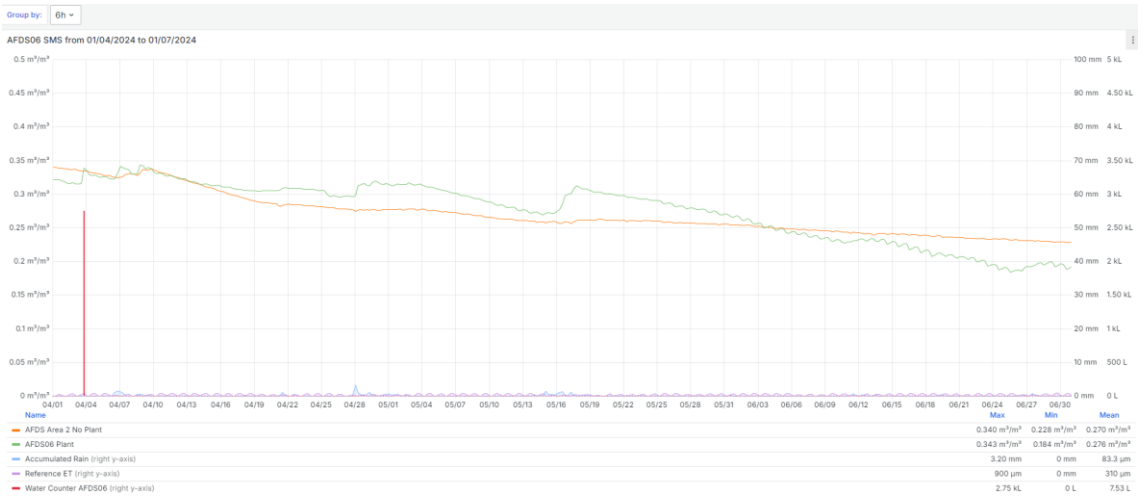


Figure 182 Soil Moisture, rain and evapotranspiration in El Pinillo, April to June 2024, in AFDS 6 sites.

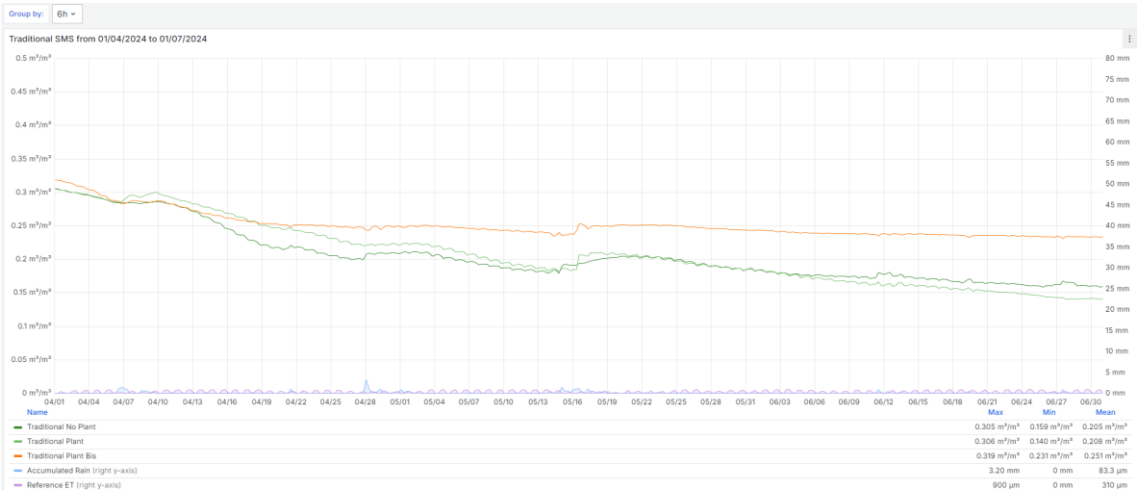


Figure 183 Soil Moisture, rain and evapotranspiration in El Pinillo, April to June 2024, in Traditional reforestation points.

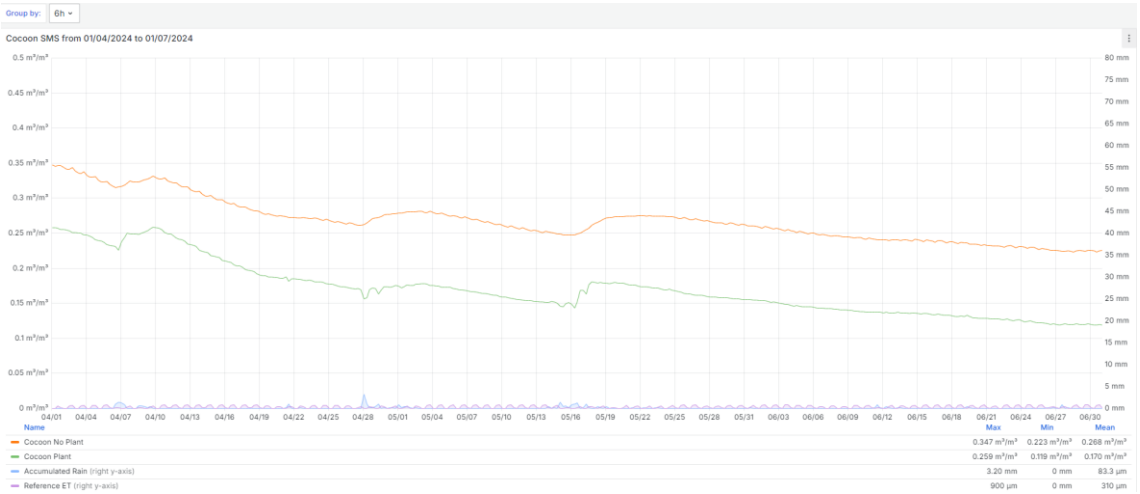


Figure 184 Soil Moisture, rain and evapotranspiration in El Pinillo, April to June 2024, in Cocoon reforestation point.

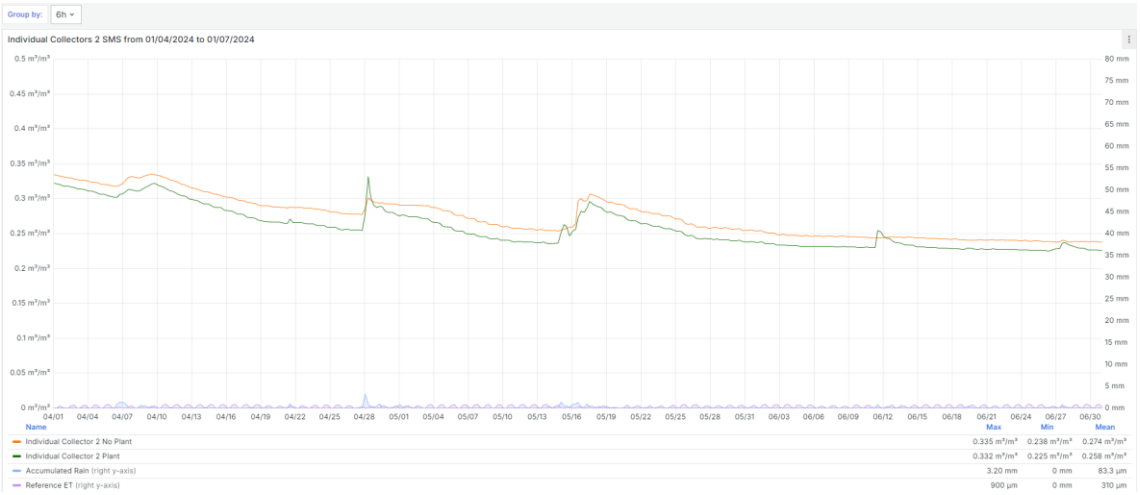


Figure 185 Soil Moisture, rain and evapotranspiration in El Pinillo, April to June 2024, in Individual collector reforestation area 2.

During this period, no irrigation events were observed on the experimental plot. The influence of precipitation during this time was noted clearly.

As was mentioned, we detected that on May 10, 2024, Biochar 2 soil moisture sensor pulse emitter was misalignment with the sun and the signal was lost.

During this period, like the other measurement points in reforestation areas, none of the points in experimental plot exceeded 0.4 m^3/m^3, indicating generally drier soil conditions. The highest soil moisture levels (> 0.35 m^3/m^3) were obtained in Biochar and Biochar 2 points. The highest average values of soil moisture (> 0.25 m^3/m^3) correspond to the points of No plant, Individual collector and Cocoon. Individual collector point continues to react rapidly to rainy events

It is worth noting that the measurement point for traditional reforestation continues to show a sharp decline in soil moisture content.

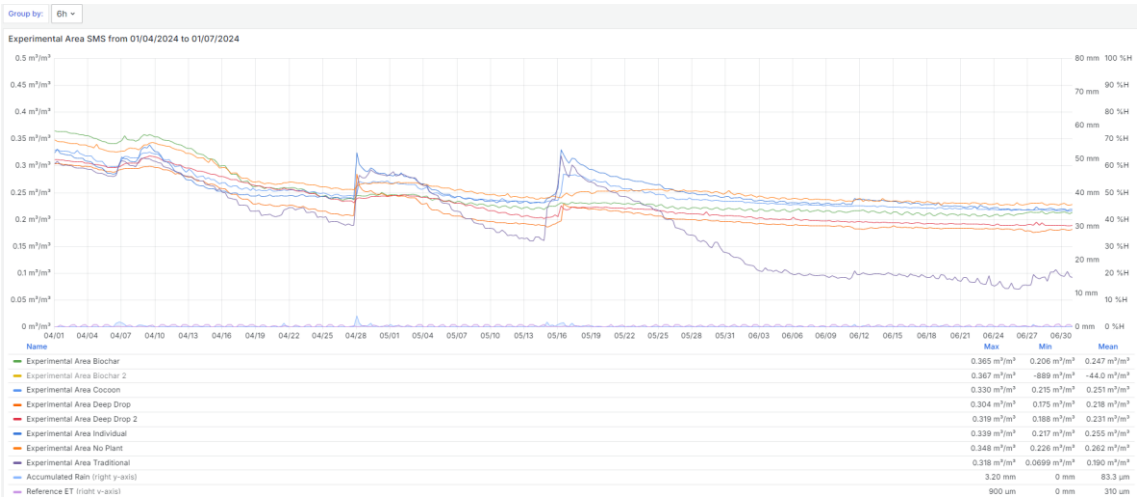


Figure 186 Soil Moisture, rain and evapotranspiration in experimental plot El Pinillo, from April to June 2024.

2024 Q3: July to September 2024

In the third quarter of 2024, we observed a predominant north wind trend, with the exception of August, where the wind was weak and south predominant direction. In the whole interval the main range of velocities was 0 to 1.3 m/s.

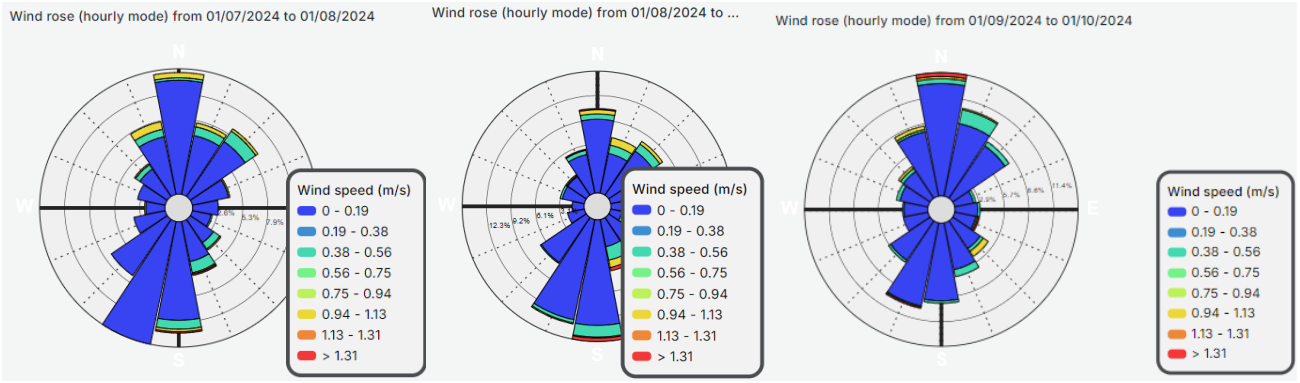


Figure 187. Wind in El Pinillo: July to September 2024 wind roses.

During this quarter, there was a total of 14.2 mm of rainfall for the area (0 mm in July, 400 µm in August and 13.8 mm in September). This quarter experienced high temperatures (with a maximum of 30.2°C, and an average of 21.7°C), an average relative humidity above 46%, and high solar irradiance. In September, there was a little rainfall, which amounted to 13.8 mm.

Figure 188 Precipitation, relative humidity and solar irradiance graph in El Pinillo, July to September 2024.

During this period, soil moisture levels in Pine area responded noticeably to the rain and fog collection event in September. The soil moisture sensors in the Elm areas also responded, though to a lesser extent.

AFDS 2 had a total discharge at July 3rd. In AFDS area 2, AFDS 4-5 and AFDS 6 had a forced complete discharge at July 3rd. Then AFDS 4-5 had an automatic complete discharge at July 22nd and AFDS 6 had an automatic complete discharge at July 18th that improve soil water content in these points.

During this period, none of the measurement points exceeded 0.4 m^3/m^3, indicating generally drier soil conditions. The highest soil moisture levels (> 0.35 m3/m3) were obtained only in AFDS 4 (below chestnut forest) during irrigation event.

The highest average values of soil moisture (> 0.26 m^3/m^3) correspond to the areas of AFDS 4, 2 and chestnut trees.

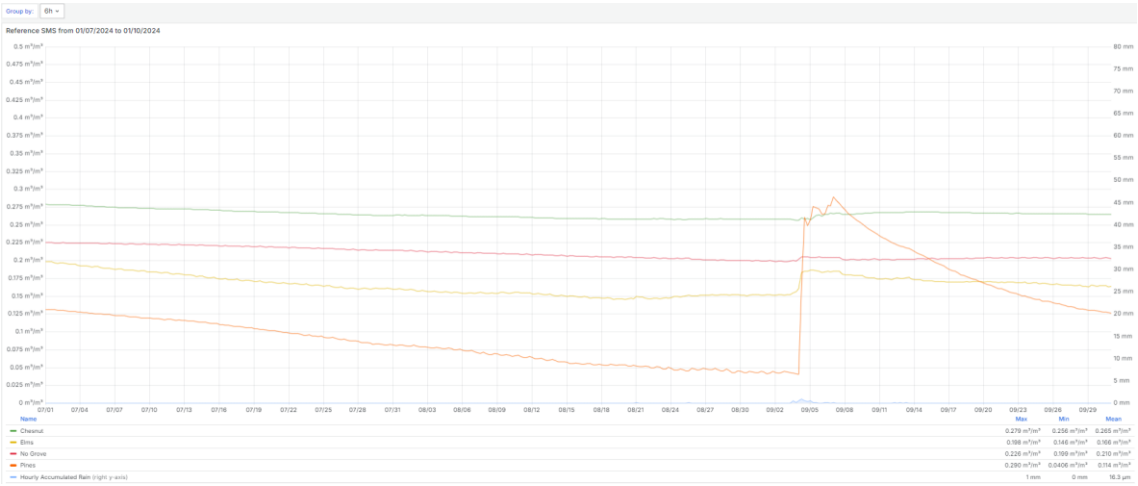


Figure 189 Soil Moisture references and rain in El Pinillo, from July to September 2024.

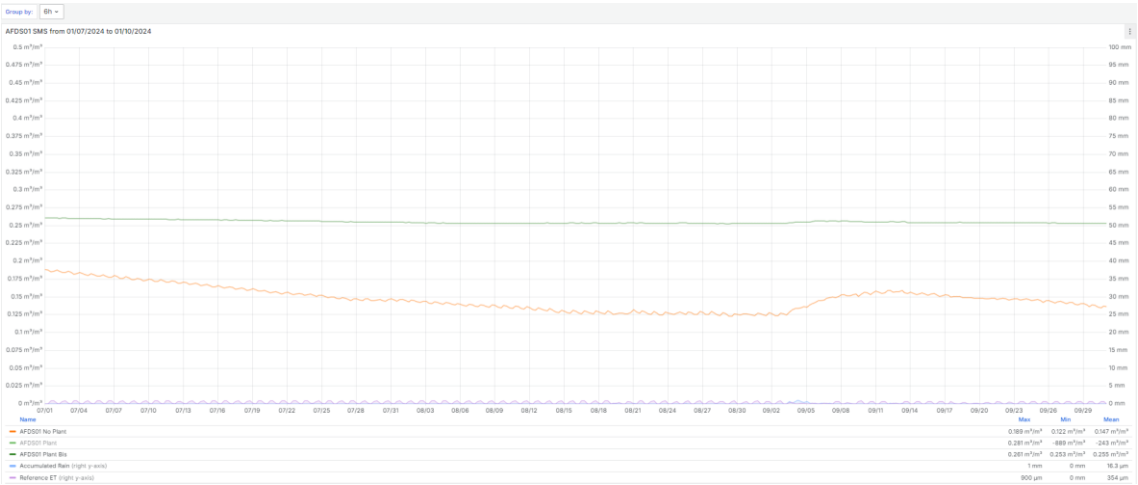


Figure 190 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September 2024, in AFDS 1 sites.

Figure 191 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September 2024, in AFDS 2 sites

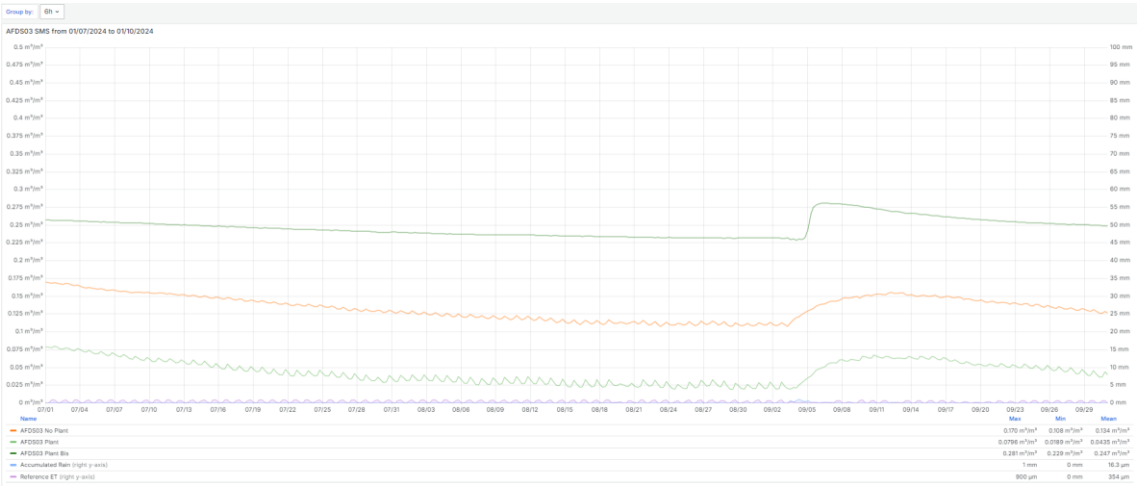


Figure 192 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September 2024, in AFDS 3 sites.

Figure 193 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September 2024, in AFDS 4 and 5 sites.

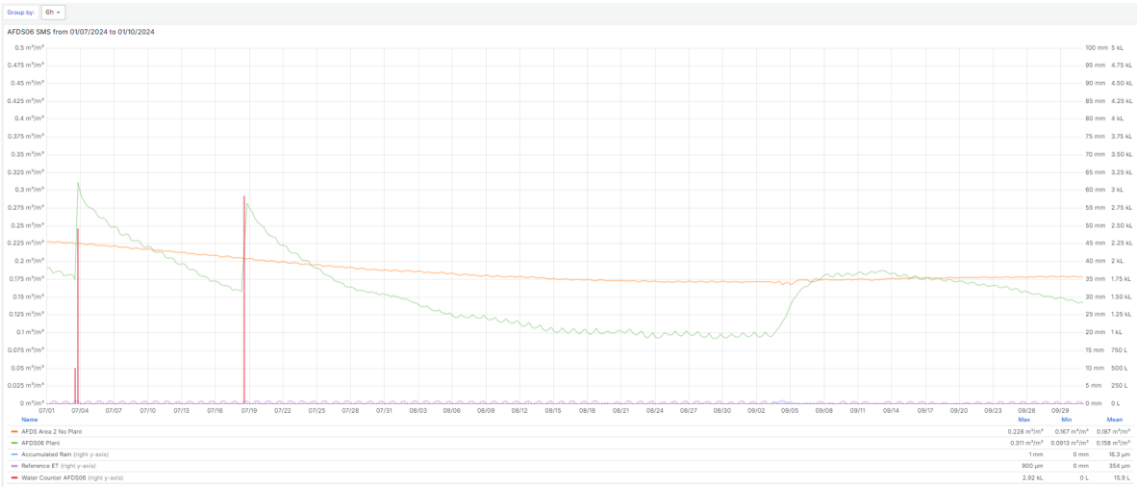


Figure 194 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September 2024, in AFDS 6 sites.

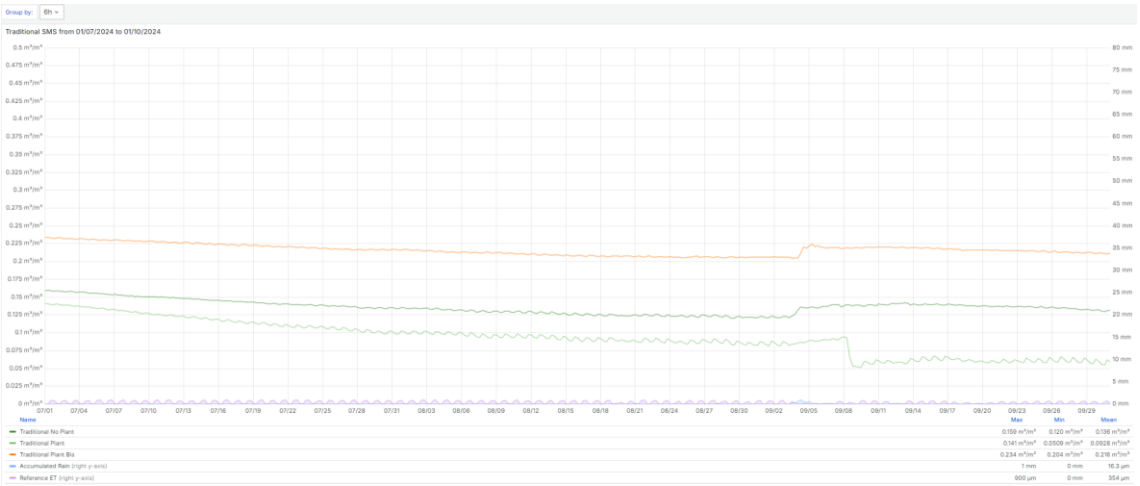


Figure 195 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September 2024, in Traditional reforestation points.

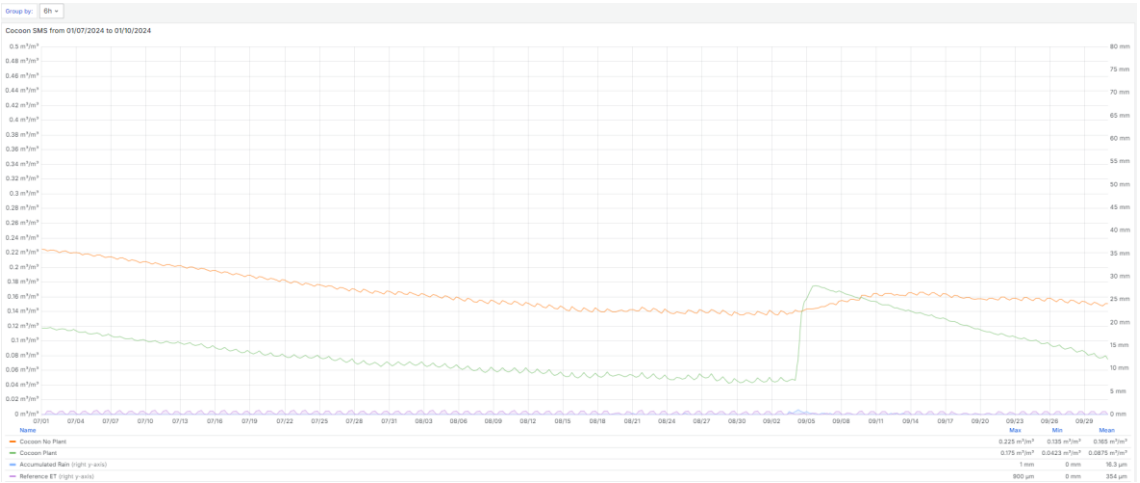


Figure 196 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September 2024, in Cocoon reforestation point.

Figure 197 Soil Moisture, rain and evapotranspiration in El Pinillo, July to September 2024, in Individual collector reforestation area 2.

During this period, an irrigation event was observed on the experimental plot (July 17th).

During this period, only Deep Drop 2 point exceeded 0.4 m^3/m^3 during the irrigation event. In the second position, the highest soil moisture levels (> 0.33 m^3/m^3) were obtained in Biochar, Biochar 2 and Cocoon points. The highest average values of soil moisture (> 0.21 m^3/m^3) correspond to the points of Cocoon, Individual collector, Biochar and Biochar 2. Individual collector point continues to react slightly to rain event in this case. It is worth noting that the measurement point for traditional reforestation continues to show a sharp decline in soil moisture content.

Figure 198 Soil Moisture, rain and evapotranspiration in experimental plot El Pinillo, from July to September 2024.

2024 Q4: October to December 2024

In the fourth quarter of 2024, we observed a predominant south wind trend, with the exception of October, where the wind was weak and north predominant direction. In the whole interval, the main range of velocities was more than 5 m/s.

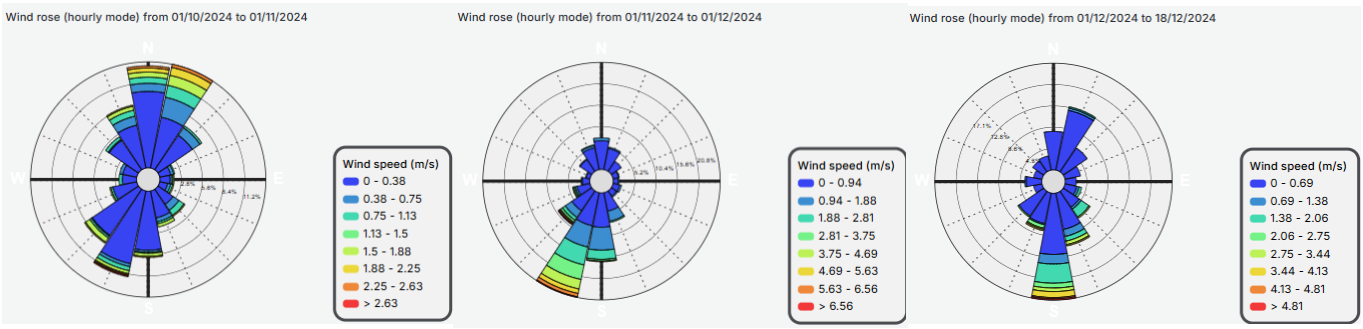


Figure 199. Wind in El Pinillo: July to September 2024 wind roses.

During this quarter, there was a total of 210 mm of rainfall for the area (110 mm in October, 6.4 mm in November and 94 mm in December). This quarter experienced mild temperatures (with a maximum of 31.4 °C, and an average of 15.5°C), an average relative humidity above 61%, and less solar irradiance.

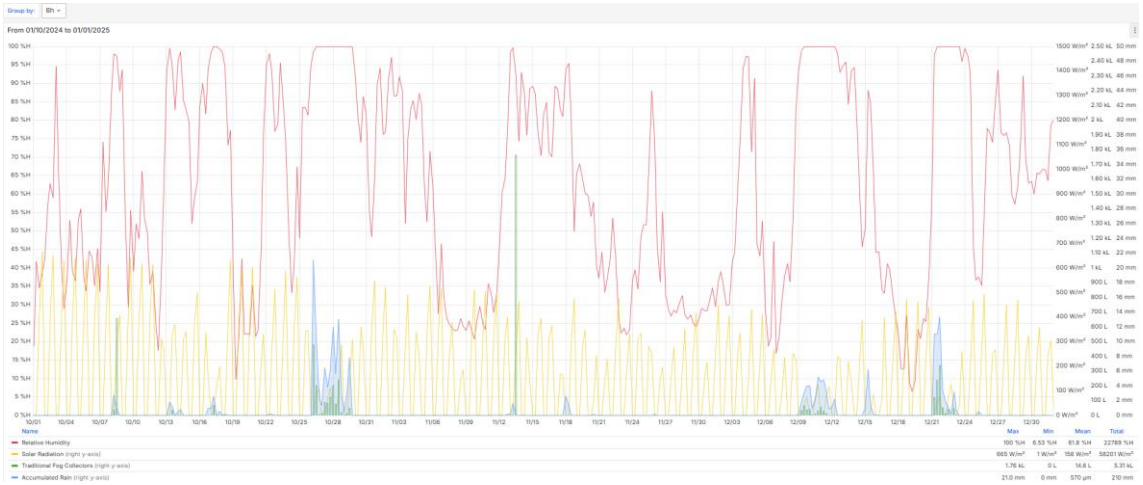


Figure 200 Precipitation, relative humidity and solar irradiance graph in El Pinillo, from October to December 2024.

During this period, soil moisture levels in Pine area responded noticeably to the rain and fog collection events. The soil moisture sensors in the chestnut area was lost in tis period. AFDS 1 and 3 were uninstalled in this period. AFDS 2 had not discharges. In AFDS area 2, AFDS 4-5 had

two complete discharges and the system was adapted for internal operation. AFDS 6 had three complete discharges in this period. During this period, none of the measurement points exceeded 0.4 m^3/m^3, indicating generally drier soil conditions.

The highest value of soil moisture (> 0.45 m^3/m^3) correspond to the areas of Individual collector 2 during a rain period on 26th of October, the second is AFDS4 (> 0.36 m^3/m^3) in the irrigation event of 6th of October.

Figure 201 Soil Moisture references and rain in El Pinillo, from October to December 2024.

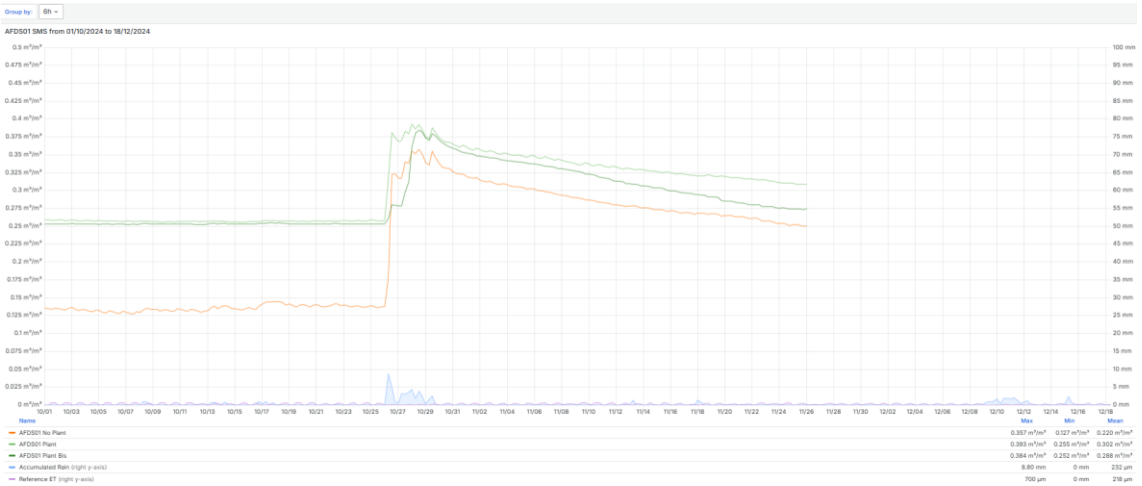


Figure 202 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to 18th of December 2024, in AFDS 1 sites.

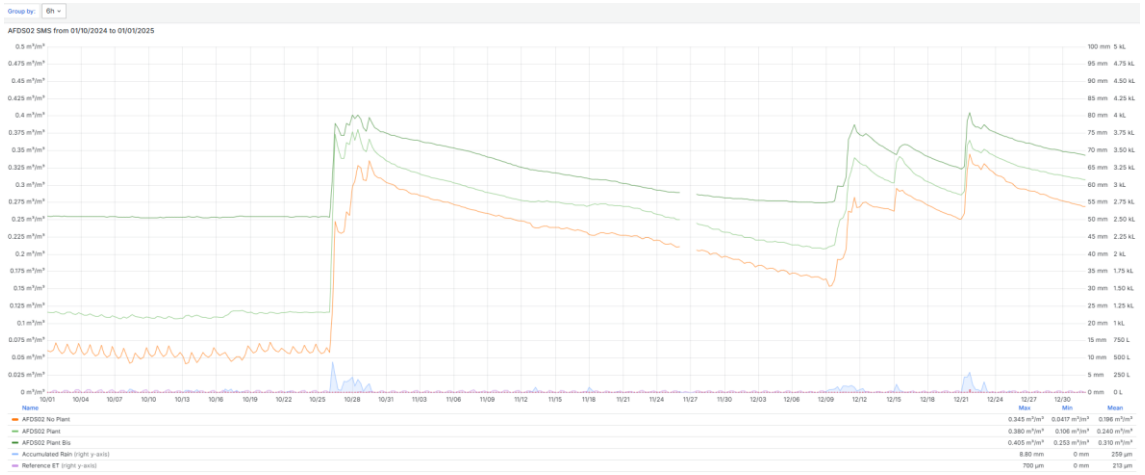


Figure 203 Soil Moisture, rain and evapotranspiration in El Pinillo from October to 18th of December 2024, in AFDS 2 sites.

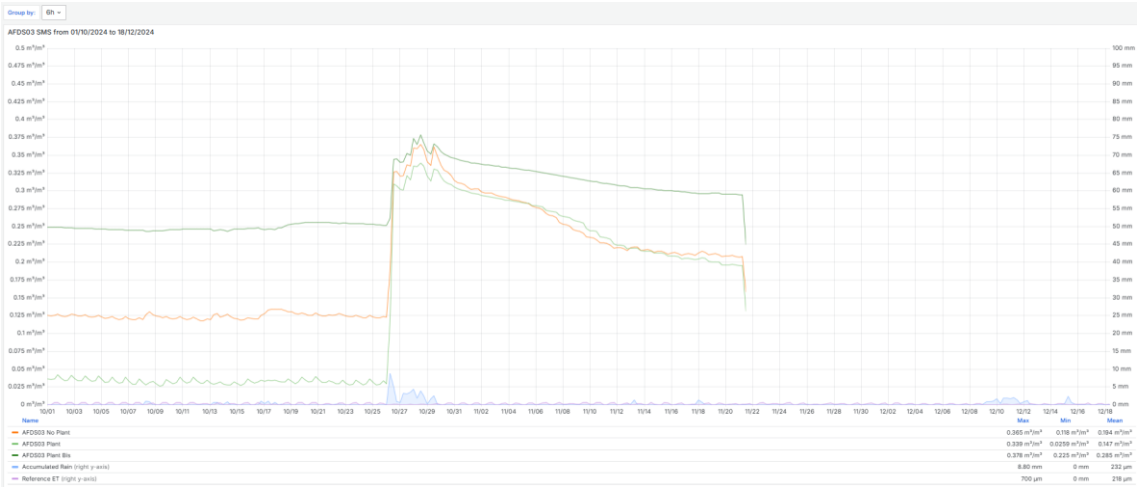


Figure 204 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to 18th of December 2024, in AFDS 3 sites.

Figure 205 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2024, in AFDS 4 and 5 sites.

Figure 206 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2024, in AFDS 6 sites.

Figure 207 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2024, in Traditional reforestation points.

Figure 208 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2024, in Cocoon

Figure 209 Soil Moisture, rain and evapotranspiration in El Pinillo, from October to December 2024, in Individual collector reforestation area 2.

During this period, none irrigation event was observed on the experimental plot. In 15th of November were installed new measure points, called Biochar Bis, Biochar 2 Bis and Individual Bis. Measure of Biochar 2 point was recovered and repaired. During this period, only Biochar 2 point exceeded 0.4 m^3/m^3 during a rain event at the end of October. In the second position, in this event (> 0.382 m^3/m^3) was obtained in Biochar (Figure 210).

Figure 210 Soil Moisture, rain and evapotranspiration in experimental plot El Pinillo, from October to December 2024.

During one of the cases (26th of October), rainfall began at 3:00 a.m., and the soil moisture content in the IFWC increased almost immediately. After 45 minutes of rainfall, an increase in soil moisture was observed at the point representing traditional reforestation, while the measurement point with biochar required 2.5 hours of light but persistent rainfall. It was not until 6.5 hours later that the soil moisture levels at the traditional reforestation and biochar points reached the level that the IFWC had achieved almost immediately (Figure 211).

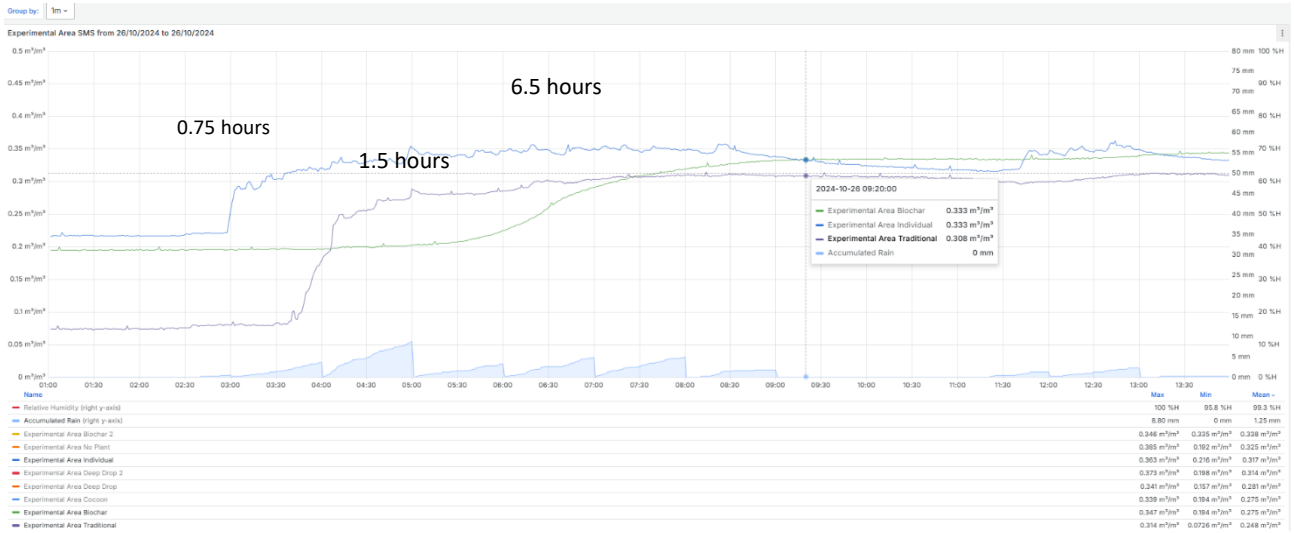


Figure 211 Difference in reaction time of soil moisture in IFWC, Biochar and Traditional.

4.3.3.4 Results and conclusions from soil moisture monitoring in El Pinillo

In general, soil moisture content fluctuates significantly with meteorological conditions and varies across reforestation systems and sites (Pines, Elms, Chestnuts, AFDS, Traditional, CCN, IFWC areas).

High temperatures and low humidity, especially in summer months, contribute to a rapid decrease in soil moisture.The area with the fastest and most fluctuating response to meteorological conditions is the *Pinus canariensis* zone, with peaks exceeding 0.35 m^3/m^3 on February 26, 2022, and dropping to 0.038 m^3/m^3 on August 13, 2022. The most stable area has been the zones under the shade of chestnut trees, with values ranging between 0.386 and 0.256 m^3/m^3.

Similarly, areas equipped with reforestation systems based on Individual Collectors show a rapid response to light rainfall and fog, with numerous peaks exceeding 0.4 m^3/m^3 throughout the study period. The irrigation networks associated with AFDS have clearly contributed to improving soil moisture content regardless of meteorological conditions. In cases of emergency irrigation, the operation can be carried out with minimal effort, as more than 400 plants, in each irrigation network, can be watered simultaneously with a simple procedure.

4.3.3.4.1 Reference soil moisture sensors

Reference sensors were placed in the Chestnut, Pines, and Elms zones to monitor soil moisture levels independently of additional irrigation systems. These areas reflect natural responses to weather conditions without any minimal irrigation.

Data from these reference sensors serve as a control, helping to compare the effects of meteorological changes on different tree species without external water input. With the data registered (n>50,000 /year) were obtained some indicators. Between these indicators is highlighted the Coefficient of Variation (CV) of the water content in the soil. (Figure 212). A CV of 10%, for example, indicates a low variability of soil moisture, as most of the data are close to the mean.

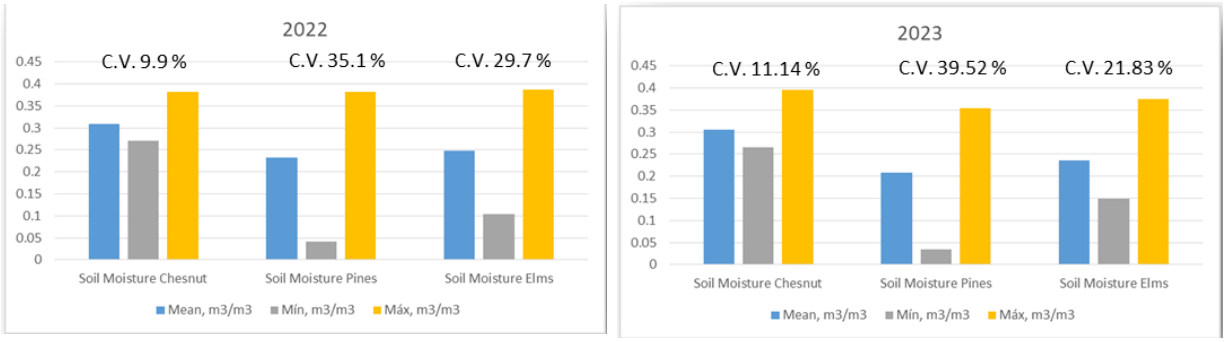


Figure 212 Indicators of reference Soil Moisture Sensors during 2022 and 2023 in El Pinillo.

Figure 213 Comparison of soil moisture content at the point of measurement in the pine and chestnut areas during 2022, 2023 and 2024.

After comparing the data from the three different reference soil moisture sensors, we can conclude that, there are some differences that can be observed:

- The Chestnut area typically exhibited more stable soil moisture retention compared to the Pines zone or the Elms. This may be attributed to the denser foliage in the Chestnut forest, which provides greater shade coverage, reduces evaporation, and contains more organic matter, allowing it to temporarily retain moisture from rainfall. During significant rainfall events, such as in Q3 2022 (Storm Hermine) and Q1 2023, the Chestnut zone responded with sharp increases in soil moisture, though it was slower to return to drier conditions compared to the Pines zone.
- The Pines reference sensor shows more rapid responses to rainfall events and fog collection compared to the Chestnut and Elms zones. The Pines zone exhibited a stronger capacity to increase soil moisture following rain or fog collection events. The Pines showed faster responses to rainfall and fog collection, although it often faced longer periods of water stress between rain events. This may be attributed to the type of trees, which likely have less dense foliage cover compared to the Chestnut. The morphology of pines allows rain droplets and fog to pass through. However, the needle structure of the Canary pine, with needles grouped in sets of three and measuring between 20 and 30 centimetres in length, enables it to intercept raindrops and fog, channelling water toward the base of each tree. Additionally, the type of soil in which the pines grow, along with the layer of fallen pine needles on the ground, may further contribute to this behaviour by creating a more favourable environment for water infiltration.
- The Elms sensor faced operational issues, particularly in Q1 2022, where battery charging was inconsistent. Corrective measures were implemented to ensure adequate solar exposure for the sensor. During rainfall periods, the Elms zone experienced modest increases in soil moisture, but these were not sustained for long. For example, after the rainfall in April 2023 and May 2023, the soil moisture levels in the Elms zone did not increase as much as in the Pines or Chestnut zones; although the CV was lower than Pines.

The results of the measurements carried out at Finca El Pinillo suggest that soil moisture measurements in the Canary pine forest can be a good indicator for detecting fog water capture conditions. The following Figures show examples where soil moisture increased at the measurement point within the pine forest, while the other points remained unaffected. (Figure 214 to Figure 219).

Figure 214 An event in May 2023 where soil moisture in the pine forest increased while the other reference points remained unchanged.

Figure 215 Weather conditions and water collection in May 2023 event.

Weather conditions just previously the increasing Pine soil moisture sensor were detected:

Period: 8 hours (0:00 – 8:00; 18/05/2023)

Total water collected: 70 litres

Total Fog Water Collector impact area: 88 m²

Water collected per m² of impact: 0.8 L/m²

Water per m² of impact per hour: 0.1 L/m²·h

Total accumulated rainfall: 1.6 mm

Rainfall per hour: 0.2 L/h

Relative humidity: 100%

Average wind speed: 1.72 m/s

Maximum wind speed: 2.8 m/s

Maximum gust speed: 5.24 m/s

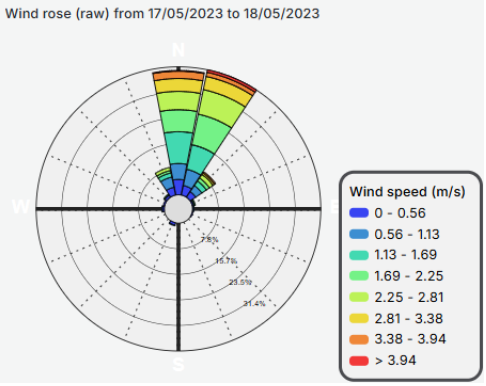


Figure 216

Figure 217 An event in September 2024 where soil moisture in the pine forest increased while the other reference points remained unchanged.

Figure 218 Weather conditions and water collection in September 2024 event.

Weather conditions just previously the increasing Pine soil moisture sensor were detected:

Period: 22 hours (10:00 – 8:00; 3 - 4/09/2024)

Total water collected: 213 litres

Total Fog Water Collector impact area: 72 m²

Water collected per m² of impact: 3 L/m²

Water per m² of impact per hour: 0.13 L/m²·h

Total accumulated rainfall: 8.4 mm

Rainfall per hour: 0.38 L/h

Relative humidity: 100%

Average wind speed: 0.03 m/s

Maximum wind speed: 0.3 m/s

Maximum gust speed: 1.3 m/s

Minimum visibility: 445 m

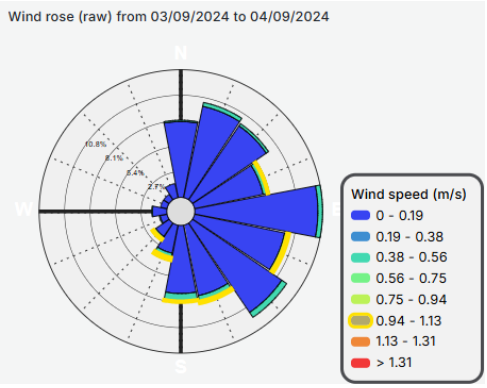


Figure 219

Chestnut and Elm sensors retaining soil moisture for a longer period than Pines may be explained by the shading and protection provided by the trees, which can reduce the amount of sunlight and wind exposure that can lead to evaporation. Additionally, the type of soil where both tree species grow may also contribute to this behaviour, as it seems to have a higher organic matter, which can improve water retention. These conditions, in addition to the increased water capture, are expected to occur as the laurisilva forest in Finca El Pinillo develops.

4.3.3.4.2 Traditional and Cocoon Reforestation Areas:

During the project, the soil moisture measurement points in the traditionally reforested areas and Cocoon did not receive any additional irrigation, with soil water content fluctuating solely in response to rainfall. Typically, more than 3 mm/day of precipitation is required to detect an increase in soil moisture at these points (Figure 220).

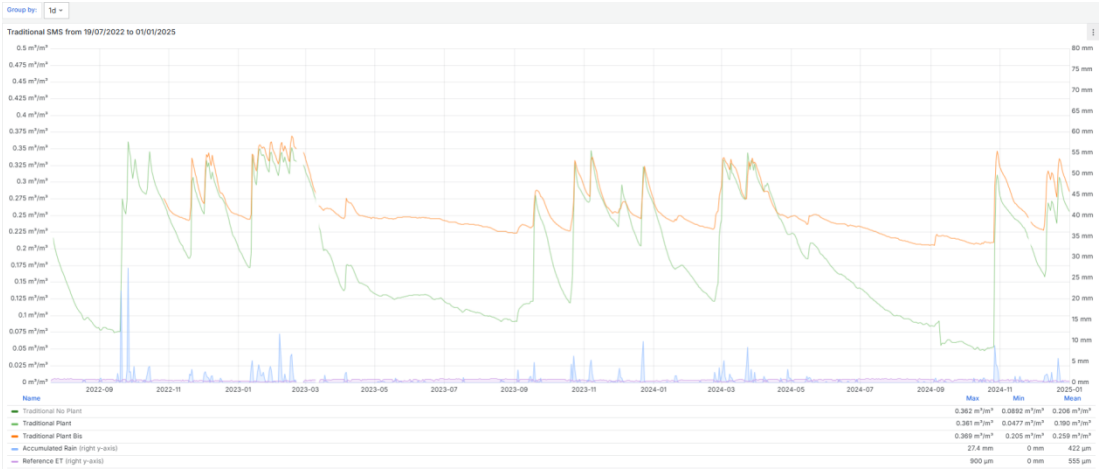


Figure 220 Evolution of soil moisture levels in traditional reforestation points.

In these conditions, the traditional reforestation is particularly vulnerable to extreme weather fluctuations, where periods of drought severely impact plant health. The lack of supplemental water sources in traditional areas results in higher mortality rates and slower growth, especially during the hotter, drier months.

In general, the Cocoon point measure showed lower soil moisture retention compared to traditional reforestation. While both systems experienced moisture loss during dry periods, the Cocoon system demonstrated a quicker and stronger response to rainfall events than the traditional reforestation method (Figure 220).

*Figure 221 After light rain events, the Cocoon system showed measurable increases in soil moisture levels compare with traditional reforestation points (**¡Error! No se encuentra el origen de la referencia.**).*

After even light rain or fog collection events, the Cocoon system showed measurable increases in soil moisture levels, while traditional areas often showed a delayed or smaller moisture increase (Figure 221). This may occur due to the fact that the Coccon system serves as an element of rainwater catchment and infiltration.

However, for future projects, it would be useful to better define the installation point and method of the soil moisture sensor to properly evaluate the Cocoon system. If the sensor is placed in the gap left by the inner cylinder, where the plant is located, it primarily measures surface moisture and may not accurately capture the moisture of the surrounding soil. This could explain the lower average readings compared to the traditional system measurement points.

The most appropriate approach seems to be installing more than one soil moisture sensor. One could be placed as in the current project, while another would be beneficial if installed beneath the Cocoon container. Therefore, further action is needed to obtain more robust conclusions.

4.3.3.4.3 Autonomous Fluid Discharge Systems (AFDS) areas

These systems were designed to automatically discharge water when a certain level of captured water is reached through a low-flow source such as a spring or fog water. This system aims to improve the resilience of reforestation efforts under water stress conditions. During the project, the AFDS systems provided critical support to reforestation zones, especially during periods of high evapotranspiration and limited rainfall, through scheduled or automatic water discharges with minimum intervention. AFDS systems were integrated with natural water sources: fog water collectors and the underground spring reservoir.

Specifically, AFDS2 was connected in February 2022 to 3 T-FWC's, that in June 2023 were transformed in 3 VFWC's. By Q1 2023, after addressing the sensor issues, AFDS2 showed signs of recovery, with several automatic discharges improving moisture levels. In Q2 2023, the system performed a partial discharge in May, leading to moderate increases in soil moisture. A full discharge in July 2023 further helped stabilize soil moisture in AFDS2's area, especially during the dry summer months. This feature proved critical during Q3 2023, allowing AFDS2 to maintain relatively stable moisture levels despite low rainfall. In September 2023, after a rainfall of 13.8 mm, AFDS2 performed well, showing a marked improvement in moisture retention. A discharge in March 2024, supported by water from fog collectors, significantly boosted moisture levels during a dry period. This discharge, along with others in April 2024, demonstrated AFDS2's growing reliability. The fog water collectors allowed AFDS2 to maintain moisture levels even when rainfall was scarce.

Compared to non-irrigated, traditional and Cocoon areas, the AFDS 1, 2, and 3 zones maintained more stable soil moisture levels, with a coefficient of variation below 10% in 2022. In 2023, the AFDS 4 -5 and 6 zone also maintained a coefficient of variation below 10%, as it began operating with automatic irrigation that year. However, the irrigation zones AFDS 1 and 3 showed a deterioration in the coefficient of variation due to the discontinuation of scheduled irrigation).

Figure 222 Indicators of AFDS Soil Moisture Sensors during 2022 and 2023 in El Pinillo.

Compared to other AFDS systems, AFDS2 initially underperformed, but technical adjustments were made to ensure its proper operation. Notably, it is the only irrigation system that remains directly connected to a fog water source (three collectors with volumetric mesh), making it valuable to maintain its long-term operation for demonstration purposes.

4.3.3.4.4 Individual fog collectors

Individual collectors are used in certain reforestation areas to enhance soil moisture by directly collecting rain or fog water and channelling it to the plants. In general, these systems show a rapid response to rainfall events, with individual collectors maintaining higher moisture levels.

In October 2021, the first sensors were installed to monitor the individual collector zones in Area 1. The two reference sensors (_plant and _no_plant) provide insights into how much moisture is retained in the soil surrounding plants and in nearby areas without direct collectors. During 2022, some water collection events were observed, along with light rainfall that was not detected by the measurement point without a plant (Figure 223).

Area 2 was later equipped with individual collectors, with additional sensors activated in Q4 2022. Sensors (_plant_bis) were installed to observe the impact on plants in the second area. In Figure 224, the soil moisture peaks detected at the measurement point equipped with an individual collector can be observed.

Figure 223 Soil moisture content evolution in Area 1 of individual collectors during 2022. Water collection events are highlighted in circles.

Figure 224 Soil moisture content evolution in Area 2 of individual collectors during 2023 and 2024.

The Individual Collectors displayed the system was able to channel the rain and fog water effectively to the plants, increasing soil moisture rapidly. This effect was more pronounced during light rain and fog events. The comparison between the _plant and _no_plant sensors showed that plants with individual collectors maintain slightly more moisture compared to surrounding areas without collectors.

Compared to traditional reforestation techniques, the individual collectors showed better moisture retention, especially during periods of light rainfall. When compared to systems like Cocoon, the individual collectors had faster response to rain events. By concentrating water directly at the plant roots, they may improve plant survival.

4.3.3.4.5 Experimental Area

The experimental plot was established to test and compare different reforestation techniques, such as individual collectors and the use of biochar under the most similar soil and shading conditions, with the aim of identifying best practices for improving soil moisture retention and plant growth under adverse conditions.

Measurements in the experimental area began in Q3 2023, with the installation of sensors to monitor soil moisture. During this period, several irrigation events and precipitation occurrences were observed, helping to record variations in moisture levels at different reforestation sites. A notable event occurred in March 2024, where a sharp drop in the moisture reading from the traditional reforestation sensor was recorded. This abrupt change of measure occurs the 11th of March 2024 at 9:55 and it reduced the quantity of measured soil moisture in 0.9 m^3/m^3. For this reason, it is preferable to compare data of Traditional point before this event.

Figure 225 Abrupt change of measure that reduced the quantity of measured soil moisture in sensor of Traditional method.

In Figure 226, the evolution of soil moisture content at different measurement points in the experimental plot can be observed. Four scheduled irrigations are highlighted: on July 10, August 10 and 29, and October 13, 2023. The remaining events where an increase in soil moisture is detected correspond to rainfall or possible water capture from fog in the case of the Individual Collector.

For the scheduled irrigations, no significant differences are detected between the different reforestation systems. When considering only scheduled irrigations, the measurement point without Biochar appears to retain a slightly higher average soil moisture content than the one with a Biochar dose (0.3 m^3/m^3 vs. 0.28 m^3/m^3 from March 9 to September 2, 2023). During the same period, the measurement points for Cocoon and Individual Collector show similar average values, around 0.26 m^3/m^3.

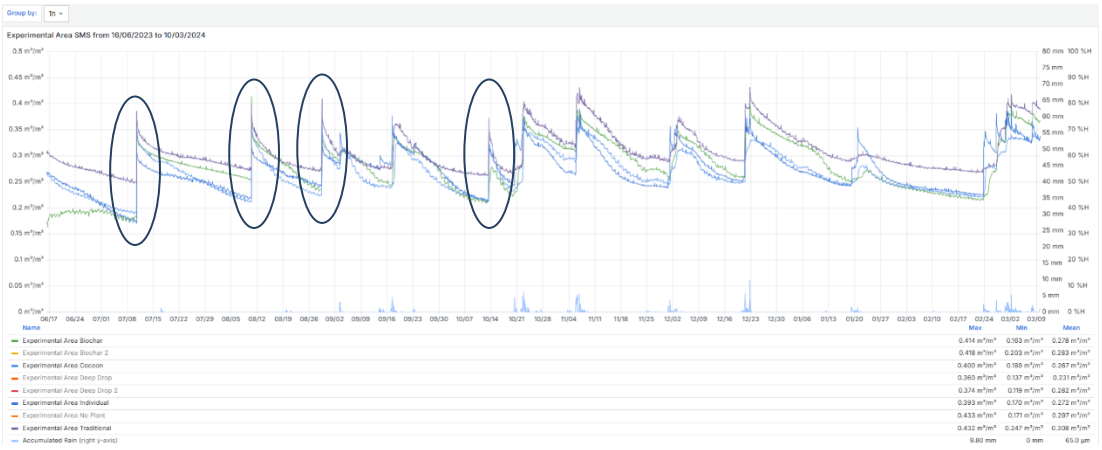


Figure 226 Soil moisture content evolution in Experimental Area from 16th of June 2023 to 10th of March 2024. Scheduled irrigations events highlighted.

On the other hand, when atmospheric precipitation events occur, even light ones, it is confirmed that the Individual Collector measurement point reacts the fastest and shows the highest peaks in soil moisture content. There are even cases of very light precipitation or probable fog water capture where the only measurement point that responds with an increase in soil moisture content is the Individual Collector (Figure 227).

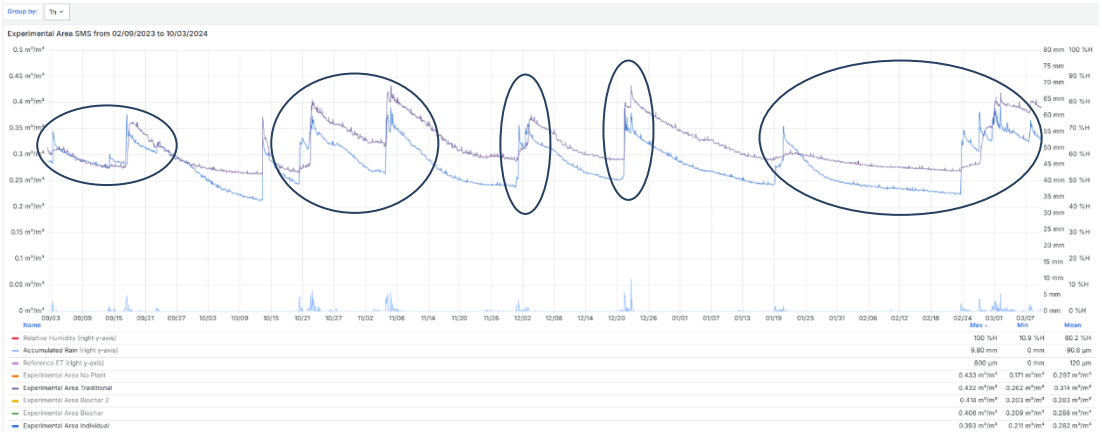


Figure 227 Traditional and Individual collector soil moisture content evolution in Experimental Area from 2nd of September 2023 to 10th of March 2024. Precipitation events with Individual collector reactions highlighted.

In Figure 228, the rapid increase in soil moisture content of the Individual Collector compared to the Cocoon is highlighted in greater detail during a light but persistent rainfall event. Figure 229 shows another event where the Individual Collector responds earlier than the other systems, increasing soil moisture content.

Finally, Figure 230 presents the evolution of soil moisture content at the Individual Collector measurement point compared to a point without planting during the period from March to May 2024. This confirms the good performance of the Individual Collector, even during very light precipitation events (or fog water capture), where the soil, in general, does not show an increase in moisture. It is also noteworthy that the Cocoon system can capture rainwater through its surface, which, in certain cases, may increase soil moisture content compared to non-planted areas. (Figure 231).

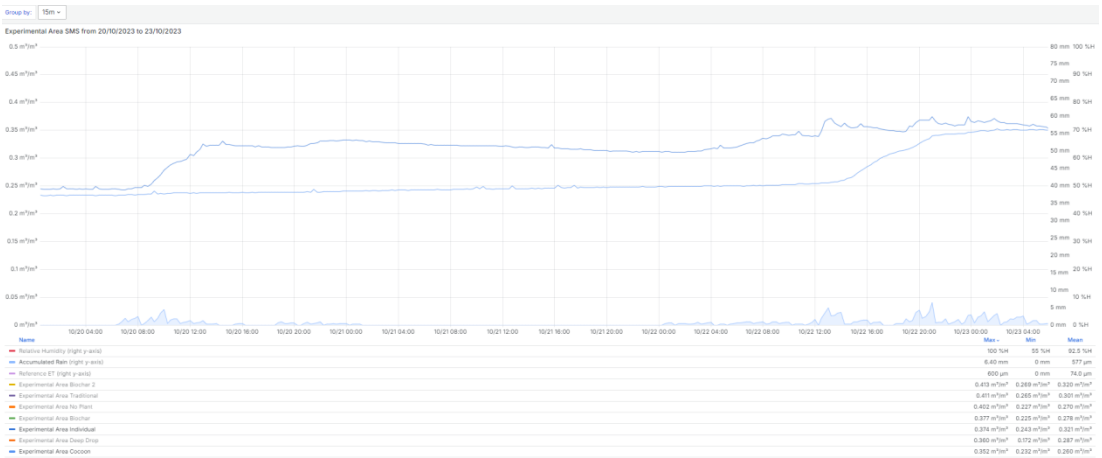


Figure 228 Individual collector reaction compare to Cocoon reaction (48 hours delayed).

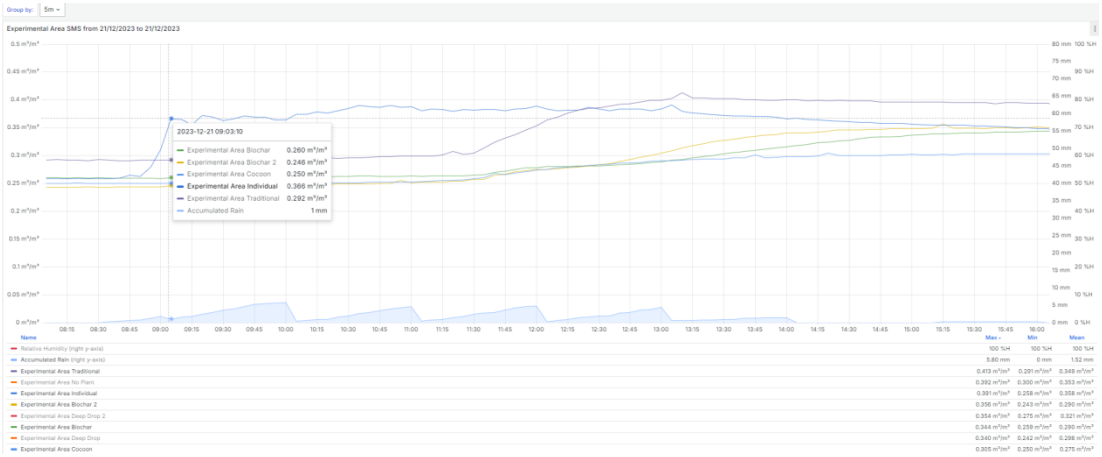


Figure 229 Individual collector reaction to weak rain compared with other reforestation systems that delay more than the 3 hours in detect soil moisture increasing.

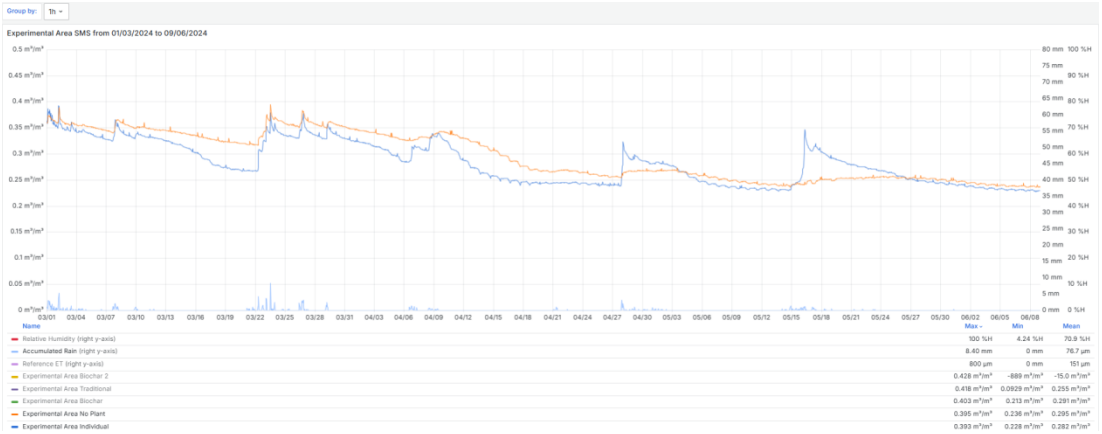


Figure 230 No plant and Individual collector soil moisture content evolution in Experimental Area from 1st of March to 9th of June 2024.

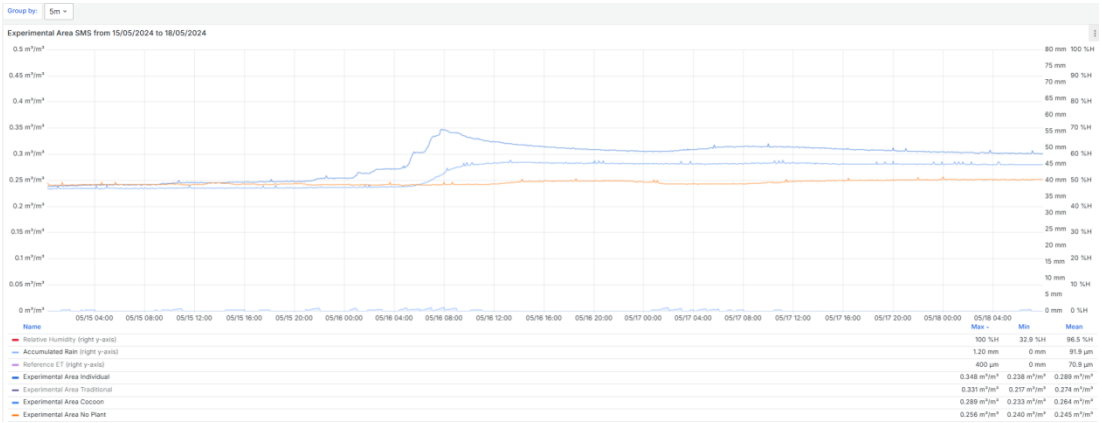


Figure 231 Individual collector, Cocoon and No plant reaction in a light and persistent rain event with fog.

Highlighted Conclusions:

Traditional Reforestation (without biochar): This measurement point recorded the highest soil moisture levels throughout all periods, particularly during irrigation events, reaching up to 0.423 m^3/m^3 in some instances.

Biochar and Biochar 2 Sensors: These points also performed well but retained slightly less moisture than the traditional reforestation point. For example, the Biochar sensor recorded a peak of 0.396 m^3/m^3.

Individual Collectors: Data suggest that this system responds more quickly to light rainfall and fog events, providing better growing conditions for plants during drought periods.

Cocoon: The Cocoon system exhibited moderate soil moisture retention compared to other systems, with a peak moisture content of 0.354 m^3/m^3. While its average moisture level (0.265 m^3/m^3) was similar to that of the Individual Collector, its response to rainfall events was **significantly slower. In this case, the soil moisture sensor was installed beneath the Cocoon.**

Deep Drop System: Designed to deliver water at greater depths, the Deep Drop system did not show significant changes in soil moisture measurements, as the water was applied beyond the detection range of the installed sensors. This suggests that sensor placement should be adjusted to better assess the impact of deep irrigation.

A longer monitoring period and an increased number of measurement points would help draw more robust conclusions.

4.3.4 Seedlings vigour and growth

Plantation monitoring is an activity that has begun from May 2022. Seedlings were measured as tagged with unique codes as identifiers, according to the monitoring protocol (A2: Environmental, social and economic indicators selection. Monitoring and evaluation protocol). The data recorded per tree is height, stem width, visual vigour, reforestation method and the species.

With no extra cost for the project, GESPLAN has developed an App to perform the monitoring. That will help the team file seedlings' reports electronically without the need for manual recording techniques. The database generated is uploaded to a server that will be revisited in consecutive years in order to update seedlings data.

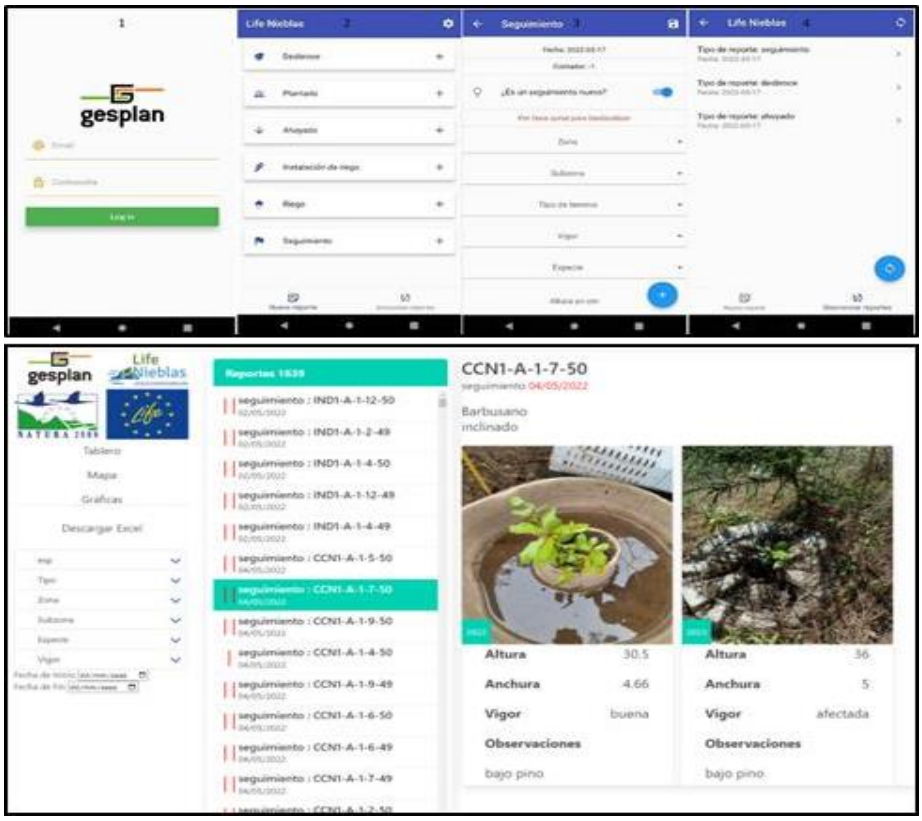


Figure 232. Image of the App interface developed for helping the data collection.

An additional advantage of this App is that technicians can geo-localise every sampled seedling which will allow them to easily find them in future visits, saving time and effort when undertaking field work. Another feature that will help technicians is the 'photo' section which provides a visual reference to support measurements taken at every monitoring run throughout

the years. Moreover, this App has been designed to also have a field work tracking system that will help field workers register their daily progress i.e. vegetation clearing, planting, ground conditioning, irrigation installation and watering.

An important effort for seedling monitoring has been done in El Pinillo plantation. More than 5000 seedlings have been labelled and monitored during the 2022-24 period, starting with 1877 in 2022, 2590 in 2023 and 681 in 2024 (Table 22). Despite the variation in soils between planting areas and meteorological conditions indifferent planting years this high number of seedlings allows to take robust conclusions on the effectiveness of the treatments. However, when analyzing the data at species level we can see that there is a low number of monitored seedlings per some treatments and planting years for which those results must be taken with caution.

Table 22 Number of monitored seedlings per year, species and planting system.

			2022		
	TR	**CCN**	**IFWC**	**AFDS**	**TOTAL**
Acebiño	13	14	2	66	95
Barbusano	51	78	88	59	276
Bencomia	29	52	65	38	184
Brezo	51	61	87	63	262
Cedro	10	6		0	16
Faya	51	65	89	49	254
Follao	45	27	66	46	184
Laurel	51	28	58	34	171
Madroño	27	6	4	19	56
Palo blanco	51	32	88	41	212
Peralillo	1	4	1		6
Tajinaste	51	4	66	40	161
TOTAL	431	377	614	455	1877
			2023		
	TR	**CCN**	**IFWC**	**AFDS**	**TOTAL**
Acebiño	5	2	2		9
Barbusano	125	46	225	85	481
Bencomia	69	34	81	36	220
Brezo	133	35	224	119	511
Cedro	25		7		32
Faya	116	31	221	151	519

Follao	24	7	60	27	118
Laurel	87	9	118	73	287
Madroño	15	4	9	13	41
Palo blanco	37	13	153	28	231
Peralillo	1	2	6	1	10
Tajinaste	22		64	40	126
Viñatigo	1	3		10	14
TOTAL	655	184	1168	583	2590

			2024		
	TR	**CCN**	**IFWC**	**AFDS**	**TOTAL**
Acebiño	2	2	0	0	4
Barbusano	7	37	0	62	106
Bencomia	7	21	0	18	46
Brezo	12	42	0	94	148
Cedro	0	1	0	0	1
Faya	8	30	0	124	162
Follao	11	17	0	19	47
Laurel	4	11	0	56	71
Madroño	4	0	0	8	12
Palo blanco	2	20	0	13	35
Peralillo	0	0	0	1	1
Tajinaste azul	13	5	0	24	42
Viñatigo	0	0	0	10	10
TOTAL	68	184	0	429	681

4.3.4.1 Physiological state: global results

Overall, the plantations established during the different campaigns (2021-22, 2022-23) showed positive results in terms of physiological state. Although mortality increased over time, most seedlings remained in good condition by 2023 and 2024— seedlings characterized by living leaves and active apexes. Notably, approximately 70% of the seedlings from the 2021-22 and 2022-23 campaigns were still in good health by 2024 (Figure 233).

Figure 233 Distribution of the seedlings according to their physiological state per planting campaign (2021-22, 2022-23, 2023-24) and year.

4.3.4.2 Physiological state per planting system

We observed a clear positive impact of the innovative planting systems— CCN, IFWC, and AFDS —compared to traditional methods (TR, Figure X). This was particularly evident in 2023, when the traditional method (TR) showed a ~60% mortality rate in the seedlings planted in the 2021-22 campaign and ~30% in the seedlings planted 2022-23, whereas all the innovative methods had much lower mortality rates. Notably, the AFDS method stood out with an exceptionally low mortality rate in 2023. In 2024 mortality rates stabilized but more than half of the seedlings were still in good condition (Figure 234).

Figure 234 Distribution of the seedlings according to their physiological state per planting method, planting campaign (2021-22, 2022-23, 2023-24) and year.

A closer analysis by species in the mid-term (seedlings planted in the 2021-22 campaign) shows that planting systems keep improving survival rates compared to TR (Figure 235).

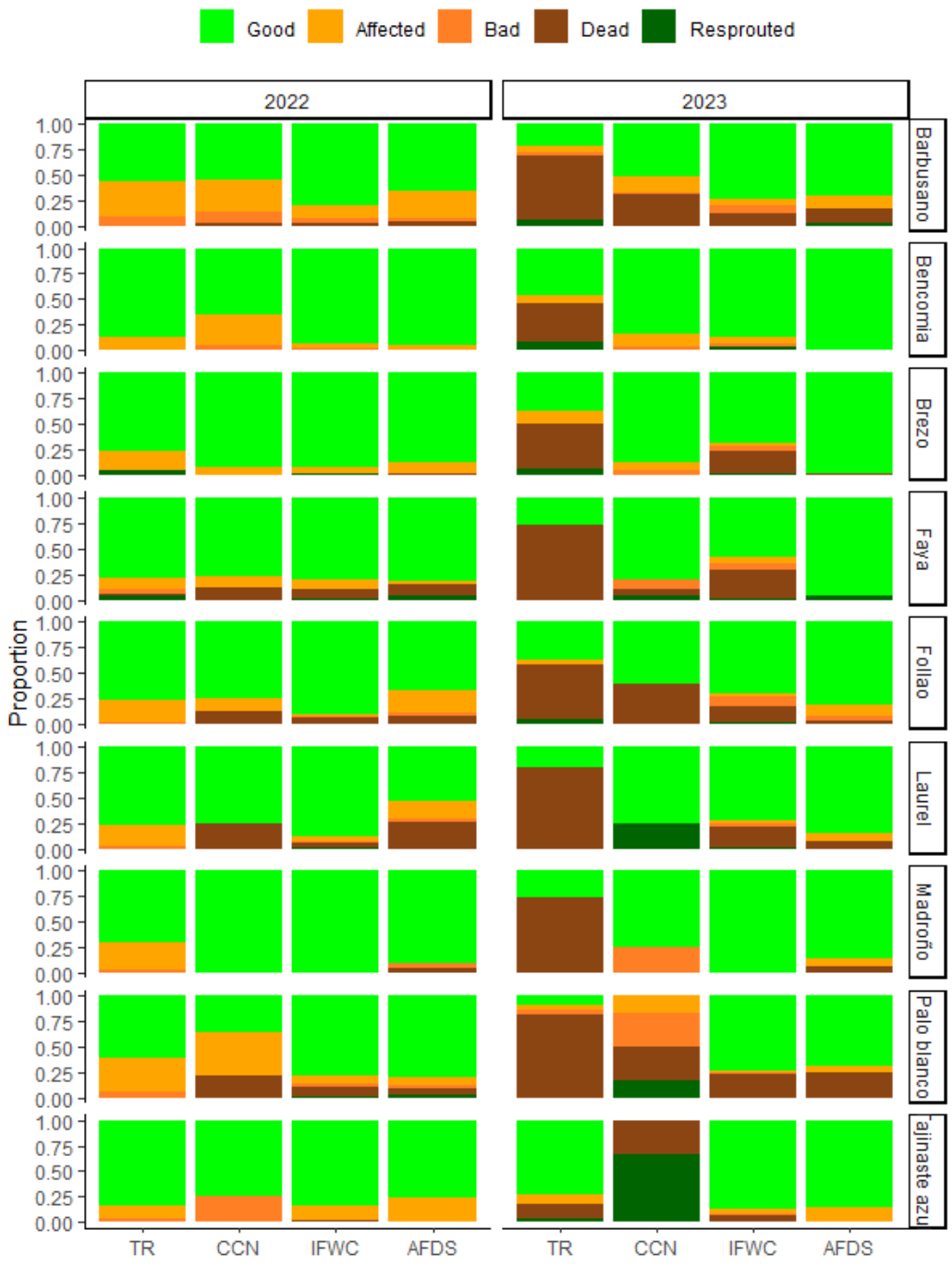


Figure 235 Distribution of the seedlings according to their physiological state per planting method, main species and year. Data from the 2021-22 campaign.

Despite the previous experiences in other sites of the island using Cocoon ecotechnology where this technology worked efficiently allowing the refilling of the bowls to improve the survival of the seedlings (Carabassa et al., 2021), the El Pinillo plantations followed a contrasting tendency. Due to the high fauna (snails, centipedes, slugs, rodents), Cocoons degrade faster, as shown by abundant holes and cracks only a few months after being installed (sometimes after the first month). This represents a high limitation for the use of this technology, since it prevents water refilling that was used in other sites that maximise seedlings survival. However, it must be considered that the conditions in this part of the island differs significantly in terms of climate and soil types: compared to Tifaracás (Carabassa et al., 2021), climate in El Pinillo is much more humid, favouring Cocoon degradation, and soils are much more organic, favouring fauna activity that at the same time affects Cocoon integrity (Figure 236).

Figure 236. Cocoon with the bowl degraded (left). Snail inside one Cocoon bowl (right). Snails seem to be one of the main promoters of Cocoon faster degradation.

Regarding the IFWC the innovation of putting a plastic net of around 1 m height to improve the water capture and reduce evapotranspiration (Figure 237) was compared to the traditional IFWC (Figure 238).

Figure 237 Innovative IFWC with 1 m high polyethylene net.

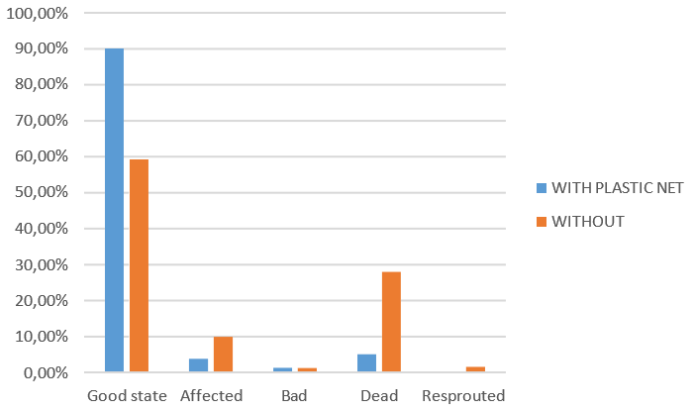


Figure 238 Physiological state of El Pinillo seedlings from the IFWC planting method (with and without plastic net).

4.3.4.3 Physiological state and growth per species

Regarding the physiological state of the seedlings per species (Figure 239) we demonstrated that the pioneer shrubs "bencomia" (Bencomia caudata), "brezo" (Erica arborea) and "tajinaste azul" (Echium callithyrsum), are the one working better, with most of the seedlings in good state. On the contrary, some other tree species are those facing greatest difficulties, with a mortality ~ 50% or greater in Acebiño, Peralillo, and Laurel (Laurus novocanariensis).

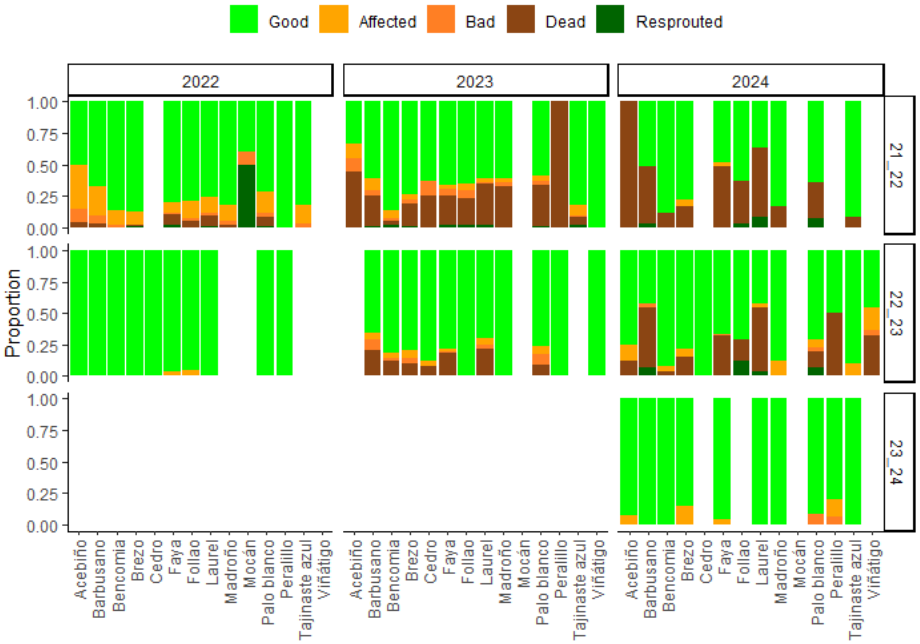


Figure 239 Distribution of the seedlings according to their physiological state per main species, planting campaigns (2021-22, 2022-23, 2023-24) and year.

In terms of growth rate, the IFWC planting system highlighted in height growth rate (Figure 240), with higher growth rates than TR. In addition, species such as Bencomia and Tajinaste azul stood out for their height growth rates, which surpassed those of other species, particularly under the IFWC (Figure 241).

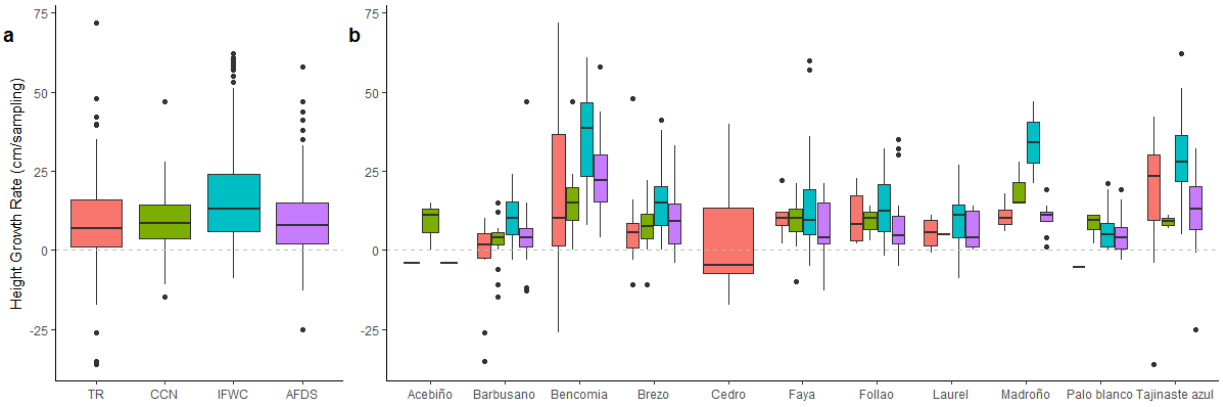


Figure 240. Height growth rate per (a) planting system, and (b) species and planting system in el Pinillo.

Width growth rate appeared to decrease under CCN compared to TR system (Figure XX). However, we believe that the observed decrease in the width in CCN may actually be an artifact of the height at which the measurements were taken, and also to the fact that seedlings planted with CCN were planted deeper than the others which provoques this lower diameter. Bencomia and Tajinaste azul stood out also for their width growth rates.

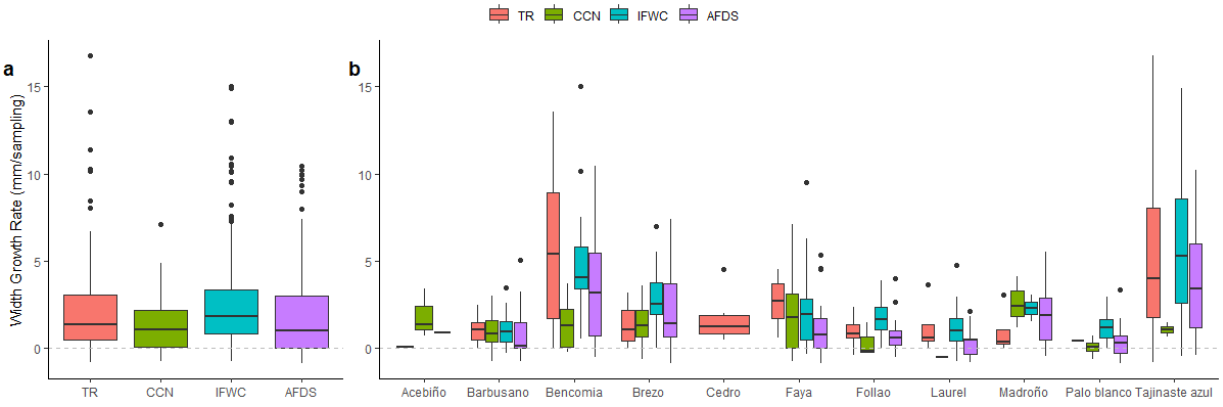


Figure 241 Width growth rate per (a) planting system, and (b) species and planting system in el Pinillo.

4.3.4.4 Physiological state per irrigation regime

According to the planting strategy, three different irrigation strategies have been established:

Strategy 1: IFWC and CCN seedlings only received the settlement irrigation (10L).

Strategy 2: all seedlings received the settlement irrigation (10L). Plants with IND and CCN treatments received 10L watering per plant, twice a month from July to September (both inclusive). Cocoon system received an additional one tank refill during summer.

Strategy 3: all plants received the settlement irrigation (10L) and 10L watering per plant, twice a month from May to September (both inclusive). Cocoon system treatment received an additional one tank refill during summer. All traditional reforestation plantations received this watering regime.

It must be considered that part of AFDS seedlings received controlled discharges according to the respective irrigation strategies and were not connected to fog water collectors (Strategy 2 and Strategy 3). The rest of the AFDS were connected to the fog water collectors and only received the settlement irrigation (10L) and two emergency waterings (one in July and one in August, 10L per plant).

For the Cocoon system the normal functioning of the device, without refilling, seems to be efficient enough and no big differences are seen between irrigation strategies (Figure 243). Taking this in consideration it seems not necessary to refill the Cocoons during summer. However, a bigger proportion of healthy seedlings are present in refilled Cocoons (68% vs 55%) and a higher survival ratio (89% vs 85%). In any case, it should be considered that refilling Cocoons seems not to be a good option since the vast majority present holes and cracks few weeks after being installed at the field (see Results per planting system section).

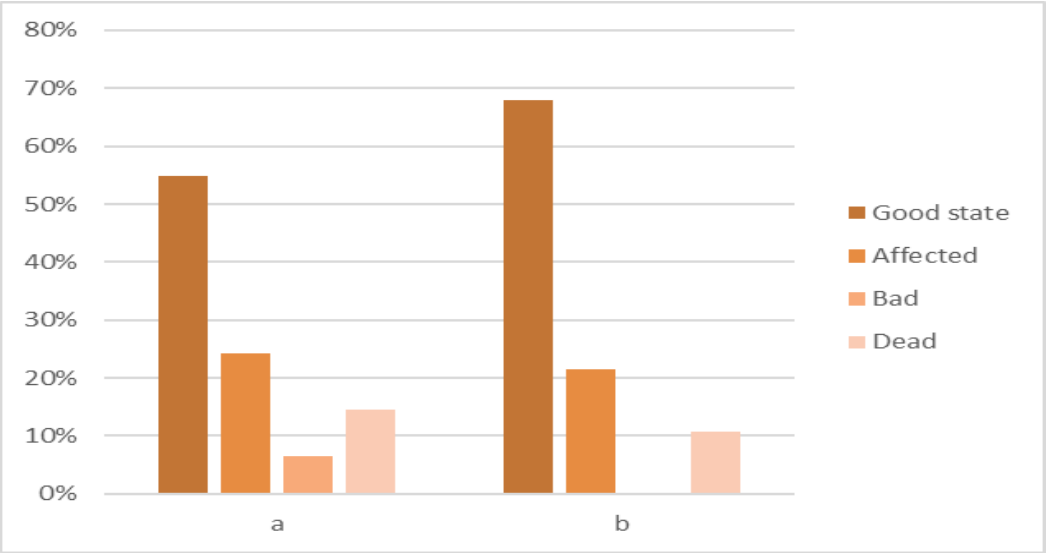


Figure 242. Distribution of seedlings planted in 2021-22 per watering regime according to their physiological state in Cocoon treatments (CCN), in February 2023. Watering regimes: without additional watering (a); refilling the Cocoon in august (b).

Regarding individual fog water collectors, as we saw with the Cocoon treatment no differences between irrigation regimes have been observed (Figure 244). However, this data must be taken with caution since the distribution of planted species is different between the different irrigation regimes areas and the soils between those areas are also different (see Preliminary report about plantations monitoring in El Pinillo (Gran Canaria). November 2022).

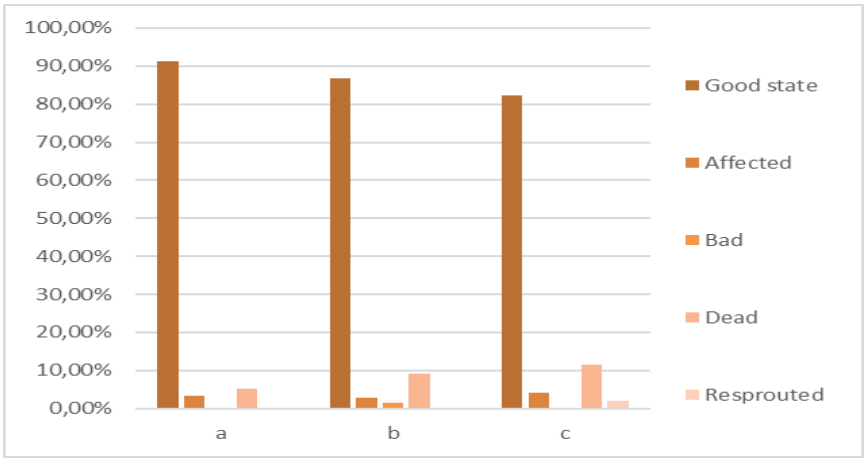


Figure 243. Distribution of seedlings planted in 2021-22 per watering regime according to their physiological state in individual fog water captors (IFWC) in February 2023. Watering regimes: without additional watering (a); watering from July to September (b); watering from May to September (b).

Regarding the AFDS treatments, we see a similar tendency than CCN and IFWC treatments, where we cannot see clear differences between irrigation regimes (Figure 245). However, as it happened with IFWC treatments, we must take this data with caution since the distribution of the species for the different irrigation regimes is different (Figure 245).

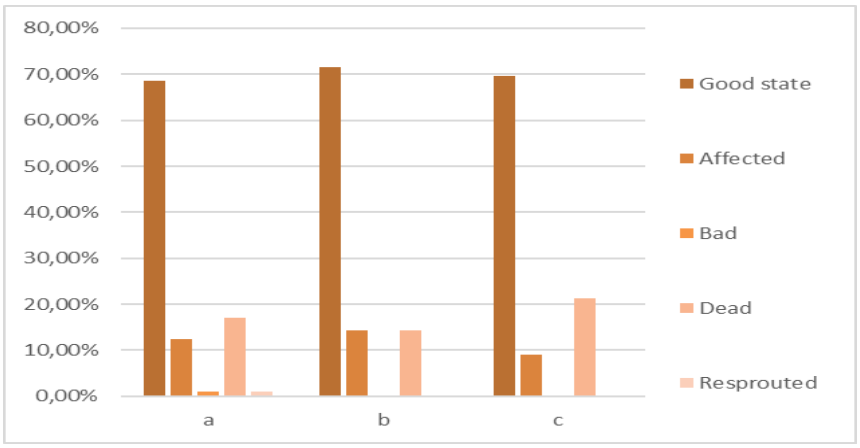


Figure 244. Distribution of monitored seedlings per watering regime according to their physiological state in autonomous discharge system plantations (AFDS). Watering regimes: watering from July to September (a); irrigation according to water fog collectors regime+two emergency waterings (b); watering from May to September (c).

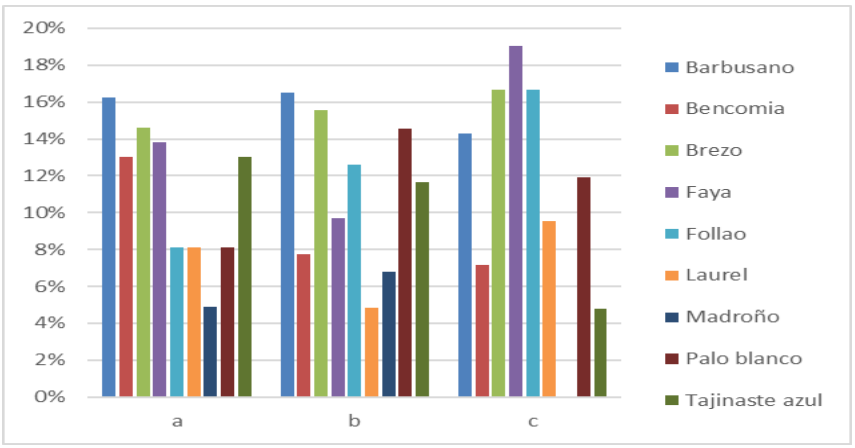


Figure 245. Distribution of monitored seedlings per watering regime and species in autonomous discharge system plantations (AFDS). Watering regimes: watering from July to September (a); irrigation according to water fog collectors regime+two emergency waterings (b); watering from May to September (c).

4.3.4.5 Seedlings vigour and growth in the experimental zone

The plantations conducted in the experimental zone yielded good results regarding the physiological state of the seedlings. In particular, the innovative planting systems CCN, IFWC,

and deep drop (DD) increased the survival rates compared to the TR method. The addition of biochar at a low rate (BC_1) did improve survival rates compared to the control of the biochar area (BC_0). However, at the highest dose (2kg, BC_2) the mortality rate above the control (BC_0) or the TR method (Figure 246). Differences between TR and BC0 treatments indicate heterogeneity in the planting area since those treatments should be equivalent for which the conclusions must be taken with caution.

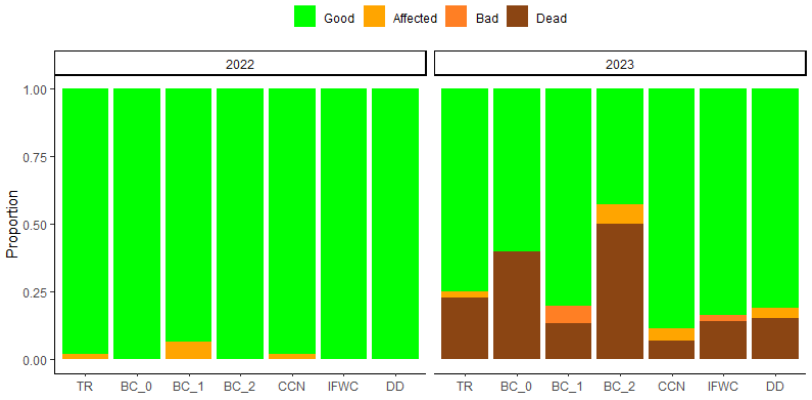


Figure 246. Distribution of the seedlings according to their physiological state along time per planting system in the experimental zone. Traditional (TR), biochar (1 kg: BC_1, and 2 kg: BC_2), cocoon (CCN), and individual fog water collectors (IFWC)

Similarly, the innovative planting systems CCN, IFWC, and DD resulted in greater height and width compared to the TR method (Figure 247). Notably, the negative effect of CCN on width growth rate observed in the El Pinillo area is not supported by the absolute width measurements in the experimental zone. This discrepancy reinforces the hypothesis that the negative effect observed in the plantations monitoring data may be an artifact of the measuring height in CCN planting systems.

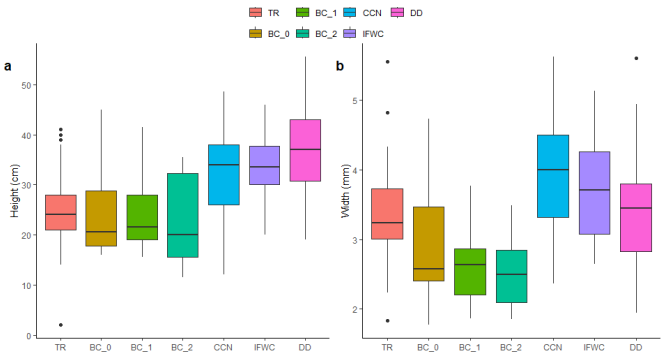


Figure 247. (a) Height and (width) of the planted seedlings per planting system in the experimental zone. Traditional (TR), biochar (1 kg: BC_1, and 2 kg: BC_2), cocoon (CCN), and individual fog water collectors (IFWC)

In addition to the regular measures of height and weight done throughout the whole experimental period, tree seedlings were excavated 2 years after planting to determine their shoot and root length and weight among the different treatments. Three individuals per treatment of *Morella faya* were sampled in the experimental area of el Pinillo, and three individuals of *Q. pyrenaica*, *Q. robur* and *Q. suber* respectively were sampled in Vouzela. Differences among treatments were tested using linear models, with the traditional (TR) treatment as the reference level.

Overall, significant differences were not found on length or weight of shoot, roots and total plant when comparing TR to the innovative planting systems, probably due to the limited number of replicates (n=3) and high variability. However, we did detect some relevant tendencies that are promising. IFWC planting systems highlighted as the most effective in promoting plant growth. IFWC consistently increased shoot, root and plant total length compared to TR (Figure 248). In terms of weight, TR exhibited strong variability, and this probably made differences among treatments less evident compared to those observed in length. However, IFWC again highlighted among the planting systems, with higher shoot, root and total plant weight than cocoon (CCN) and the addition of biochar (BC_1 and 2, Figure 249). On the other hand, CCN slightly increased shoot length compared to TR (Figure 248) but not root or total length, and either improved weight variables (Figure 249).

In contrast, we did not detect a beneficial effect on plant growth related to the addition of biochar (Figure 248-Figure 249). Similarly, the addition of biochar did not improve vigor (Figure 246), height or width (Figure 247) compared to the control (BC_0) in the same particular area (zone BC). Interestingly, this suggests that, at least in this context, water management was more beneficial for plant growth than the addition of an organic amendment.

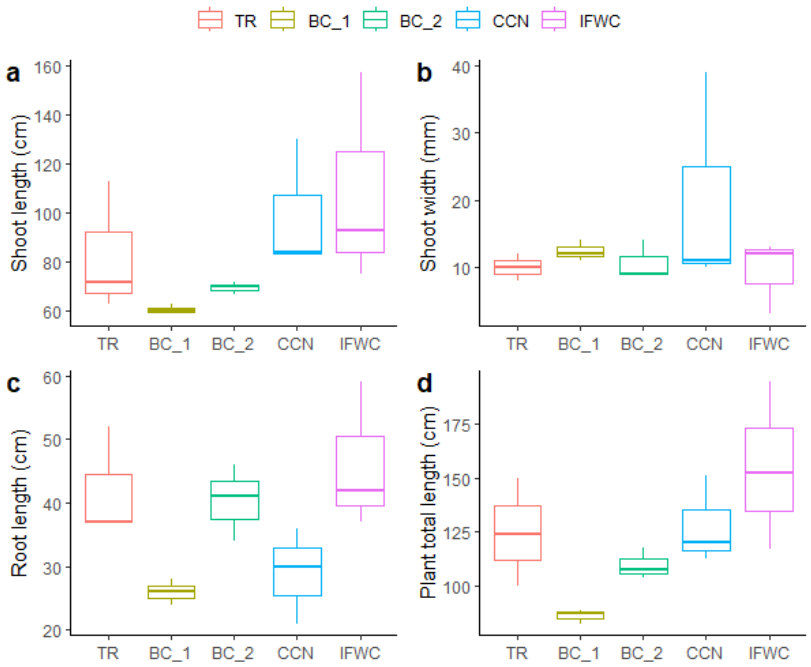


Figure 248 (a) Shoot length, (b) shoot width, (c) root length, and (d) plant total length per planting system: traditional (TR), biochar (1 kg: BC_1, and 2 kg: BC_2), cocoon (CCN), and individual fog water collectors (IFWC). Boxplots show the interquartile range (IQR: Q1-Q3), whiskers extend to the farthest data points within 1.5 times that IQR, and dots display outliers outside this range.

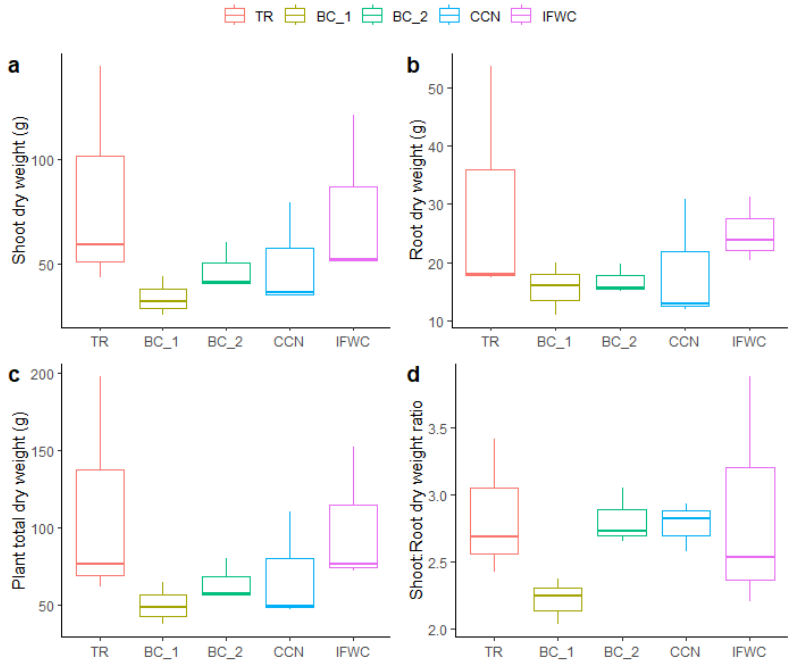


Figure 249 (a) Shoot dry weight, (b) root dry weight, (c) plant total dry weight, and (d) shoot: root dry weight ratio per treatment. Traditional (TR), biochar (1 kg: BC_1, and 2 kg: BC_2), cocoon (CCN), and individual fog water collectors (IFWC). Boxplots show the interquartile range (IQR: Q1-Q3), whiskers extend to the farthest data points within 1.5 times that IQR, and dots display outliers outside this range.

Traditional (TR)

Biochar (BC)

Cocoon (CCN)

Individual fog water collectors (IFWC)

Figure 250 Root systems of unearthed Mirica faya seedlings, 2 years after planting.

4.3.5 Passive restoration

Similarly to Vouzela and Cerregal, the passive restoration in el Pinillo was followed using Sentinel-2 satellite images (10 m/pixel) from years 2018, 2020, 2023 and 2024. Satellite images showed how shrubs and tree cover decreased from 2018 to 2020, after the forest fire of October 2019. Moreover, bare soil drastically increased form a 1.6 to 14.3%, especially in the north of the study area (Figure 251 and Table 23). During the first years after the fire, an increase in herbaceous vegetation could be observed, despite in the last year this herbaceous vegetation has been replaced by shrubs like *Adenocarpus foliolosus,* a typical pioneer specie that dominates in the first stadiums after forest fire. Notably, by 2024 tree and shrub cover was even higher than before the fire event, as well as bare soil was minimum (0.1%).

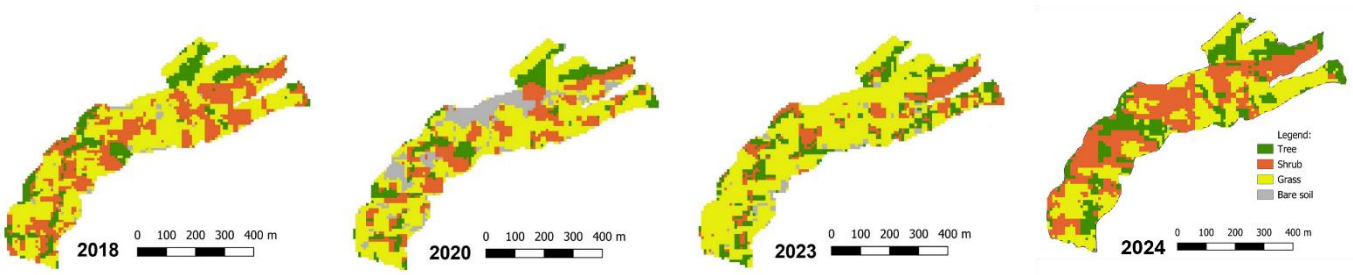


Figure 251 Land cover classification Sentinel-2 (10m/pixel) images from 2018-2024.

Table 23 Land covers percentages according to the classification of Sentinel-2 images from 2018-2024.

Year	Tree (m^2)	Shrub (m^2)	Grass (m^2)	Bare soil (m^2)	TOTAL	Tree (%)	Shrubs (%)	Grass (%)	Bare soil (%)
2018	43093	91869	145227	4666	284855	15.1	32.3	51.0	1.6
2020	40294	65657	138101	40803	284855	14.1	23.0	48.5	14.3
2023	48522	47674	180006	8653	284855	17.0	16.7	63.2	3.0
2024	63586	100556	99530	257	263929	24.1	38.1	37.7	0.1

Several inventories were realized to follow species abundance along the different areas and planting systems, see and overview of the inventories areas in in Figure 252.

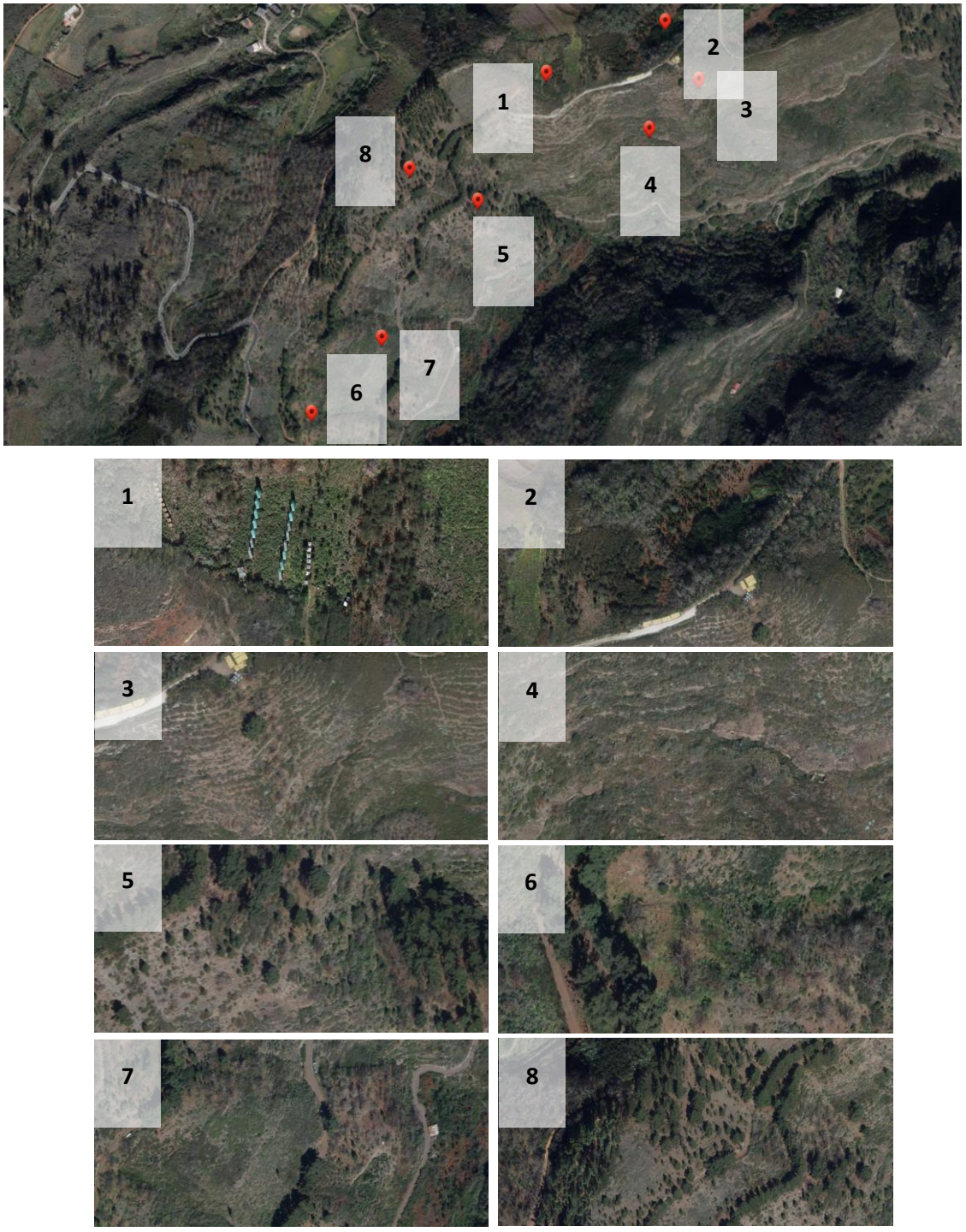


Figure 252. Overview of the location of plant species inventories. Source: https://catalogo.idecanarias.es, accessed on January 2024. Ortofoto flight dates 11/02/2024-20/03/2024. (1) TR1, (2) IWFC1, (3) AFDS1, IND1, (4) AFDS1, (5) CCN1-2, (6) IWFC2, (7) AFDS2, IND2, (8) CCN1-2.

Table 24 Plant species abundance in el Pinillo planting area in February 2025. Species abundance is categorized as follows: 1, testimonial; 2, low frequency (<25% soil cover); 3, high frequency (25%-75% soil cover); and 4, dominant (>75% soil cover).

SPECIES	Sub-site 1 TR1	Sub-site 2 IWFC1	Sub-site 3 AFDS1	Sub-site 4 AFDS1	Sub-site 5 CCN1, CCN2	Sub-site 6 IWFC2	Sub-site 7 AFDS2, IWFC2	Sub-site 8 CCN1, CCN2
Adenocarpus foliolosus **(codeso)**	2	2	4	2	2	1	1	2
Agave americana **(pitera)**	1			1			1	
Artemisia thuscula **(incienso canario)**							2	
Arundo donax **(caña común)**	1			1				
Carduus tenuiflorus **(cardo)**	3	1	2	4	1		2	1
Castanea sativa **(castaño)**					1	1	1	2
Oxalis pes-caprae **(trebolina)**								
Pinus canariensis **(pino canario)**					2	2	1	3
Pteridium aquilinum **(helecho común)**								
Pteridium aquilinum **(helecho común)**		4						
Rubus ulmifolius **(zarza)**	2			2		1	1	1
Sonchus acaulis **(cerraja o lechuga)**	4	1	1	2	1	1	1	
Ulmus minor **(olmo)**			1					

Conclusions and final remarks

Continuous monitoring and evaluation of innovative planting systems in El Pinillo (Gran Canaria, Spain), Carregal do Sal, and Vouzela (Viseu, Portugal) provided valuable insights into their effectiveness for ecosystem reforestation. Overall, the systems significantly improved seedling survival rates. Notably, Cocoon (CCN) stood out in both Portuguese sites, achieving the highest survival and growth rates, with approximately 75% and 80% of seedlings in good condition two years after planting in Carregal do Sal and Vouzela, respectively. However, Cocoon system presents some limitations in El Pinillo soils due to an abundant soil fauna activity that damages the bowl very fast resulting in the loss of water. In any case, Cocoon system also presents good results in El Pinillo, always better than traditional systems.

The Individual Fog Water Collectors (IFWC) performed the best in El Pinillo, with survival rates reaching ~75% even three years after planting. Notably, the addition of a polyethylene net significantly enhanced the system's water capture efficiency, minimized evaporation, and enhanced plant growth. This innovation makes the IFWC system particularly well-suited for remote areas where water availability is limited, and irrigation is impractical. However, the performance of IFWC was not so good in Portuguese plantations. This could be due to the fact that fog conditions are less favorable for which the quantity of water captured by the collectors is reduced. Also, it must be considered that in most of the IFWC no plastic net was installed, that could significantly reduce the capacity of the collector to retain water. In this sense, the use of IFWC with plastic net is strongly recommended in future uses of this device in order to improve its functionality.

In general, we observe that passive restoration is working well in El Pinillo and Vouzela, despite plantations and vegetation management is needed in order to guide it to the targeted habitats. In El Pinillo, successional shrub Adenocarpus foliolosus dominates in almost all areas, despite also some ruderal species like Carduus tenuiflorus and Sonchus acaulis could be observed. In Vouzela, passive restoration is strongly conditioned by the performance of invasive species, mainly eucalyptus (Eucalyptus globulus) and acacia (Acacia melanoxylon, Acacia caudata). In this area management strategies played a crucial role in ecosystem recovery. Repeated clearings effectively reduced invasive species, allowing native vegetation to establish and thrive, despite this could be reinforced with other practices like grazing, with proper protection of seedlings to prevent damage. In this sense, areas with IFWC could be available for testing this solution in Vouzela. Additionally, the incorporation of plant debris served as a valuable organic matter

input, enhancing soil fertility and nutrient cycling, further supporting long-term ecosystem restoration.

In Carregal do Sal the wildfire that affected the area in September 2024 burned all the plantations and the accompanying vegetation. The fire was of high severity in some of the areas despite in others some patches of unaffected vegetation remain. Specific actions for the restoration of native vegetation should be taken in the context of Post-LIFE plan, in the line of those carried out during the implementation of this project. In this sense, the use of Cocoons and IFWC with plastic net is recommended, avoiding the use of IFWC without net. Despite using those techniques, the adoption of emergency waterings during the first summer must be planned in order to avoid high mortality ratios in case of severe drought.

Despite those problems in Carregal do Sal, making a global analysis of the habitat restoration actions carried out in the project, we can conclude that considering the extremely dry conditions during the project years and the challenges that had to be faced, the results obtained from the testing of the innovative planting techniques are remarkably successful and provide practical guidelines for future reforestation efforts.

Acknowledgements

Cabildo de Gran Canaria and Heredad de Aguas de Arucas y Firgas for their continuous support during field visits and research on this action.

Supplementary materials

DOCUMENTO DE ACOMPAÑAMIENTO
COMPOST JUAN GRANDE

TIPO DE FERTILIZANTE:

Nombre comercial: COMPOST JUAN GRANDE
Nº Registro: F0003580/2020
Tipo producto: ENMIENDA ORGÁNICA COMPOST

RIQUEZAS GARANTIZADAS:

NITRÓGENO (N) TOTAL	1,7 – 2,3
NITRÓGENO (N) ORGÁNICO	1,7 – 2,0
CARBONO (C) ORGÁNICO	30,0 – 35,0
MATERIA ORGÁNICA TOTAL	60,0 – 80,0

CONTENIDO Y CARÁCTERÍSTICAS:

RELACIÓN C/N	< 20
HUMEDAD MÁXIMA	40%
EL 90% DE LAS PARTÍCULAS PASAN POR LA MALLA DE	25 mm
ÁCIDOS HÚMICOS	2,5 – 5,0
pH	6,0 – 8,0
CONDUCTIVIDAD ELÉCTRICA (dS)	5,0 – 11,0

- CONTENIDO EN METALES PESADOS INFERIOR A LOS LÍMITES AUTORIZADOS PARA LA CLASIFICACIÓN TIPO "A" (ANEXO V DEL R.D. 506/2013)
- LOS MÉTODOS DE ANÁLISIS EMPLEADOS SON LOS ESTABLECIDOS EN EL ANEXO VI DEL R.D. 506/2013
- CONDICIONES DE ENTREGA: A GRANEL

RECOMENDACIONES:

P102 – MANTENER FUERA DEL ALCANCE DE LOS NIÑOS
P270 – NO COMER, BEBER, NI FUMAR DURANTE SU UTILIZACIÓN
Para su aplicación al suelo en agricultura, se deberá seguir los códigos de buenas prácticas agrarias.
No se recomienda el uso para cervas de ganado

CANTIDAD: _____ kg

LOTE: _____

FABRICANTE:

U.T.E. COMPLEJO AMBIENTAL JUAN GRANDE (ECOPARQUE G.C. SUR)

DIRECCIÓN:

CARRETERA GC-502, km 6,5 (C.P. 35107) – SAN BARTOLOMÉ DE TIRAJANA – GRAN CANARIA – LAS PALMAS – ESPAÑA

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U-76334675

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CORREO ELECTRÓNICO:

MATERIAS PRIMAS:

- RESIDUOS BIODEGRADABLES (RESTOS VEGETALES 20 02 01) 70-80%
- MATERIALES INADECUADOS PARA EL CONSUMO O LA ELABORACIÓN (02 06 01) 5-10%
- LODOS DE DIGESTIÓN DEL TRATAMIENTO AERÓBICO DE RESIDUOS MUNICIPALES (19 08 05) 10-20%

El COMPOST JUAN GRANDE es una enmienda orgánica obtenida mediante la fermentación aeróbica de residuos de naturaleza orgánica. Tiene el aspecto de una "tierra" muy suelta de tacto esponjoso y ligeramente húmedo, color pardo-negruzco y de olor neutro. Con una temperatura estable. Entre sus virtudes destacan la ausencia de impurezas (vidrio, plásticos), patógenos y semillas de malas hierbas. Mejora la textura y estructura de los suelos, la actividad microbiana y el crecimiento de las plantas. Una ENMIENDA DE SUELO es un producto fertilizante que añadido a la tierra de cultivo mejora sus propiedades físicas, químicas y biológicas.

ALMACENAR EN LUGAR SECO Y VENTILADO.

INFORMACIÓN DEL TRANSPORTE

FECHA:

ORIGEN: PLANTA DE COMPOSTAJE – ECOPARQUE G.C. SUR

EXPEDIDOR: U.T.E. COMPLEJO AMBIENTAL JUAN GRANDE – U-76334675

CLIENTE/DESTINATARIO(*):

DESTINO (*Marcar la casilla que corresponda): ☐ Finca agrícola (Referencia catastral: _____) ☐ Jardinería pública (Municipio: _____) ☐ Vivero ☐ Otros: _____

CLIENTE/DESTINATARIO (*):

TRANSPORTISTA (*):

EXPEDIDOR	CLIENTE / DESTINATARIO	TRANSPORTISTA

(*) A cumplimentar por el cliente/destinatario antes de la retirada

Figure S1 Analytical characterization of the compost added in El Pinillo plantations.

Figure S2 Pictures from the seedlings processing for the shoot/root measures in Vouzela.

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